Delegating to the Machine: How People Hand Work to AI, and Whether Equity Markets Priced It

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Table of contents

Abstract **4**

Executive brief **4**

1 Introduction **6**

2 Related literature **9**
2.1 Tasks and exposure measurement . . . 9
2.2 Generative AI at work . . . 10
2.3 Algorithm aversion, appreciation and trust calibration . . . 11
2.4 Asset pricing of technology exposure and text-based measurement . . . 11
2.5 The Anthropic Economic Index . . . 12

3 Data and measurement **12**
3.1 The AEI releases . . . 12
3.2 Measurement regimes . . . 13
3.3 The harmonised panel . . . 14
3.4 Tasks, occupations and the O*NET-SOC harmonisation . . . 14
3.5 OEWS weights and the timing of the inputs . . . 14
3.6 The industry crosswalk . . . 15
3.7 The exposure indices . . . 15
3.8 The zero-assignment assumption . . . 16
3.9 The "could" benchmarks . . . 16
3.10 Market data . . . 17
3.11 Filings for the talk index . . . 17

4 Pre-registration and methods **17**
4.1 The protocol . . . 17
4.2 RQ1: behaviour . . . 18
4.3 RQ2: the bridge . . . 20
4.4 RQ3: markets . . . 20
4.5 H6: talk and use . . . 22
4.6 Robustness reporting rules . . . 22
4.7 Departures from the spec . . . 22

5 Behaviour: who delegates, for what, and is it changing? (RQ1) 23
5.1 The anatomy of collaboration . . . 25
5.2 H1: education, wages and the automation share . . . 27
5.3 H2: measured within-task automation across releases . . . 27
5.4 H3: validation in high-error-cost occupations . . . 29
5.5 H4: geography . . . 29
5.6 The RQ1 decisions together . . . 30

6 Bridge: industry exposure, "could" versus "does" (RQ2) 30
6.1 Three industry indices . . . 31
6.2 Could versus does . . . 31
6.3 What the correlation shares, and what it assumes . . . 32
6.4 Where "could" and "does" part . . . 33
6.5 Robustness rows for RQ2 . . . 33

7 What the market evidence can and cannot yet say (RQ3 and H6) 34
7.1 The confirmatory test of H5 and H5b . . . 34
7.2 The fixed wording . . . 35
7.3 Intervals, the R13 MDE and the secondary null . . . 35
7.4 Secondary inference . . . 36
7.5 The influence of Softw . . . 36
7.6 Exploratory market analyses . . . 36
7.7 Talk versus use (H6, exploratory) . . . 38
7.8 What a future window would need . . . 39

8 Robustness 39
8.1 What the rows vary . . . 40
8.2 Reporting rules . . . 41
8.3 RQ1 . . . 41
8.4 Market rows . . . 42
8.5 H6 . . . 42
8.6 Where the rows are . . . 42

9 Limitations 42

10 Conclusion 44

References 45

Statements 49

A Pre-registration 50
A.1 Signature (PAP §0.1) . . . 50
A.2 Decision register (PAP §0.3) . . . 50
A.3 Code binding (PAP §9.2) . . . 52

B Deviations 52
B.1 Summary of the PAP §11 entries . . . 52
B.2 PAP §11, verbatim . . . 57

B.3 PAP §10: corrections to the design spec, verbatim . . . 87

C Disclosures **88**
C.1 Before the freeze (PAP §0.4) . . . 88
C.2 After the freeze: facts recorded in the Plan 2–4 ledgers . . . 89

D Robustness tables **92**
D.1 Robustness appendix: summary (PAP §8.2) . . . 93
D.2 Implementation choices where the PAP is silent . . . 124
D.3 R1: robustness row (PAP §8.2 R1) . . . 132
D.4 R2: robustness row (PAP §8.2 R2) . . . 135
D.5 R3: robustness row (PAP §8.2 R3) . . . 138
D.6 R4: robustness row (PAP §8.2 R4) . . . 141
D.7 R5: robustness row (PAP §8.2 R5) . . . 170
D.8 R6: robustness row (PAP §8.2 R6) . . . 171
D.9 R7: robustness row (PAP §8.2 R7) . . . 172
D.10 R8: robustness row (PAP §8.2 R8) . . . 173
D.11 R9: robustness row (PAP §8.2 R9) . . . 204
D.12 R10: robustness row (PAP §8.2 R10) . . . 228
D.13 R11: robustness row (PAP §8.2 R11) . . . 232
D.14 R14: robustness row (PAP §8.2 R14) . . . 235
D.15 R15: robustness row (PAP §8.2 R15) . . . 238
D.16 R16: robustness row (PAP §8.2 R16) . . . 241
D.17 R17: robustness row (PAP §8.2 R17) . . . 296
D.18 R18: robustness row (PAP §8.2 R18) . . . 298
D.19 R19: robustness row (PAP §8.2 R19) . . . 299
D.20 R20: robustness row (PAP §8.2 R20) . . . 300
D.21 R21: robustness row (PAP §8.2 R21) . . . 303
D.22 R22: robustness row (PAP §8.2 R22) . . . 304
D.23 R23: robustness row (PAP §8.2 R23) . . . 315
D.24 R24: robustness row (PAP §8.2 R24) . . . 318
D.25 R25: robustness row (PAP §8.2 R25) . . . 321
D.26 R26: robustness row (PAP §8.2 R26) . . . 323
D.27 R27: robustness row (PAP §8.2 R27) . . . 326
D.28 R28: robustness row (PAP §8.2 R28) . . . 327
D.29 R29: robustness row (PAP §8.2 R29) . . . 331
D.30 R30: robustness row (PAP §8.2 R30) . . . 334
D.31 R31: robustness row (PAP §8.2 R31) . . . 337
D.32 R32: robustness row (PAP §8.2 R32) . . . 338
D.33 R33: robustness row (PAP §8.2 R33) . . . 339
D.34 R34: robustness row (PAP §8.2 R34) . . . 342
D.35 R35: robustness row (PAP §8.2 R35) . . . 343
D.36 R36: robustness row (PAP §8.2 R36) . . . 344
D.37 R37: robustness row (PAP §8.2 R37) . . . 347
D.38 R38: robustness row (PAP §8.2 R38) . . . 347

E Data and replication **350**
E.1 Raw data (`data/raw/MANIFEST.json`) . . . 350

E.2 Environment . . . 357
E.3 Build targets (`Makefile`) . . . 357
E.4 Seed . . . 360
E.5 Replication . . . 360

F Supplementary tables and figures **360**
F.1 Data and measurement . . . 361
F.2 Behaviour (RQ1) . . . 364
F.3 Bridge (RQ2) . . . 370
F.4 Market evidence (RQ3 and H6) . . . 376
F.5 Limitations . . . 391
F.6 Supplementary figures . . . 392

2026-10-24

Abstract

We ask how people delegate to a language model and whether industry returns track it. Anthropic Economic Index releases 2025-02-10 to 2026-06-26, conversations matched to O*NET tasks, are aggregated to occupations, countries and 49 Fama–French industries under a pre-registered protocol (pap-v1). H1, less delegation in high-education, high-wage occupations, is not supported: its null is not rejected at 0.01875. H2, delegation rising within tasks, is not supported: the release-level co-condition fails (wild $p = 0.927$ against 0.0125), although the task-clustered trend is 4.98 pp per year [2.71, 7.25], $p < 0.001$; descriptively, a single step at a change of classified-conversation base. H3, more checking where errors are costly, is not supported: its null is not rejected. H4, country differences beyond task mix, is supported. Industry delegation and human-rated theoretical exposure have Spearman $\rho = 0.716$ [0.527, 0.839], partly through shared weights and zero-assigned unpublished occupations. As the market test's minimum detectable effect exceeds D11's 5 pp threshold, market evidence is reported as what it can and cannot yet say: H5 is not rejected and H5b is not rejected (H0: $b = \bar{b}$; intervals [-2.76, 6.68] and [-1.86, 8.79]; 80% power only against a departure of at least 7.43 and 8.38 pp per SD; verbatim in Section 7). Employment weights postdate formation, so RQ2, H5 and H5b are not strictly real-time (R32 is real-time).

Executive brief

What the index measures

The Anthropic Economic Index (AEI) publishes how conversations with one provider's model spread across the tasks of the US O*NET occupational database, and how each conversation divides the work: in an *automation* pattern the user hands the task over, and in an *augmentation* pattern the user works through it with the model. This paper calls the automation share *delegation*. It uses the releases from 2025-02-10 to 2026-06-26, links tasks to occupations and occupations to the 49 Fama–French industries, and tests hypotheses fixed in a protocol registered as pap-v1 on 2026-10-21. AEI data are conversations matched to O*NET tasks. Nothing here says anything about job

risk, displacement or reassurance about the labour market.

Who delegates, and is it changing? (RQ1)

The first hypothesis, that delegation is lower in occupations requiring more education or paying higher wages (H1), is not supported: its null is not rejected at 0.01875. Within task families the automation share differs by -0.53 percentage points per year of education (interval -1.68 to 0.62), and the wage coefficient points the opposite way from the prediction without being significant at its level. Under the pre-registered robustness row R24 r5 the H1 test falls below its level, which changes no decision. The second, that delegation rises within the same task across releases (H2), is not supported: the release-level condition fails (wild-bootstrap $p = 0.927$, level 0.0125), although the task-clustered trend is 4.98 percentage points per year (interval 2.71 to 7.25, $p < 0.001$). As context, that rise is one step between two releases, then a fall, at a change in how conversations were classified, so classifier or platform drift cannot be told apart from behaviour; the balanced comparison also leaves out 890 tasks holding 5.39 per cent of usage (descriptive). The third, that users check the model's work more where errors are costly (H3), is not supported: its null is not rejected, and the estimate is -0.18 percentage points of the checking share (interval -1.10 to 0.75). The fourth, that countries differ in delegation beyond the mix of tasks their users bring (H4), is supported: holding task mix fixed, the automation share differs by -1.19 percentage points per unit of the log (a hundred log points) of the Anthropic Usage Index, usage relative to working-age population (interval -1.98 to -0.40). Usage and income are strongly correlated across countries, so their separate contributions are hard to tell apart, though the joint test is not.

Could versus does (RQ2)

RQ2 is descriptive and carries no test. Across the 49 industries, the rank correlation between delegation exposure and the human-rated theoretical exposure scores of Eloundou and co-authors is 0.716 (bootstrap interval 0.527 to 0.839; Figure 5). Both measures use the same occupation-to-industry employment weights, so part of the agreement is shared occupational mix. Occupations the index does not publish are assigned zero usage by assumption, not measured as zero; dropping them instead, in the pre-registered robustness row R3, lowers the correlation most. The employment weights (May 2024) were published on 2025-04-02, after the 2025-03-27 formation date of the industry measures, as were the O*NET 29.3 task links, so these are out of sample in AEI data only and not strictly real-time; the strict real-time row R32 is run for the market tests only.

Did markets price it? (RQ3)

At 80 per cent power the main test's minimum detectable effect, 7.43 percentage points of annualised risk-adjusted return per standard deviation of exposure, exceeds rule D11's threshold of 5 points, confirmed with that effect in view, so the market evidence is reported as what it can and cannot yet say. For H5, that more delegation-exposed industries earned risk-adjusted returns departing from the historical pattern from 2025-04-01 to 2026-08-31, the fixed wording reads: H5 is not rejected (H0: b = bbar (the 1990-2022 pattern for this exposure vector)). Interval [-2.76, 6.68]

pp per 1 SD; the test had 80% power only against a departure from H0 of at least 7.43 pp per 1 SD (MDE at Holm level 0.025). For H5b, that delegation exposure carries information beyond theoretical exposure, it reads: H5b is not rejected (H0: b = bbar (the 1990-2022 pattern for this exposure vector)). Interval [-1.86, 8.79] pp per 1 SD; the test had 80% power only against a departure from H0 of at least 8.38 pp per 1 SD (MDE at Holm level 0.025). Both decisions (H5 not rejected, H5b not rejected) rest on employment weights published inside the return window and O*NET links postdating formation, so these tests are out of sample in AEI data only and not strictly real-time; R32 is the strict real-time row and R1 is look-ahead.

What this does not show

The index records conversations on one provider's platforms, classified by an unstated method that may drift across releases; it does not sample people at work and says nothing about the prospects of any occupation. The associations are descriptive, not statements of what would follow if delegation changed. The market non-rejections do not show that exposure is unrelated to returns; their power clauses say what this short window could detect. Exploratory analyses, H6 among them, are not findings.

1 Introduction

Most quantitative work on how generative AI bears on the labour market starts from a measure of what a language model *could* do. Occupations and their tasks are scored for theoretical exposure, by human raters or by a model, and those scores are aggregated to occupations, industries and regions (Eloundou et al. 2024; Felten et al. 2021b; Webb 2020). Such scores describe capability, not use. They are silent on whether anyone actually brings a given task to a model and, if so, on how the work is shared between person and machine. The Anthropic Economic Index (AEI) offers a different kind of record (Handa et al. 2025; Massenkoff, Lyubich, Sacher, et al. 2026). Its data are conversations matched to O*NET tasks, published in successive releases from 2025-02-10 to 2026-06-26 (Table 5; Section F). Each release reports not only which tasks appear in observed usage but also the *collaboration pattern* of those conversations: whether the user hands the task over or works through it with the model. That second dimension is a behavioural variable, and it is the one this paper studies. It is also a record of usage on a single provider's platforms, not of the population of people at work, and every statement below is about that usage; the scope and the drift of the platform across releases are discussed in Section 9.

The AEI classifies conversations into five collaboration patterns and groups them into two buckets. *Automation* covers the directive and feedback-loop patterns, in which the user asks for a finished product or relays results back for the model to act on; *augmentation* covers task iteration, learning and validation, in which the user refines, studies or checks the work (Handa et al. 2025). We call the automation share *delegation* and the augmentation share *collaboration*, and we read the distinction through the psychology of trust in automation. Parasuraman and Riley (1997) describe how people use, misuse and disuse automated systems, and Lee and See (2004) argue that appropri-

ate reliance requires trust calibrated to a system's actual capability. Work on algorithm aversion and appreciation finds that the willingness to defer to an algorithm varies with the stakes, with whether the algorithm has been seen to err, and with the kind of judgement involved (Dietvorst et al. 2015; Logg et al. 2019). These literatures suggest where delegation should be lower (tasks and occupations where errors are costly or human judgement is prized), where checking should be more common, and how reliance might change as users gain experience. The RQ1 hypotheses are written as tests of those suggestions on observed usage rather than on survey responses or laboratory choices (Section 2).

The paper makes three contributions. First, it documents the behavioural structure of delegation: in which occupations and for which tasks work is handed to the model, how that differs across countries, and whether the measured pattern changes across releases (Section 5). Second, it turns task-level usage and collaboration shares into industry-level exposure measures that keep *usage* apart from *delegation* and *augmentation*, built for the 49 Fama–French industries through occupation and industry crosswalks (Section 3), and benchmarks them against theoretical exposure scores to ask how far "could" and "does" are apart (Section 6). Third, it tests whether industries' risk-adjusted returns over a 17-month window depart from their historical pattern in line with observed delegation, and beyond theoretical exposure (Section 7). Along the way the paper builds the measurement apparatus that the AEI's changing release formats require: a harmonised panel of releases, a record of the measurement regime behind each one (Table 6; Section F), and a replication package that rebuilds every number from the raw downloads (Section E).

The analysis was pre-registered. The protocol, tagged pap-v1 in the repository and dated 2026-10-21, fixes every confirmatory specification, decision rule and robustness row before the confirmatory outcomes were read; it is reproduced in Section A and summarised in Section 4. There are two confirmatory families. For RQ1 the family-wise level of 0.05 is split: H2 is tested on its own at 0.0125, and H1, H3 and H4 form a Holm family at 0.0375 (Holm 1979). For RQ3, H5 and H5b form a Holm family whose step levels are 0.025 and 0.05. H5 and H5b are tested against the frozen historical pattern of cross-industry slopes for the same exposure vector, estimated on windows that end before the market seal date, rather than against zero; the test against zero is kept as a labelled secondary result. The historical slopes are not centred at zero, and a test against zero over-rejects across the historical windows, so a rejection here means a departure from that pattern, a narrower claim than the approved wording of H5 (Section 4); the recentring was proposed after those pre-seal rejection rates had been seen, as Section C records. RQ2 is descriptive and carries no test, and H6 is exploratory. Every departure from the protocol after the tag is listed in Section B, and the disclosures are in Section C.

The protocol also contains a rule about what the market evidence can bear. The pre-registered power analysis gives a minimum detectable effect (MDE) for the primary H5 test of 7.43 pp of annualised alpha per standard deviation of exposure, at the first Holm step and on the frozen historical null scale. Decision D11 states that if this MDE exceeds 5 pp per standard deviation, RQ3 is reported as "what the market evidence can and cannot yet say", and the paper is led by RQ1 and RQ2. The MDE exceeds the threshold, so the rule fires. The threshold was written into the project

plan before the power analysis was first run, but it became binding only when the protocol was signed, by which time the MDE was known; a reader should treat it as a threshold confirmed with the MDE in view (Section 4). The market tests are still run and reported in full, with the MDE beside each result (Section 7).

For RQ1 (Section 5, Table 1), H1 is not supported: the null that either coefficient is zero is not rejected at 0.01875 (intersection–union p = 0.361). Within task families, the automation share differs by -0.53 pp per year of required education (CI [-1.68, 0.62], p = 0.361) and by 4.40 pp per unit of log wage (a hundred log points; CI [0.07, 8.72], nominal p = 0.046). The wage coefficient is opposite in sign to H1's prediction and is not significant at its Holm level, so it neither supports nor contradicts H1; the pre-registered robustness row R24 r5 takes the H1 p-value below its Holm level with both coefficients negative, and changes no decision. H2 is not supported: the release-level co-condition fails. The within-task trend in the automation share over r2–r5 is 4.98 pp per year (task-clustered CI [2.71, 7.25], $p < 0.001$), but the release-level wild bootstrap gives p = 0.927 against a level of 0.0125. The measured rise is a single r2-to-r3 step followed by a fall (Figure 3); leaving out r2 reverses the sign of the trend, to -6.55 pp per year (a post-result diagnostic, PAP §11 D-3; exploratory; not a decision; release-level p in Table 14); and the r2-to-r3 step coincides with a change in the base of classified conversations, so classifier or platform drift cannot be separated from behaviour. Rows R22 (`api_r3_r5`) and R23 run the other way, and the balanced-panel rule removes 890 tasks holding 5.39 % of usage (descriptive). H3 is not supported: the null that γ is zero is not rejected (Holm stopped at H1), with γ = -0.18 pp of validation share in high-error-cost occupations (CI [-1.10, 0.75], p = 0.707). H4 is supported: the joint Wald test of the two country-level coefficients gives $p < 0.001$ at 0.0125. Conditional on task mix, the automation share differs by -1.19 pp per unit of log AUI (a hundred log points; CI [-1.98, -0.40]) and by -0.92 pp per unit of log GDP per working-age person (also a hundred log points; CI [-1.61, -0.23]). The two regressors are strongly correlated across countries, since both share the working-age population denominator, so b_1 and b_2 are only weakly separable one by one, although their joint test is not.

RQ2 is descriptive (Section 6). Across the Design B industries, the Spearman correlation between delegation exposure and the human-rated exposure of Eloundou et al. (2024) is 0.716 (industry-bootstrap CI [0.527, 0.839]; Table 2, Figure 5). Industries rated most exposed also tend to show the most delegation, and the industries whose ranks differ most between the two measures are listed in Table 18 (Section F). Two qualifications sit beside this number. The correlation inherits the same OEWS occupation-to-industry weights on both sides, so part of it is common occupational mix; and it depends on the D3 assumption, under which occupations unpublished in the AEI are assigned zero usage by assumption rather than measured as zero, with R3 lowering it most, to 0.568 (Section D). As to timing, Design B's OEWS May 2024 weights were released on 2025-04-02, after the 2025-03-27 formation date and inside the return window, and the O*NET 29.3 links postdate formation; Design B is out of sample in AEI data only and is not strictly real-time. The strict real-time row, R32, uses OEWS May 2023 weights and is run for the market tests only (Section 8).

Under D11, RQ3 is reported as what the market evidence can and cannot yet say (Section 7, Table 3). H5 is not rejected, and H5b is not rejected. Over the return window from 2025-04-01 to 2026-08-

31, the cross-industry slope of annualised alpha on delegation exposure is 1.96 pp per standard deviation (p = 0.8934 against the frozen historical pattern), and 3.46 pp per standard deviation once theoretical exposure is controlled for (p = 0.3906). In the pre-registered wording: H5 is not rejected (H0: b = bbar (the 1990-2022 pattern for this exposure vector)). Interval [-2.76, 6.68] pp per 1 SD; the test had 80% power only against a departure from H0 of at least 7.43 pp per 1 SD (MDE at Holm level 0.025). H5b is not rejected (H0: b = bbar (the 1990-2022 pattern for this exposure vector)). Interval [-1.86, 8.79] pp per 1 SD; the test had 80% power only against a departure from H0 of at least 8.38 pp per 1 SD (MDE at Holm level 0.025). The robustness rows that move the H5 slope most are R7 (dropping Hardw, Softw and Chips), R12 (ranks) and R5 (equal-weighted returns), and for H5b they are R12, R7 and R6 (FF30); all are reported as robustness and none changes a decision (Section 8). The timing caveat stated for RQ2 applies here too: Design B is out of sample in AEI data only and is not strictly real-time. Under R32, the strict real-time row, the slopes are 2.20 and 3.72 pp per standard deviation (Section D), and R1 is reported throughout as look-ahead. A test with this MDE can detect only a large departure from the historical pattern, and the non-rejections are statements about what this window can and cannot resolve. Exploratory: H6 compares how much industries discuss AI in their 10-K filings with how much they are observed to use it, and sits outside every testing family (Section 7).

The paper proceeds as follows. Section 2 reviews the literatures on exposure measurement, on generative AI at work, on trust in automation and on the pricing of technology exposure. Section 3 describes the AEI releases, the crosswalks and the construction of the industry exposure indices. Section 4 sets out the pre-registration, the testing families and the inference for each hypothesis. Section 5 reports the behavioural results, Section 6 the comparison of observed and theoretical exposure, and Section 7 the market evidence together with the exploratory analyses. Section 8 summarises the pre-registered robustness rows, whose full tables are in Section D. Section 9 discusses what the design cannot establish, and Section 10 concludes.

2 Related literature

This paper draws on four bodies of work that rarely meet: the task-based economics of technology and the exposure scores built from it; field and experimental studies of generative AI at work; the psychology of reliance on algorithms and automation; and the asset-pricing literature on how markets value a firm's exposure to a technology or a risk. Each strand closes with the point at which this paper departs from it, and the section ends with the paper's data source, the Anthropic Economic Index (AEI); results begin in Section 5.

2.1 Tasks and exposure measurement

The task approach treats an occupation as a bundle of tasks and asks which of them a technology can perform. Autor et al. (2003) classify work by its routine and non-routine content and argue that computer capital substitutes for routine tasks while complementing non-routine analytical and interactive ones, so that a technology's reach is read at the level of the task rather than the

job. Acemoglu and Restrepo (2019) formalise the same unit: automation moves tasks from labour to capital, while the creation of new tasks returns work to labour, and the net balance depends on both margins. The task, not the occupation, is therefore the natural place to measure what a technology does.

A second generation of papers turns that framework into exposure scores. Webb (2020) measures exposure by the textual overlap between patents and the task descriptions of occupations. Felten et al. (2021b) build the AI Occupational Exposure (AIOE) index by linking measured progress in a set of AI applications to the abilities that O*NET records for each occupation, and aggregate it to industries and regions. Eloundou et al. (2024) rate O*NET tasks by whether access to a large language model, alone or with software built on it, could substantially reduce the time a task takes, using both human annotators and a model as raters. These measures share a design choice: they describe what the technology *could* do to a task, judged from text, abilities or rubrics, before or without observing its use.

This paper takes the task as its unit as well, but measures what is observed rather than what is possible. Its behavioural outcomes are shares of conversations matched to O*NET tasks, and its industry indices aggregate observed usage, delegation and augmentation through occupational employment weights (Section 3). The theoretical scores enter as benchmarks: AIOE as the pre-period control in the market tests and the Eloundou ratings as the "could" side of the descriptive comparison in Section 6.

2.2 Generative AI at work

A fast-growing empirical literature measures what happens when generative AI is introduced into real work. Brynjolfsson et al. (2025) study the staggered introduction of a generative AI assistant to customer-support agents at a large firm and find productivity gains that are largest for less experienced agents. Noy and Zhang (2023) randomise access to a chatbot among professionals performing writing tasks and find that access shortened the time taken and raised the rated quality of the output. Dell'Acqua et al. (2023) run a field experiment with management consultants and describe a "jagged technological frontier": on tasks inside the frontier AI assistance improved performance, while on a task chosen to lie outside it, participants using AI were more likely to reach an incorrect answer. They also describe distinct ways of working with the model, some participants dividing tasks between themselves and the model and others interleaving their work with it closely.

These studies measure outcomes for a defined population, a firm or an experimental sample, under controlled or quasi-random assignment. They are informative about productivity in particular settings and silent on how use is distributed across the economy's many tasks. This paper reverses the trade-off. It has no outcome measure and no assignment; it observes, across a very wide range of tasks and several releases, *how* conversations are conducted, in particular whether the task is handed over or worked through jointly. The collaboration mode that the field studies observe for a few tasks is, in the AEI, recorded for all of them. The paper accordingly makes no productivity claim and no claim about the people behind the conversations; its RQ1 results (Section 5) describe

usage only.

2.3 Algorithm aversion, appreciation and trust calibration

The behavioural hypotheses of RQ1 come from the literature on when people rely on automated judgement, which PAP §1 names for each hypothesis. Dietvorst et al. (2015) show that people who see an algorithm make errors become less willing to use it than a human forecaster, even when the algorithm is the more accurate of the two. Logg et al. (2019) report the opposite tendency in other settings: lay participants weighted advice more when it was attributed to an algorithm than to a person, a pattern that weakened among experts. Reliance on algorithms is thus neither uniformly low nor uniformly high; it varies with the decision maker and the setting.

The human-factors literature frames the same question as one of calibration. Parasuraman and Riley (1997) distinguish use, misuse (over-reliance), disuse (under-use) and abuse of automation. Lee and See (2004) define appropriate reliance as trust that matches the system's capability, and describe trust as something that develops with experience of the system's performance. H1 reads the aversion literature at the level of tasks: delegation should be lower for tasks that require more education and in higher-wage occupations, where the stakes of a task are higher. H2 reads the calibration literature over time: within a task, delegation should rise across releases as experience with the model accumulates. H3 reads the misuse literature: the validation pattern, in which the person checks the model's output, should be more prevalent in occupations where errors are costly.

The departure here is one of scale and of what is observed. The aversion and appreciation studies are laboratory or online experiments with controlled stimuli and measured accuracy; the calibration literature is largely about operators of specific automated systems. This paper observes neither the accuracy of the model nor the beliefs of the user. It asks only whether the prevalence of delegation and validation across tasks, occupations and releases follows the pattern these theories predict, and it tests each prediction under a pre-registered decision rule (Section 4; Section A). A pattern consistent with a theory is not evidence of its mechanism, a limit stated in Section 9.

2.4 Asset pricing of technology exposure and text-based measurement

A finance literature asks whether markets price a firm's exposure to a new technology. Eisfeldt et al. (2023) construct firm-level exposure to generative AI from the occupational composition of firms' workforces, using task-level exposure assessments, and study how firm values moved around the public release of ChatGPT; their exposure measure, like those above, describes what the technology could do. Return tests of this kind rest on the standard tools of empirical asset pricing: the five-factor model of Fama and French (2015) for risk adjustment, which this paper extends with the momentum factor of Carhart (1997), and the cross-sectional regressions of Fama and MacBeth (1973) for relating characteristics to average returns. A parallel literature measures firm exposure from text. Hassan et al. (2019) build a firm-level measure of political risk from the share of earnings-call discussion devoted to political topics, an approach that treats what a firm's managers say as a measurable signal of what the firm faces.

This paper's market tests differ in three ways. Their exposure variable is built from observed delegation rather than theoretical capability, and H5b asks whether it carries information beyond a theoretical score (AIOE) that predates the release of ChatGPT on 2022-11-30. Their unit is the industry, because AEI usage is matched to tasks and occupations and reaches firms only through employment weights. And they are confirmatory: the specification, the null, the inference and the decision rule were fixed in the pre-analysis plan before any post-seal industry return was read by code, which closes for the post-seal test the search over specifications that can inflate apparent findings in return studies (Romano and Wolf 2005; Bailey and López de Prado 2014), although the null was recentred after pre-seal rejection rates were seen (Section 4; Section C). The text-based strand enters through H6, which is exploratory: it compares what industries' 10-K filings say about AI with what their tasks' observed usage shows, in the spirit of Hassan et al. (2019), and the comparison is reported without a rejection claim (Section 7).

2.5 The Anthropic Economic Index

The AEI is the paper's source of observed usage. Handa et al. (2025) describe the method: a privacy-preserving analysis of millions of conversations with Claude, in which each conversation is matched to O*NET tasks and classified by its collaboration pattern, distinguishing automative use, where the task is handed to the model, from augmentative use, where the person iterates with it, learns from it or checks it. Later releases extend the same measurement, among them reports on uneven geographic and enterprise adoption (Appel et al. 2025), on economic primitives (Appel et al. 2026) and on learning curves (Massenkoff, Lyubich, McCrory, et al. 2026), and the latest release used here (Massenkoff, Lyubich, Sacher, et al. 2026). This paper uses the published releases as inputs under a frozen plan. AEI data are conversations matched to O*NET tasks: they measure usage of Anthropic's models, not workers, jobs or outcomes, and Section 3 sets out how the releases are harmonised and what that harmonisation can and cannot support.

3 Data and measurement

Four sources enter the confirmatory analysis: the Anthropic Economic Index (AEI) releases, the O*NET task and occupation files, the Bureau of Labor Statistics Occupational Employment and Wage Statistics (OEWS) and the Kenneth French data library; the "could" benchmarks of RQ2 and the SEC EDGAR filings of H6 are descriptive or exploratory. The tables cited here come from the data audit, which reports coverage, missingness and measurement regimes only and relates no delegation measure to any outcome. Raw files, retrieval times and hashes are listed in Section E.

3.1 The AEI releases

AEI publishes aggregated, privacy-filtered counts of Claude conversations classified to O*NET task statements, with the share in each collaboration pattern (Handa et al. 2025; Appel et al. 2025, 2026; Massenkoff, Lyubich, McCrory, et al. 2026; Massenkoff, Lyubich, Sacher, et al. 2026). This paper uses all six releases published before the pre-registration: r1 on 2025-02-10, r2 on 2025-03-27, r3

on 2025-09-15, r4 on 2026-01-15, r5 on 2026-03-24 and r6 on 2026-06-26. Table 5 (Section F) gives each release's date, source, data window, geography levels and taxonomies. Two sources appear: `claude_ai`, the consumer product, in every release, and `1p_api`, Anthropic's first-party API, from r3 onwards. From r3 the `claude_ai` files add countries and subregions.

The releases arrive in three file layouts: separate tables keyed by task name (r1, r2); a shared facet layout with one row per value, geography, platform, facet and variable (r3 to r5); and calendar-month extracts in a long metric layout that adds published SOC occupation rows and the O*NET work-activity hierarchy (r6).

The unit of observation throughout is a conversation, classified by a model and matched to a task. AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market. Delegation by task, occupation, industry or country always means the share of matched conversations in the automation patterns, weighted as stated, never anything about the people who hold those occupations.

3.2 Measurement regimes

Table 6 (Section F) lists, for each release and source, the platform scope, the classifier, the collaboration taxonomy, the suppression construct and the occupation basis. Four regime changes matter.

The first is platform scope: the consumer scope widens from the Free and Pro plans to include Max and then, in r6, the Cowork product; the API source enters at r3, and r6's API scope excludes Claude Code (Table 6; Section F). The second is the classifier: no release file names the model that assigns conversations to tasks and patterns, so a change of classifier can be neither observed nor separated from a change in behaviour. The third is the collaboration taxonomy: r1 uses the earlier taxonomy, `collab_v1`, and publishes only an overall split, with no task-level patterns; r2 to r6 use `collab_v2`, with the same five pattern labels throughout. Following the AEI definition, automation is the sum of the `directive` and `feedback_loop` patterns and augmentation the sum of `task_iteration`, `validation` and `learning`. Both are renormalised over classified conversations, so the primary delegation measure, `automation_pct`, is defined identically in every release from r2 to r6 (D1).

The fourth is suppression: AEI withholds cells below its privacy thresholds, in a way that changes twice (Table 6; Section F). In r2 each task carries a `filtered` share, and the patterns are renormalised over the remainder. In r3 to r5, small task-by-pattern cells are folded into a `not_classified` row alongside a `none` row. In r6 cells below the thresholds are simply absent, there is no `not_classified` row, and the patterns are published as shares of all conversations, which this paper renormalises. The base of classified conversations therefore differs between r2 and the later releases, and task-level pattern completeness is not comparable between r3 to r5 and r6. H2 therefore measures the change in measured within-task automation, including any change of platform, classifier or suppression (Section 5). The occupation basis changes as well: r1 to r5 occupations are derived from task usage, while r6 publishes SOC occupation rows directly, which this paper uses for r6 (D9).

3.3 The harmonised panel

The parsers write one long table indexed by release, period, source, geography, taxonomy, node and metric, with one value per cell, and keep an unpublished value distinct from a zero: a cell AEI does not publish has no row, is never set to zero and is filled by no later step. A group of shares that sums to less than the whole is recorded as unpublished mass, not as a measured absence of usage. Checks in code bound shares within each node and geography, validate task and occupation codes against the O*NET and SOC lists, and pin each release's row count. The zero assignment in the industry indices, discussed below, is the one place where unpublished occupations are given a value.

3.4 Tasks, occupations and the O*NET-SOC harmonisation

AEI identifies tasks by their statement text. Each statement is linked to an O*NET Task ID through the O*NET 29.3 task file (National Center for O*NET Development 2025), the earlier statements shipped with r2 and the O*NET occupation crosswalk, and every occupation code is resolved to the O*NET-SOC 2019 taxonomy, the vintage of the OEWS weights and of AEI's own r6 occupation rows. A statement with more than one candidate Task ID is allocated to the union of its candidates' occupations, and a task shared by several occupations gives each an equal part of its usage (D4). The equal split conserves usage and uses no employment information, keeping the numerator of the usage intensity defined below independent of its denominator, at the cost that occupations sharing many statements can differ widely in intensity through allocation alone. An employment-proportional split is pre-registered as robustness row R11 (Section 8).

3.5 OEWS weights and the timing of the inputs

Occupation weights come from OEWS May 2024 (U.S. Bureau of Labor Statistics 2025), at national level for each occupation's share of employment and at four-digit NAICS level for each industry's occupation mix. AEI occupations are joined to OEWS detailed codes by exact match, by rollup where OEWS publishes only the aggregate, and by a hybrid step for codes that O*NET-SOC splits and OEWS merges. Exact matching and rollup alone leave 0.66 % of AEI usage without an OEWS occupation; after the hybrid step none is left, and the occupations AEI reaches hold 92.61 % of OEWS employment. The remaining 7.39 % of employment sits in OEWS occupations with no AEI r6 occupation. Those occupations are unpublished in AEI, not measured as zero.

The timing of these inputs requires a correction to the design (PAP §10, S1). Design B, the confirmatory exposure vector, takes its AEI usage from release r2 alone, published on 2025-03-27, its formation date. Its OEWS May 2024 weights were released on 2025-04-02, after that formation date and inside the return window that opens on 2025-04-01. Its task links come from O*NET 29.3, which was also released after formation, although its task statements are byte-identical to those of O*NET 29.2 (clarification C-74 on R32, Section B). Design B is therefore out of sample in its AEI data only; it is not strictly real-time in its occupation weights and task links (PAP §2.1). Both are slow-moving structural inputs that carry no return information, but the claim of real-time construction

does not hold for them. Robustness row R32 rebuilds Design B from inputs public at formation, the OEWS May 2023 weights and the O*NET 29.2 task file; since the task files are identical, R32 differs from the primary only in its weights. Row R1, which uses the OEWS May 2025 weights published during the return window, is labelled look-ahead wherever it appears. Both rows are reported in Section 8 and Section D, and this timing caveat applies to every RQ2 correlation in Section 6 and to every H5 and H5b number in Section 7.

3.6 The industry crosswalk

The market tests use the 49 Fama–French industry portfolios (FF49), which are defined by SIC codes, while OEWS industries are NAICS codes. Each four-digit NAICS industry is mapped to SIC codes through the chain of Census concordances (U.S. Census Bureau 2022), and SIC codes are mapped to FF49 industries by the ranges of the French library's industry definitions. Where a NAICS code maps to several SIC codes, its employment is split equally among them (D23). This departs from the design specification's employment-weighted crosswalk, as PAP §10 and Section B disclose: the Census concordances carry no employment weights, and the equal split is the vector on which the power analysis was run and which was frozen at the tag, so its weights are structural, not employment-based. Row R33 is the specification's crosswalk, weighted by the employment of each NAICS industry's finer children, and row R20 gives each NAICS code wholly to one SIC code (Section 8). Management of companies is excluded from FF49, because through holding-company SIC codes it would otherwise set the occupation mix of the finance industry (D7); row R15 redistributes it in proportion to mapped private employment. Table 7 (Section F) reports, by design, the range and median of the share of each industry's employment that carries a delegation value.

3.7 The exposure indices

The exposure indices follow the design specification as frozen in the pre-analysis plan (PAP §2.1). For occupation o, let s_o be its share of AEI usage, π_o its share of national employment, and a_o and g_o the automation and augmentation shares of its classified conversations. Usage, delegation and augmentation intensity are

$$U_o = \frac{s_o}{\pi_o}, \qquad D_o = U_o\, a_o, \qquad A_o = U_o\, g_o.$$

For industry j, with e_{oj} the employment of occupation o in j and O_j the occupations of j that carry a value, the industry index for each intensity $X \in \{U, D, A\}$ is the employment-weighted sum

$$E_j^X = \sum_{o \in O_j} w_{oj}\, X_o, \qquad w_{oj} = \frac{e_{oj}}{\sum_{o'} e_{o'j}}.$$

Each index is standardised across the J industries present, where $J = 49$:

$$z_j^X = \frac{E_j^X - \bar{E}^X}{\sigma^X}, \qquad \bar{E}^X = \frac{\sum_k E_k^X}{J},$$

with σ^X the population standard deviation of E^X across those industries (divisor J). The three standardised indices are `z_usage`, `z_delegation` and `z_augmentation`; H5 uses `z_delegation`, and H5b adds the AIOE benchmark beside it (Section 4).

Three exposure vectors are built from these formulas. Design B uses release r2 alone (`claude_ai`, the only r2 source) and is the confirmatory vector. `A_r2` is numerically the same vector, labelled exploratory because it serves the exploratory Design A analyses. `A_pooled` averages r2 to r6 `claude_ai` with equal release weights; it uses releases published during and after the return window, so it contains look-ahead and is labelled so wherever it appears (PAP §10, S3). Table 7 (Section F) shows that the Design B `z_delegation` distribution is strongly right-skewed: its largest value, for Softw, is 4.739 SD, against 1.977 SD for the next industry, which is why rows R12 (ranks) and R13 (dropping that industry) are co-primary (Section 4).

3.8 The zero-assignment assumption

An OEWS occupation with no AEI usage enters the indices with U_o, D_o and A_o set to zero (D3). This is an assumption, not a measurement: AEI leaves these occupations unpublished. Under Design B, occupations holding 12.14 % of OEWS employment are unpublished in r2 and assigned zero by assumption, as for `A_r2` (12.14 %); under `A_pooled`, which matches more tasks, the share unpublished and assigned zero by assumption is 6.34 %. These differ from the 7.39 % of employment with no AEI r6 occupation, which refers to r6's published occupations, while Design B rests on r2, which matches fewer tasks (PAP §10, S5). The occupations concerned are mostly hands-on trades (D3). Dropping them instead would treat their unpublished usage as equal to their industry's mean, which would overstate exposure in the industries that employ them most. Robustness row R3 drops them from the industry sums (Section 8). The RQ2 correlations depend on this choice, and R3 is the row that lowers them most (Section 6).

3.9 The "could" benchmarks

RQ2 compares the observed indices with two measures of what language models could do. The AI Occupational Exposure index (AIOE) of Felten et al. (2021b, 2021a) is published on SOC codes of an earlier vintage. It is carried to O*NET-SOC 2019 through the O*NET occupation crosswalk: an earlier code takes the successors of all its O*NET children, a split copies the value and a merge takes the simple mean (D6). Its weakness is that some earlier codes are copied to several successors with different content. The published BLS SOC crosswalk (U.S. Bureau of Labor Statistics 2018), with the same copy and mean rule, is robustness row R14. The second benchmark is the occupation-level exposure of Eloundou et al. (2024, 2023), already on O*NET-SOC codes. Its headline rating is the human annotators' β, chosen because it does not depend on a language model's assessment of language-model exposure; the GPT-4 β of that study (`dv_rating_beta`) is reported beside it

in every RQ2 table (D8), and row R17 makes it the headline. Both benchmarks are descriptive (Section 6) and enter no confirmatory test, except that AIOE is the control in H5b.

The benchmarks pass through the same OEWS weights, the same crosswalk and the same standardisation as the AEI indices, giving `z_aioe`, `z_eloundou` and `z_eloundou_gpt4`. Both sides of every RQ2 comparison thus share the occupation-to-industry weights (Section 6).

3.10 Market data

Industry returns are the French library's daily value-weighted FF49 portfolio returns, and the factors are the five Fama–French factors and momentum (French 2026; Fama and French 2015; Carhart 1997). French revises these files retroactively, so only the frozen files whose hashes are listed in Section E are used. The market data end on 2026-08-31, the last date of the frozen daily files and of the 17-month confirmatory return window that opens on 2025-04-01. The pre-registration was frozen, as tag pap-v1, before any code read an industry return dated on or after the seal date, 2022-11-30 (Section C); the history before the seal supplies the null scale described in Section 4.

3.11 Filings for the talk index

H6, which is exploratory, compares how often firms in an industry mention AI in their 10-K filings with the industry's AEI usage exposure. Its data come from SEC EDGAR (U.S. Securities and Exchange Commission 2026). For each industry the talk index is the share of distinct 10-K filers whose 10-K filed in 2024, between 2024-01-01 and 2024-12-31 and so before Design B's formation date, is returned by EDGAR full-text search for at least one primary term of the pre-registered AI dictionary (D12). The denominator is the count of filers of the same forms in the EDGAR full index. Two inputs are fixed at retrieval, not at the filing year. Each filer's industry is its current SIC code in the EDGAR submissions file (C-40), retrieved at 2026-10-23T01:19:58Z and applied to every filing year; and the numerator uses the phrase matching of the SEC's full-text search as implemented on the retrieval date (PAP §6). Filers whose SIC falls in no FF49 range are left unmapped and reported, never assigned to an industry. All H6 results are reported as exploratory in Section 7.

4 Pre-registration and methods

4.1 The protocol

The analysis follows a pre-analysis plan signed and frozen on 2026-10-21 as the annotated git tag pap-v1, on commit 6a16b1ad052977650b90c8062ceeeab9ac6572e1; its signature, decision register and code binding are in Section A. For every confirmatory hypothesis the plan fixes the outcome, regressors, sample, estimator, inference and decision rule, and it labels every other analysis secondary, robustness or exploratory. Every confirmatory result is reported, a non-rejection as prominently as a rejection, with a confidence interval and a two-sided p-value; for a directional hypothesis (H1, H2, H3) the decision requires both significance and the stated sign.

Two protocols govern the first read of outcomes. Under P-H5, the first read of industry returns dated on or after the seal date, 2022-11-30, is a single run of `aideleg.confirm_h5`, whose git blob hash the plan records; the script verifies its binding to the tag, an unchanged source tree and the hashes of its frozen inputs, and an attempt marker, created before the first post-seal read, records a failure as an attempt and blocks a rerun. Its output and the marker are committed unedited, alone in one commit, and every other post-seal read happens later. P-RQ1 is the weaker counterpart for H1–H4, whose estimators were written after the tag: the code was fixed only at its first run on real outcomes, a single logged commit with unedited output, and any later change to it is a deviation. Under the demotion rule, a deviation adopted after a confirmatory result has been seen demotes the hypothesis it touches to exploratory, and the pre-registered specification is still reported as primary beside it. Both runs are recorded in Section C.

The plan's own disclosures (PAP §0.4, Section C) record what was seen before the freeze: aggregate factor rows from the sealed period, displayed once, unused and uninformative about the slope; a retracted audit figure on renormalisation; RQ2 descriptives and design diagnostics, including the ranking and skew of the Design B scores and their correlations with the benchmarks, although no delegation measure had been related to the H1–H4 regressors and no post-seal industry return had been read by code; the pre-seal slopes and their frozen means, with centring of the market null recommended after the uncentred in-sample rejection rates had been seen; general knowledge of markets after the seal, which is why Design A is exploratory; the event rule's exceptions, chosen with the source list in view; and a confirmatory script tested on synthetic data only. The appendix also lists every fact recorded in the working ledgers after the freeze.

PAP §11 records every later change, and Section B lists each entry with its date and its "Results seen" field: additions of data under D20, clarifications, and three deviations. D-1 corrects the secondary H6 talk measure: two 10-K primary documents truncated in transfer were fetched again and the measure rerun, with values before and after recorded (ratified by Ajai, C-55). D-2 aligns an H6 helper's default phrase list with its documented list, so that robustness downloads cannot enter the primary numerator; the primary outputs are unchanged. D-3 adds labelled post-result influence diagnostics for H1, H2 and H6 that change no confirmatory or committed estimate. All three followed confirmatory results; none alters a confirmatory specification, and the H6 analyses they touch are exploratory by design.

4.2 RQ1: behaviour

RQ1's family-wise level, 0.05, is split under D22: H2 is tested alone at 0.0125, and {H1, H3, H4} by Holm's step-down procedure (Holm 1979) at family-wise 0.0375. Each hypothesis's level depends on the order of the family p-values (Table 1). The outcome is the task automation share (D1; Section 3).

H1 and H3. Both use the r6 Claude.ai cross-section averaged over its two months (D13), on (task, occupation) rows within O*NET Intermediate Work Activity fixed effects (D14). H1 is

$$y_{io} = \beta_e \, \mathrm{educ}_i + \beta_w \log w_o + \mu_{f(i)} + \varepsilon_{io},$$

the task's automation share on its mean years of education and the log OEWS median wage of the occupation; H3 is $v_{io} = \gamma \, \mathrm{err}_o + \mu_{f(i)} + \varepsilon_{io}$, the validation share on an indicator of legal, health-practitioner and regulated-finance occupations (D15). Both are weighted least squares, each task's usage split across its occupations and families, on 1684 rows (1142 tasks, 388 occupations) for H1 and 1689 rows (1147 tasks, 391 occupations) for H3. Standard errors are clustered by occupation and by task (Cameron et al. 2011),

$$\hat{V} = \hat{V}_{\mathrm{occ}} + \hat{V}_{\mathrm{task}} - \hat{V}_{\mathrm{occ} \cap \mathrm{task}},$$

with small-sample corrections and t inference on one fewer degree of freedom than the smaller cluster count. The non-PSD rule is fixed in advance: a covariance block of the tested coefficients with a negative eigenvalue is rebuilt with its negative eigenvalues set to zero, a tested coefficient whose variance is then zero takes the larger of its two one-way cluster standard errors, and each use is reported (Section B). H1 is conjunctive, so its p-value is the intersection–union maximum of the two coefficient p-values (D17), and it is supported only if Holm rejects and both coefficients are negative; H3 needs a Holm rejection and positive γ.

Effective estimands (P13). The outcome and education are constant within a task; only the wage varies across a task's occupation rows. β_e is thus the usage-weighted, within-family association between a task's automation share and its education requirement. β_w compares tasks through their allocation-weighted mean log wage; the within-task wage spread enters the estimator's denominator but not its numerator, which attenuates β_w by the between-task share of the within-family wage variance, 1.00. γ is attenuated likewise, by 1.00. H1 and H3 describe how tasks differ, weighted by the occupations they are allocated to; they do not compare workers.

H2. Under D22, H2 is tested as "measured within-task automation rises across releases", because neither the unstated classifier nor the platform scope is held fixed. It uses the balanced panel of tasks published in all four releases r2 to r5 (D2), 1315 tasks and 5260 rows, with task fixed effects, fixed weights and a linear trend in years since the r2 window midpoint:

$$y_{it} = \alpha_i + \beta \, \tau_t + \varepsilon_{it}.$$

The trend is identified only from four release cross-sections, and a task-clustered p ignores shocks common to a release, such as a classifier change, so H2 is supported only if the task-clustered p is at or below 0.0125, β is positive, and a restricted wild cluster bootstrap over releases also gives p at or below that level. The bootstrap uses Webb's six-point weights (Webb 2023; Cameron et al. 2008) and enumerates all 1296 weight vectors, so it involves no random draws. A randomisation over the 24 orderings of release labels is descriptive only, and release fixed effects relative to r2 are plotted in Figure 3.

H4. H4 is an r6 country cross-section of 113 countries (D16), each variable averaged over the two months before logs are taken:

$$y_c = a + b_{\text{AUI}} \log \text{AUI}_c + b_{\text{inc}} \log \text{inc}_c + b_{\text{mix}} M_c + e_c, \qquad M_c = \sum_g \bar{s}_{cg}\, \bar{a}_g,$$

where y is the overall automation share, AUI the Anthropic Usage Index, inc GDP per working-age person, and M weights each Generalised Work Activity's global automation share by its share of the country's usage. Estimation is unweighted least squares with HC3 covariance. H4 states no direction, so its p-value is the joint Wald test that b_AUI and b_inc are both zero (D17); it is supported if Holm rejects.

4.3 RQ2: the bridge

RQ2 is descriptive. Across the 49 Fama–French industries it reports Spearman correlations between the AEI usage, delegation and augmentation indices (Section 3) and the theoretical benchmarks, AIOE (Felten et al. 2021b) and the ratings of Eloundou et al. (2024) (human-rated β as headline, D8). Each carries a percentile interval from an industry bootstrap of 10000 draws with the project seed, 20261021, for Design B and, labelled look-ahead, for Design A_pooled. No test is attached (Section 6).

4.4 RQ3: markets

Primary specification. H5 and H5b use the Design B exposure vector, formed from release r2 alone, published on 2025-03-27. Each industry's annualised alpha is the intercept of a daily regression of excess value-weighted returns on the five Fama–French factors and momentum (Fama and French 2015; Carhart 1997; French 2026) over the window from 2025-04-01 to 2026-08-31 (17 months). The unweighted cross-sectional regressions are

$$\hat{\alpha}_j = a + b\, z_j^{\text{del}} + e_j \qquad \text{and} \qquad \hat{\alpha}_j = a + b\, z_j^{\text{del}} + \delta\, z_j^{\text{aioe}} + e_j,$$

with b in pp of annualised alpha per SD of delegation exposure. Design B is out of sample in its AEI data but not strictly real-time (PAP §10, S1; Section 3); row R32 uses the May 2023 weights, public at formation.

Why randomisation inference is not primary. The spec named randomisation inference over permutations of exposure; run as specified on 378 overlapping pre-seal windows, where later exposure can bear no relation to alphas, it rejected at nominal 0.025 in 13.8 % of windows for H5 and 14.8 % for H5b; the Freedman–Lane scheme for H5b (Freedman and Lane 1983; Winkler et al. 2014) rejected in 5.6 % scored by the pivotal t and 6.1 % scored by the coefficient. Because industries with similar exposure co-move, permutation, which treats industries as exchangeable, understates the null spread of the slope. Randomisation inference with 10,000 permutations is kept as secondary, printed with its documented size and never used for a decision.

The historical-null test (D10) and its null (D21). For variant v at Holm level L,

$$T_v = \frac{\hat{b}_v - b_v^\circ}{s_v}, \qquad c_{v,L} = \max\left(t_\nu^{(L)},\ q_v^{(L)}\right),$$

where s is the standard deviation of the same slope for the frozen exposure vector over the 378 overlapping windows of 1990-2022; $\nu = 21$, allowing for the overlap; and the critical is the larger of the two-sided t critical and the empirical quantile q of the centred historical distribution. With the frozen values the t critical binds at both steps, 2.41 at 0.025 and 2.08 at 0.05; s is 2.27 pp for H5 and 2.56 pp for H5b. Under D21 the null value b° is $\bar{b}$, the frozen mean historical slope for the same vector (1.65 pp for H5, 1.22 pp for H5b), because the zero null was oversized on the historical windows. A rejection says the slope departs from the 1990-2022 pattern for this vector, a narrower claim than the approved wording; the zero null is reported as secondary. Because $\bar{b}$ is not zero, the $\bar{b}$ null makes a positive finding harder to reach and a negative one easier than the zero null does, a property of the test's geometry, not an interpretation of the sign of $\bar{b}$. Holm over {H5, H5b} tests the smaller p at 0.025 and the other at 0.05 only after a first rejection (Table 3).

The R13 co-condition. R12 (ranks) and R13 (dropping Softw) are co-primary because the Design B scores are skewed (2.571, with Softw at 4.739 SD). The claim that delegation exposure is priced requires the hypothesis and its R13 variant both to reject under the same null and Holm level, R13 with its own scale and critical; otherwise the fixed wording is that the result is not robust to dropping Softw, which does not attribute the result to one industry: R13's larger scale leaves it less power, with minimum detectable effects of 10.46 pp (H5) and 12.15 pp (H5b).

Intervals, wording and power. Each result carries two intervals, the estimate plus or minus each Holm-level critical times s. A non-rejection is worded exactly as the script prints it (P14): the null, the interval, and that the test had 80 % power only against departures of at least the MDE, which uses the test's own scale and first-step critical:

$$\mathrm{MDE}_v = \left(c_{v,L^*} + t_\nu^{[\pi]}\right) s_v,$$

with L* = 0.025 and the t quantile at power π, so that the MDE is 3.2729 times s: 7.43 pp for H5 and 8.38 pp for H5b. Under D11, if the H5 MDE exceeds 5 pp per SD, RQ3 is reported as "what the market evidence can and cannot yet say" and the paper is led by RQ1–RQ2. As P9 records, the threshold is not in the spec: committed in the plan file before the power analysis first ran, it became binding only when Ajai confirmed it after the MDE was known, so it is a threshold confirmed with the MDE in view. With a primary MDE of 7.43 pp the rule fires (Section 7).

Design A. The spec formed Design A from the earliest release, but r1 (2025-02-10) has no task-level collaboration split, so correction S3 uses r2 (2025-03-27), which makes the spec's Design A, `A_r2`, numerically identical to Design B. `A_pooled`, built from r2 to r6, is an addition that uses releases published during and after the return window and is labelled look-ahead. Every Design A analysis, with returns from 2022-11-30, is exploratory, as are the event studies, terciles and Fama–MacBeth

regressions; none carries family-wise error control, and each is labelled in the same sentence wherever it appears (Section 7).

4.5 H6: talk and use

H6 is exploratory (D18). An industry's talk index is the share of its filers in the EDGAR full index (U.S. Securities and Exchange Commission 2026) whose 10-K filed in 2024 (2024-01-01 to 2024-12-31, all public before formation) the SEC's full-text search returns for at least one primary term of the fixed AI dictionary (D12; retrieval semantics in Section 3). Divergence is the Spearman correlation of talk with Design B usage exposure, with an industry-bootstrap interval; the gap, the standardised residual of talk on usage exposure, enters the market regression under recomputed historical-null inference. Every H6 p-value is nominal and outside any testing family (Section 7).

4.6 Robustness reporting rules

The rules of PAP §8.1 bind every robustness row (Section 8, Section D): every row is reported in full; none changes a confirmatory decision except R13, the co-condition above; rows are never summarised by counting them by significance and carry no error control across rows; market rows recompute s, $\bar{b}$ and q for their exposure vector or sample by the same pre-seal procedure, and RQ1 rows use the inference of the hypothesis they vary. Every market robustness read followed the single confirmatory run.

4.7 Departures from the spec

PAP §10.2 lists the departures from the design spec, reproduced in order in Section B beside the corrections of fact S1 to S5 (PAP §10.1):

- the historical-null test (D10) replaces randomisation inference as primary for H5 and H5b, because of its documented size; randomisation inference is secondary;
- the wild cluster bootstrap is not used for H5 and H5b, which have no cluster structure, but is used across releases for H2 (D22);
- the industry crosswalk splits equally (D23) rather than by employment; the spec's version is R33;
- H1 standard errors are clustered by occupation and task rather than occupation alone; the spec's version is R38;
- H5 and H5b are tested against the frozen historical slope $\bar{b}$ rather than zero (D21), a narrower claim than the approved wording, which Ajai approved;
- H2 is reworded as "measured within-task automation rises" and tested outside the RQ1 Holm family with the release-level co-condition (D22), on r2 to r5 (D2);
- the spec's model-launch dates are fixed by an objective rule over a named source (D19).

5 Behaviour: who delegates, for what, and is it changing? (RQ1)

This section reports RQ1 in the order the pre-registration fixes (Section 4): first a descriptive anatomy of how conversations divide between the automation and augmentation patterns, then the four confirmatory hypotheses. AEI data are conversations matched to O*NET tasks, so every number below describes how those conversations are distributed across tasks, occupations and countries; none describes the people who hold the occupations. RQ1's family-wise α of 0.05 is split as registered: H2 is tested on its own at 0.0125, and {H1, H3, H4} by Holm at family-wise 0.0375. Table 1 gathers every confirmatory coefficient with its unit, standard error, family p-value, Holm level and decision, and Figure 1 draws the same estimates with their confidence intervals. Each hypothesis is reported below with its decision first, then its estimate, interval and p-value, whatever the decision.

Table 1: Confirmatory RQ1 coefficients with units, standard errors, family p-values, Holm levels and decisions (PAP §4.0 D22)

hypothesis	coefficient	tested	unit	estimate	SE	95% CI	p	df	family p	level	status	P13 attenuation
H1	human_education_years_mean	yes	pp of automation per year of education	-0.5348	0.5849	[-1.6848, 0.6152]	0.3611	387	0.3611	0.01875	not_rejected	—
H1	log_wage	yes	pp of automation per unit of log wage (= 100 log points)	4.3952	2.1993	[0.0711, 8.7192]	0.0464	387	0.3611	0.01875	not_rejected	0.9978
H2	trend	yes	pp per year	4.9792	1.1586	[2.7063, 7.2520]	1.85e-05	1314	1.85e-05	0.0125	not_rejected	—
H3	error_cost	yes	pp of validation share	-0.1763	0.4689	[-1.0983, 0.7456]	0.7071	390	0.7071	0.0375	not_rejected	1.0000

hypothesis	coefficient	tested	unit	estimate	SE	95% CI	p	df	family p	level	status	P13 attenuation
H4	log_aui	yes	pp of automation per unit of log AUI (= 100 log points)	-1.1920	0.4041	[-1.9840, -0.4000]	0.0032	normal	4.83e-24	0.0125	supported	—
H4	log_income	yes	pp of automation per unit of log GDP per working-age person (= 100 log points)	-0.9208	0.3533	[-1.6133, -0.2283]	0.0092	normal	4.83e-24	0.0125	supported	—
H4	mix	control	pp per pp	1.1684	0.4210	[0.3432, 1.9935]	0.0055	normal	—	—	—	—

Source: `docs/results/rq1/confirmatory_table.md`, committed unedited; the numbers are as printed there.

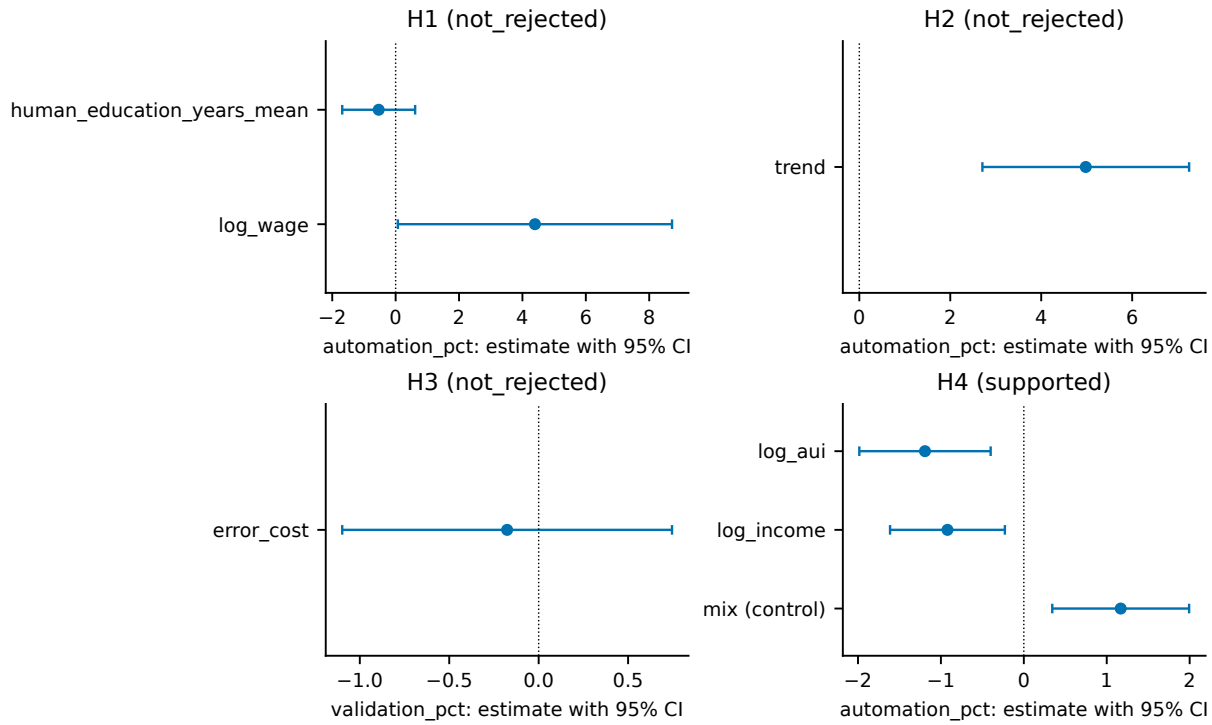


Figure 1: Confirmatory RQ1 coefficients with confidence intervals and each hypothesis's decision (PAP §4).

5.1 The anatomy of collaboration

Figure 2 and Table 8 (Section F) show the five collaboration patterns by SOC major group in r6 `claude_ai`. They are computed over published cells only, on AEI's published occupation basis (D9), and carry no test or p-value. Usage is concentrated in Computer and Mathematical occupations, which hold 24.70 per cent of published major-group usage, and in that group the automation patterns (directive plus feedback loop) account for 63.89 per cent of classified conversations. The next largest groups lean the other way. Arts, Design, Entertainment, Sports and Media (13.73 per cent of usage) has an augmentation share of 64.00 per cent. Educational Instruction and Library (12.77 per cent of usage) has 57.09 per cent. The highest automation shares belong to groups with little usage: Protective Service at 75.42 per cent and Installation, Maintenance and Repair at 68.76 per cent, with usage shares of 0.35 and 0.61 per cent. Legal occupations (65.07 per cent augmentation) and Healthcare Practitioners and Technical occupations (58.43 per cent) sit on the augmentation side. The validation pattern, the outcome of H3, is a small part of every group's conversations (Table 8).

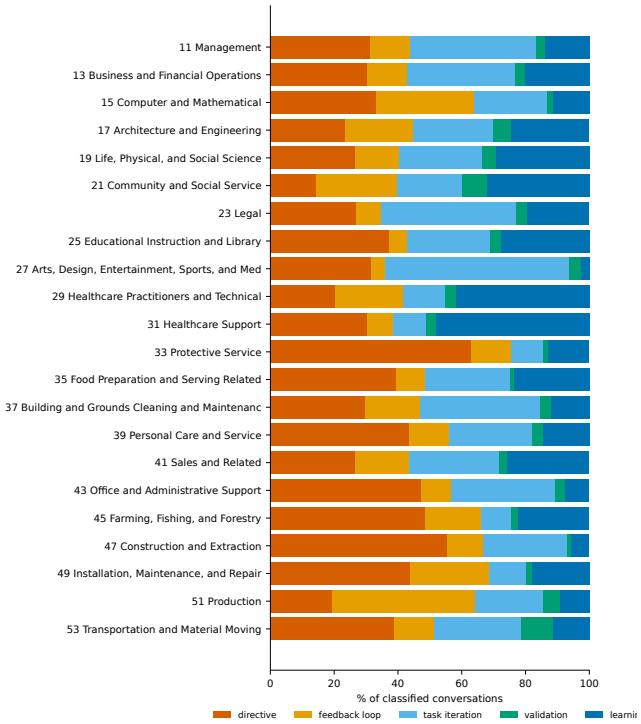


Figure 2: Collaboration patterns by occupation major group, r6 claude_ai (descriptive).

By wage, usage and employment are distributed very differently (Figure 9, Table 9; Section F). The top decile of the OEWS median wage holds 27.13 per cent of conversations matched to OEWS occupations against 9.07 per cent of employment. The bottom decile holds 2.32 per cent against 9.98 per cent. The automation share does not move monotonically with the decile: it is 36.65 per cent in the bottom decile, 54.48 in the second, 39.38 in the third and 56.29 in the top decile. This raw gradient pools all task families. H1 asks the narrower, within-family question, so the two need not agree.

By the education a task requires (Figure 10, Table 10; Section F), usage sits mostly in the two lowest bins, which hold 45.38 and 38.45 per cent of usage. The highest bin holds 0.02 per cent. The automation share is 48.28 per cent in the lowest bin, 55.86 per cent in the middle bin and 37.11 per cent in the second-highest bin, so the descriptive education profile has no single direction either. Tasks published without an education value carry 0.00 per cent of usage: they are the tasks published with zero usage.

Usage is also concentrated across tasks (Figure 11, Table 11; Section F). In r6 `claude_ai`, the largest 54 tasks hold half of the usage over 2821 published matched tasks; in r6 `1p_api` the figure is 43 tasks over 2426. The as-published Gini coefficients and top-share columns are not comparable across releases. r6 publishes usage to coarser precision and keeps its zero-usage tasks in the count. Table 12 therefore puts every release on r6's rounding and drops zero-usage tasks. Even there the suppression regimes differ by release (PAP §5), so differences between releases are not read as changes in the conversations themselves.

5.2 H1: education, wages and the automation share

H1 is not supported at the pre-registered level: the null that β_e or β_w is zero is not rejected at 0.01875, the Holm level at which H1 was tested (Table 1). Within O*NET Intermediate Work Activities, β_e is -0.53 pp of automation per year of education (confidence interval [-1.68, 0.62], p = 0.361). β_w is 4.40 pp of automation per unit of log wage, that is per unit of the log rather than per log point (interval [0.07, 8.72], p = 0.046). H1's intersection–union p-value, the larger of the two, is 0.361. The sample has 1684 rows, 1142 task clusters and 388 occupation clusters. β_e has the predicted sign, but its interval includes zero. β_w is opposite in sign to H1's prediction and is not significant at its Holm level. Under PAP §0.2 and clarification C-21 on mixed signs, it therefore neither supports H1 nor contradicts it, and its nominal p-value is not read as evidence at any registered level. The P13 attenuation factor beside β_w is 1.00. It is the between-task share of the within-family wage variation, so the spread of wages across a task's occupations removes little of the identifying variation. Both coefficients compare tasks, weighted by the occupations they are allocated to, and do not compare workers (Section 4).

Two pre-registered robustness rows are named beside this decision (Section 8, Section D). R24 r5 re-estimates H1 on release r5 `claude_ai`. Under it, H1's p-value, 0.011, falls below H1's Holm level with both coefficients negative: β_e is -1.11 and β_w is -5.55. As a robustness row it changes no decision (PAP §8.1). R23 replaces the automation share with the mean AI-autonomy score, the other delegation measure that H1 names, in autonomy scale points per unit of each regressor. PAP §4.1 prints a disagreement statement only if each R23 coefficient is opposite in sign to the primary coefficient and nominally significant (PAP §11, C-77). The education coefficient is opposite in sign, at 0.03 (p < 0.001). The log-wage coefficient has the same sign as the primary, at 0.03 (p = 0.177). The condition therefore does not hold, and the statement is not printed.

The following are a post-result diagnostic (PAP §11 D-3); exploratory; not a decision. Table 13 (Section F) re-estimates H1 with groups of occupations left out. Leaving out SOC major group 15 (Computer and Mathematical, 25.81 per cent of H1's row weight) moves β_e to -0.01 and β_w to 0.34, with an H1 p-value of 0.978. Leaving out the largest occupation, 15-1299 (8.49 per cent of the weight), moves β_w to 1.50. Leaving out the three largest (15-1299, 25-4022, 27-3043; 17.10 per cent) gives β_e of -1.06, β_w of 2.75 and an H1 p-value of 0.096. β_w thus rests largely on the computing occupations. Among the pre-registered robustness rows, those that move the coefficients most are as follows. R26 with GWA fixed effects gives β_e 0.23 and β_w 5.57; R26 with DWA fixed effects gives -0.52 and 2.15. Under R24 r4 and R24 r5, β_w changes sign, to -3.60 and -5.55. R24 `r6_1p_api` gives 0.38 and 1.36. R10's upper bound gives -0.86 and 3.98 (Section D).

5.3 H2: measured within-task automation across releases

H2 is not supported at the pre-registered level: the release-level co-condition fails. β is 4.98 pp per year (interval [2.71, 7.25], task-clustered p < 0.001) on the balanced panel of 1315 tasks and 5260 task-release rows over r2–r5. That meets the task-clustered condition and has the predicted sign. The release-level wild cluster bootstrap, over all 1296 enumerated weight vectors, gives p

= 0.927, above 0.0125. The release-label randomisation over 24 orderings gives p = 0.708 and is descriptive only. The trend is identified from four release cross-sections. As registered (D22), the co-condition is demanding and may be conservative: only a trend that is large relative to release-to-release variation can meet it (Section 4). β is the change in measured within-task automation. It includes any change in AEI's classifier, which no release names, and in platform scope (D2, D22; PAP §5). The estimate is therefore not a change in behaviour alone.

The event-style plot (Figure 3) shows where the trend comes from. Relative to r2, the release coefficients are 8.45 pp for r3 (interval [6.96, 9.93]), 5.27 pp for r4 ([3.21, 7.32]) and 5.22 pp for r5 ([3.20, 7.23]). The measured rise is one r2-to-r3 step followed by a fall. That step coincides with a change in the base of classified conversations. r2's shares are renormalised without `filtered`, while r3–r5 exclude `not_classified` and `none` (PAP §5). Classifier or platform drift therefore cannot be separated from behaviour. The next results are a post-result diagnostic (PAP §11 D-3); exploratory; not a decision. In Table 14 (Section F), leaving out r2 reverses the sign of β to -6.55 pp per year (task-clustered p < 0.001, wild p = 0.204). Leaving out r3, r4 or r5 instead gives 5.95, 5.22 and 8.44. Two pre-registered robustness rows run the other way. R22 `api_r3_r5`, the `1p_api` series over r3–r5, gives -9.51 pp per year ([-12.75, -6.27]). R23's H2 part, the autonomy score over r4–r6, gives -1.24 autonomy scale points per year ([-1.40, -1.09]). The latter is labelled confounded with the r6 suppression-regime change (Section D).

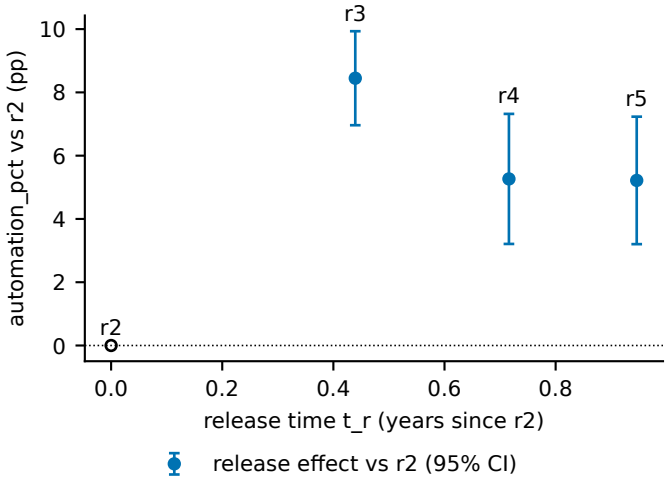


Figure 3: Within-task automation-pattern share by release, the H2 release effects with confidence intervals (PAP §4.2).

The balanced panel also has a cost, reported here as descriptive (Table 14; Section F). The D2 rule keeps only tasks published in all four releases. It removes 890 tasks that are published in some but not all of them, holding 5.39 per cent of the unbalanced panel's task weight. R22's unbalanced row, which keeps them, gives 5.00 pp per year ([2.72, 7.27]; Section D).

5.4 H3: validation in high-error-cost occupations

H3 is not supported at the pre-registered level: the null that γ is zero is not rejected; Holm stopped at H1, and p exceeds even H3's level, 0.0375 (Table 1). Within O*NET Intermediate Work Activities, γ is -0.18 pp of validation share (interval [-1.10, 0.75], p = 0.707). The sample has 1689 rows, 1147 task clusters and 391 occupation clusters. The error-cost occupations are the legal, health-practitioner and regulated-finance codes fixed in D15. The point estimate is negative, opposite to H3's predicted direction, and its interval includes zero. Under PAP §0.2 it therefore neither supports nor contradicts the hypothesis. The P13 attenuation factor is 1.00, the between-task share of the error-cost indicator's within-family variation. The pre-registered rows that vary H3 are reported in full in Section 8, and no generated file ranks them. R27, with the continuous consequence-of-error measure, gives -0.16 ([-0.57, 0.25]). R31, with H1's covariates added, gives -0.20 ([-1.02, 0.62]). R38, with occupation-clustered errors, gives -0.18 (p = 0.702). R10's upper bound gives -2.52 ([-6.26, 1.21]).

5.5 H4: geography

H4 is supported: Holm over {H1, H3, H4} at family-wise 0.0375 rejects H4 at its first step, at level 0.0125, with a joint HC3 Wald p < 0.001 (Table 1). The sample is 113 countries in r6 `claude_ai`. b_1 is -1.19 pp of automation per unit of log AUI (interval [-1.98, -0.40], p = 0.003). b_2 is -0.92 pp of automation per unit of log GDP per working-age person (interval [-1.61, -0.23], p = 0.009). Both log coefficients are per unit of the log, not per log point. H4's direction was left open, and the negative signs are a description, not a registered prediction. Conditional on the task-mix control, countries with a higher usage index and higher income per working-age person have a lower share of classified conversations in the automation patterns. The two regressors share the working-age population in their denominators and are strongly correlated across countries. b_1 and b_2 are therefore only weakly separable one at a time, although the joint Wald test on which H4 is decided is not affected. The task-mix control's coefficient is 1.17 pp per pp ([0.34, 1.99]); it is a control and is not tested. Under the pre-registered robustness row R28 for US states, the coefficients are -0.71 for log AUI (p = 0.240) and -2.61 for log income (p = 0.083). The other R28 rows are in Section D.

The shift-share maps split each geography's gap from the global automation share into a task-mix part and a within-task part. They are descriptive, with no test attached (Figure 4; Figure 12 in Section F). Across countries, the within-task part spans a wider range than the mix part. In the within-task map, Western European countries, Canada and Australia sit below the global share, and several countries in sub-Saharan Africa and South Asia sit above it. Grey marks a country that is not published, and hatching marks one not published in both months; neither is ever shown as zero. Figure 13 and Figure 14 (Section F) draw the same split for US states. The states are not part of the H4 sample, and their regression is the R28 row above.

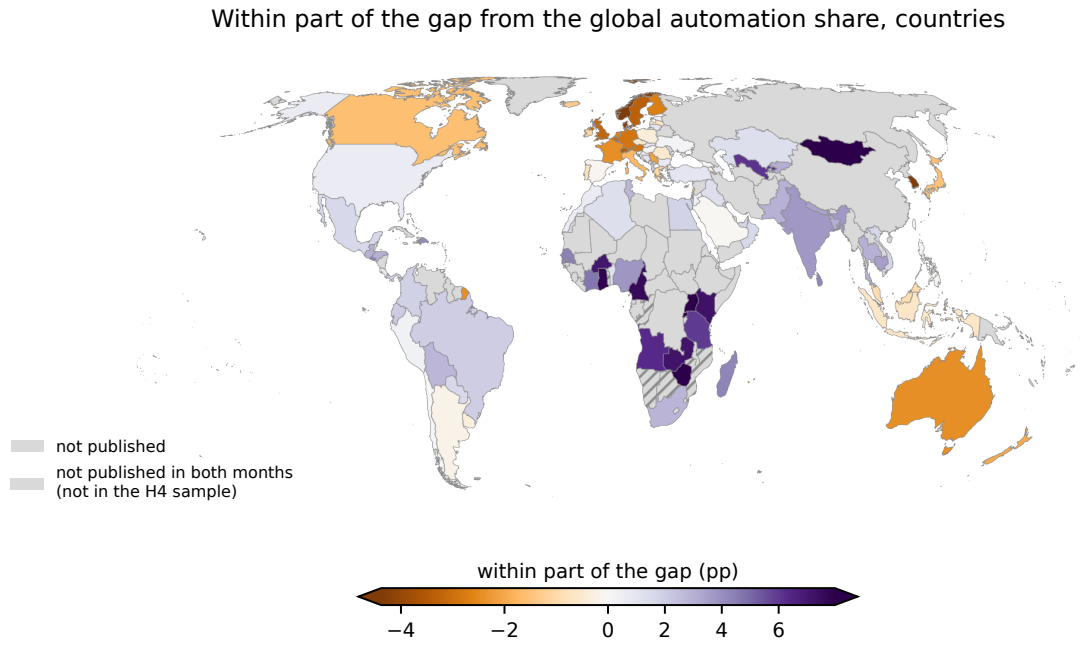


Figure 4: Within part of the gap from the global automation share, by country (descriptive).

5.6 The RQ1 decisions together

Table 1 records the Holm sequence over {H1, H3, H4}. H4 is tested first, at 0.0125, and is supported. H1 is tested next, at 0.01875, where its null is not rejected, so the procedure stops there; H3's level is 0.0375. H2 is tested alone at 0.0125 and is not supported. A hypothesis that is not supported at the pre-registered level is not evidence that its coefficient is zero. The intervals above show which values the data leave open. For β_w and for H2 these include sizeable positive associations. The post-result diagnostics above (PAP §11 D-3; exploratory; not a decision) show that β_w depends on the computing occupations and that the H2 trend depends on the r2 release. How H1–H4 respond to every pre-registered robustness row is set out in Section 8, and the measurement limits behind H2 are discussed in Section 9.

6 Bridge: industry exposure, "could" versus "does" (RQ2)

RQ2 asks a descriptive question: across the 49 FF49 industries, how far does the ranking by observed usage of the model stand from the ranking by theoretical exposure? Nothing in this section is a hypothesis test or a decision (PAP §2.0). The AEI side of each comparison is built from conversations matched to O*NET tasks, carried to occupations and then to industries; the "could" side is a published exposure score carried to the same industries. Definitions are in Section 3 and Section 4; the values are for Design B.

6.1 Three industry indices

The three indices, `z_usage`, `z_delegation` and `z_augmentation`, share one construction with OEWS May 2024 weights (Section 3): delegation and augmentation exposure are usage exposure split by collaboration mode, and they sit close to each other in rank. Softw, Books, Aero, Guns and Hardw head all three rankings, and Meals, Rtail, Cnstr and Coal sit at the foot of all three (Table 15, Table 16, Table 17; Section F).

The distribution of the industry scores is strongly skewed. Softw holds the largest delegation z-score, 4.739 SD, against 1.977 SD for the next industry and -0.834 SD at the bottom; the skewness of the 49 delegation scores is 2.571, and that of the usage scores 2.284 (Table 7; Section F). Softw's benchmark exposures are also high (Table 16), but on the AEI side it stands much further from the rest of the cross-section. A rank correlation suits such a distribution, since it reads Softw as first and no more (the market tests' ranks and drop-Softw rows answer the same skew, Section 7).

6.2 Could versus does

Table 2 reports the Spearman correlation of each AEI index with each benchmark, with industry-bootstrap percentile intervals over 10000 draws. The headline benchmark is the Eloundou et al. human-annotator β (Eloundou et al. 2024), with the GPT-4 β (`z_eloundou_gpt4`) beside it in every row (PAP D8). The headline correlation of delegation exposure with the human β is 0.716 [0.527, 0.839]; with the GPT-4 β beside it, it is 0.815 [0.676, 0.900]. Against the Felten, Raj and Seamans AIOE (Felten et al. 2021b) it is 0.763 [0.594, 0.866]. AIOE reaches the current SOC codes through the D6 crosswalk (Section 3), which R14 replaces with the published BLS crosswalk (U.S. Bureau of Labor Statistics 2018). Every value here is Design B with OEWS May 2024 weights, which are not strictly real-time (S1, set out below).

Table 2: Could versus does: Spearman correlation of AEI exposure with the benchmark exposures across FF49 industries, Design B, primary (descriptive; PAP §2.0)

aei	benchmark	n	spearman	ci_low	ci_high	n_draws_used
z_usage	z_aioe	49	0.809	0.661	0.895	10000
z_usage	z_eloundou	49	0.768	0.600	0.873	10000
z_usage	z_eloundou_gpt4	49	0.843	0.721	0.913	10000
z_delegation	z_aioe	49	0.763	0.594	0.866	10000
z_delegation	z_eloundou	49	0.716	0.527	0.839	10000
z_delegation	z_eloundou_gpt4	49	0.815	0.676	0.900	10000
z_augmentation	z_aioe	49	0.825	0.692	0.901	10000
z_augmentation	z_eloundou	49	0.783	0.624	0.883	10000
z_augmentation	z_eloundou_gpt4	49	0.850	0.736	0.915	10000

Source: `docs/results/rq2/could_vs_does.md`, committed unedited; the numbers are as printed there.

For usage exposure the correlations are 0.768 [0.600, 0.873] with the human β, 0.843 [0.721, 0.913] with the GPT-4 β and 0.809 [0.661, 0.895] with AIOE; for augmentation exposure they are 0.783 [0.624, 0.883], 0.850 [0.736, 0.915] and 0.825 [0.692, 0.901] (Table 2; Design B, OEWS May 2024 weights, not strictly real-time, S1). Within each benchmark, delegation exposure has the lowest point correlation and augmentation exposure the highest; within each AEI index, the human β gives the lowest and the GPT-4 β the highest, with AIOE between them. These orderings describe point estimates only: the intervals are wide for 49 industries, overlap throughout the table and do not test one correlation against another.

Figure 5 plots delegation exposure against each benchmark, one point per industry; most industries lie along a common increasing band. Softw sits apart at the top right of every panel; the industries that sit off the band in the other direction are discussed below.

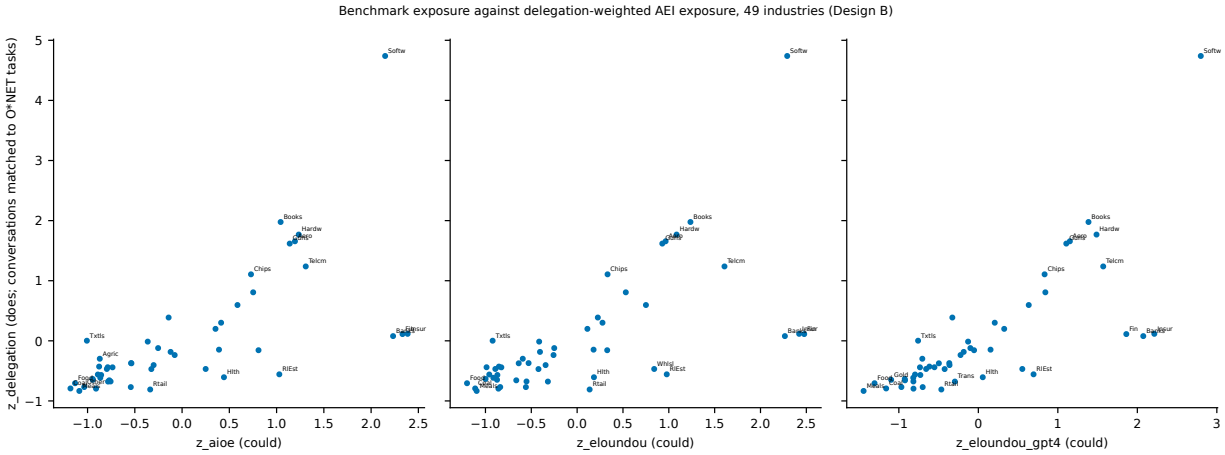


Figure 5: Could versus does: benchmark exposure against delegation-weighted AEI exposure across FF49 industries, Design B (descriptive; PAP §2.0).

6.3 What the correlation shares, and what it assumes

Two features of the construction bear on how far apart "could" and "does" appear. First, both sides of every ρ inherit the same OEWS occupation-to-industry weights: the AEI indices and the AIOE and Eloundou benchmarks are each aggregated to industries with the same matrix (PAP §11). Two industries that employ similar occupations therefore receive similar scores on both sides, whatever the occupation-level agreement between usage and theoretical exposure, so part of each correlation in Table 2 is common occupational mix rather than agreement between the two kinds of measure. Second, ρ depends on the D3 zero assumption. Occupations with no AEI match are unpublished in AEI and are assigned zero usage by assumption (D3); they are not measured as zero. They hold 12.14% of OEWS employment under Design B (Section 3). R3 drops them instead, and R3 lowers ρ most among the RQ2 robustness rows: the headline correlation of delegation exposure with the human β falls to 0.568 [0.321, 0.754] (Section D; Design B, OEWS May 2024 weights, not strictly real-time, S1).

The timing caveat S1 applies to every Spearman in this section (PAP §10 S1). Design B's OEWS

May 2024 weights were released on 2025-04-02, after the 2025-03-27 formation date and inside the return window, and the O*NET 29.3 task links also postdate formation, although their task text is byte-identical to that of 29.2 (PAP §11, C-74). Design B is out of sample in its AEI data only; it is not strictly real-time (PAP §2.1). The strict real-time row R32 uses OEWS May 2023 weights; it is a market row and reports no RQ2 Spearman, so no RQ2 value under strict real-time inputs exists (Section 7, Section D). R1, with OEWS May 2025 weights, is labelled look-ahead; its headline correlation is 0.724 [0.542, 0.846].

6.4 Where "could" and "does" part

Table 18 (Section F) lists the industries whose rank on delegation exposure differs most from their rank on the human β, with the GPT-4 β and AIOE versions in Table 19 and Table 20 (Design B, OEWS May 2024 weights, not strictly real-time, S1). The industries that rank higher on "could" than on "does" are trade, real estate, health, transport and finance: Rtail, RlEst, Whlsl, Hlth and Trans head the list against the human β, Rtail and RlEst head it against all three benchmarks, and Fin, Banks and Insur sit near the top of each theoretical ranking but in the upper middle of the delegation ranking. The industries that rank higher on "does" than on "could" are mostly manufacturing: against the human β the list ends with Txtls and Autos, preceded by FabPr, Steel, Mach and BldMt, and Txtls ends it against all three benchmarks. With the correlations, the picture is broad agreement in the ordering of industries, with sizeable disagreement for some service and finance industries on one side and several manufacturing industries on the other.

The delegation index also carries measurement uncertainty from AEI's suppression of small cells. Table 21 and Figure 15 (Section F) report each industry's delegation exposure between its lower and upper suppression bounds. Relative to the index level, the bounds are widest for Other, Rtail, Coal and Cnstr, which lie at the bottom of the delegation ranking, and narrowest for Softw, Txtls, Hardw and Books. Rtail thus shows both a large positive rank gap and one of the widest relative bounds; the tables record both and attach no interpretation to their coincidence.

Exploratory and look-ahead: Design `A_pooled`, which pools later releases (PAP §10 S3), gives a correlation of delegation exposure with the human β of 0.770 [0.597, 0.872] and with the GPT-4 β of 0.855 [0.734, 0.923] (Table 22; Section F). Its zero-assigned share under D3 is 6.34% of employment, again unpublished in AEI and assigned zero by assumption, not measured as zero.

6.5 Robustness rows for RQ2

The RQ2 rows are robustness, not decisions (PAP §8.1; Section 8, Section D). For the headline pair, delegation exposure against the human β, R1 (look-ahead) and R3 are given above; R2 (wage-bill weights) 0.786 [0.629, 0.886]; R11 (employment-weighted task allocation, D4) 0.721 [0.533, 0.845]; R20 (the `majority` crosswalk mode in place of `split`) 0.714 [0.524, 0.835]; and R33 (the employment-weighted NAICS to SIC crosswalk, D23) 0.708 [0.515, 0.829]. Each except R1 is a Design B construction under the S1 timing caveat above; R1 is look-ahead.

Three rows leave the headline pair at its primary value, for reasons the PAP records. R14 changes

only the AIOE crosswalk (D6), so it moves the AIOE column, to 0.766 [0.597, 0.867] for delegation exposure, and leaves the Eloundou columns at 0.716. R15 redistributes NAICS head-office employment by the same fraction of every industry, which leaves every rank, and so every Spearman, at the primary value of 0.716 (PAP §11, C-70). R17 makes the GPT-4 β the headline and changes no value: delegation exposure against the GPT-4 β stays at 0.815 (PAP D8; Section D).

7 What the market evidence can and cannot yet say (RQ3 and H6)

The title of this section was fixed before any post-seal industry return was read by code, by the pre-registered plausibility rule D11 (Section 4; Section A), which compares the minimum detectable effect (MDE) of the primary H5 test with a threshold of 5 pp of annualised alpha per SD of exposure. At 80% power and Holm's first-step level of 0.025, the primary MDE is 7.43 pp per SD. It exceeds the threshold, so the rule fires: RQ3 is reported under this title, and the paper is led by RQ1 and RQ2 (Section 5, Section 6). The threshold was written into the plan before the power analysis first ran but became binding only when confirmed with the MDE in view (PAP §7). The MDE is printed beside every confirmatory market null below; no MDE is computed for the exploratory ones.

7.1 The confirmatory test of H5 and H5b

Table 3 reproduces the record of the single confirmatory run (Section 4), which used all 49 FF49 industries (dropped for a missing return: none) and 356 factor days from 2025-04-01 to 2026-08-31. The primary null is D21's $b = \bar{b}$ and the statistic $(\hat{b} - \bar{b})/s$ (Section 4). For H5, $\hat{b} = 1.958$ pp per SD against $\bar{b} = 1.650$ and $s = 2.271$, so $T = 0.136$ and $p = 0.8934$. For H5b, which holds the AIOE benchmark (Felten et al. 2021b) in the regression, $\hat{b} = 3.461$ against $\bar{b} = 1.217$ and $s = 2.560$, so $T = 0.877$ and $p = 0.3906$. Neither p reaches the first-step level, so Holm's procedure (Holm 1979) stops there: H5 is not rejected, and H5b is not rejected.

Table 3: Confirmatory test of H5 and H5b and their co-primary R12 and R13 specifications, as recorded by the single confirmatory run (PAP §3.3; D10, D21; Holm over H5 and H5b)

specification	b	s	bbar	b / s	(b - bbar) / s	p primary	p secondary	c_0.025	c_0.05
H5	1.958	2.271	1.650	0.862	0.136	0.8934	0.3982	2.4138	2.0796
H5b	3.461	2.560	1.217	1.352	0.877	0.3906	0.1908	2.4138	2.0796
H5 R12 (ranks)	-1.065	2.211	1.317	-0.482	-1.077	0.2936	0.6350	2.4138	2.0796
H5b R12 (ranks)	-2.386	2.982	0.539	-0.800	-0.981	0.3379	0.4326	2.4138	2.0796
H5 R13 (drop Softw)	1.655	3.195	1.666	0.518	-0.003	0.9974	0.6098	2.4138	2.0796
H5b R13 (drop Softw)	3.634	3.713	1.068	0.979	0.691	0.4972	0.3389	2.4138	2.0796

Source: `docs/results/confirmatory_h5.md`, committed unedited; the numbers are as printed there.

Every number in Table 3 carries the S1 timing caveat (PAP §10): Design B is out of sample in its AEI data only, not strictly real-time. Its OEWS May 2024 weights were released on 2025-04-02, after the formation date of 2025-03-27 and inside the return window, and its O*NET 29.3 task links also postdate formation, although their task text is byte-identical to that of O*NET 29.2 (PAP §11, C-74). R32, the strict real-time row on the OEWS May 2023 weights public at formation, gives $\hat{b} = 2.20$ for H5 (interval [-2.67, 7.08], p = 0.843) and 3.72 for H5b ([-1.75, 9.19], p = 0.374). R1, on the later OEWS May 2025 weights, is look-ahead: 1.96 for H5 and 3.51 for H5b (Section 8).

The co-primary specifications share the run and the table. R12 replaces exposure by its standardised ranks. R13 drops Softw, the industry with the largest Design B `z_delegation` (4.739 SD). Under R12 both slopes are negative: -1.065 for H5 (T = -1.077, p = 0.2936) and -2.386 for H5b (T = -0.981, p = 0.3379). Under R13 they are 1.655 (T = -0.003, p = 0.9974) and 3.634 (T = 0.691, p = 0.4972), each on its own, larger scale (s = 3.195 and 3.713). The claim that delegation exposure is priced in this window beyond the historical pattern required a primary rejection together with an R13 rejection at the same Holm level. No primary is rejected, so the co-condition is not met for either hypothesis.

7.2 The fixed wording

The script's P14 wording of each non-rejection (PAP §3.3), verbatim from the record behind Table 3, with the S1 caveat above (R32):

> H5 is not rejected (H0: b = bbar (the 1990-2022 pattern for this exposure vector)). Interval [-2.76, 6.68] pp per 1 SD; the test had 80% power only against a departure from H0 of at least 7.43 pp per 1 SD (MDE at Holm level 0.025).
>
> H5b is not rejected (H0: b = bbar (the 1990-2022 pattern for this exposure vector)). Interval [-1.86, 8.79] pp per 1 SD; the test had 80% power only against a departure from H0 of at least 8.38 pp per 1 SD (MDE at Holm level 0.025).

The paper does not paraphrase these sentences into a statement that the market did not price delegation. Their null is narrower than the approved hypothesis (Section 4): a rejection would have said that the slope departs from what this exposure vector showed over 1990-2022, not that it differs from zero.

7.3 Intervals, the R13 MDE and the secondary null

Two intervals are reported for each hypothesis, $\hat{b} \pm c \cdot s$ at each Holm level. For H5 they are [-2.76, 6.68] pp per SD at 0.05 and [-3.52, 7.44] at 0.025. For H5b they are [-1.86, 8.79] and [-2.72, 9.64]. The co-conditioned claim has R13's larger MDE at the first-step level, 10.46 pp for H5 and 12.15 pp for H5b, so a pricing claim robust to dropping Softw had 80% power only near those values (Table 3).

The zero-slope statistic from the same run is a secondary null, not part of the decision: T = 0.862 and p = 0.3982 for H5, beside an MDE of 7.43 pp per SD (the primary's s and first-step critical),

and T = 1.352 and p = 0.1908 for H5b, beside an MDE of 8.38 pp. Both carry the S1 caveat (R32).

The robustness rows that move $\hat{b}$ most are R7 (dropping Hardw, Softw and Chips), R12 (ranks) and R5 (equal-weighted returns) for H5, and R12, R7 and R6 (FF30) for H5b; their values are in Section 8 and Section D. Each carries the S1 caveat and none changes a decision (PAP §8.1).

7.4 Secondary inference

Randomisation inference, oversized on the pre-seal windows (Section 4), is kept as secondary, each p printed with its documented size at a nominal 0.025. With 10,000 permutations each, permuting `z_delegation` gives p = 0.3287 for H5 (size 13.8%), and permuting it with `z_aioe` held in place gives p = 0.0942 for H5b (size 14.8%, oversized even under exchangeability). The Freedman–Lane scheme (Freedman and Lane 1983; Winkler et al. 2014) gives p = 0.2099 for H5b scored by the pivotal t (size 5.6%) and 0.2052 scored by the coefficient (size 6.1%). Conventional OLS with HC3 standard errors gives, for reference, 1.958 (SE 1.629) for H5 and 3.461 (SE 1.638) for H5b. None of these enters the decision.

7.5 The influence of Softw

Table 23 (Section F) reports Softw's influence on the confirmatory fits, post-result diagnostics that PAP §3.3 (P11) places outside the decision. Softw's leverage is 0.4788 in the H5 fit and 0.5185 in the H5b fit. Its DFBETAS on b are 0.1475 and -0.0621, and its Cook's distances are 0.0116 and 0.0017. The industries with the largest Cook's distance in each fit, listed in Table 24, are Gold and Hardw, not Softw. This agrees with the R13 estimates lying close to the primaries.

7.6 Exploratory market analyses

Exploratory: Design A reads returns from the market seal, 2022-11-30, to 2026-08-31 (PAP §8.3, D19). `A_r2` uses release r2 alone and is numerically identical to Design B, so the Design B rows below are the `A_r2` rows over a sub-window, not independent evidence. `A_pooled`, which is look-ahead, pools releases r2 to r6, of which r3 to r6 were published during and after the return window (PAP §10); its largest `z_delegation`, again Softw, is 4.320 SD. Design A is exploratory because general knowledge of post-seal market moves cannot be removed (PAP §0.4); nothing here tests H5 or H5b or carries error control.

Exploratory: the event studies use the D19 rule. The events are ChatGPT, which the design specification names, and every flagship model announcement by OpenAI, Anthropic or Google's AI labs in a named source over the period the rule fixes (Table 25; Section F). Day zero is the first trading day on or after the announcement; Table 26 lists moved dates and overlapping windows. The rule's exceptions (excluding Gemma and Gemini Ultra, adding ChatGPT by name, accepting an OpenAI release's absence from the source) were chosen with the source's list, and so the dates, in view (PAP §0.4). Each industry's cumulative abnormal return (CAR) over an FF5+UMD fit (Fama and French 2015; Carhart 1997) is regressed across industries on exposure, with HC3 standard errors.

Exploratory: Table 27 and Figure 16 (Section F) report every event, horizon and design. For ChatGPT, the `A_r2` slope is -0.32 pp of CAR per SD at the shorter horizon (CI [-0.79, 0.16], nominal p = 0.192) and 0.51 at the longer one ([-0.58, 1.60], nominal p = 0.357). The p-values are nominal and single-test, and may be oversized because abnormal returns are correlated across industries on a common date.

Exploratory: the tercile portfolios go long the highest-exposure third of FF49 industries and short the lowest, with membership fixed at formation, monthly capitalisation weights and equal-weighted legs beside them (PAP §8.3, D19). They report CAPM, FF3 and FF5+UMD alphas (Fama and French 1993) with Newey–West errors (Newey and West 1987, 1994), stationary-bootstrap intervals (Politis and Romano 1994), and the probabilistic (PSR) and deflated (DSR) Sharpe ratios (Bailey and López de Prado 2012, 2014). They are measurement devices that express the cross-sectional slope as a return spread, not strategies; the costed versions show the spread's fragility, not a tradable return (Table 29; Section F).

Exploratory: for Design B (secondary) with cap-weighted legs, the annualised mean high-minus-low return is 15.49 pp (bootstrap CI [-5.18, 34.08], nominal p = 0.170). The FF3 alpha is 12.91 pp and the FF5+UMD alpha 0.89 pp ([-8.78, 10.56], p = 0.842). That spread across factor models rests on 17 monthly returns and is model-fragile (Table 28; Section F). The PSR is 0.91 and the DSR 0.75; with equal-weighted legs they are 0.46 and 0.21. Table 4 gives them for `A_r2` and for `A_pooled` (look-ahead). The DSR treats every variant as an independent trial, which overstates the number of independent trials: B is nested in `A_r2`, and `A_r2` and `A_pooled` hold nearly the same portfolios (Figure 17).

Table 4: Deflated Sharpe ratios of the tercile portfolios (exploratory)

variant	monthly SR	Sharpe (annualised)	DSR
B (secondary), cap	0.3373	1.168	0.7473
B (secondary), ew	-0.0267	-0.093	0.2123
A_r2 (exploratory), cap	0.2372	0.822	0.6777
A_r2 (exploratory), ew	0.0970	0.336	0.3147
A_pooled (exploratory, look-ahead), cap	0.2367	0.820	0.6764
A_pooled (exploratory, look-ahead), ew	0.0916	0.317	0.3021

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Exploratory: Fama–MacBeth regressions (Fama and MacBeth 1973) of monthly industry excess returns on exposure, log size, log book-to-market and momentum give a mean exposure slope of 0.23 pp per month per SD for Design B over 17 months (CI [-0.40, 0.87], nominal p = 0.448; Table 40; Section F). For `A_r2` over 45 months it is 0.21 ([-0.03, 0.45], p = 0.087; Table 41). For `A_pooled`, look-ahead, it is 0.17 ([-0.06, 0.40], p = 0.144; Table 42).

Exploratory: the comparison with Eisfeldt, Schubert and Zhang (Eisfeldt et al. 2023), who report higher returns for more exposed firms, asks only for the sign of the CAR slope on `z_aioe` around the ChatGPT release, before `z_delegation` is added. At the FF49 level, which is not their firm-level design, the slope in the first step is -0.14 pp of CAR per SD of `z_aioe` (CI [-0.92, 0.65], nominal p

= 0.736; Table 43; Section F). The sign is negative and the interval contains zero, so their sign is not reproduced at this level of aggregation. Table 44 adds `z_delegation` for `A_r2` and for `A_pooled` (look-ahead).

7.7 Talk versus use (H6, exploratory)

Exploratory: H6 (PAP §6, D18) belongs to no testing family. For the gap, the talk index T (Section 4) uses filings dated 2024-01-01 to 2024-12-31, public before formation. Pooled across all filers, the share rises from 0.082 in the first filing year to 0.375 in 2024 and 0.516 in the last (Table 45, Figure 18). Over the same years the share of filers with no FF49 industry rises from 11.92% to 15.81%, so part of that trend is composition; the mapped-only series runs from 0.093 to 0.591.

Exploratory: across industries, the Spearman correlation of T with Design B `z_usage` (conversations matched to O*NET tasks) is 0.361, industry-bootstrap CI [0.050, 0.612] (Table 46, Figure 6). Its p (t-approximation, n minus two degrees of freedom) is 0.011, nominal and exploratory (PAP §11 D-3; Section B), a post-result diagnostic in no testing family. Two inputs are fixed at retrieval, not in 2024: each filer's industry is its current SIC from the EDGAR submissions file, retrieved at 2026-10-23T01:19:58Z and applied to every filing year (PAP §11, C-40), and the numerator uses the SEC's phrase matching as implemented on the retrieval date (PAP §6).

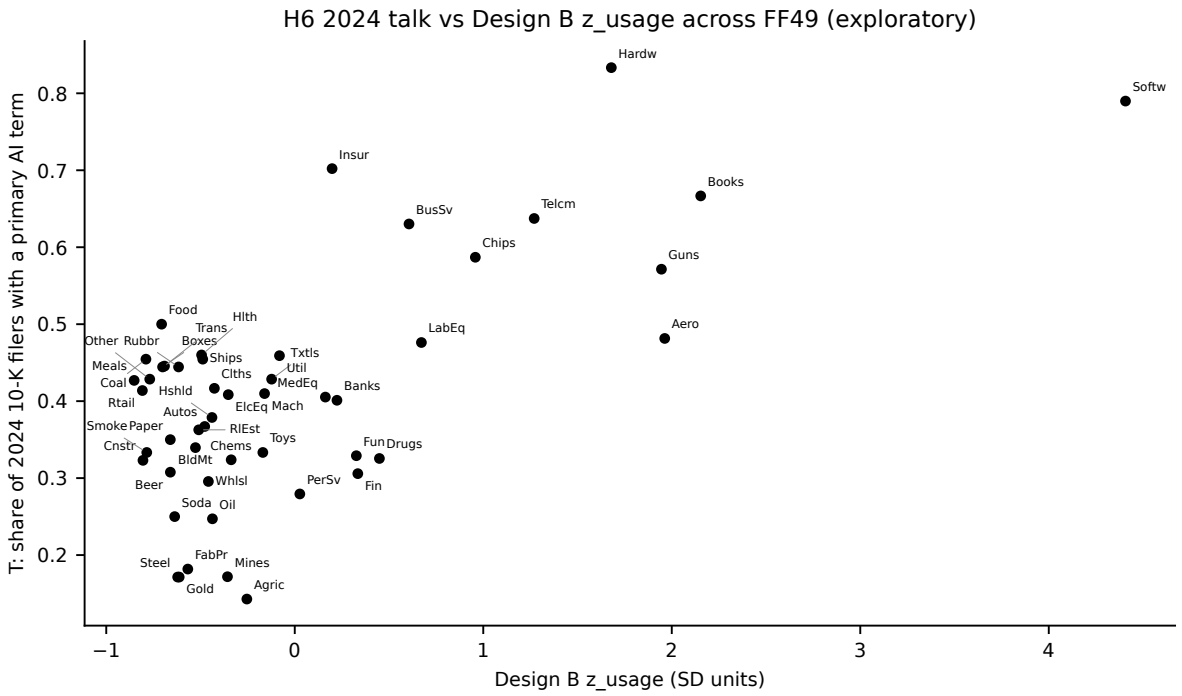


Figure 6: H6 talk index against Design B `z_usage` across FF49 industries for the formation filing year (exploratory).

Exploratory: the leave-one-industry-out Spearman in Table 47 is a post-result diagnostic (PAP §11 D-3); exploratory; not a decision. Leaving out Softw gives 0.320 ([0.001, 0.585]); leaving out Hardw,

0.322 ([-0.006, 0.588]), and BusSv, 0.323 ([-0.002, 0.590]), give intervals crossing zero; every leave-out is in the table. The R29 parts whose intervals also cover zero are the change from the first filing year, float weighting and the pre-formation filings (Section 8).

Exploratory: the gap (Table 48) in place of `z_delegation`, under recomputed historical-null inference (Section 4; Table 49), gives $\hat{b}$ = -2.40 pp of annualised alpha per SD of gap (interval [-6.48, 1.68]; T = -1.28, nominal p = 0.215 against b = $\bar{b}$; against the secondary null of a zero slope, T = -1.22, nominal p = 0.235); no MDE computed (exploratory); in no testing family; Design B's S1 caveat applies.

7.8 What a future window would need

The MDE is 3.2729 × s, with s the spread of the historical slope across 17-month windows (2.27 for H5, 2.56 for H5b; Section 4); for the H5 MDE to reach the 5 pp threshold, a window's s would have to fall to that threshold divided by 3.2729, and a claim robust to dropping Softw would need the same of R13's scale (MDEs 10.46 and 12.15 pp). What the evidence cannot yet say is whether this exposure vector is priced by a departure smaller than 7.43 and 8.38 pp per SD (Table 3).

8 Robustness

The rows (PAP §8.2) and the plan's defaults were fixed before any confirmatory result; their §11 clarifications followed, several after their own rows ran (Section B).

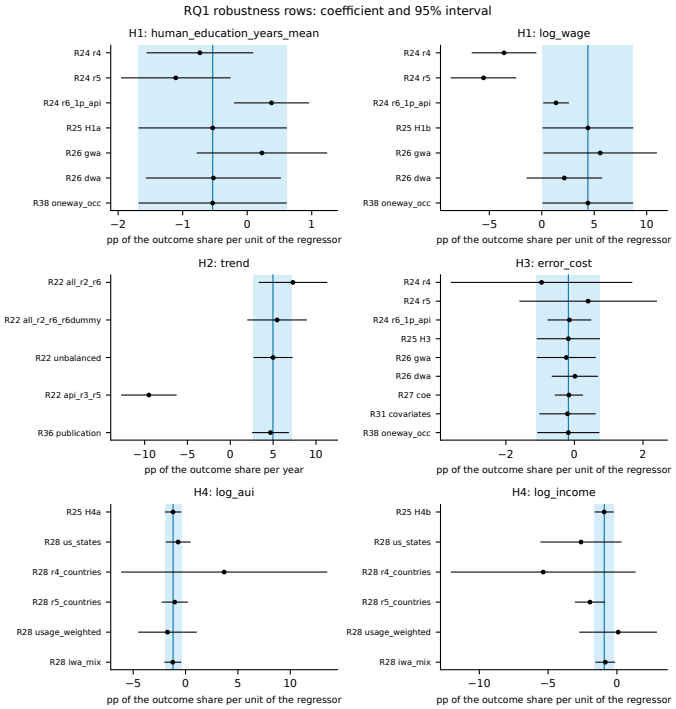


Figure 7: Robustness rows for the RQ1 coefficients beside the primary estimate, with intervals; every row reported (PAP §8.1, §8.2).

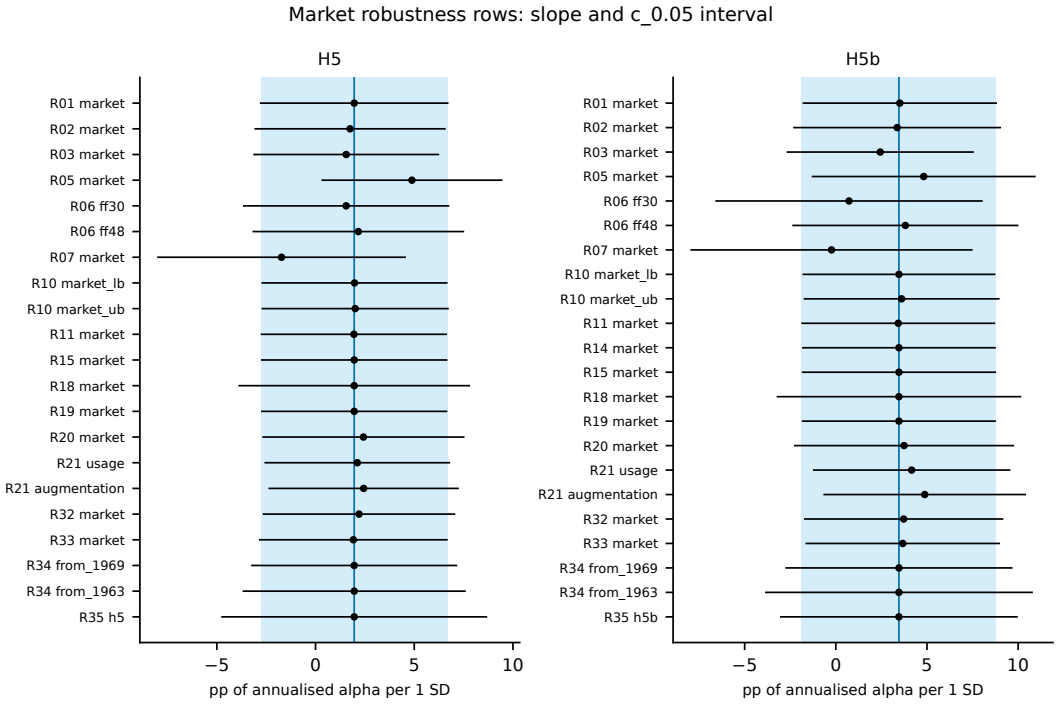


Figure 8: Robustness rows for H5 and H5b beside the primary estimate, with each row's interval; every row reported, no error control across rows (PAP §8.1, §8.2).

8.1 What the rows vary

Construction rows change how the exposure measure is built: the later OEWS May 2025 weights (R1, look-ahead for H5 and H5b), wage-bill weights (R2), dropping rather than zero-assigning occupations without AEI usage (R3, D3), the API source for Design `A_pooled` (R4), the suppression bounds in place of the point measure (R10), employment-weighted task allocation (R11), standardised ranks (R12), the AIOE crosswalk (R14), the headquarters industry redistributed (R15), the r6 task-allocated basis (R16), the GPT-4 Eloundou β (R17), the majority crosswalk mode (R20), other indices in place of `z_delegation` (R21), strict real-time inputs (R32) and the employment-weighted NAICS-to-SIC crosswalk (R33). Sample rows change which industries or days enter: FF30 and FF48 granularity (R6), dropping Hardw, Softw and Chips (R7), excluding the large-cap rally year in Design A (R8), placebo event dates (R9) and dropping Softw (R13). The returns row uses equal-weighted industry returns (R5). Inference rows change the null scale or the statistic: non-overlapping windows (R18), the root mean square around the null value (R19), earlier-starting windows (R34), the studentised statistic (R35) and one-way occupation clustering (R38). RQ1 rows vary the behavioural hypotheses (R22 to R28, R31, R36 and R37), and the H6 rows vary the filing sample and the dictionary (R29, R30).

8.2 Reporting rules

The rules are those of PAP §8.1, fixed in `pap-v1` and set out in Section 4. Every row was run and is reported; Figure 7 and Figure 8 set the RQ1, H5 and H5b rows beside the primaries, and the summary table of Appendix D (Section D) sets each row's estimate beside the primary's. Only R13, a co-condition of the H5 and H5b claim, bears on a decision; no row, R21 included, carries error control across rows, and rows are not tallied. Every market robustness interval here, for H5, H5b and the H6 gap–return slope, is $\hat{b} \pm c \cdot s$ with c the row's critical at 0.05 (R35 uses its own scale and t_{21} critical); no row file prints the interval at the first Holm level, 0.025. No row-level MDE is computed, so this section reports estimates, intervals and p and calls no market row not rejected, which would need the P14 form of Section 7.

8.3 RQ1

H1 is not supported: the null that β_e or β_w is zero is not rejected at 0.01875 (Section 5). Under R24 r5 the H1 p-value, 0.011, falls below that level with both coefficients negative (β_e = -1.11, β_w = -5.55); it changes no decision. The other rows that move the H1 coefficients furthest are R24 r4, where β_w changes sign to -3.60; R24 `r6_1p_api` (β_e = 0.38, β_w = 1.36); R26 with GWA fixed effects (β_e = 0.23, β_w = 5.57) and with DWA fixed effects (β_e = -0.52, β_w = 2.15); and R10 at the upper bound (β_e = -0.86, β_w = 3.98).

H2 is not supported: the release-level co-condition fails. Its measured rise is one r2-to-r3 step followed by a fall, leaving out r2 reverses the sign of β (-6.55; Table 14; post-result diagnostic, PAP §11 D-3; exploratory; not a decision), and the step coincides with a change of the classified-conversation base, so classifier or platform drift cannot be separated from behaviour (Section 5). The rows that run the other way are R22 on the API source over r3 to r5 (-9.51, interval [-12.75, -6.27]) and R23 with `ai_autonomy_mean` (-1.24, interval [-1.40, -1.09]). In the primary's direction lie R22's all-releases series (7.31), labelled confounded with the r6 suppression-regime change, R22's unbalanced panel (5.00) and R36 with publication dates (4.69); R37, descriptive, gives a within-r6 change of -0.13.

H3 is not supported: the null that γ is zero is not rejected, and Holm stopped at H1 (Section 5). No generated file ranks its rows: R10 at the upper bound, -2.52 (interval [-6.26, 1.21]); R27, the continuous Consequence of Error, -0.16 pp per standard deviation; R31, with H1's covariates, -0.20; and R38, -0.18 (p = 0.702).

H4 is supported (Section 5), and its rows are not ranked either. R25 tests the coefficients separately in a five-member Holm family: p = 0.003 for b_1 and p = 0.009 for b_2. R28 changes units and weights: US states (b_1 = -0.71, b_2 = -2.61), r4 countries (b_1 = 3.68, interval [-6.16, 13.53]; b_2 = -5.36), r5 countries (b_1 = -1.03, b_2 = -1.96), usage weights (b_1 = -1.73, b_2 = 0.09) and the IWA-level mix (b_1 = -1.22, b_2 = -0.84).

8.4 Market rows

For H5, $\hat{b} = 1.96$ pp per standard deviation (decision and P14 wording in Section 7); the rows that move $\hat{b}$ furthest are R7, dropping Hardw, Softw and Chips (-1.72, interval [-8.02, 4.57], p = 0.430); R12, standardised ranks (-1.065); and R5, equal-weighted returns (4.88, interval [0.30, 9.47], p = 0.212). For H5b, $\hat{b} = 3.46$; the rows that move it furthest are R12 (-2.386), R7 (-0.24, interval [-7.98, 7.50], p = 0.971) and R6 at FF30 (0.72, interval [-6.61, 8.05], p = 0.827).

Every H5 and H5b number carries the S1 caveat of Section 3 and Section 7: Design B's May 2024 OEWS weights and O*NET 29.3 links postdate the 2025-03-27 formation date, so Design B is out of sample in AEI data only, not strictly real-time. R32, the strict real-time row (May 2023 weights), gives 2.20 for H5 (interval [-2.67, 7.08]) and 3.72 for H5b (interval [-1.75, 9.19]); R1, labelled look-ahead, gives 1.96 and 3.51.

8.5 H6

H6 is exploratory throughout (Section 7). Its divergence Spearman is $\rho = 0.361$ (interval [0.050, 0.612]). Its two retrieval-dated inputs are set out in Section 7. The R29 parts whose intervals cover zero are the change in talk (0.081, interval [-0.213, 0.368]), the float-weighted filers (0.190, interval [-0.121, 0.467]) and the pre-formation filings (0.267, interval [-0.036, 0.525]). R30's narrow and broad dictionaries give 0.361 and 0.364. The exploratory gap–return slope is -2.40 pp per standard deviation of the gap in the primary and -3.37 under the change in talk (interval [-6.97, 0.23]); no MDE computed (exploratory).

8.6 Where the rows are

Appendix D (Section D) prints each row file unedited, the summary table, and the implementation choices where the PAP is silent, grouped by the §11 clarification each cites. It also records the reproduction tolerance for the row files (PAP §11, C-80).

9 Limitations

Every behavioural quantity in this paper is a share of conversations matched to tasks (Section 3): it describes usage of one assistant on one platform, not the conduct of the people employed in an occupation. The wording rule is therefore a substantive limit, binding RQ1 most, where H1–H4 relate usage shares to occupational wages, education, error costs and country income (Section 5): the regressors describe occupations and countries, while the outcome describes only the conversations that reached the platform from them. Usage is also concentrated. In a post-result diagnostic (PAP §11 D-3; exploratory; not a decision), leaving out SOC major group 15, which carries 25.81% of the H1 weight, gives $\beta_w = 0.34$ (p = 0.802) and $\beta_e = -0.01$ (p = 0.978) (Table 13; Section F). The robustness rows that move the H1 coefficients most, R26 `gwa` and `dwa`, R24 r4 and r5, R24 `r6_1p_api` and R10 `ub`, are reported in Section 8.

No AEI release names the model that classifies its conversations, and the platform scope widens across releases (Table 6; Section F). A change in the classifier, or in the conversations it sees, between two releases cannot be distinguished from a change in how people work with the model; this is the first open risk of the data audit, which the published files cannot resolve (Section 3). It bears on every comparison across releases, above all H2 and the exploratory, look-ahead `A_pooled` index, which averages releases measured under different regimes (Section 6).

Within published tasks, AEI withholds cells below its thresholds; the primary measure renormalises over the classified patterns, and the R10 bounds count the suppressed mass as all automation or as none. Across occupations, the width of those bounds on the automation share has a median of 3.97 pp and an interquartile range of 2.65 to 7.31 pp over 512 occupations (Table 50; Section F). Occupations with no AEI usage are a separate case: they are unpublished in AEI, not measured as zero, and are assigned zero by assumption (D3). Under Design B this covers 12.14% of employment (Section 3). The RQ2 correlations depend on that assumption, and R3, which drops those occupations instead, lowers them most (Section 6).

Two further construction choices are equal splits that use no employment information. A task statement shared by several occupations gives each an equal part of its usage (D4): this keeps the usage numerator independent of its employment denominator but produces dispersion within groups of related occupations that share tasks, an open risk of the data audit (Section 3). The preregistered alternative, R11, splits by employment, which removes the dispersion by construction rather than measuring how much of it is an artefact (Section 8). Likewise, the NAICS-to-FF49 concordance splits a NAICS code equally over its SIC codes (D23), so the industry weights are structural rather than employment-based, a departure from the spec's employment-weighted crosswalk (PAP §10.2), which is row R33 (Section 8). The same OEWS occupation-to-industry weights enter both sides of every RQ2 comparison, so part of each correlation is common occupational mix (Section 6).

The time dimension of H2 is four release cross-sections, r2 to r5 (Section 4). Under task fixed effects the trend is identified only across releases, which is why the release-level wild bootstrap is a co-condition. The measured within-task trend is 4.98 pp per year, but H2 is not supported: the release-level co-condition fails at 0.0125 (Section 5). The rise is one r2 to r3 step followed by a fall (Figure 3). Leaving out r2 reverses the sign of β, to -6.55 pp per year, in the post-result diagnostic of PAP §11 D-3 (exploratory; not a decision; Table 14; Section F). The r2 to r3 step coincides with the change in the classified-conversation base, so classifier or platform drift cannot be separated from behaviour, and R22 `api_r3_r5` and R23 run the other way (Section 8). The balanced panel removes 890 tasks carrying 5.39% of usage (descriptive; Table 14).

The market test is a cross-section of 49 FF49 industries, and one of them dominates it. Design B delegation exposure places Softw at 4.739 SD, against 1.977 SD for the next industry, and Softw's leverage is 0.4788 in the H5 regression and 0.5185 in H5b (Table 23; Section F). Its DFBETAS and Cook's distance on the slope are nonetheless small, and the R13 estimates lie close to the primaries; R12 (ranks) and R13 (dropping Softw) are co-primary for this reason, R13 with larger MDEs (Section 7).

The confirmatory returns span one 17-month window, 2025-04-01 to 2026-08-31, a single draw of the cross-section of industry alphas, for which the H5 MDE of 7.43 pp per standard deviation exceeds the D11 threshold of 5 pp (Section 7).

Design B is out of sample in AEI data only, not strictly real-time (S1). Its OEWS May 2024 weights were released on 2025-04-02, after the formation date of 2025-03-27 and inside the return window, and its O*NET 29.3 task links also postdate formation, although their task text matches 29.2 (PAP §11, C-74; Section B). R32 rebuilds the index from inputs public at formation (OEWS May 2023, O*NET 29.2), giving H5 and H5b slopes of 2.20 and 3.72 pp per SD (Section D); R1, on OEWS May 2025, is look-ahead (Section 8). R32 is a market row only, so no RQ2 correlation exists under strictly real-time inputs.

The seal kept code from reading industry returns dated on or after 2022-11-30 until the protocol was tagged (pap-v1, 2026-10-21), but it cannot remove what the researchers, or the language model assisting them, know from public commentary about broad market moves since then. The protection is procedural: the confirmatory test, its exposure vector, null values and script were frozen before any post-seal industry return was read by code, and Design A is exploratory for this reason (Section C). Pre-tag choices were made with that knowledge; no procedure can show it left them untouched.

The H6 talk index counts filers whose 10-K filings EDGAR full-text search returns for a quoted phrase. Its matching semantics are those the SEC implemented on the retrieval date, which this paper does not control; the case, whitespace and hyphen rules bind only the secondary measure. Each filer's FF49 industry is its current SIC from the EDGAR submissions file retrieved at 2026-10-23T01:19:58Z, applied to every filing year, so both inputs are fixed at retrieval rather than in the filing year (Section 7). The secondary measure needed a correction to input data after results were seen, logged as deviation D-1 in PAP §11 (Section B): two truncated 10-K documents were fetched again. H6 is exploratory, and its sensitivity to single industries is the post-result leave-one-industry-out diagnostic (PAP §11 D-3; exploratory; not a decision; Table 47; Section F).

10 Conclusion

RQ1 asked whether delegation varies with stakes, over time, with error costs and across countries (Section 5, Table 1). H1 is not supported: the null that the education or the wage coefficient is zero is not rejected at 0.01875, with an H1 p-value of 0.361 and β_e = -0.53 pp of automation per year of education (-1.68 to 0.62); β_w is opposite in sign to H1's prediction and is not significant at its Holm level. Under the robustness row R24 r5 that p-value (0.011) falls below that level with both coefficients negative, which changes no decision. H2 is not supported: the release-level co-condition fails, with task-clustered $p < 0.001$ beside wild-bootstrap $p = 0.927$ at 0.0125. The measured rise is one r2-to-r3 step followed by a fall, leaving out r2 reverses the sign of β (-6.55 pp per year; post-result diagnostic, PAP §11 D-3; exploratory; not a decision; Table 14), and the step coincides with a change of the classified-conversation base, so classifier or platform drift cannot be separated from behaviour; R22 `api_r3_r5` and R23 run the other way. The balanced panel removes 890 tasks hold-

ing 5.39 per cent of usage (descriptive). H3 is not supported: the null that γ is zero is not rejected (Holm stopped at H1), with γ = -0.18 pp of validation share (-1.10 to 0.75; p = 0.707). H4 is supported: the joint Wald test rejects at 0.0125 ($p < 0.001$), with b_1 = -1.19 pp per unit of log AUI and b_2 = -0.92 pp per unit of log GDP per working-age person, a unit of a log being a hundred log points. The two regressors are strongly correlated across countries, so each coefficient is weakly separable on its own, although the joint test is not.

RQ2 is descriptive (Section 6). Across the 49 industries, delegation exposure's Spearman correlation with the human-rated Eloundou benchmark is 0.716 (interval 0.527 to 0.839) (Table 2). "Could" and "does" order the industries similarly, with qualifications. Both sides share the same OEWS occupation-to-industry weights, so part of the correlation is common occupational mix; and the correlation depends on the D3 assumption that occupations unpublished in AEI carry zero usage, which R3 relaxes, lowering it most (0.568; Section D). Design B is out of sample in AEI data only: its May 2024 OEWS weights were released on 2025-04-02, after the 2025-03-27 formation date and inside the return window, and its O*NET 29.3 links postdate formation, so it is not strictly real-time; R32 is market-only.

For RQ3 the D11 rule fires: the primary MDE of 7.43 pp per standard deviation exceeds the plausibility threshold of 5 pp, a threshold confirmed with the MDE in view, so the market evidence is reported as what it can and cannot yet say (Section 7). Over the 17-month window the results read, in their fixed wording (Table 3): H5 is not rejected (H0: b = bbar (the 1990-2022 pattern for this exposure vector)). Interval [-2.76, 6.68] pp per 1 SD; the test had 80% power only against a departure from H0 of at least 7.43 pp per 1 SD (MDE at Holm level 0.025). H5b is not rejected (H0: b = bbar (the 1990-2022 pattern for this exposure vector)). Interval [-1.86, 8.79] pp per 1 SD; the test had 80% power only against a departure from H0 of at least 8.38 pp per 1 SD (MDE at Holm level 0.025). The two decisions (H5 not rejected, H5b not rejected) rest on the Design B inputs whose timing is stated above. Under the strict real-time inputs of R32 (May 2023 weights), the H5 and H5b slopes are 2.20 and 3.72 pp (Section D); R1 is look-ahead. Neither non-rejection says that exposure is unrelated to returns. Each says that this window could detect only a departure from the historical pattern at least as large as its MDE.

Only a more powerful test, not another reading of this one, could change the market conclusion. Under the pre-registered test the MDE is 3.2729 times s, the across-window spread of the historical slopes (2.27 pp for H5, 2.56 pp for H5b), so a future window would lead the paper with RQ3 only if its own s brought the MDE, here 7.43 and 8.38 pp, to the 5 pp threshold or below. Because R13 is a co-condition of the claim, its MDEs, 10.46 pp and 12.15 pp, would have to fall as well. Until such a window exists, the evidence on how the model is used rests on RQ1 and RQ2.

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Statements

Use of AI tools. This study was carried out with Claude, a language model made by Anthropic, running in Claude Code. Under the author's direction, Claude wrote the code, ran the analyses, drafted the pre-analysis plan (with a recommended option for each decision in its register, all of which the author accepted at signing, Section A) and this paper, and ran the red-team review; separate model instances implemented and reviewed each task. The author set the research questions and hypotheses, signed the pre-analysis plan, approved every deviation from it (Section B) and takes responsibility for the content. No number in the paper is typed: each is generated by the code of the replication package (Section E). The AEI data are conversations with Claude matched to O*NET tasks, so the assistant is a product of the firm whose data and product are studied. Anthropic did not fund, review or approve this study, and the confirmatory tests were fixed by the tagged protocol and its guards before their outcomes were read (Section 4), not by the assistant's judgement afterwards.

Declaration of interests. The author has no employment, funding, consulting or equity relationship with Anthropic, OpenAI or any other AI developer, and holds no position in the industry

portfolios studied beyond broad index funds.

Licences and data. The code is released under the MIT License and the text, results and figures under CC BY 4.0. The raw data keep their sources' terms, which the replication package's README quotes with each source's attribution; the frozen raw archive is deposited as that README describes, with the files whose sources state no licence available on request (Section E). This paper includes information from the O*NET Database by the U.S. Department of Labor, Employment and Training Administration (USDOL/ETA), used under the CC BY 4.0 licence; the author has modified all or some of this information, and USDOL/ETA has not approved, endorsed or tested these modifications.

A Pre-registration

A.1 Signature (PAP §0.1)

Signed: Ajai Upadhyaya, 2026-10-21.

Ajai's statement at signing: "I accept all 23 recommendations in the pre-analysis plan, including D21 (the $b = \bar{b}$ null) and D22 (the α split). Sign it, freeze it and tag pap-v1."

Adopted as recommended: D1–D23. That includes D11 (the 5 pp plausibility threshold, which fires), D21 (the H5/H5b confirmatory null is H0: $b = \bar{b}$, with $b = 0$ secondary; Ajai explicitly approved this narrower claim than the original H5 wording) and D22 (H2 leaves the RQ1 Holm family; α is split, with H2 at 0.0125 and {H1, H3, H4} by Holm at 0.0375).

The signed plan is the git tag `pap-v1`, an annotated tag on commit `6a16b1ad052977650b90c8062ceeeab9ac6572e1` (2026-10-21). Every later change to the plan is an entry of PAP §11 (appendix: Deviations).

A.2 Decision register (PAP §0.3)

ID	Decision	Recommended	Section
D1	Primary delegation measure	renormalised `automation_pct`; R7 bounds as robustness	§2.1
D2	H2 definition-stable metrics and series	`automation_pct`, r2–r5 `claude_ai`, balanced panel, linear trend	§5
D3	Occupations with no AEI match	zero usage (assumption); `drop` as robustness	§2.1
D4	Task-to-occupation allocation	equal split; employment-weighted split as robustness	§2.1
D5	Rank-based robustness for the skew	standardised ranks (R12), drop Softw (R13), both co-primary in reporting	§2.1
D6	AIOE vintage crosswalk	O*NET-detail union (as built); BLS SOC 2010→2018 as robustness	§2.1

ID	Decision	Recommended	Section
D7	NAICS 5511	excluded; pro-rata redistribution as robustness	§2.1
D8	Eloundou rating	human β headline, GPT-4 β beside it (descriptive only)	§2.1
D9	r6 occupation basis	published SOC; task-allocated as robustness	§2.1
D10	Confirmatory inference for H5/H5b	historical-null scale s, critical $\max(t_{21}, q)$, Holm, R13 co-condition; RI secondary	§3.3
D11	5 pp plausibility rule	rule fires (primary MDE 7.43 pp)	§7
D12	H6 AI-term dictionary	the list and matching rules in §6.1	§6.1
D13	RQ1 cross-sectional sample (H1, H3)	r6 `claude_ai`, both months averaged	§4.1
D14	Task family fixed effects	O*NET IWA via Tasks-to-DWAs + DWA Reference (acquired)	§4.1
D15	H3 error-cost indicator	SOC-code list (legal, health practitioners, regulated finance)	§4.3
D16	H4 specification	r6 country cross-section; log AUI, log GDP per working-age person, task-mix control	§4.4
D17	One p-value per multi-coefficient hypothesis	H1 intersection–union; H4 joint Wald	§4.0
D18	H6 status and talk index	exploratory; EDGAR full-text search, 2024 filings, full-index denominator	§6
D19	Exploratory market parameters	objective event rule, windows, terciles with monthly cap reweighting	§8.3
D20	Data acquired after the freeze	robustness-only inputs listed, pre-specified	§9.4
D21	H5/H5b confirmatory null	$b = \bar{b}$ (frozen 1990–2022 mean slope); $b = 0$ secondary	§3.3
D22	H2 inference and family	outside the RQ1 Holm family; release-level wild bootstrap co-condition; α split: H2 at 0.0125, {H1, H3, H4} Holm at 0.0375	§4.2

ID	Decision	Recommended	Section
D23	NAICS → SIC → FF49 crosswalk	equal split (departure from the spec); employment-weighted as robustness	§2.1

A.3 Code binding (PAP §9.2)

- Code commit the plan binds to, HEAD at generation: `75a5eed5af191f7ca89ee09cd1efdedeab4f34bd` ("docs(pap): P-H5 adds the untracked-file check (e); §9 regenerated"), from `git rev-parse HEAD` when this block was generated.
- `src/aideleg/confirm_h5.py` git blob: `f581264ae9c982ff6a84eb1d1e2f465d18544144` (`git rev-parse HEAD:src/aideleg/confirm_h5.py`), with `PRIMARY_NULL = "bbar"`.
- `src/aideleg/power.py` git blob: `a353a7c2683d0cc2e9663e17d3835055b3bb63fc` (`git rev-parse HEAD:src/aideleg/power.py`).

B Deviations

B.1 Summary of the PAP §11 entries

One row per entry of PAP §11, in document order: additions under D20 (`A-`), clarifications (`C-`) and deviations (`D-`), with the entry's date and its "Results seen" field as written ("not stated" where the entry has none).

id	kind	date	results seen
A-1	addition	2026-10-22	not stated
A-2	addition	2026-10-22	not stated
A-3	addition	2026-10-22	not stated
A-4	addition	2026-10-23	not stated
A-5	addition	2026-10-23	not stated
A-6	addition	2026-10-23	not stated
A-7	addition	2026-10-23	not stated
A-8	addition	2026-10-23	not stated
A-9	addition	2026-10-23	not stated
A-10	addition	2026-10-23	not stated
A-11	addition	2026-10-23	not stated
A-12	addition	2026-10-23	not stated
A-13	addition	2026-10-23	not stated
A-14	addition	2026-10-23	not stated
A-15	addition	2026-10-23	not stated
A-16	addition	2026-10-23	not stated
A-17	addition	2026-10-23	not stated
A-18	addition	2026-10-23	not stated

id	kind	date	results seen
A-19	addition	2026-10-23	not stated
A-20	addition	2026-10-23	not stated
A-21	addition	2026-10-23	not stated
A-22	addition	2026-10-23	not stated
A-23	addition	2026-10-23	not stated
A-24	addition	2026-10-23	not stated
A-25	addition	2026-10-23	not stated
A-26	addition	2026-10-23	not stated
A-27	addition	2026-10-23	not stated
A-28	addition	2026-10-23	not stated
A-29	addition	2026-10-23	not stated
A-30	addition	2026-10-23	not stated
A-31	addition	2026-10-23	not stated
A-32	addition	2026-10-23	not stated
A-33	addition	2026-10-23	not stated
A-34	addition	2026-10-23	not stated
A-35	addition	2026-10-23	not stated
A-36	addition	2026-10-23	not stated
A-37	addition	2026-10-23	not stated
A-38	addition	2026-10-23	not stated
A-39	addition	2026-10-23	not stated
A-40	addition	2026-10-23	not stated
A-41	addition	2026-10-23	not stated
A-42	addition	2026-10-23	not stated
A-43	addition	2026-10-23	not stated
A-44	addition	2026-10-23	not stated
A-45	addition	2026-10-23	not stated
A-46	addition	2026-10-23	not stated
A-47	addition	2026-10-23	not stated
C-1	clarification	2026-10-22	no
C-2	clarification	2026-10-22	no
C-3	clarification	2026-10-22	no
C-4	clarification	2026-10-22	no
C-5	clarification	2026-10-22	no
C-6	clarification	2026-10-22	no
C-7	clarification	2026-10-22	no
C-8	clarification	2026-10-22	no
C-9	clarification	2026-10-22	no
C-10	clarification	2026-10-22	no
C-11	clarification	2026-10-22	no

id	kind	date	results seen
C-12	clarification	2026-10-22	no
C-13	clarification	2026-10-22	no
C-14	clarification	2026-10-22	no
C-15	clarification	2026-10-22	no
C-16	clarification	2026-10-22	no
C-17	clarification	2026-10-22	no
C-18	clarification	2026-10-22	no
C-19	clarification	2026-10-22	no
C-20	clarification	2026-10-22	no
C-21	clarification	2026-10-22	no
C-22	clarification	2026-10-22	no
C-23	clarification	2026-10-22	no
C-24	clarification	2026-10-22	no
C-25	clarification	2026-10-22	no
C-26	clarification	2026-10-22	no
C-27	clarification	2026-10-22	no
C-28	clarification	2026-10-22	no
C-29	clarification	2026-10-22	no
C-30	clarification	2026-10-22	no
C-31	clarification	2026-10-22	no
C-32	clarification	2026-10-22	RQ1 confirmatory results seen
C-33	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-34	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-35	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-36	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-37	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-38	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-39	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-40	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-41	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-42	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-43	clarification	2026-10-22	RQ1 confirmatory results seen; H5/H6 not yet run
C-44	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; event-study results seen
C-45	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; event-study results seen
C-46	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; event-study and tercile results seen
C-47	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; event-study, tercile and Fama–MacBeth results seen

id	kind	date	results seen
C-48	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; event-study, tercile and Fama–MacBeth results seen; H6 not yet run
C-49	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; event-study, tercile and Fama–MacBeth results seen; H6 not yet run
C-50	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; event-study, tercile and Fama–MacBeth results seen; H6 not yet run
C-51	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; event-study, tercile and Fama–MacBeth results seen; H6 not yet run
C-52	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; event-study, tercile, Fama–MacBeth and H6 talk/gap results seen
C-53	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; event-study, tercile, Fama–MacBeth and H6 talk/gap results seen; H6 secondary not yet run
C-54	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; all Plan 3 exploratory and H6 results seen
C-55	clarification	2026-10-23	RQ1 and H5 confirmatory results seen; all Plan 3 exploratory and H6 results seen
C-56	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-57	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-58	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-59	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-60	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-61	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-62	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-63	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-64	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

id	kind	date	results seen
C-65	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-66	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-67	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-68	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-69	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-70	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-71	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
C-72	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no R2 row run. Before this entry the implementation computed max |Δ z_delegation| against the primary under this rule (0.607) and under the alternative below (0.415); no H5, H5b or RQ2 estimate was computed under either
C-73	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no R14 row run
C-74	clarification	2026-10-23	RQ1 and H5 confirmatory results and all Plan 3 results seen; no R32 row run
C-75	clarification	2026-10-23	RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 8 rows including R35 (d2da8f1) seen
C-76	clarification	2026-10-23	RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 10 rows R4, R8, R9, R16 (92e12c3) seen
C-77	clarification	2026-10-23	RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 11 rows R10, R23, R24, R26, R27, R31, R38 (25480d9) seen
C-78	clarification	2026-10-23	RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 11 rows R10, R23, R24, R26, R27, R31, R38 (25480d9) seen
C-79	clarification	2026-10-23	RQ1 and H5 confirmatory results, all Plan 3 results, the Task 11 rows, and the Task 12 rows R22, R23, R36, R37 (f8ea9d1) seen
C-80	clarification	2026-10-24	RQ1 and H5 confirmatory results, all Plan 3 results, and every Plan 4 row R1–R38 seen

id	kind	date	results seen
C-81	clarification	2026-10-24	RQ1 and H5 confirmatory results, all Plan 3 results, every Plan 4 row R1–R38, the D-3 diagnostics and the Task 16 red-team findings seen
C-82	clarification	2026-10-24	yes: RQ1 and H5 confirmatory results, all Plan 3 results, all Plan 4 robustness rows, the D-3 diagnostics and the Task 16 and Task 31 red-team findings
C-83	clarification	2026-10-24	yes: all confirmatory, Plan 3 and Plan 4 results and both red teams seen
C-84	clarification	2026-10-25	yes: RQ1 and H5 confirmatory results, all Plan 3 results, all Plan 4 robustness rows and diagnostics, and both red-team rounds
D-1	deviation	2026-10-23	RQ1 and H5 confirmatory results seen; event-study, tercile, Fama–MacBeth, H6 talk/gap and H6 secondary results seen
D-2	deviation	2026-10-23	RQ1 and H5 confirmatory results, all Plan 3 results, and Plan 4 rows R1–R28 seen; R29/R30 run, uncommitted
D-3	deviation	2026-10-24	yes: RQ1 and H5 confirmatory results, all Plan 3 results, all Plan 4 robustness rows, and the Task 16 red-team findings

B.2 PAP §11, verbatim

None at freeze.

Any change after `pap-v1` to a hypothesis, specification, sample, inference, decision rule, frozen input or the confirmatory script is recorded here with its date, the reason, Ajai's approval, and whether results had been seen at the time. A change made after a confirmatory result has been seen demotes that hypothesis to exploratory, and the pre-registered specification is still reported as the primary result (§0.5).

B.2.1 Additions (D20)

B.2.2 A-1. Addition: `ne_admin0_50m` (2026-10-22)

- Type: addition under D20 (outside §9.4; descriptive; not a deviation)
- Used by: descriptive choropleths (spec §5.2); no confirmatory or robustness input
- URL: https://naciscdn.org/naturalearth/50m/cultural/ne_50m_admin_0_countries.zip
- Retrieved (UTC): 2026-10-22T14:41:41Z; bytes: 799734; SHA-256: `5fed433373581fa648920435f937d95f2d3c0200e067409c6478dcdf1b853139`

B.2.3 A-2. Addition: `census_states_20m` (2026-10-22)

- Type: addition under D20 (outside §9.4; descriptive; not a deviation)
- Used by: descriptive choropleths (spec §5.2); no confirmatory or robustness input

- URL: https://www2.census.gov/geo/tiger/GENZ2023/shp/cb_2023_us_state_20m.zip
- Retrieved (UTC): 2026-10-22T14:41:41Z; bytes: 186432; SHA-256: `0fd2d6562708ff8182c00d5d25b5556d049ecf2794d97b89ed2dac4d5e9e2c8d`

B.2.4 A-3. Addition: `ff3_monthly` (2026-10-22)

- Type: addition under D20 (outside §9.4; the PAP names FF3; FF5 2×3 SMB is not FF3 SMB; not a deviation)
- Used by: Plan 3 Task 6 tercile FF3 alphas
- URL: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Research_Data_Factors_CSV.zip
- Retrieved (UTC): 2026-10-22T21:55:15Z; bytes: 13065; SHA-256: `593f4fbef03181bc0b22ff6292f217689dd1fb79355049ca905d95b262040a66`

B.2.5 A-4. Addition: `sec_formidx_2019q1` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2019/QTR1/form.idx
- Retrieved (UTC): 2026-10-23T01:14:27Z; bytes: 46348993; SHA-256: `88901ad5ec510ebf9c74f58f50fc94bfe44c2ccd2bab94680864a5a1a6323ff2`

B.2.6 A-5. Addition: `sec_formidx_2019q2` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2019/QTR2/form.idx
- Retrieved (UTC): 2026-10-23T01:14:28Z; bytes: 38937900; SHA-256: `64a108ce2f798cbf1ef23bc7a24ff15a88d4bd19b2f80acd80765f8751baf68c`

B.2.7 A-6. Addition: `sec_formidx_2019q3` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2019/QTR3/form.idx
- Retrieved (UTC): 2026-10-23T01:14:29Z; bytes: 30960533; SHA-256: `2a2b2ac46d85e10b7f9ef04b2acb1ab3fc3eba939a625e208ea42a6dca26da8f`

B.2.8 A-7. Addition: `sec_formidx_2019q4` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2019/QTR4/form.idx
- Retrieved (UTC): 2026-10-23T01:14:29Z; bytes: 32230060; SHA-256: `a8e6d873ff24cd31ea8036e990274e9e18d045cfbfc8c7c07d85f94a4cab9e28`

B.2.9 A-8. Addition: `sec_formidx_2020q1` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2020/QTR1/form.idx
- Retrieved (UTC): 2026-10-23T01:14:30Z; bytes: 50685481; SHA-256: `593cbf371d422fe526408a74142f00e1e13f22efe63e4b71b48011e077f9af2a`

B.2.10 A-9. Addition: `sec_formidx_2020q2` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2020/QTR2/form.idx
- Retrieved (UTC): 2026-10-23T01:14:31Z; bytes: 41419860; SHA-256: `038d6eb07e7fca5b37b8d8f3e0f899f16e1e3cc3a3a38c22a5ffb0ed4fdbec50`

B.2.11 A-10. Addition: `sec_formidx_2020q3` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2020/QTR3/form.idx
- Retrieved (UTC): 2026-10-23T01:14:32Z; bytes: 34590653; SHA-256: `d378841b0abbe2d2161d8c9bbd3cb2b99bb4d386eee13118c17f21bae57bbc5f`

B.2.12 A-11. Addition: `sec_formidx_2020q4` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2020/QTR4/form.idx
- Retrieved (UTC): 2026-10-23T01:14:33Z; bytes: 36414292; SHA-256: `84aa7452907f241582e01db0d589314af070f434ee0e1884a2e3f14960a56142`

B.2.13 A-12. Addition: `sec_formidx_2021q1` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2021/QTR1/form.idx
- Retrieved (UTC): 2026-10-23T01:14:34Z; bytes: 56365441; SHA-256: `78b6719feb4d8da28f7c2f1ca2a40de5f31d6c71339215d90e1b666ca23938d0`

B.2.14 A-13. Addition: `sec_formidx_2021q2` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2021/QTR2/form.idx
- Retrieved (UTC): 2026-10-23T01:14:36Z; bytes: 46825572; SHA-256: `9bddf421721ec657e66ebbce9e460a62e31bfe2a609f1411a602dd0a3b892d4f`

B.2.15 A-14. Addition: `sec_formidx_2021q3` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2021/QTR3/form.idx
- Retrieved (UTC): 2026-10-23T01:14:37Z; bytes: 39349433; SHA-256: `8c8ef9f8be2acdc39b73b5ffeb262c27ffc804a56de3caefdbf43282560ce3d6`

B.2.16 A-15. Addition: `sec_formidx_2021q4` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2021/QTR4/form.idx

- Retrieved (UTC): 2026-10-23T01:14:38Z; bytes: 40299160; SHA-256: `025be2086f7e01181f1b6560d3731e65268eca1a7e4e8511a2ef2eae2d436d67`

B.2.17 A-16. Addition: `sec_formidx_2022q1` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2022/QTR1/form.idx
- Retrieved (UTC): 2026-10-23T01:14:39Z; bytes: 55621633; SHA-256: `3fa246248e37398a3690a83b0730690a9a332db515b734480326a1bc8b5d98ee`

B.2.18 A-17. Addition: `sec_formidx_2022q2` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2022/QTR2/form.idx
- Retrieved (UTC): 2026-10-23T01:14:39Z; bytes: 45498636; SHA-256: `6b72ce7a562225bc06c255b9a99ff3035017e291080ac521ea75ae0fd4c7fee0`

B.2.19 A-18. Addition: `sec_formidx_2022q3` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2022/QTR3/form.idx
- Retrieved (UTC): 2026-10-23T01:14:40Z; bytes: 37094765; SHA-256: `bce3e54373776e8038aa0007a37cc1680e12c98ca405d8318a2ae12d194ef7a9`

B.2.20 A-19. Addition: `sec_formidx_2022q4` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2022/QTR4/form.idx
- Retrieved (UTC): 2026-10-23T01:14:41Z; bytes: 35987632; SHA-256: `e4fa292fcd6e15eb9d0f4552613ebc189ea3b166a8a3b5b550641785f69ec1c2`

B.2.21 A-20. Addition: `sec_formidx_2023q1` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2023/QTR1/form.idx
- Retrieved (UTC): 2026-10-23T01:14:43Z; bytes: 54362089; SHA-256: `0f11bf75dd8c43d66a9bb9dd7498fab0cd18316d468f6a941418358efc84d895`

B.2.22 A-21. Addition: `sec_formidx_2023q2` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2023/QTR2/form.idx
- Retrieved (UTC): 2026-10-23T01:14:44Z; bytes: 48017880; SHA-256: `de6bea77f33fd87c9489ae0cf5936d902159797aeaf733d851eea006737894c1`

B.2.23 A-22. Addition: `sec_formidx_2023q3` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2023/QTR3/form.idx
- Retrieved (UTC): 2026-10-23T01:14:45Z; bytes: 39411989; SHA-256: `58e7d1100b2f7f6ba4733d0e82b42f0fa9589c651b5c30682c037895ccc448c7`

B.2.24 A-23. Addition: `sec_formidx_2023q4` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2023/QTR4/form.idx
- Retrieved (UTC): 2026-10-23T01:14:46Z; bytes: 39211372; SHA-256: `a6fde5999151322606a428cc2dd5b40a7abe5d263d5ddef4b7e108cbb7b73cf0`

B.2.25 A-24. Addition: `sec_formidx_2024q1` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2024/QTR1/form.idx
- Retrieved (UTC): 2026-10-23T01:14:47Z; bytes: 57770533; SHA-256: `5f91755d27d18df08a2f8497c7b092174285e30b24ea8297b0ec5e17daa0a553`

B.2.26 A-25. Addition: `sec_formidx_2024q2` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2024/QTR2/form.idx
- Retrieved (UTC): 2026-10-23T01:14:48Z; bytes: 49302228; SHA-256: `a213f3926c23261d493f88956d23ea20ebf98ac9751a78452b01bc12aeba07f8`

B.2.27 A-26. Addition: `sec_formidx_2024q3` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2024/QTR3/form.idx
- Retrieved (UTC): 2026-10-23T01:14:49Z; bytes: 42269441; SHA-256: `2c6e48b5c6a26450366436a2677589dd62bc59a5608c3c8e1ca071c51953fc56`

B.2.28 A-27. Addition: `sec_formidx_2024q4` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2024/QTR4/form.idx
- Retrieved (UTC): 2026-10-23T01:14:50Z; bytes: 44336440; SHA-256: `a94001eef171b013dc7971dd59e51763fa8555a870ea81b66f1fca087ec205f0`

B.2.29 A-28. Addition: `sec_formidx_2025q1` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2025/QTR1/form.idx

- Retrieved (UTC): 2026-10-23T01:14:52Z; bytes: 53059957; SHA-256: `becca6a54875c22d1621310c6b3ad69f9927f0b0b882ab1e1c24cb860a992d89`

B.2.30 A-29. Addition: `sec_formidx_2025q2` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2025/QTR2/form.idx
- Retrieved (UTC): 2026-10-23T01:14:53Z; bytes: 51759696; SHA-256: `78a2722d7d483bfe45496179a84c37bf46bcf27ba2183a463adf05f92844cf92`

B.2.31 A-30. Addition: `sec_formidx_2025q3` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2025/QTR3/form.idx
- Retrieved (UTC): 2026-10-23T01:14:54Z; bytes: 44912705; SHA-256: `924d6992695d0337af57d79cbb64b7957520f8c1e24f59de9523d02f62ab6faf`

B.2.32 A-31. Addition: `sec_formidx_2025q4` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/full-index/2025/QTR4/form.idx
- Retrieved (UTC): 2026-10-23T01:14:54Z; bytes: 43155988; SHA-256: `8555ad43d2a491e1280060c7fa7954eae0b18db5ce414684674e69cd77265ddb`

B.2.33 A-32. Addition: `sec_submissions_bulk` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 denominator and SIC (D18), Plan 3 Task 10
- URL: https://www.sec.gov/Archives/edgar/daily-index/bulkdata/submissions.zip
- Retrieved (UTC): 2026-10-23T01:19:58Z; bytes: 1570500398; SHA-256: `b6534bf3efaca6ea2a0a9509deb9eb4c612cb9aa12d4913fdb641a526cc4b033`

B.2.34 A-33. Addition (set): `sec_fts/` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 numerator (D18), Plan 3 Task 11
- Records: 285; retrieved (UTC) from 2026-10-23T01:35:03Z to 2026-10-23T01:46:09Z
- Set digest: `3e6b069bfe3b2248b4acd25ddf5bdbfe231c26ae82d1f0e792cb77061cc96638`

B.2.35 A-34. Addition (set): `sec_docs_index` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: H6 secondary measure, Plan 3 Task 13 (INDEX.jsonl lines; digest = SHA-256 of INDEX.jsonl)
- Records: 47946; retrieved (UTC) from 2026-10-23T02:20:56Z to 2026-10-23T09:30:11Z
- Set digest: `b3380262e475a3bc888b2257dfc6d8a1025dc08c48b16b56afc57849eabb7153`

B.2.36 A-35. Addition (set): `sec_docs_index` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)

- Used by: H6 secondary measure, Plan 3 Task 13 (INDEX.jsonl lines; digest = SHA-256 of INDEX.jsonl); supersedes A-34 after 2 truncated documents were fetched again and their INDEX lines replaced in place (D-entry of the same date)
- Records: 47946; retrieved (UTC) from 2026-10-23T02:20:56Z to 2026-10-23T09:58:41Z
- Set digest: `c416d6bcba7e41cf92d042d2ec5b09cea75ecd8bae5861f201044953ce3b5fed`

B.2.37 A-36. Addition: `bls_soc_2010_2018` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R14
- URL: https://www.bls.gov/soc/2018/soc_2010_to_2018_crosswalk.xlsx
- Retrieved (UTC): 2026-10-23T15:27:35Z; bytes: 46630; SHA-256: `f3a847561562d3e5a30eb848f2902a5f7b02e9c48b3d7f2cc8879899fbc242a7`

B.2.38 A-37. Addition: `onet_29_3_work_context` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R27
- URL: https://www.onetcenter.org/dl_files/database/db_29_3_text/Work%20Context.txt
- Retrieved (UTC): 2026-10-23T15:27:42Z; bytes: 35975227; SHA-256: `dbe428983db68eef798bc6684f28cbb37bd42f485a5de26dfafe67fed7d99ecb`

B.2.39 A-38. Addition: `oews_2025_nat` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R1
- URL: https://www.bls.gov/oes/special-requests/oesm25nat.zip
- Retrieved (UTC): 2026-10-23T15:27:43Z; bytes: 279525; SHA-256: `b5855a37f3e03e779f6fbf173d3bbc94aeeff33426aef32fa95ce6d025bab1af`

B.2.40 A-39. Addition: `oews_2023_nat` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R32
- URL: https://www.bls.gov/oes/special-requests/oesm23nat.zip
- Retrieved (UTC): 2026-10-23T15:27:43Z; bytes: 271583; SHA-256: `8ce5e8277c3ba68a19f468fec9b4da21170593d5ad9693a813f1afc04c80efa2`

B.2.41 A-40. Addition: `oews_2023_ind` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R32
- URL: https://www.bls.gov/oes/special-requests/oesm23in4.zip
- Retrieved (UTC): 2026-10-23T15:27:44Z; bytes: 31801148; SHA-256: `2da9b5426eba1a3a58aa9eab26accbd3b81c3b48b1a6084dd1a0b73c60a49c06`

B.2.42 A-41. Addition: `onet_29_2_tasks` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R32
- URL: https://www.onetcenter.org/dl_files/database/db_29_2_text/Task%20Statements.txt
- Retrieved (UTC): 2026-10-23T15:27:43Z; bytes: 2759559; SHA-256: `fecdcda81a4b8a28e750a6c75ed8bf2d0a9b71186e5be37eec8dc6cef7985d1a`

B.2.43 A-42. Addition: `ff30_daily` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R6
- URL: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/30_Industry_Portfolios_daily_CSV.zip
- Retrieved (UTC): 2026-10-23T15:27:44Z; bytes: 2597362; SHA-256: `04a7a74fcd7c12e90461718e5ba8490e99d18c3e9323182b6c7a7b320e6cd34a`

B.2.44 A-43. Addition: `ff48_daily` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R6
- URL: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/48_Industry_Portfolios_daily_CSV.zip
- Retrieved (UTC): 2026-10-23T15:27:45Z; bytes: 4098912; SHA-256: `95b65f678bfe12bedd32d60495b017482311de91c0e1f467d56762535e5e56e5`

B.2.45 A-44. Addition: `ff_siccodes30` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R6
- URL: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/Siccodes30.zip
- Retrieved (UTC): 2026-10-23T15:27:45Z; bytes: 9244; SHA-256: `eddab10956ef97c8e5cb867ddf5c89e8126e75ea5cb889b2607941c52a2b6272`

B.2.46 A-45. Addition: `ff_siccodes48` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R6
- URL: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/Siccodes48.zip
- Retrieved (UTC): 2026-10-23T15:27:45Z; bytes: 9551; SHA-256: `f37edffc024fe7b91b794dd933244430da2de212df2549dc448c0dfcca45d740`

B.2.47 A-46. Addition: `sec_public_float_cy2024q2` (2026-10-23)

- Type: addition under D20 (outside §9.4; R29 cap proxy; not a deviation)
- Used by: R29
- URL: https://data.sec.gov/api/xbrl/frames/dei/EntityPublicFloat/USD/CY2024Q2I.json
- Retrieved (UTC): 2026-10-23T15:27:45Z; bytes: 578973; SHA-256: `d5ef310add8635a1f18c0362fcf6c10c11e830cad9549e615c8a59c622996c75`

B.2.48 A-47. Addition (set): `sec_fts/` (2026-10-23)

- Type: addition under D20 (pre-specified in §9.4; not a deviation)
- Used by: R29, R30
- Records: 22; retrieved (UTC) from 2026-10-23T23:35:45Z to 2026-10-23T23:36:05Z
- Set digest: `8081d937c805cf51313b3cf4a6dc17e1a73ecc4e8f26575c491ef31be62ae2af`

B.2.49 Clarifications (pre-result)

B.2.50 C-1. Clarification: H1/H3 occupation cluster (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 4 (plans committed at a7fb9bd)

- Results seen: no
- Rule: The occupation cluster is the soc6 code; two soc6 codes sharing an OEWS code are separate rows and clusters with the same wage.

B.2.51 C-2. Clarification: Variable-level drops (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 6 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Each estimator drops rows missing its own variables only (outcome, then predictors or indicator, then covariates), reported as percent of total task usage without renormalising; H3 with no covariates needs no OEWS code, wage or education.

B.2.52 C-3. Clarification: Zero variance after PSD clipping (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 5 (plans committed at a7fb9bd)
- Results seen: no
- Rule: A tested variance $\leq$ 1e-12 $\times$ the larger one-way variance after clipping is replaced by the larger one-way variance.

B.2.53 C-4. Clarification: R7 bounds in H1/H3 rows (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 4 (plans committed at a7fb9bd)
- Results seen: no
- Rule: `automation_share_all_lb/_ub` are computed per period with that release's suppression components, then D13-averaged.

B.2.54 C-5. Clarification: Wage deciles (descriptive) (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 13 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Employment-weighted deciles of `a_median` (top-coded at $239,200; suppressed excluded): decile = clip(ceil(10·(cum_emp − tot_emp/2)/total_emp), 1, 10), occupations sorted ascending.

B.2.55 C-6. Clarification: Education bins (descriptive) (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 13 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Bins <12, 12–14, 14–16, 16–18, ≥18 years, left-closed; tasks without an education value reported as "not published".

B.2.56 C-7. Clarification: Concentration shares (descriptive) (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 13 (plans committed at a7fb9bd)
- Results seen: no
- Rule: The top-x% share is the usage share of the $\lceil x\% \cdot n \rceil$ largest tasks.

B.2.57 C-8. Clarification: RQ2 bootstrap draws (descriptive) (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 15 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Spearman ρ by average-tie ranks then Pearson; a draw with a constant vector is excluded and counted; percentile 95% CI; one `default_rng(SEED)` drawn in design order B, A_pooled.

B.2.58 C-9. Clarification: Task-to-family union (D14) (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 3 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Task-to-family mapping (D14): a conversation matched to a task statement is assigned the sorted, de-duplicated union of IWA/GWA/DWA families over ALL Task IDs carrying that statement text (not only the smallest candidate ID), consistent with the occupation allocation (ruling R6). A task whose candidates have no Tasks-to-DWAs link has no family and is dropped from the family-level analysis, with the dropped weight reported.

B.2.59 C-10. Clarification: Two-way zero-variance fallback (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 5 (plans committed at a7fb9bd)
- Results seen: no
- Rule: (i) The fallback to the larger one-way cluster variance also applies when the tested block was not clipped: a tested variance ≤ ZERO_VARIANCE_RTOL × the larger one-way variance, with ZERO_VARIANCE_RTOL = 1e-12, is replaced by the larger one-way variance. (ii) If both one-way variances are also zero, the rule gives no SE and the run raises (STOP).

B.2.60 C-11. Clarification: P13 attenuation factor (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 6 (plans committed at a7fb9bd)
- Results seen: no
- Rule: The factor reported beside β_w (H1) and γ (H3) is $\Sigma\, w \cdot \bar{r}^2 \,/\, \Sigma\, w \cdot r^2$. r is the Frisch–Waugh–Lovell residual of the regressor: it is residualised by WLS, with the estimation weights, on the IWA fixed effects and on the estimator's other regressors (education for log wage in H1; the covariates for the indicator in H3 with covariates, R31). $\bar{r}$ is the weighted mean of r within (task, IWA). It is the between-task share of the variation β is identified from, and $E[\hat{\beta}] = \beta \times$ factor when the outcome is built on the task's allocation-weighted mean regressor.

B.2.61 C-12. Clarification: H3 own variables (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 6 (plans committed at a7fb9bd)
- Results seen: no
- Rule: H3 without covariates drops rows only for a missing `validation_pct` or `error_cost`. It requires no OEWS code, wage or education, because PAP §4.1 says H3 "applies the same rule to its own variables".

B.2.62 C-13. Clarification: no_wage_row drop reason (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 6 (plans committed at a7fb9bd)
- Results seen: no

- Rule: `wage_status` determines the drop reason for `log_wage`: `no_oews_code` gives `no_oews_code`; `suppressed` gives `wage_suppressed`; `no_wage_row` (an OEWS code that is not in the May 2024 national wage table) gives `no_wage_row`; any other null gives `missing_log_wage`. Each drop is reported in % of total r6 task usage.

B.2.63 C-14. Clarification: γ-only non-PSD block under R31 (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 6 (plans committed at a7fb9bd)
- Results seen: no
- Rule: In H3 with covariates, the PAP §4 non-PSD rule is applied to γ's block alone ("γ for H3"), so the covariates never change γ's inference. The covariate coefficients are reported from a second call on the same design whose tested block is the covariates; point estimates are identical. That call's non-PSD diagnostics are in `diagnostics["covariates_psd"]` and in `notes`.

B.2.64 C-15. Clarification: H2 wild-bootstrap granularity (disclosure) (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 7 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Disclosure, no rule change: with G = 4 releases and Webb six-point weights, full enumeration has $6^4 = 1{,}296$ draws, and the six constant Webb vectors always reproduce |t|, so p_wild $\geq 6/1296 \approx 0.0046$. Near-ties from vectors that tilt the trend slightly give p_wild of about 0.009–0.042 even for strong synthetic trends, so the 0.0125 co-condition can fail an unambiguous trend. This is a property of the PAP's design (D22); the rule is unchanged and the granularity is reported with the H2 result.

B.2.65 C-16. Clarification: H4 global share and m_global offset (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 8 (plans committed at a7fb9bd)
- Results seen: no
- Rule: `global_automation` is AEI's published GLOBAL overall `automation_pct`, as the equal-weighted two-month mean. `m_global` (the mix formula applied to the global shares) $\neq$ `global_automation` in general, so `mix_part` carries the common offset (m_global − global_automation). The shift-share output prints that offset alongside.

B.2.66 C-17. Clarification: H4 carried variables (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 8 (plans committed at a7fb9bd)
- Results seen: no
- Rule: `usage_pct`, `validation_pct` and `ai_autonomy_mean` are averaged over the published months, with no completeness rule. R28's `usage_pct` weight may therefore come from a single month.

B.2.67 C-18. Clarification: H4 exclusion reason (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 8 (plans committed at a7fb9bd)
- Results seen: no
- Rule: A geography failing several tests is listed under the first one, in the order missing_month, missing_income (only when `require_income`), missing_mix.

B.2.68 C-19. Clarification: H4 income and AUI (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 8 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Income is null when GDP or the working-age population is missing or ≤ 0. A non-positive AUI raises `ValueError`, and there is no new exclusion reason for it.

B.2.69 C-20. Clarification: H4 mix (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 8 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Negative `pct` rows are ignored. `missing_mix` applies when no node the geography publishes has a published $\bar{a}$.

B.2.70 C-21. Clarification: H1 mixed-sign wording (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 9 (plans committed at a7fb9bd)
- Results seen: no
- Rule: H1 is Holm-rejected, but one coefficient has the wrong sign (> 0). Status `rejected_mixed_sign`, rejected=True, not supported. Text: "H1: the null is rejected, but {name} has the opposite sign, which contradicts the hypothesis, so H1 is not supported ({coef_lines})." If more than one coefficient is named: "{a} and {b} have the opposite sign". With both coefficients of the wrong sign the status is `contradicts`.

B.2.71 C-22. Clarification: Zero estimate (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 9 (plans committed at a7fb9bd)
- Results seen: no
- Rule: H1 or H3 is Holm-rejected, or H2 meets both p conditions, but a tested estimate is exactly 0. Status `rejected_zero_estimate`, rejected=True, not supported. Text: "{H}: the null is rejected, but the estimate of {name} is exactly 0, so the hypothesis is not supported ({coef_lines})."

B.2.72 C-23. Clarification: H2 failed conditions (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 9 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Each failed p condition is appended to the not-rejected template: "H2 is not rejected (H0: $\beta = 0$). {coef_lines}. Not met: {fail1}; {fail2}." A non-finite p is worded "task-clustered p is not finite" or "release-level wild bootstrap p is not finite".

B.2.73 C-24. Clarification: Exact levels (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 9 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Holm and H2 levels are compared exactly on the printed decimals of α and p (`Fraction(repr(x))`), so p = 0.0125 meets the level 0.0375/3.

B.2.74 C-25. Clarification: NaN p (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 9 (plans committed at a7fb9bd)
- Results seen: no
- Rule: A NaN p is never rejected and sorts last in Holm.

B.2.75 C-26. Clarification: Non-finite tested estimate, SE or p (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 9 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Status `not_supported_nonfinite`, rejected=False, not supported, for any of H1–H4. Text: "{H} is not supported: {name} has a non-finite {fields} ({coef_lines})." Several coefficients are joined with ";". It takes precedence over the Holm, H2 and sign rules.

B.2.76 C-27. Clarification: H1 NaN p_family (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 9 (plans committed at a7fb9bd)
- Results seen: no
- Rule: p_H1 = NaN if either tested p is NaN (never the max of the remaining one). Holm then sorts it last and never rejects it.

B.2.77 C-28. Clarification: H4 decision text (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 9 (plans committed at a7fb9bd)
- Results seen: no
- Rule: The H4 decision text lists the two tested coefficients only (b_1 log AUI, b_2 log income), not the task-mix control.

B.2.78 C-29. Clarification: Robustness rows R6, R32, R35 (organisational) (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 10 (plans committed at a7fb9bd)
- Results seen: no
- Rule: Robustness rows R6, R32 and R35 move from "plan 3" to Plan 4, with no specification change.

B.2.79 C-30. Clarification: H1/H3 task unit, zero-weight drop, r6 automation measure, H3 sample (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2 Task 10 (plans committed at a7fb9bd)
- Results seen: no
- Rule: (a) Task unit and cluster: the H1/H3 task unit and cluster is the panel node_id (the O*NET Task ID for r6), not the statement text. Two r6 IDs sharing one statement stay separate tasks and clusters: one such pair with positive weight (nodes 976 and 14630), about 0.02% of r6 task usage. (b) Zero-weight drop: 1,611 of the 2,823 r6 task nodes are dropped as `zero_weight`, because r6 task pct is published rounded to 0.01 and those nodes show 0.00; this follows from D13, as the usage-weighted mean is undefined at zero usage. (c) r6 automation measure: the H1–H4 automation outcome in r6 is AEI's published automation bucket. 41 r6 global task-month cells publish all five patterns but no bucket; they are null and skipped (0.033% of usage). Where

both exist, the bucket and the renormalised pattern sum differ by ≤ 0.034 pp. (d) H3 sample: the H3 sample does not require wage, education or an OEWS code (1,689 vs H1's 1,684 rows); this reading follows the H3 missing-values row ("H3 applies the same rule to its own variables"), and C-12 is the related entry.

B.2.80 C-31. Clarification: Disclosure: a unit-test artefact in docs/results/rq1 before the P-RQ1 run (2026-10-22)

- Type: clarification (pre-result); the PAP is silent or ambiguous here and no PAP rule changes
- Fixed by: Plan 2, Task 10
- Results seen: no
- Rule: On 2026-10-22 at 19:47:28 UTC, while recording a failing test for the panel-hash refusal, a unit test called confirm.run against the real output directory with the clean-tree check stubbed (HEAD recorded as 'abc') and load_inputs stubbed to raise before reading any data. It created an attempt marker, a log and a markdown file under docs/results/rq1 containing only the stub's traceback. No data were loaded and no outcome was read, so it is not a P-RQ1 attempt. The files were never committed; they were removed and a copy is kept outside the repository. The test fixture now redirects the output directory to a temporary path so no test can write the real one.

B.2.81 Clarifications (post-result)

B.2.82 C-32. Clarification: H4 task-mix control reproduces to about 1e-14 (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 2 final fix wave (I1)
- Results seen: RQ1 confirmatory results seen
- Rule: H4's task-mix control (`h4._mix`) sums over a polars group_by whose row order is not fixed, so `mix` reproduces only to about 1e-14 between runs; this is far below any reported precision and does not affect the committed P-RQ1 output; figures are built from values rounded to 10 significant digits.

B.2.83 C-33. Clarification: Event-study cross-sectional inference (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 5 (plans committed at a7fb9bd)
- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: Each CAR-on-exposure regression is OLS with an intercept and HC3 SEs, nominal and single-test; randomisation inference is not used (PAP §3.2 documents its size).

B.2.84 C-34. Clarification: Tercile ties and size (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 6 (plans committed at a7fb9bd)
- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: k = n // 3 over industries with non-null exposure; ties at a cut broken by ascending `ff49`.

B.2.85 C-35. Clarification: French firm-size timing (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 2, 6, 7 (plans committed at a7fb9bd)

- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: French does not document the timing of 'Average Firm Size'; it is determined empirically from pre-seal rows ($\leq$ 2022-11-29, non-July months): corr(size growth, prior-month VW return) = 0.956 vs corr(size growth, same-month VW return) = 0.084 (median per industry 0.972 vs 0.090), so row t holds ME at the end of t − 1 (SIZE_ME_OFFSET = 1). Cap weights for month t and FM log_size use the row whose ME is the prior month-end, i.e. the row dated t − (1 − SIZE_ME_OFFSET) months (for SIZE_ME_OFFSET = 1, the row dated t). Cap = average firm size × number of firms, both from that row. Script: scripts/french_size_timing.py.

B.2.86 C-36. Clarification: FF3 factors (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 6 (plans committed at a7fb9bd)
- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: FF3 alphas use Mkt-RF, SMB and HML of `F-F_Research_Data_Factors_CSV.zip` (`ff3_monthly`), never FF5 columns.

B.2.87 C-37. Clarification: Tercile bootstrap and DSR (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 6 (plans committed at a7fb9bd)
- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: The stationary bootstrap (block 3, 10,000 draws) is applied to the Sharpe ratio and the mean, one draw set per variant, one `default_rng(SEED)` consumed in `VARIANTS` order; the DSR is taken over the variants run (six in the primary run).

B.2.88 C-38. Clarification: Fama–MacBeth characteristics (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 7 (plans committed at a7fb9bd)
- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: log BE/ME = ln("Sum of BE / Sum of ME") with French's July–June timing; a month with n $\leq$ regressors + 1 complete industries is skipped and counted.

B.2.89 C-39. Clarification: Exhibit-only CIKs (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 11 (plans committed at a7fb9bd)
- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: A CIK whose only hit is in an exhibit of a 10-K (`file_type` not in FORMS) counts in `n_talk`; such CIKs are also reported separately.

B.2.90 C-40. Clarification: SIC source (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 10 (plans committed at a7fb9bd)
- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: SIC = the current SIC of the EDGAR bulk submissions file.

B.2.91 C-41. Clarification: Secondary measure years (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 13 (plans committed at a7fb9bd)
- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: The secondary talk measure covers all D18 filings, 2019–2025.

B.2.92 C-42. Clarification: Gap–return test (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 12 (plans committed at a7fb9bd)
- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: The gap–return association uses the H5 form (no control) with the D21 null and nominal p.

B.2.93 C-43. Clarification: H6 `all` (2026-10-22)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 12 (plans committed at a7fb9bd)
- Results seen: RQ1 confirmatory results seen; H5/H6 not yet run
- Rule: `python -m aideleg.h6 all` = build + talk + secondary + summary from saved raw files, with no network; acquisition is the separate `h6-data` target.

B.2.94 C-44. Clarification: Event-study defaults: returns, HC3 reference, event-window drops (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 5 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results seen; event-study results seen
- Rule: Three defaults Task 5 fixed where PAP §8.3 D19 is silent: (1) industry returns are the daily value-weighted FF49 returns, following §3.1; (2) each CAR-on-exposure regression's HC3 95% CI and p use the normal reference (statsmodels cov_type="HC3" default), matching Plan 2's H4; (3) an industry missing any return in the event window [0, max k] is dropped at every horizon of that event (as is one missing any return in [-250, -21]). These were fixed in code before any event-study output was read and are unchanged. After the outputs were seen, one label sentence in event_study.md was corrected (A_pooled pools r2-r6, as exposure.POOLED_RELEASES says, not r3-r6); no estimate was affected.

B.2.95 C-45. Clarification: Tercile defaults: leg returns, H-L excess, empty legs, inference details (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 6 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results seen; event-study results seen
- Rule: Defaults Task 6 fixed where PAP §8.3 D19 and C-34..C-37 are silent, fixed in code before any tercile output was read: (1) both cap and EW legs are built from the FF49 monthly value-weighted industry returns (following §3.1); EW legs weight those industries equally (French's EW industry returns are not used); (2) H-L = high minus low is a zero-investment return, so no rf is subtracted from it in the alpha regressions, the mean or the Sharpe ratio; (3) an industry with a missing return leaves that month's leg under both weightings, one

with a missing cap leaves the cap leg only, the weights renormalise over the rest, and a leg with no valid member in a month makes the run refuse (no month is dropped); factor rows are matched to the H-L months by date and the run refuses if any month or value is missing; (4) alpha 95% CIs use the t distribution with T - k - 1 df, matching the t-based p of the Newey-West regression (Bartlett kernel, lag floor(4 (T/100)^(2/9)), no small-sample correction); (5) the mean is reported in annualised pp (x 12) with a Newey-West SE and t-based p (intercept-only regression) and the stationary-bootstrap percentile CI; the bootstrap statistics are the annualised Sharpe (monthly mean / sd with ddof 1, x sqrt 12) and the annualised mean, computed on the same index draw, equal-tailed 95% percentile intervals; (6) PSR(0) uses the gross monthly Sharpe ratio, and each variant's DSR is its PSR against the expected maximum of the gross (zero-cost) monthly Sharpe ratios of the variants run (variance across variants with ddof 1); (7) costed mean and Sharpe are annualised as the gross ones; (8) in run_ terciles a WINDOWS key left out of months uses its whole window, and a month outside the window makes the run refuse. The annual French characteristics are not used (cap weights use the monthly firm-size rows only).

B.2.96 C-46. Clarification: Fama–MacBeth defaults: returns and rf, characteristics, momentum, exposure, inference, units (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 7 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results seen; event-study and tercile results seen
- Rule: Defaults Task 7 fixed where PAP §8.3 D19 and C-35/C-38 are silent, fixed in code before any Fama–MacBeth output was read: (1) the dependent variable is the FF49 monthly value-weighted industry return (following §3.1) minus the rf of the FF5 monthly file, matched by date, in percent; a month with no return row or no rf makes the run refuse; (2) exposure is the design's z_delegation as given, not re-standardised per month or after drops; one specification is run, [1, z, log size, log BE/ME, 12-2 momentum], for (B, B window), (A_r2, A window) and (A_pooled, A window), with the tercile windows; (3) log size = ln(average firm size) of the row whose ME is the prior month end (C-35), log BE/ME = ln(Sum of BE / Sum of ME) of the row for year y used July y to June y+1 (C-38; the row labelled y is BE for fiscal y-1 over ME at December y-1); a non-positive, missing or NaN size or BE/ME (or a missing or NaN return, or a null exposure) removes that industry from that month's regression only, and such industry-months are listed; (4) momentum = prod over calendar months t-12..t-2 of (1 + r/100) - 1, a decimal, from the same value-weighted industry returns (full history before the window), null unless all 11 returns are present; (5) each month is an unweighted OLS across the complete industries, characteristics in raw logs (not cross-sectionally standardised); a month with n <= 5 complete industries is skipped and counted, a rank-deficient month or fewer than 2 usable months makes the run refuse; (6) each coefficient is the time-series mean of the monthly slopes with the Newey-West SE of an intercept-only regression (Bartlett kernel, lag floor(4 (T/100)^(2/9)), no small-sample correction), t-based p and a 95% CI from the t distribution with T - 1 df; the intercept is estimated but not reported; n_mean is the mean number of complete industries over the months used; (7) units: the slope on z is pp per month per 1 SD, shown x 12 (annualised pp per 1 SD) beside it in the doc and per month in the CSV; the other slopes are pp per month per unit of log size, of log BE/ME and of the decimal cumulative return.

B.2.97 C-47. Clarification: Fama–MacBeth drop-list label: non-positive BE/ME (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 7 review fix (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results seen; event-study, tercile and Fama–MacBeth results seen
- Rule: After the Fama–MacBeth outputs were seen, the drop list in fama_macbeth.md was changed to label a size or BE/ME that is present but <= 0 apart from an absent one (data.beme gained keep_nonpositive, used only for this label); Smoke, whose Sum of BE / Sum of ME is negative in every year used, now reads 'BE/ME <= 0 (all months)' instead of 'BE/ME (all months)'. The exclusion rule (C-46 (3)) is unchanged; no estimate changed:

fama_macbeth.csv is byte-identical to commit 575b093 and only that label differs in fama_macbeth.md.

B.2.98 C-48. Clarification: H6 dictionary matching defaults (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 9 review fix, before any H6 output was produced
- Results seen: RQ1 and H5 confirmatory results seen; event-study, tercile and Fama–MacBeth results seen; H6 not yet run
- Rule: Defaults for the D12 matching rules where the PAP is silent, fixed in code before any H6 output was produced: (a) the contents of ix:header, script and style elements are dropped; (b) every other tag is replaced by a space; (c) the acronym gate is a raw, case-insensitive occurrence of a GATE term (with the space/hyphen rule, and plural rule for plural terms), independent of the terms argument and of overlap resolution, so 'generative artificial intelligence' alone gates; (d) primary terms and acronyms are resolved jointly, longest span then leftmost, non-overlapping; (e) a space in a term matches any run of whitespace and hyphens, including U+2010 and U+2011, and soft hyphens (U+00AD) and zero-width spaces (U+200B) are deleted in cleaning; (f) acronym boundaries are Unicode letters and digits (an underscore is not a boundary character); (g) n_ words is the number of non-whitespace tokens (§+) after cleaning. Plain-text documents go through clean_text with html=False (note for Task 13).

B.2.99 C-49. Clarification: H6 denominator defaults: forms, year, co-registrants, SIC member, unmapped, index layout (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 10, before any H6 output was produced (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results seen; event-study, tercile and Fama–MacBeth results seen; H6 not yet run
- Rule: Defaults Task 10 fixed for the H6 denominator where PAP §6 D18 and C-40 are silent, fixed in code before any H6 output was produced: (1) a filing counts if its full-index Form Type equals 10-K, 10-K405 or 10-KT exactly (10-K/A, 10-K405/A, 10-KT/A and NT 10-K are excluded); (2) year = the calendar year of the full index's Date Filed, as D18 says (filed in calendar years 2019-2025), and all of 2025 is used (the 2025-03-27 cut is R29 only); (3) a filing that form.idx lists under several CIKs (co-registrants) counts for each listed CIK, and a CIK with several such filings in a year counts once in that year (D18: distinct CIKs); every filing is kept in filers.parquet; (4) SIC = int of the sic field of member CIK{cik:010d}.json of the bulk submissions.zip (C-40: the current SIC, applied to every year); the -submissions-NNN.json members are not read; a CIK with no member, or whose sic is absent or empty, has no SIC; (5) no SIC, or a SIC in no Siccodes49 range, leaves ff49 null: such CIKs are counted under ff49 = null in the denominator and reported per year (no SIC vs SIC outside every range), never assigned to an industry; (6) form.idx rows are read at their measured column offsets 0/17/79/91/103, which differ from the header labels' offsets 0/12/74/86/98, and the parser refuses a header or row off that layout (all 28 quarters 2019-2025 parse).

B.2.100 C-50. Clarification: H6 full-text-search numerator defaults: year, forms, documents, co-registrants, matching to filers, bisection, page integrity (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 11, before any H6 output was produced (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results seen; event-study, tercile and Fama–MacBeth results seen; H6 not yet run
- Rule: Defaults Task 11 fixed for the D18 numerator where PAP §6 D18, C-39 and C-49 are silent, fixed in code

before any H6 output was produced (D18 fixes the quoted phrases, the form types, filing dates in year y, FTS phrase semantics on the retrieval date and one MANIFEST record per response): (1) the 15 phrases are dictionary.fts_phrases (the 12 primary terms and the plurals of items 4, 7 and 11), each sent as q="phrase" with forms=10-K,10-K405,10-KT and dateRange=custom, one query range per calendar year y (startdt y-01-01, enddt y-12-31; the server filters file_date with gte/lte, both ends inclusive); a hit's year is the calendar year of its FTS file_date; (2) FTS returns one hit per document of a filing; a hit counts only if its filing's form (_source.form) equals 10-K, 10-K405 or 10-KT exactly: 10-K/A and the other forms the server returns (it filters on root_forms) are dropped, counted and printed per phrase and year; any document of a counted filing counts (C-39), and a CIK is exhibit-only in year y if none of its hits that year is in a document whose file_type is in FORMS; (3) one row per (document, CIK in _source.ciks), CIKs read as int of the zero-padded strings; each co-registrant CIK counts once per year, mirroring C-49 (3); a hit with no CIK gives no row and is counted and printed; rows are deduplicated by (document _id, CIK), and a CIK found by several phrases, documents or pages counts once per (year, ff49) (D18: distinct CIKs); (4) a hit CIK counts in year y if it is one of year y's full-index filers (matched by (CIK, year), not by accession number), in that filer's ff49, null ff49 for unmapped filers (never an industry); hit CIKs that are not year y's filers are listed by out_of_denominator and never counted; the numerator has the denominator's grid (FF49 1..49 and null for every year), zeros where no hit; (5) a range is capped when total.relation is not eq or total.value >= 10000; it is split at mid = start + (end - start) // 2 into [start, mid] and [mid + 1, end], so each boundary day is in exactly one half; a single day still capped raises FtsCapError (a stop); (6) every page of a range must report page 0's total and the distinct document _ids over its pages must equal that total, otherwise FtsPageError (a stop; no partial range is used); (7) the retrieval date of D18 is the interval from the first to the last retrieved_utc of the sec_fts/ records, logged in the §11 set addition; hits_from_saved rebuilds the frame from the saved leaf ranges only (exact page 0 below the cap, all pages present).

B.2.101 C-51. Clarification: H6 talk index, gap, Spearman CI and gap–return defaults (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 12, before any H6 output was produced (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results seen; event-study, tercile and Fama–MacBeth results seen; H6 not yet run
- Rule: Defaults Task 12 fixed where PAP §6 D18, C-42 and C-49/C-50 are silent, fixed in code before any H6 output was produced: (1) T_j, zT, the gap OLS, Spearman and the gap–return regression use the 49 FF49 industries only; the unmapped ff49 = null denominator row (C-49 (5)) enters none of them; its count and share of filers are stated per year in talk_index.md and for 2024 in gap.md; (2) talk = n_talk / n_filers per (year, FF49), null when n_filers = 0; an FF49 industry with no 2024 filer has T undefined: it is listed and excluded from zT, the OLS, Spearman and the gap–return regression (no imputation); (3) z_usage is Design B (release r2 only, as §3.1) z_usage from the hash-checked rq3.data.exposures(), used as given (not re-standardised over the industries with T); zT = (T - mean) / population SD (ddof 0) across the industries with T; the OLS of zT on [1, z_usage] is unweighted; gap = residual / population SD (ddof 0) of the residuals; the ranking is gap descending, ties by industry name; (4) Spearman uses average ranks for ties; the 95% CI resamples industries with replacement as (T_j, z_usage_j) pairs, n = the industries with T, 10,000 draws from one default_rng(SEED) integer draw of shape (draws, n); a draw with a constant T or z_usage is undefined, dropped and counted; the CI is the equal-tailed percentile interval of the defined draws (numpy quantile, linear interpolation, rq3.stats.percentile_ci); rho itself is computed on the full sample; (5) pooled talk per year = the sum of n_talk over the sum of n_filers over every row including the unmapped row, i.e. the share of all distinct 10-K filer CIKs of the year (not a mean of industry shares); the mapped-only share is reported beside it; (6) the gap–return alphas are rq3.inference.confirmatory_alphas over confirm_h5.WINDOW (the Design B window, 2025-04-01 to 2026-08-31, FF5+UMD on FF49 daily VW), refused unless the recomputed H5/H5b slopes equal the committed record (check_matches_confirmatory); the industries are those with a gap, in return-column order; vector_

test(alphas, gap, names, control=None) with its default historical matrix (the pre-seal power.historical_ windows on the kept columns); an industry with no alpha is dropped by vector_test and gap is not re-standardised after that drop (as §3.1 for z_delegation); b, s, $\bar{b}$ and the CI are reported x 100 in pp of annualised alpha per 1 SD of gap, with T and nominal p for both D21 nulls ($b = \bar{b}$ primary, $b = 0$ secondary) from t_21 and CI = $b \pm \max(t_21, q) \times s$; (7) organisational: until Task 13 the secondary subcommand and acquire –docs print that they are not implemented and do nothing, so all = build + talk + secondary + summary completes; summary.md links every other h6 doc and carries no estimate.

B.2.102 C-52. Clarification: H6 presentation: talk-vs-usage figure labels and 2024 counts in the gap ranking (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 12 review fix (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results seen; event-study, tercile, Fama–MacBeth and H6 talk/gap results seen
- Rule: After the H6 talk and gap outputs were seen (commit 929a094), two presentation-only changes were made at the reviewer's request: (1) fig_h6_talk_vs_usage places its industry labels with a deterministic greedy placement (candidate offsets around each point, the first whose box overlaps no placed label and no marker, a thin leader line when a label sits away from its point), so labels in the dense cluster no longer collide (Steel and Gold were unreadable); the plotted points, axes and title are unchanged; (2) the gap.md ranking table gains two columns, the 2024 talk filers (n_talk) and filers (n_filers) of each industry, read from the same talk table, so a reader can see ranks that rest on few filers (e.g. Coal 5/11, Boxes 4/9). No estimate, sample, specification or default (C-51) changed: talk_index.csv and gap.csv are byte-identical to commit 929a094, and the only change in the docs is the two added table columns in gap.md.

B.2.103 C-53. Clarification: H6 secondary measure defaults: documents, cleaning, error bodies, skipped filings, means, 2024 table, agreement check (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 Task 13, in code before the secondary measure was run (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results seen; event-study, tercile, Fama–MacBeth and H6 talk/gap results seen; H6 secondary not yet run
- Rule: Defaults Task 13 fixed for the D18 secondary measure where PAP §6 D18, §6.1 D12, C-41 and C-48 are silent, fixed in code before the secondary measure was run: (1) one primary document per accession number of filers.parquet (every D18 filing 2019-2025, C-41): its primaryDocument is read from the bulk submissions.zip, member CIK{cik:010d}.json filings.recent then each filings.files[] page (an earlier listing wins), trying the filing's CIKs in ascending order and taking the first that lists the accession with a non-empty primaryDocument; that CIK's EDGAR path gives the URL and data/raw/sec/docs/{cik}/{adsh}/{doc}.gz; a co-registrant filing gives the same document, counts and rate to each of its CIKs (as C-49 (3)); the document is the SEC's copy on the retrieval date (the span is in the §11 set addition); (2) a filing with no listed or an empty primaryDocument, a .pdf, or a name ending in anything other than .htm, .html or .txt is skipped; a document not saved after the retries (HTTP 404 and the like; logged in ERRORS.jsonl) is not downloaded; a document with zero words (\S+ tokens after cleaning) has no rate; all four are left out of every mean (no imputation, never zero) and counted per year in secondary_talk.md; rated filings under 2,000 words are kept and counted; (3) .htm/.html documents, inline XBRL included, go through clean_text(html=True) exactly as C-48 (ix:header dropped, every other tag, ix tags included, blanked, so the text of tagged facts stays); .txt documents through clean_text(html=False) (C-48 note); bodies are decoded as UTF-8, else Latin-1; (4) a 200 body of at most 64 KiB that contains 'Request Rate Threshold Exceeded', 'Undeclared Automated Tool', 'Your Request Originates' or 'Access Denied' (case-insensitive) is an SEC error page: never saved or indexed; it, an HTTP 403, and a 429 or

5xx that persists through fetch's retries are retried after 60 s, and 3 such answers in a row stop the acquisition (BLOCKED); requests go through acquire.http (contact UA, 0.1 s SEC spacing) plus a 0.05 s pause, one process; (5) bytes and SHA-256 in INDEX.jsonl are of the uncompressed body; the .gz has mtime 0; the MANIFEST record sec_docs_index is INDEX.jsonl itself (its retrieved_utc the last document's); the acquisition refuses below 30 GiB free; (6) filing rate = 10,000 x (primary hits + gated acronym hits) / n_words with dictionary.count_terms (D12, C-48); firm-year rate = the mean over the CIK's rated filings of the year (year = the filers.parquet filing year, C-49 (2)); industry rate = the equal-weight mean over the CIKs of the (year, FF49) cell with a firm-year rate; unmapped CIKs form the ff49 = null row, are counted and enter no industry, no industry mean and no Spearman; an industry with no rated CIK has no rate (listed for 2024, — in the table); (7) series by year: filings, rated filings, rated CIKs, the equal-weight mean and median of firm-year rates over every rated CIK (unmapped included), the equal-weight mean of the FF49 industry rates over the industries with a rate, and the share of rated CIKs with a primary-term hit; (8) the 2024 table lists the 49 FF49 industries with their rated-CIK count, 2024 rate and T_j (talk.t_index); Spearman(secondary 2024, T_j) is over the industries with both defined, average ranks for ties (talk._rank_corr), descriptive with no CI or p-value (Task 13 brief); (9) reporting step carried from the Task 12 review: a per-year agreement table over the distinct (year, CIK) filer-years of filers.parquet between the FTS indicator (the CIK is counted in edgar.numerator that year) and the text indicator (some rated primary document of the CIK that year has at least one primary-term hit; main indicator 'any primary term'); counts both / FTS only / text only / neither and agreement = (both + neither) / filer-years with a rated document; filer-years with no rated document are counted apart (with how many of them are FTS talk); the number of filer-years where adding gated acronyms changes the text indicator is stated; reporting only, it checks whether the 2023 to 2024 change in FTS talk is an FTS artefact and changes no estimate; (10) document counts are computed in parallel processes and joined back by accession, so the outputs do not depend on the order of work; outputs are secondary_talk.md, secondary_talk.csv (the (year, FF49) grid with the null row) and fig_h6_secondary_by_year.

B.2.104 C-54. Clarification: Plan 3 final review: disclosures and process record (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 final whole-branch review fix wave (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results seen; all Plan 3 exploratory and H6 results seen
- Rule: (a) Presentation-only doc changes after every Plan 3 result had been seen (code commit cf27c32): terciles.md and fama_macbeth.md gain a Caveats section generated from the run (Design B and Design A_r2 exposures are numerically identical, PAP §8.3 D19, so the B rows are a 17-month sub-window of A_r2 with the same membership; the DSR's N = 6 overstates the number of independent trials, B being nested in A_r2 and A_r2 close to A_pooled, monthly SRs read from the computed frame; B's FF5+UMD alpha rests on T = 17 with 7 parameters, 10 residual df, NW lag 2, so the CAPM/FF3/FF5+UMD spread is model-fragile, and PSR(0) is a normal approximation on 17 months); event_study.md gains a caveat that cross-sectional correlation on common event dates also affects the HC3 SEs, so the nominal p-values may be oversized; rq3 and h6 docs print a p-value below 1e-4 as '< 0.0001' (CSVs keep full precision); influence.md says 'post-result diagnostics'; talk_index.md and gap.md state, from filers.parquet, that the unmapped filers with a SIC outside Siccodes49 are mostly SIC 6189 asset-backed issuers and that the unmapped share rises 11.92% to 15.81% (2019 to 2025), so the all-filers pooled trend is partly composition and the mapped-only column isolates it; gap_returns.md takes its t degrees of freedom from power.T_DF and report.py imports power.PLAUSIBILITY_PP (both unchanged in value); summary writes docs/results/rq3/RECORD_COMMIT (the confirmatory record's commit, read only without .git). No estimate, sample, specification or default changed: every CSV under docs/results and every figure under docs/paper/figures is byte-identical to b5d137b after make rq3 and make h6. The strict coverage contract of h6.edgar.hits_from_saved and the no-.git fallbacks of rq3.guard and rq3.report compute no estimate. (b) Process record: C-44, C-47 and C-52 were post-result presentation changes filed as C (not D) entries because no estimate changed; commits 01738b6, 527e2ba and 297ed3b were post-result code refactors/hardening with byte-identical outputs and no entry; C-45's defaults were committed in the same commit as the tercile outputs

(b12cb06) and C-50 in the same commit as the FTS acquisition (8fd155e), so their 'before' timing rests on the implementer's report; C-49 was committed after the denominator counts were printed (no default depended on them); C-35's range should read 'through 2022-10' (the script stops there); D-1 was filed without Ajai's approval under a controller ruling and awaits his ratification. No earlier entry is edited.

B.2.105 C-55. Clarification: Ajai's approval of D-1 (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 3 close-out (Ajai's ratification in chat, 2026-10-23)
- Results seen: RQ1 and H5 confirmatory results seen; all Plan 3 exploratory and H6 results seen
- Rule: Ajai approved D-1 (the re-fetch of two truncated 10-K primary documents, H6 secondary measure) in chat on 2026-10-23, after the before and after values in D-1 were reported to him. D-1 itself is unchanged (§11 is append-only); this entry is the approval its 'Approved by Ajai' field records as not sought at filing. No other entry, specification or estimate changes.

B.2.106 C-56. Clarification: Unbalanced H2 panel (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 12 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: The unbalanced panel (R22) keeps every (task, release) row with a non-null outcome for tasks with such rows in at least two of the listed releases (Plan 2's `build_h2_panel(balanced=False)`); there is no other unbalanced rule.

B.2.107 C-57. Clarification: R10 for H3 (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 11 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: H3's bounds use `validation_share_all_lb = v·(100 − s)/100` and `_ub = lb + s`, with s the release's suppression share (the components of `occupations.SUPPRESSION[release]`), computed per period and then D13-averaged, as for the automation bounds.

B.2.108 C-58. Clarification: R27 standardisation (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 11 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: Consequence of Error = the mean `Data Value` (`CX`) over O*NET-SOC codes within soc6, z-scored across the distinct sample occupations (population SD); rows without a value are dropped and reported as `missing_coe`.

B.2.109 C-59. Clarification: R28 r4/r5 countries (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 13 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- Rule: r4/r5 publish no country usage index or GWA rows: log AUI = ln(usage_pct_c / wap_c) (its scale moves only the intercept); s_cg from the country's `onet_task` pct aggregated to GWAs through the D14 files (equal split over a task's GWAs), renormalised over GWAs with an ā_g; ā_g = the same release's global task automation aggregated to GWA by D13 weights; then Plan 2's `estimate_h4`.

B.2.110 C-60. Clarification: R9 scope (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 10 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: Placebo day-0 dates are the 14 event month-days in 2017, 2018, 2019 and 2021 (56); slopes on A_r2 `z_delegation` at horizons 5 and 20 via Plan 3's `run_events`; per horizon, the actual events' mean slope and its empirical percentile among the 56 placebo slopes, descriptive.

B.2.111 C-61. Clarification: R29 denominator (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 14 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: `pre_formation_2025` counts filings dated 2025-01-01 to 2025-03-26 in both numerator and denominator; `float_weighted` = Σ float·1[talks] / Σ float over 2024 filers with a CY2024Q2I public float (the R29 cap proxy).

B.2.112 C-62. Clarification: R30 lists (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 14 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: The plural rule applies to the primary list only; `broad` is computed for 2024 (T_j's year) only.

B.2.113 C-63. Clarification: R7 (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 8 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: Dropping Hardw, Softw and Chips does not re-standardise z (as R13).

B.2.114 C-64. Clarification: R18 (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 4 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: `nonoverlap_max`: s = the largest of the 17 offset sds; $\bar{b}$ stays the 378-window mean; q is recomputed with the new s.

B.2.115 C-65. Clarification: Exposure-variant benchmarks (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes

- Fixed by: Plan 4 Task 5 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set (R1, R2, R6, R15, R20, R32, R33); occupation-level variants (R3, R11, R16) keep the primary benchmark; `1p_api` pools r3–r6 for A_pooled.

B.2.116 C-66. Clarification: Wage-bill weights (R2) (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 6 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: `A_MEAN #` is set to $239,200 as for the median; `*/**` give a null wage bill excluded from the shares; R2 changes w_oj only (the U_o denominators stay national employment shares).

B.2.117 C-67. Clarification: Employment-weighted crosswalk (R33) (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 6 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: Children without published employment share the residual $\max(E_4 - \Sigma \text{ published}, 0)$ equally; with no published child, the primary equal split.

B.2.118 C-68. Clarification: RQ2 variant bootstrap (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 7 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: A variant's Spearman CIs use Plan 2's `bootstrap_indices(n)["B"]` draw set, so the primary rows reproduce.

B.2.119 C-69. Clarification: H4 parity metric (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 2 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: The adapter's H4 parity check (`tests/robustness/test_upstream.py`) compares `b` for `log_aui` and `log_income` and the joint F statistic at rel 1e-12, and `p_family` through −log10 p at rel 1e-12, because H4's `mix` is order-dependent at about 1e-14 (C-32), which moves a p of 4.8e-24 by about 1e-12 relative. No estimate, specification or committed output changes.

B.2.120 C-70. Clarification: R15 is identical to the primary by construction (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 7 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: R15 (hq="redistribute", D7) is run as the PAP specifies: NAICS 5511's own occupation mix is added to each FF49 industry with weight proportional to that industry's mapped private employment (c = emp_5511 /

mapped private employment = 2.02%). Because the weight is the same fraction of every industry, each industry index moves by the same affine map E_j' = (E_j + c·E_5511)/(1 + c), so the population-SD z (z_delegation, z_usage, z_aioe), every Spearman rank and therefore every H5, H5b and RQ2 estimate are identical to the primary (max |Δz| 8.9e-16; same §9.3 vector hash); only coverage and index levels change. R15 is still run and reported (§8.1) with this note. No non-proportional allocation is substituted: that would be a new specification.

B.2.121 C-71. Clarification: R20 on the 46 industries the majority rule assigns (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 7 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run
- Rule: Under the frozen majority rule (each NAICS code to the FF49 industry with the largest weight; ties to the lowest industry number) Guns (26), Gold (27) and Boxes (40) receive no NAICS code. R20 therefore reports on the 46 industries it assigns, with z re-standardised over those 46 (as the variant's own industry set, like R6); the market part removes the three from the window and from the 378-window history before the null scales are computed (the market runner's drop list).

B.2.122 C-72. Clarification: R2 replaces employment by the wage bill in every industry weight (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 6 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no R2 row run. Before this entry the implementation computed max |Δ z_delegation| against the primary under this rule (0.607) and under the alternative below (0.415); no H5, H5b or RQ2 estimate was computed under either
- Rule: R2's wage bill (TOT_EMP × A_MEAN, C-66) replaces OEWS employment in both weights of the industry aggregation: the occupation shares within each OEWS NAICS code, and the code weight (crosswalk weight × code total) that combines codes into an FF49 industry. w_oj is therefore o's share of FF49 industry j's crosswalk-weighted wage bill, as in the primary it is o's share of j's employment (§2.1). The benchmarks are re-aggregated with the same weights (C-65), and emp_covered_share is wage-bill weighted. U_o keeps national employment shares (C-66). Not used: wage-bill shares within codes combined with employment code weights, which is no industry's wage-bill share.

B.2.123 C-73. Clarification: R14 maps AIOE's 19-1020 by its O*NET successor (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 6 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no R14 row run
- Rule: AIOE's 19-1020 (from O*NET-SOC 2010 19-1020.01 Biologists) is not a SOC 2010 detailed code and is absent from the BLS SOC 2010→2018 crosswalk; it is the only AIOE code the BLS file lacks. It is mapped by its O*NET-SOC 2010→2019 successor 19-1029 (Biological Scientists, All Other; May 2024 employment 59,710, 0.04% of detailed employment), where it is averaged with AIOE's own 19-1029 (0.752 and 1.010 give 0.881), exactly as in the primary mapping. Every other AIOE code uses the BLS crosswalk.

B.2.124 C-74. Clarification: R32's O*NET part is identical to the primary (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 6 (plans committed at a7fb9bd)

- Results seen: RQ1 and H5 confirmatory results and all Plan 3 results seen; no R32 row run
- Rule: The O*NET 29.2 Task Statements file (db_29_2_text, retrieved 2026-10-23) is byte-identical to the 29.3 file used by the primary (SHA-256 fecdcda81a4b8a28e750a6c75ed8bf2d0a9b71186e5be37eec8dc6cef7985d1a, 2,759,559 bytes; latest task Date 08/2024). R32's task index therefore equals the primary's, and R32 differs from the primary only through the OEWS May 2023 weights (oesm23in4, oesm23nat). It is still run and reported as specified (§8.1) with this note. That 29.3 left task statements unchanged is inferred from the byte identity and the task dates, not from O*NET's release notes.

B.2.125 C-75. Clarification: R35 interval (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 8 review (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 8 rows including R35 (d2da8f1) seen
- Rule: R35 reports the interval $\hat{b} \pm t_{21}$,0.975·κ·SE_Σ,2025–26 (pp): the §3.3 P14 interval form with R35's scale κ·SE_Σ in place of s and R35's t_{21} critical, i.e. the set of null values b_0 that T_stud does not reject at level 0.05. It is presentation only: T_stud, its p-value (t_{21}), the criticals and the scale are as PAP §8.2 R35 specifies and do not change; R35.csv is unchanged. As for every Plan 4 market row, the single reported interval uses c_0.05.

B.2.126 C-76. Clarification: R9 label and percentile convention; Design A DSR trial counts (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 10 review (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 10 rows R4, R8, R9, R16 (92e12c3) seen
- Rule: Presentation only; no estimate, sample, specification or default changes, and every numeric column of R04, R08, R09 and R16 .csv stays byte-identical to commit 92e12c3. (1) The applies_to field of R9's 112 placebo records is changed from 'placebo' (a part name that fell through to the field) to 'event studies', the PAP §8.2 R9 and registry value R9's summary records already carry. (2) The 'empirical percentile' of C-60 is 100 × (number of the 56 placebo slopes ≤ the actual mean slope) / 56 at each horizon, descriptive with no p; this convention was fixed in code (0d884ed) before R9 ran and is recorded here. With these data the strict (<) and midrank conventions give the same values (h5 60.71, h20 57.14) because no placebo slope ties the actual mean; an (n+1) denominator, not used, would give 59.65 and 56.14. (3) The deflated Sharpe ratio in a Design A row is computed against the tercile variants that row runs (2 for R4 and R16, 4 for R8; the primary used 6, C-37), so those DSR values are not comparable with the primary's; the rows' notes say so.

B.2.127 C-77. Clarification: R23's disagreement condition (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 11 (rule in rq1.r23_disagrees and its tests at cf00271, before R23 ran; recorded after; plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 11 rows R10, R23, R24, R26, R27, R31, R38 (25480d9) seen
- Rule: PAP §4.1's 'if its coefficients have the opposite sign and are significant at nominal 0.05' is read as: each of R23's two H1 coefficients (education, log wage) has the sign opposite to the corresponding committed primary (automation_pct) H1 coefficient and a two-sided $p < 0.05$ under H1's inference, mirroring H1's conjunctive decision (D17: both coefficients). With these data the condition does not hold and the sentence is not printed:

education +0.0258 against the primary −0.5348, $p < 0.001$; log wage +0.0284 against +4.3952, $p = 0.177$. Readings not used: (i) any one coefficient opposite in sign and significant would print the sentence (education qualifies); (ii) signs taken against H1's predicted negative direction instead of the primary estimates give the same result under this rule and would also print under (i). R23.md states each coefficient's sign comparison and p for either outcome of the check; R23.csv is unchanged.

B.2.128 C-78. Clarification: RQ1 robustness tables label their coefficients (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 11 review (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 11 rows R10, R23, R24, R26, R27, R31, R38 (25480d9) seen
- Rule: Presentation only; every numeric column of R10, R23, R24, R26, R27, R31 and R38 .csv stays byte-identical to commit 25480d9 (R10's market lines to d2da8f1). The RQ1 rows' .md tables name each coefficient (e.g. education, log wage, p_H1); the units sentence states each row's own outcome unit (R23: ai_autonomy_ mean scale points); the notes say n_dropped is not reported for RQ1 rows because Plan 2 records drops as usage shares (C-2). R38's p_H1 is H1's D17 operator (the larger of the two coefficient p-values) applied to R38's one-way occupation-cluster p-values, which is how the brief's 'p_H1 = res.p_family' is read under se_key (§8.1: an RQ1 row uses the inference of the hypothesis it varies).

B.2.129 C-79. Clarification: R37 zero-weight tasks; R23 H2 estimand label (2026-10-23)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 12 (R37 rule in rq1.within_r6 and its tests at 59242bf, before R37 ran; recorded after; plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results, all Plan 3 results, the Task 11 rows, and the Task 12 rows R22, R23, R36, R37 (f8ea9d1) seen
- Rule: (1) R37 leaves out r6 tasks whose mean pct over April and May is 0 (r6 task pct is published rounded to 0.01 and shows 0.00 in both months), as D13 and C-30 (b) do for H1/H3 and as Plan 2's H2 builder does (a zero-usage task-release has no D13 value). Such tasks carry no weight in the WLS fit; n_tasks, G_task and the df (G_task − 1) count tasks with positive weight. With these data 1,141 tasks are left out and 1,133 kept. A task-month whose metric is published but whose pct is null counts as pct 0 in the fixed task weight. (2) Presentation only: R23's H2 record names its estimand in ai_autonomy_mean scale points per year (it said pp), and R23.md's label line separates the H1 and H2 labels by part; every numeric column of R23.csv stays byte-identical to f8ea9d1.

B.2.130 C-80. Clarification: Reproduction tolerance for robustness CSVs (2026-10-24)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 15 (plans committed at a7fb9bd)
- Results seen: RQ1 and H5 confirmatory results, all Plan 3 results, and every Plan 4 row R1–R38 seen
- Rule: A re-run of the Plan 4 rows reproduces every .md byte for byte, but three CSVs differ in the last float digits from run to run: R28 (relative ~9e-12; Plan 2's bound rq1/h4._mix sums after an unordered group_by, C-32), R15 (~1e-15; the crosswalk's hq_shares unordered group_by sum) and R29 float_weighted (~2e-15; an unordered group_by sum). These are summation-order noise, not a code or input change, and no printed number moves. Rule: a regenerated robustness CSV counts as reproducing the committed one when every non-numeric cell is identical and every numeric cell agrees within relative 1e-9 (absolute 1e-12 near zero); in that case the committed bytes are kept, so a re-run leaves the tree clean. A .md must still match byte for byte.

The same rule applies to these CSVs in the fresh-clone check (Task 33). No estimate, specification or committed output changes; the bound Plan 2 code is not edited.

B.2.131 C-81. Clarification: Task 16 red-team label fixes (presentation only) (2026-10-24)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 16 (red-team resolution)
- Results seen: RQ1 and H5 confirmatory results, all Plan 3 results, every Plan 4 row R1–R38, the D-3 diagnostics and the Task 16 red-team findings seen
- Rule: Presentation only; no estimate, sample, inference or decision changes. (1) RQ1 coefficient units (src/aideleg/results/rq1.py UNITS; regenerated by make results-rq1: the unit column of docs/results/rq1/confirmatory_table.csv and the unit text of confirmatory_table.md and decisions.md): a coefficient on a natural log is read per unit of the log, so log_wage is 'pp of automation per unit of log wage (= 100 log points)', log_aui 'pp of automation per unit of log AUI (= 100 log points)' and log_income 'pp of automation per unit of log GDP per working-age person (= 100 log points)', the D16 regressor; PAP §4.1 says 'per log point' for β_w, and this label states that unit exactly; the confirmatory_rq1.* records are untouched (red-team cross C6 on rq1.H1.beta_w, rq1.H4.b1, rq1.H4.b2). (2) docs/redteam/headlines.csv: the six RQ2 z_aioe keys cite §2.1 D6 (the AIOE crosswalk) instead of D8, and the three quantities above carry the new units (red-team rq2 C2). (3) docs/pap/power.md (its generated block) and docs/audit/tables/ (make audit) are outputs that Task 33's fresh-clone compare list must include; both were regenerated byte-identical in scratch by the referee (red-team cross C1 on d11.primary_mde and d3.zero_assigned_pct).

B.2.132 C-82. Clarification: Reproduction tolerance extended to the D-3 diagnostics CSVs (2026-10-24)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 31 review (plans committed at a7fb9bd)
- Results seen: yes: RQ1 and H5 confirmatory results, all Plan 3 results, all Plan 4 robustness rows, the D-3 diagnostics and the Task 16 and Task 31 red-team findings
- Rule: The C-80 rule (a regenerated CSV reproduces the committed one when every non-numeric cell is identical and every numeric cell agrees within relative 1e-9, absolute 1e-12 near zero; a .md must match byte for byte) also applies to docs/results/diagnostics/*.csv. A re-run of the D-3 diagnostics differs only in the last float digit of h2_influence.csv's usage_share_pct (an unordered group_by sum, as in C-80); no printed number moves. No estimate, specification or committed output changes.

B.2.133 C-83. Clarification: Record corrections: C-81's unit wording; the Plan 4 file amended after the confirmatory runs (2026-10-24)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: Plan 4 Task 34 review (plans committed at a7fb9bd; amended at 358c021)
- Results seen: yes: all confirmatory, Plan 3 and Plan 4 results and both red teams seen
- Rule: (1) C-81 says the new RQ1 unit label states the PAP's unit exactly; it corrects it: PAP §4.1's 'per log point' read literally is per 0.01 of the natural log, while the coefficients are per unit of the natural log (a hundred log points), which is what the generated tables now say. No estimate changes. (2) The C-entries cite 'plans committed at a7fb9bd'. The Plan 4 file was amended once more at 358c021 (Plan 3's final fix wave, after both confirmatory runs and before any Plan 4 robustness read); that amendment changed interfaces, test fixtures and guard notes only (hits_from_saved strict/saved_ranges, guard fixture bytes, the R9 placebo name, R4/R8 bootstrap-draw notes, zero_assigned_lines reuse) and no default, specification or row.

B.2.134 C-84. Clarification: Decisions Ajai delegated at the Plan 4 stop (2026-10-25)

- Type: clarification (post-result); results had been seen; the PAP is silent or ambiguous here and no PAP rule or confirmatory code changes
- Fixed by: No file other than this section changes for (1)-(3). Recorded at the Plan 4 stop after commit bf9e2a7.
- Results seen: yes: RQ1 and H5 confirmatory results, all Plan 3 results, all Plan 4 robustness rows and diagnostics, and both red-team rounds
- Rule: Ajai, in chat, 2026-10-24, delegating the open decisions: 'I believe in you to make the best possible decisions for these questions given the context of this project and my behavior'. The controller decided, and nothing here changes an estimate, a sample, a test or a decision: (1) W-24 is accepted: the abstract, capped at 200 words, states both market non-rejections in one sentence carrying every P14 element (the null b = b-bar, the interval and the MDE power clause) and points to section 7, while the brief and the body quote the P14 wording verbatim; (2) the practice of filing post-result changes that alter no estimate (labels, wording, tolerances, record corrections) as clarifications rather than deviations is ratified, with every such entry marked post-result and its results-seen field stated, and any change to an estimate, sample or inference still a deviation needing approval; (3) D-3's leave-out of the top 3 occupations by weight, one step beyond the H1 leave-outs Ajai listed, is ratified as part of D-3: it is labelled post-result and exploratory and changes no committed estimate.

B.2.135 Deviations after the freeze

B.2.136 D-1. Deviation: H6 secondary: two truncated 10-K primary documents fetched again (data-integrity correction) (2026-10-23)

- Type: deviation (needs Ajai's approval; demotion rule of §0.5 if a confirmatory result was seen)
- Reason: Task 13 review found saved primary documents cut off mid-file. Cause: acquire.fetch skips its Content-Length check when the response has a Content-Encoding, and SEC serves gzip, so a body truncated in transfer was saved and indexed. A completeness check is now applied in secondary.download_documents (no change to fetch.py): an .htm/.html body must end with
, then only whitespace, comments or EDGAR's / closers, or else with EDGAR's closer (287 whole bodies whose filers' HTML lacks
, accepted and listed); a .txt body with or (all 24 do), else accepted with the reason stated; a failing body is requested again, 3 attempts, then logged in ERRORS.jsonl and never indexed. An offline scan of all 47,946 saved bodies flagged exactly 2: 0000950170-22-006874 (IronNet, CIK 1777946, irnt-20220131.htm, 62,263 of 3,880,619 bytes, ended mid-XBRL context) and 0000950170-22-005238 (Stryve Foods, CIK 1691936, snax-20211231.htm, 125,262 of 3,397,287 bytes, ended mid-attribute). Both were fetched again through the normal path (contact UA, SEC throttle) and are whole; their INDEX.jsonl lines were replaced in place (one line per key) and sec_docs_ index re-recorded (A-35 supersedes A-34). No specification, sample or default of C-53 changed; this corrects input data only.
- Before: INDEX.jsonl SHA-256 b3380262e475a3bc888b2257dfc6d8a1025dc08c48b16b56afc57849eabb7153, 138,657,694,780 bytes; series 2022: mean over CIKs 0.2736, mean over industries 0.20825 (other years unchanged); 2022 rated filings under 2,000 words 67; agreement 2022: both 1333, FTS only 5, text only 0, neither 6329, agreement 0.999 (FTS-only total 2019-2025: 32); industry rates: Softw 2022 1.4628 (no 2024 rate moved); Spearman(secondary 2024, T_j) 0.5667 (n 49)
- After: INDEX.jsonl SHA-256 c416d6bcba7e41cf92d042d2ec5b09cea75ecd8bae5861f201044953ce3b5fed, 138,664,785,161 bytes; series 2022: mean over CIKs 0.2740, mean over industries 0.20837; 2022 rated filings under 2,000 words 65; agreement 2022: both 1334, FTS only 4, text only 0, neither 6329, agreement 0.999 (FTS-only total 31, of which 30 exhibit-only); industry rates: Softw 2022 1.4686 (no 2024 rate moved); Spearman(secondary 2024, T_j) 0.5667 (unchanged to 16 digits, n 49)
- Results seen: RQ1 and H5 confirmatory results seen; event-study, tercile, Fama–MacBeth, H6 talk/gap and H6 secondary results seen
- Approved by Ajai: not sought: the controller ruled this a data-integrity correction to proceed with, logged in full (Task 13 review); to be relayed to Ajai

B.2.137 D-2. Deviation: hits_from_saved default phrases (2026-10-23)

- Type: deviation (needs Ajai's approval; demotion rule of §0.5 if a confirmatory result was seen)
- Reason: The default of Plan 3's edgar.hits_from_saved(phrases=None) loaded every saved sec_fts/ phrase, although its docstring and Plan 4's contract say fts_phrases(). R30's new automation and robotics responses for 2024 (A-47, 22 records) would have entered the primary H6 numerator on an offline h6 build (h6/**main**.py build calls hits_from_saved() with no phrases): hits_from_saved() returned 22,545 rows instead of the 20,581 of fts_hits.parquet, and two tests failed (test_upstream::test_h6_talk_reproduces_committed, test_edgar_fts::test_hits_from_saved_strict_default_equals_the_derived_parquet). Fix: one line in Plan 3's h6/edgar.py. Process fact: during the verification make h6 run, a stray git stash removed the edgar.py fix from the working tree for a few seconds and it was restored at once with git stash pop; fts_hits.parquet was written after the restore and was checked afterwards (20,581 rows, 15 phrases, equal to hits_from_saved()).
- Before: hits_from_saved(phrases=None) filtered the saved ranges to every saved phrase (wanted = None); with the A-47 records an offline h6 build would add automation/robotics hits to the primary numerator (22,545 rows); the committed H6 outputs were built before A-47 and used the 15 phrases (20,581 rows).
- After: hits_from_saved(phrases=None) filters to fts_phrases() (the 15 phrases); hits_from_saved() equals fts_hits.parquet (20,581 rows); make -o data/derived/exposure_ff49.parquet h6 regenerates docs/results/h6 and docs/paper/figures byte-identical (git status empty). Tests: new test_hits_from_saved_default_excludes_extra_saved_phrases; the last assertion of test_hits_from_saved_strict_default_equals_the_derived_parquet amended from set(saved_ranges().phrase) == set(fts_phrases()) to set(fts_phrases()) <= set(saved_ranges().phrase), because A-47 adds phrases to the MANIFEST; its parquet equality unchanged.
- Results seen: RQ1 and H5 confirmatory results, all Plan 3 results, and Plan 4 rows R1–R28 seen; R29/R30 run, uncommitted
- Approved by Ajai: Ajai, in chat, 2026-10-23: 'I approve D-2: make hits_from_saved(phrases=None) mean fts_phrases(), as its docstring says. Log it in §11 with my approval, confirm the primary H6 outputs are byte-identical, then commit Task 14 and continue Plan 4.'

B.2.138 D-3. Deviation: Post-result influence diagnostics (2026-10-24)

- Type: deviation (needs Ajai's approval; demotion rule of §0.5 if a confirmatory result was seen)
- Reason: the Task 16 red-team (referee files .superpowers/sdd/2026-10-21-plan-4-paper/redteam/rq1.csv and rq3_h6.csv) found influence and uncertainty that no generated file reports: C8 on rq1.H1.beta_e and rq1.H1.beta_w (a read-only referee leave-out moved beta_w from 4.40 to 0.34, p 0.80, without SOC major group 15 and to 1.50, p 0.37, without occupation 15-1299, and beta_e to -0.015, p 0.98, without SOC 15); C8 on rq1.H2.beta (leaving out r2 gave beta -6.55); C7 on the four H2 keys (the balanced panel drops the tasks of R22's unbalanced panel, 2205 vs 1315, and neither their count nor their usage share is printed); C8 and C11 on the three h6.spearman keys (no leave-one-industry-out sensitivity is generated, and the Spearman has a CI but no p, PAP §0.2)
- Before: no generated file reported these sensitivities; H6 Spearman had a CI but no p
- After: three generated post-result diagnostics, docs/results/diagnostics/h1_influence.{csv,md} (estimate_h1 on the primary task rows leaving out each SOC major group, the top occupation by weight and the top 3), h2_influence.{csv,md} (estimate_h2 leaving out each release of the primary balanced panel, and the tasks and usage share the balancing removes) and h6_influence.{csv,md} (the 2024 T vs Design B z_usage Spearman leaving out each FF49 industry, and a nominal exploratory two-sided p for the primary Spearman by the t-approximation with n - 2 df), written by uv run python -m aideleg.robustness –diagnostics (src/aideleg/robustness/diagnostics.py) through Plan 2's and Plan 3's estimators, each labelled exploratory; not a decision; no confirmatory or committed estimate changes
- Results seen: yes: RQ1 and H5 confirmatory results, all Plan 3 results, all Plan 4 robustness rows, and the Task 16 red-team findings
- Approved by Ajai: "I approve D-3: add the labelled post-result influence diagnostics (H1 leave-out by SOC major group and top occupation, H2 leave-one-release-out and the balanced-panel drop count, H6 leave-one-industry-out Spearman and a nominal exploratory p). No confirmatory or committed estimate

changes. Log it in", in chat, 2026-10-24

B.3 PAP §10: corrections to the design spec, verbatim

The spec is not edited. Where it is wrong on a fact, or where this plan departs from it, the item is listed here with the correction this plan applies.

B.3.1 10.1 Corrections of fact

- **S1. §5.3, real-time weights.**
  - *Spec:* OEWS May 2024 is "the latest data before the first market-test formation date".
  - *Correction:* OEWS May 2024 was released on **2025-04-02**, after the Design B formation date (2025-03-27) and one trading day into the return window. The vintage public at formation was May 2023. The O*NET 29.3 task statements used to link AEI tasks to occupations were released in May 2025, also after formation.
  - *Consequence:* Design B is out of sample in its AEI data, but not strictly real-time in its occupation weights and task links. Strict-real-time variant R32 [facts §1; §2.1].
- **S2. §5.4.1, the start of AI autonomy.**
  - *Spec:* "Mean AI autonomy (from release 3)".
  - *Correction:* `ai_autonomy_mean` is published from **r4** (2026-01-15), so it covers r4–r6 [MET; audit, "Decisions the audit forces", item 2].
- **S3. §5.4, Design A's release.**
  - *Spec:* Design A uses exposure "built from the *earliest* release only (Feb 2025)", i.e. r1 (2025-02-10).
  - *Correction:* r1 has only an aggregate collaboration split, with no task- or occupation-level automation (as spec §5.4.1 itself says). The earliest release that can form a delegation index is r2 (2025-03-27). So A_r2 is the spec's Design A, numerically identical to Design B. This plan's A_pooled (r2–r6) is an addition. It uses releases published during and after the return window, so it contains look-ahead and is labelled so.
- **S4. The 5 pp plausibility threshold.**
  - *Spec:* §5.4 says "implausibly large" and gives no number.
  - *Correction:* the 5 pp figure comes from plan 1 (Task 15, Step 1, item 7), committed before the power analysis ran. It is confirmed by Ajai after the MDE is known (§7, D11). Earlier wording in `power.md` that called it "the spec's" threshold has been corrected.
- **S5. §4, the 16.6% of employment with no AEI match.**
  - *Spec:* 16.6% is the complement of the "83.4% of US employment" in the spec's preliminary join test, run before the matching rules (rollup, hybrid codes) were built.
  - *Correction:* the audited figures, which supersede it, measure three different things:
    - **7.39%** of May 2024 employment is in OEWS occupations with no AEI r6 published occupation, after matching [audit §6];
    - **12.14%** is zero-assigned under Design B (r2, which has fewer matched tasks), the figure that matters for the confirmatory test;
    - **6.34%** is zero-assigned under A_pooled.

B.3.2 10.2 Departures from the spec (each a proposed decision or a stated change)

- **D-1. §5.4.1, H5/H5b inference.** RI is replaced as primary by the historical-null test (D10) because of its documented size (§3.2). RI remains secondary.
- **D-2. §5.4, the wild-cluster bootstrap.** Not used for H5/H5b, which have no cluster structure (§3.2). It is used for H2 (D22).
- **D-3. §5.3, the crosswalk.** The spec says "employment-weighted crosswalk". The implementation splits equally (D23). The spec's variant is R33.
- **D-4. §5.2, H1 standard errors.** One-way clustering by occupation becomes two-way clustering by occupation and task (§4). This is intended as a conservative change; the spec's one-way version is R38.

- **D-5. §2, H5 null.** H5/H5b are tested against the frozen historical slope $\bar{b}$ rather than zero (D21), which is a narrower claim than the approved wording. This needs Ajai's approval.
- **D-6. §2 and §5.2, H2.** H2 is reworded as "measured within-task automation rises", tested outside the RQ1 Holm family at $\alpha = 0.0125$ (the Holm family {H1, H3, H4} at 0.0375), with a release-level co-condition (D22). Its primary span is r2–r5 rather than "Dec 2024 → May 2026" (D2).
- **D-7. §5.4, event dates.** The spec's "pre-registered model-launch dates" are fixed by an objective rule over a named source (D19).

C Disclosures

C.1 Before the freeze (PAP §0.4)

What was seen before this plan was frozen:

1. **Task 3 incident.** While inspecting the French file layout, the Task 3 implementer subagent displayed aggregate monthly FF5 factor rows for 2026-05..2026-08 and the momentum row for 2026-08. These are aggregate factors, not industry returns. They were not used or recorded, and they carry no information about the cross-sectional slope of industry alphas on exposure (H5/H5b). The controller ruled it not contamination (ledger, Task 3); it is disclosed here as required.
2. **Retracted figure.** The Task 6 finding that renormalising task automation over published patterns inflates it by "7–9 points" (r3 57.8 vs 51.1) is **retracted**. It came from an uncommitted scratch script. Re-derived on the committed panel, the gap is 51.20 vs 51.07 (0.13–0.35 points for `claude_ai`, 1.11–1.86 for `1p_api`) [audit §4]. No decision in this plan relies on the retracted figure.
3. **Exposure descriptives seen before the freeze.** The researchers saw: the top and bottom FF49 industries by Design B `z_delegation` (top Softw 4.739, next 1.977; bottom Meals, Rtail, Cnstr); its skewness (2.571); the within-group dispersion of U in SOC 15-125x; the Spearman correlations of the "could" benchmarks with `z_delegation` and `z_usage` across FF49 (AIOE 0.763 / 0.809, Eloundou human β 0.716 / 0.768, GPT-4 β 0.815 / 0.843); Pearson(`z_aioe`, `z_delegation`) = 0.671; the correlations of `z_delegation` with its R7 bounds (≥ 0.9997); the allocated-vs-published r6 SOC correlations [audit §5, §8, §9]; the leverage and OLS weight of every industry in the H5/H5b regressions [power, leverage table]. These are RQ2 descriptives and design diagnostics. No delegation measure has been related to wage, education, error-cost or country variables (H1–H4), and no industry return dated on or after 2022-11-30 has been read by code.
4. **Pre-seal market history.** The power analysis estimated FF5+UMD alphas and cross-sectional slopes on 1990–2022 windows (returns read to 2022-11-29; in Task 15b also windows from 1963-07 and 1969-07). Printed: slope dispersions, RI rejection rates, the D10 rule's in-sample rejection rates (centred and uncentred), empirical centred quantiles, kurtosis, leverage, and the **frozen mean slopes $\bar{\mathbf{b}}$** for H5, H5b and their R12/R13 variants. The referee ruled that freezing $\bar{b}$ is harmless because it is pre-seal and the exposure vector is fixed (ruling on C3).

The frozen values therefore show the level and sign of the mean historical slope. They are not interpreted anywhere in this plan. The recommendation in D21 (centre the null on $\bar{b}$) was made **after** the uncentred in-sample rejection rates (7.1% for H5, 3.2% for H5b at nominal 2.5%) had been seen.

5. **General knowledge of the sealed period.** The seal stops code from reading post-2022-11-29 returns. It cannot remove what the researchers (or the language model assisting them) know from public commentary about broad market moves in 2023–2026, for example the AI-related rally in large technology stocks. This is why Design A is exploratory and why the confirmatory test (§3) is fixed here, together with its exposure vector, its frozen null values and the script that computes it (§9.3), before any post-seal return is read.
6. **D19's event-rule exceptions were chosen with the source list in view** (Gemma and Gemini Ultra excluded, ChatGPT added, GPT-4o absent from the source); see §8.3.
7. **The confirmatory script exists and has been tested on synthetic data only.** `aideleg.confirm_h5` was written before the tag. Its tests monkeypatch the seal and use synthetic returns. It has not been run on real data, and `docs/results/` does not exist.

C.2 After the freeze: facts recorded in the Plan 2–4 ledgers

Each fact below happened after the tag `pap-v1` (2026-10-21) and is recorded in one of the working ledgers of Plans 2–4 or in an entry of PAP §11: what was seen, when, by whom, and the consequence. The ledgers are:

Plan 2: `docs/superpowers/ledgers/2026-10-21-plan-2-behaviour-ledger.md`
Plan 3: `docs/superpowers/ledgers/2026-10-21-plan-3-markets-ledger.md`
Plan 4: `docs/superpowers/ledgers/2026-10-21-plan-4-paper-ledger.md`

Dates are §11 entry dates (UTC) or the dates of the commits named. "The controller" is the agent session that ran each plan; "an implementer" or "a referee" is a subagent it dispatched. No fact here is interpreted.

- **Machine restart (Plan 2).** What: the first dispatch of Plan 2 Task 1 was interrupted by a machine restart, with no changes made. When: 2026-10-22. Who: the controller. Consequence: the task was dispatched again. Source: Plan 2 ledger, line 23.
- **Stray unit-test artefact before the RQ1 run.** What: while a failing test for the panel-hash refusal was being recorded, a unit test called `confirm.run` against the real `docs/results/rq1` with the clean-tree check stubbed (HEAD recorded as "abc") and `load_inputs` stubbed to raise before reading any data; it wrote an attempt marker, a log and a markdown file holding only the stub's traceback. No data were loaded and no outcome was read. When: 2026-10-22, 19:47:28 UTC. Who: a Plan 2 Task 10 implementer. Consequence: the files were never committed and were removed; a copy is kept under `docs/pap/evidence/c-31`; the test fixture now redirects the output directory; disclosed as §11 C-31 (commit 3ef3fb7). Source: Plan 2 ledger, lines 86–87; §11 C-31.
- **RQ1 single run (P-RQ1).** What: the single logged first run of H1–H4 (`make rq1-confirm`) on real outcomes; statuses as printed: H1 not_rejected, H2 not_rejected, H3 not_rejected, H4

supported. When: 2026-10-22 (output commit 75e8956, run at HEAD edface2). Who: the Plan 2 controller. Consequence: the output was committed unedited (4 files); from this point every RQ1 confirmatory result had been seen, and later §11 entries say so in their "Results seen" field. Source: Plan 2 ledger, lines 89–92.

- **C-32 corrected in place after filing.** What: §11 C-32 (the H4 task-mix control reproduces to about 1e-14) was first filed as a pre-result clarification although the RQ1 confirmatory results had been seen; the Plan 2 final fix-wave re-review found this. When: 2026-10-22. Who: the Plan 2 final reviewer (finding) and the controller (ruling). Consequence: paplog gained the "Clarifications (post-result)" subsection and C-32 was corrected in place on the unmerged branch: an edit to an already-filed §11 entry. Source: Plan 2 ledger, line 116; §11 C-32.
- **H5/H5b single run (P-H5).** What: the single confirmatory run of `aideleg.confirm_h5` from a worktree at `pap-v1`, outcome A (exit 0, both files written); the fixed wording printed for H5 and H5b is quoted in the ledger. When: 2026-10-22 (commit ebe59be). Who: the Plan 3 controller. Consequence: the output and attempt marker were committed unedited (2 files); from this point the H5 and H5b confirmatory results had been seen. Source: Plan 3 ledger, lines 18–22.
- **Plan 3 results-seen points.** What: Plan 3's §11 entries record what had been seen when each was filed: "RQ1 confirmatory results seen; H5/H6 not yet run" (C-33 to C-43, before the H5 run), then, after it, "RQ1 and H5 confirmatory results seen" together with event-study results (C-44, C-45), event-study and tercile results (C-46), event-study, tercile and Fama–MacBeth results (C-47 to C-51), H6 talk/gap results (C-52, C-53), H6 secondary results (D-1), and all Plan 3 exploratory and H6 results (C-54, C-55). When: 2026-10-22 to 2026-10-23. Who: the Plan 3 controller and implementers. Consequence: each entry prints its state in its "Results seen" field (Ruling P3-R2). Source: Plan 3 ledger, line 8; §11 C-33 to C-55.
- **Event-study real run executed twice.** What: Plan 3 Task 5's real event-study run was executed twice after its output had been seen; the second run fixed one sentence of a .md file (A_pooled = r2–r6), and the CSVs were byte-identical. When: 2026-10-22. Who: a Plan 3 Task 5 implementer. Consequence: reported as a concern and accepted at review (the sentence is correct). Source: Plan 3 ledger, lines 46 and 48.
- **Process record C-54(b) and later entries filed after their runs.** What: C-54(b) records that C-44, C-47 and C-52 were post-result presentation changes filed as C entries; that commits 01738b6, 527e2ba and 297ed3b were post-result refactors or hardening with byte-identical outputs and no entry; that C-45's defaults were committed in the same commit as the tercile outputs (b12cb06) and C-50 in the same commit as the full-text-search acquisition (8fd155e), so their "before" timing rests on the implementer's report; that C-49 was committed after the denominator counts were printed; and that D-1 awaited Ajai's ratification. In Plan 4, C-75 (R35 interval) was filed after R35 had run and C-79 (R37 zero-weight tasks; R23 H2 estimand label) after R37 and R23 had run. When: 2026-10-22 to 2026-10-23. Who: the Plan 3 and Plan 4 controllers. Consequence: each is presentation or a rule fixed in code before the run; no estimate changed. Source: §11 C-54; Plan 4 ledger, lines 62 and 87; §11 C-75, C-79.
- **`git checkout -- src` by a fix agent.** What: a Plan 3 Task 4 fix agent ran `git checkout`

`-- src` to undo an accidental `ruff format`. When: 2026-10-22. Who: a Plan 3 Task 4 implementer. Consequence: the tree was clean afterwards; the concurrent Task 5 agent was warned to re-check its tracked edits. Source: Plan 3 ledger, line 44.

- **H6 denominator counts printed before C-49; one unrecorded index read.** What: the H6 denominator counts were printed at build time before C-49 was committed; one ad-hoc 2024Q1 `form.idx` request (about 300 KB) was used to read the file layout, then discarded and not recorded in the MANIFEST. When: 2026-10-22. Who: a Plan 3 Task 10 implementer. Consequence: no default changed afterwards and none depends on the counts; disclosed in the process record C-54(b). Source: Plan 3 ledger, line 83; §11 C-54.
- **The Mac sleeping during the 10-K download.** What: while the 47,946 10-K primary documents for the H6 secondary measure were being downloaded (retrieved 2026-10-23 02:20:56 UTC to 09:30:11 UTC, §11 A-34), the controller saw from the INDEX.jsonl retrieval stamps a near-stall from about 02:30 to 03:40 UTC (588 documents in 02:2x, then 9–20 per 10 minutes until 03:4x) while the downloader's own elapsed clock showed about 13 minutes; the controller read this as the Mac sleeping. When: 2026-10-23. Who: the Plan 3 controller (observation); a Plan 3 Task 13 implementer (download). Consequence: the agent then held `caffeinate`, and no system settings were changed; the download finished with 0 documents skipped and 0 errors; reported to Ajai in chat at the time. Source: Plan 4 ledger, line 147 ("Controller record (for disclosure sourcing)"); Plan 3 ledger, line 104; §11 A-34.
- **Truncated 10-K documents (D-1).** What: two saved primary documents were cut off mid-file (IronNet 2022, 62,263 of 3,880,619 bytes; Stryve Foods 2021, 125,262 of 3,397,287 bytes), because `acquire.fetch` skips its Content-Length check when SEC serves gzip. When: found 2026-10-23 at the Task 13 review. Who: the Plan 3 Task 13 reviewer. Consequence: both were fetched again whole, the H6 secondary measure was rerun, A-35 supersedes A-34, and D-1 records the before and after values (only 2022 values moved; the 2024 Spearman is unchanged). D-1 was filed under a controller ruling without Ajai's approval. Source: Plan 3 ledger, lines 105–109; §11 D-1.
- **Ajai ratifies D-1 (C-55).** What: Ajai approved D-1 in chat after its before and after values were reported. When: 2026-10-23 (commit be917b6). Who: Ajai. Consequence: recorded as §11 C-55; D-1 itself is unchanged. Source: Plan 4 ledger, line 4; §11 C-55.
- **R2 weight differences computed before C-72.** What: before C-72 was filed, the implementation computed max $|\Delta$ z_delegation$|$ against the primary under the adopted R2 rule (0.607) and under the alternative (0.415); no H5, H5b or RQ2 estimate was computed under either. When: 2026-10-23. Who: a Plan 4 Task 6 implementer. Consequence: disclosed in C-72's "Results seen" field. Source: Plan 4 ledger, line 49; §11 C-72.
- **R23 disagreement reading (C-77).** What: R23's condition was read as both H1 coefficients opposite in sign to the primary and significant at 0.05 (committed with its tests at cf00271 before the run); after the run, the "any one coefficient" reading was seen to print the sentence (education $+0.0258$ against the primary -0.5348, $p < 0.001$; log wage $+0.0284$ against $+4.3952$, $p = 0.177$). When: 2026-10-23. Who: the Plan 4 Task 11 implementer and reviewer, and the controller. Consequence: the pre-run reading was kept; the alternative is disclosed in C-77;

R23.md states each coefficient's sign comparison; relayed to Ajai in chat at the time (2026-10-23, as information, with no approval needed). Source: Plan 4 ledger, lines 80 and 148; §11 C-77.

- **H6 robustness responses and `hits_from_saved` (D-2).** What: after R29/R30 ran (uncommitted), the default of Plan 3's `hits_from_saved(phrases=None)` was found to load every saved `sec_fts/` phrase, so the new automation and robotics records (A-47) would enter the primary H6 numerator on an offline rebuild (22,545 rows instead of 20,581); two parity tests failed. When: 2026-10-23. Who: a Plan 4 Task 14 implementer. Consequence: Task 14 stopped; Ajai approved D-2 in chat; the one-line fix was applied and the primary H6 outputs were verified byte-identical. Source: Plan 4 ledger, lines 97–101; §11 D-2.
- **Stray `git stash` during D-2's verification.** What: during the verification `make h6` run, a stray `git stash` removed the D-2 fix from the working tree for a few seconds; it was restored at once with `git stash pop`. When: 2026-10-23. Who: a Plan 4 Task 14 implementer. Consequence: `fts_hits.parquet` was written after the restore and checked afterwards (20,581 rows, 15 phrases, equal to `hits_from_saved()`); the stash list was empty. Source: Plan 4 ledger, line 99; §11 D-2.
- **Summation-order differences in three robustness CSVs (C-80).** What: an offline re-run of every Plan 4 row reproduced every .md byte for byte and 33 of 36 CSVs; R15 (about 1e-15), R28 (about 9e-12) and R29 float_weighted (about 2e-15) differed in the last float digits from run to run (unordered polars `group_by` sums). When: 2026-10-24. Who: a Plan 4 Task 15 implementer. Consequence: the controller filed C-80 (a CSV reproduces when non-numeric cells are equal and numeric cells agree within relative 1e-9, absolute 1e-12), restored the three committed CSVs, and the writer keeps the committed bytes when equivalent. Source: Plan 4 ledger, lines 105–107; §11 C-80.
- **Task 16 red team.** What: four hostile referees checked 708 headline–challenge pairs (59 headline keys): 577 holds, 87 fails, 44 not applicable; no fail concerned a confirmatory decision. When: 2026-10-23 to 2026-10-24. Who: four referee subagents dispatched by the Plan 4 controller. Consequence: label fixes (C-81), 63 binding wording requirements (`docs/redteam/wording_requirements.md`) and the D-3 diagnostics. Source: Plan 4 ledger, lines 114–122.
- **Influence diagnostics (D-3).** What: the red-team referees' read-only leave-out checks showed influence that no generated file reported (for example H1 β_w from 4.40 to 0.34 without SOC major group 15; H2 β -6.55 without r2). When: 2026-10-24. Who: Task 16 referees; approved by Ajai in chat. Consequence: three labelled post-result diagnostics under `docs/results/diagnostics/`; no confirmatory or committed estimate changed. Ajai's approval message was cut at 300 characters; the recorded intent (log it in §11 with the approval and continue) is the request the message answered. Source: Plan 4 ledger, lines 120–123; §11 D-3.

D Robustness tables

D.1 Robustness appendix: summary (PAP §8.2)

Generated by `uv run python -m aideleg.robustness --summary`. Do not edit by hand.

Every row is reported whatever its result. No row changes a confirmatory decision (R13 is the confirmatory script's co-condition, reported in `confirmatory_h5.md`), no row carries error control across rows, and rows are not summarised by counting (PAP §8.1).

Every market interval of the robustness rows (H5, H5b and the H6 gap–return slope) is the c_0.05 interval $\hat{b} \pm$ c_0.05·s, with s and c recomputed for the row (R35: its own scale and t_{21} critical, PAP §11 C-75); P14 also defines the c_0.025 interval $\hat{b} \pm$ c_0.025·s, which the row files do not print (both criticals are in each market record).

Any prose that calls a market robustness result a non-rejection uses the fixed P14 wording of `docs/results/confirmatory_h5.md` verbatim (PAP §3.3 P14).

Each line puts a row's estimate beside the primary's, both formatted as in the row files (pp and RQ1 coefficients to 2 decimals, correlations to 3); the R12 and R13 lines quote `confirmatory_h5.md` at its printed precision. Primaries are read from the committed files: H5 and H5b from `confirmatory_h5.md` and the H6 rows from `h6/gap.md` and `h6/gap_returns.md` (as printed, then formatted); RQ1 from `rq1/confirmatory_rq1.json`, only for records with the primary's outcome and units; RQ2 from `rq2/could_vs_does.csv` (Design B, or A_pooled for the A_pooled rows); Design A from `rq3/terciles.csv`, `rq3/fama_macbeth.csv` and `rq3/event_study.csv`; the r6 descriptives from `rq1/anatomy_wage_decile.csv`. Intervals, statistics, p-values and notes are in the row files. Implementation choices where the PAP is silent are listed in implementation_choices.md.

row	applies to	label	PAP source	part	estimand	variant	primary	file
R1	H5	robustness; look-ahead	spec §5.6; P16	market	slope of annualised FF5+UMD alpha on z_delegation	1.96	1.96	R01.md
R1	H5b	robustness; look-ahead	spec §5.6; P16	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.51	3.46	R01.md
R1	RQ2	robustness; look-ahead	spec §5.6; P16	rq2	spearman z_eloundou vs z_usage	0.779	0.768	R01.md
R1	RQ2	robustness; look-ahead	spec §5.6; P16	rq2	spearman z_eloundou vs z_delegation	0.724	0.716	R01.md
R1	RQ2	robustness; look-ahead	spec §5.6; P16	rq2	spearman z_eloundou vs z_augmentation	0.783	0.783	R01.md
R1	RQ2	robustness; look-ahead	spec §5.6; P16	rq2	spearman z_eloundou_gpt4 vs z_usage	0.846	0.843	R01.md
R1	RQ2	robustness; look-ahead	spec §5.6; P16	rq2	spearman z_eloundou_gpt4 vs z_delegation	0.818	0.815	R01.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R1	RQ2	robustness; look-ahead	spec §5.6; P16	rq2	spearman z_eloundou_gpt4 vs z_augmentation	0.849	0.850	R01.md
R1	RQ2	robustness; look-ahead	spec §5.6; P16	rq2	spearman z_aioe vs z_usage	0.813	0.809	R01.md
R1	RQ2	robustness; look-ahead	spec §5.6; P16	rq2	spearman z_aioe vs z_delegation	0.768	0.763	R01.md
R1	RQ2	robustness; look-ahead	spec §5.6; P16	rq2	spearman z_aioe vs z_augmentation	0.824	0.825	R01.md
R2	H5	robustness	spec §5.6	market	slope of annualised FF5+UMD alpha on z_delegation	1.75	1.96	R02.md
R2	H5b	robustness	spec §5.6	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.36	3.46	R02.md
R2	RQ2	robustness	spec §5.6	rq2	spearman z_eloundou vs z_usage	0.814	0.768	R02.md
R2	RQ2	robustness	spec §5.6	rq2	spearman z_eloundou vs z_delegation	0.786	0.716	R02.md
R2	RQ2	robustness	spec §5.6	rq2	spearman z_eloundou vs z_augmentation	0.822	0.783	R02.md
R2	RQ2	robustness	spec §5.6	rq2	spearman z_eloundou_gpt4 vs z_usage	0.879	0.843	R02.md
R2	RQ2	robustness	spec §5.6	rq2	spearman z_eloundou_gpt4 vs z_delegation	0.868	0.815	R02.md
R2	RQ2	robustness	spec §5.6	rq2	spearman z_eloundou_gpt4 vs z_augmentation	0.878	0.850	R02.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R2	RQ2	robustness	spec §5.6	rq2	spearman z_aioe vs z_usage	0.847	0.809	R02.md
R2	RQ2	robustness	spec §5.6	rq2	spearman z_aioe vs z_delegation	0.817	0.763	R02.md
R2	RQ2	robustness	spec §5.6	rq2	spearman z_aioe vs z_ augmentation	0.855	0.825	R02.md
R3	H5	robustness	D3	market	slope of annualised FF5+UMD alpha on z_delegation	1.55	1.96	R03.md
R3	H5b	robustness	D3	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_ aioe	2.44	3.46	R03.md
R3	RQ2	robustness	D3	rq2	spearman z_ eloundou vs z_ usage	0.667	0.768	R03.md
R3	RQ2	robustness	D3	rq2	spearman z_ eloundou vs z_ delegation	0.568	0.716	R03.md
R3	RQ2	robustness	D3	rq2	spearman z_ eloundou vs z_ augmentation	0.648	0.783	R03.md
R3	RQ2	robustness	D3	rq2	spearman z_ eloundou_gpt4 vs z_usage	0.771	0.843	R03.md
R3	RQ2	robustness	D3	rq2	spearman z_ eloundou_gpt4 vs z_delegation	0.710	0.815	R03.md
R3	RQ2	robustness	D3	rq2	spearman z_ eloundou_gpt4 vs z_ augmentation	0.757	0.850	R03.md
R3	RQ2	robustness	D3	rq2	spearman z_aioe vs z_usage	0.727	0.809	R03.md
R3	RQ2	robustness	D3	rq2	spearman z_aioe vs z_delegation	0.643	0.763	R03.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R3	RQ2	robustness	D3	rq2	spearman z_aioe vs z_ augmentation	0.720	0.825	R03.md
R4	RQ2	robustness; exploratory	spec §5.6	rq2	spearman z_ eloundou vs z_ usage	0.772	0.846	R04.md
R4	RQ2	robustness; exploratory	spec §5.6	rq2	spearman z_ eloundou vs z_ delegation	0.769	0.770	R04.md
R4	RQ2	robustness; exploratory	spec §5.6	rq2	spearman z_ eloundou vs z_ augmentation	0.777	0.815	R04.md
R4	RQ2	robustness; exploratory	spec §5.6	rq2	spearman z_ eloundou_gpt4 vs z_usage	0.846	0.888	R04.md
R4	RQ2	robustness; exploratory	spec §5.6	rq2	spearman z_ eloundou_gpt4 vs z_delegation	0.831	0.855	R04.md
R4	RQ2	robustness; exploratory	spec §5.6	rq2	spearman z_ eloundou_gpt4 vs z_ augmentation	0.858	0.876	R04.md
R4	RQ2	robustness; exploratory	spec §5.6	rq2	spearman z_aioe vs z_usage	0.798	0.867	R04.md
R4	RQ2	robustness; exploratory	spec §5.6	rq2	spearman z_aioe vs z_delegation	0.788	0.813	R04.md
R4	RQ2	robustness; exploratory	spec §5.6	rq2	spearman z_aioe vs z_ augmentation	0.814	0.858	R04.md
R4	A_pooled	robustness; exploratory	spec §5.6	index	records without a primary counterpart (in the row file)	see file	—	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	alpha_CAPM A_ pooled cap	4.66	4.60	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	alpha_FF3 A_ pooled cap	5.02	5.07	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	alpha_ FF5+UMD A_ pooled cap	3.79	3.66	R04.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R4	terciles	robustness; exploratory	spec §5.6	terciles	mean A_pooled cap	7.58	7.92	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	sharpe A_pooled cap	0.82	0.82	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	psr A_pooled cap	0.95	0.95	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	dsr A_pooled cap	0.91	0.68	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	alpha_CAPM A_ pooled ew	2.33	0.52	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	alpha_FF3 A_ pooled ew	1.96	0.86	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	alpha_ FF5+UMD A_ pooled ew	0.95	-0.41	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	mean A_pooled ew	3.02	2.66	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	sharpe A_pooled ew	0.39	0.32	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	psr A_pooled ew	0.77	0.72	R04.md
R4	terciles	robustness; exploratory	spec §5.6	terciles	dsr A_pooled ew	0.67	0.30	R04.md
R4	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM z A_pooled	0.20	0.17	R04.md
R4	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM log_size A_ pooled	-0.09	-0.07	R04.md
R4	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM log_beme A_ pooled	-0.01	0.00	R04.md
R4	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM mom A_ pooled	0.23	0.24	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope ChatGPT h5 A_ pooled	-0.33	-0.39	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope ChatGPT h20 A_ pooled	0.43	0.26	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope ChatGPT h9 none [z_aioe in z_aioe]	-0.13	-0.13	R04.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope ChatGPT h9 A_ pooled [z_aioe in z_aioe+z_ delegation]	-0.17	0.01	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope ChatGPT h9 A_ pooled [z_ delegation in z_ aioe+z_ delegation]	0.05	-0.21	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope GPT-4 h5 A_pooled	0.65	0.66	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope GPT-4 h20 A_pooled	0.46	0.41	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope PaLM 2 h5 A_pooled	1.18	1.22	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope PaLM 2 h20 A_pooled	2.04	2.19	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Claude 2 h5 A_ pooled	0.43	0.45	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Claude 2 h20 A_ pooled	0.52	0.39	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Claude 2.1 h5 A_ pooled	0.01	-0.04	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Claude 2.1 h20 A_pooled	0.17	0.12	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Gemini 1.0 h5 A_ pooled	0.14	0.15	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Gemini 1.0 h20 A_pooled	0.02	-0.04	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Gemini 1.5 h5 A_ pooled	-0.63	-0.68	R04.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Gemini 1.5 h20 A_pooled	-0.93	-1.09	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Claude 3 h5 A_ pooled	-0.13	-0.24	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Claude 3 h20 A_ pooled	-0.50	-0.70	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Claude 3.5 h5 A_ pooled	0.40	0.40	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Claude 3.5 h20 A_pooled	-0.21	-0.22	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope OpenAI o1 h5 A_ pooled	-0.07	-0.06	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope OpenAI o1 h20 A_pooled	-0.10	-0.14	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Gemini 2.0 h5 A_ pooled	-0.07	-0.19	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Gemini 2.0 h20 A_pooled	-0.12	-0.22	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Claude 3.7 h5 A_ pooled	-0.20	-0.22	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Claude 3.7 h20 A_pooled	-0.27	-0.40	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope GPT-4.5 h5 A_ pooled	0.17	0.12	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope GPT-4.5 h20 A_ pooled	-0.56	-0.63	R04.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Gemini 2.5 h5 A_pooled	-0.51	-0.46	R04.md
R4	event studies	robustness; exploratory	spec §5.6	events	CAR slope Gemini 2.5 h20 A_pooled	-1.20	-1.12	R04.md
R5	H5	robustness	spec §5.6	market	slope of annualised FF5+UMD alpha on z_delegation	4.88	1.96	R05.md
R5	H5b	robustness	spec §5.6	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	4.82	3.46	R05.md
R6	H5	robustness	spec §5.6; P10	ff30	slope of annualised FF5+UMD alpha on z_delegation	1.55	1.96	R06.md
R6	H5b	robustness	spec §5.6; P10	ff30	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	0.72	3.46	R06.md
R6	H5	robustness	spec §5.6; P10	ff48	slope of annualised FF5+UMD alpha on z_delegation	2.17	1.96	R06.md
R6	H5b	robustness	spec §5.6; P10	ff48	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.81	3.46	R06.md
R7	H5	robustness	spec §5.6	market	slope of annualised FF5+UMD alpha on z_delegation	-1.72	1.96	R07.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R7	H5b	robustness	spec §5.6	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	-0.24	3.46	R07.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_CAPM A_r2 cap	3.83	4.64	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_FF3 A_r2 cap	6.99	5.12	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_FF5+UMD A_r2 cap	5.50	3.71	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	mean A_r2 cap	7.89	7.97	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	sharpe A_r2 cap	0.73	0.82	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	psr A_r2 cap	0.89	0.95	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	dsr A_r2 cap	0.79	0.68	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_CAPM A_r2 ew	-0.48	0.00	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_FF3 A_r2 ew	1.84	0.86	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_FF5+UMD A_r2 ew	1.90	-0.14	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	mean A_r2 ew	2.87	2.65	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	sharpe A_r2 ew	0.34	0.34	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	psr A_r2 ew	0.71	0.74	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	dsr A_r2 ew	0.55	0.31	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_CAPM A_pooled cap	3.78	4.60	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_FF3 A_pooled cap	6.91	5.07	R08.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_FF5+UMD A_pooled cap	5.43	3.66	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	mean A_pooled cap	7.83	7.92	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	sharpe A_pooled cap	0.73	0.82	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	psr A_pooled cap	0.89	0.95	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	dsr A_pooled cap	0.79	0.68	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_CAPM A_pooled ew	-0.47	0.52	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_FF3 A_pooled ew	1.24	0.86	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	alpha_FF5+UMD A_pooled ew	1.37	-0.41	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	mean A_pooled ew	2.51	2.66	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	sharpe A_pooled ew	0.28	0.32	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	psr A_pooled ew	0.67	0.72	R08.md
R8	terciles	robustness; exploratory	spec §5.6	terciles	dsr A_pooled ew	0.51	0.30	R08.md
R8	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM z A_r2	0.21	0.21	R08.md
R8	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM log_size A_r2	0.08	-0.08	R08.md
R8	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM log_beme A_r2	0.22	0.01	R08.md
R8	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM mom A_r2	-0.08	0.22	R08.md
R8	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM z A_pooled	0.16	0.17	R08.md
R8	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM log_size A_pooled	0.10	-0.07	R08.md
R8	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM log_beme A_pooled	0.21	0.00	R08.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R8	Fama-MacBeth	robustness; exploratory	spec §5.6	fm	FM mom A_ pooled	-0.07	0.24	R08.md
R9	event studies	robustness; exploratory	spec §5.6	placebo	records without a primary counterpart (in the row file)	see file	—	R09.md
R9	event studies	robustness; exploratory	spec §5.6	summary	records without a primary counterpart (in the row file)	see file	—	R09.md
R10	H5	robustness	D1	market_lb	slope of annualised FF5+UMD alpha on z_delegation_ lb	1.98	1.96	R10.md
R10	H5b	robustness	D1	market_lb	slope of annualised FF5+UMD alpha on z_delegation_ lb, controlling for z_aioe	3.46	3.46	R10.md
R10	H5	robustness	D1	market_ub	slope of annualised FF5+UMD alpha on z_delegation_ ub	2.01	1.96	R10.md
R10	H5b	robustness	D1	market_ub	slope of annualised FF5+UMD alpha on z_delegation_ ub, controlling for z_aioe	3.61	3.46	R10.md
R10	H1	robustness	D1	h1_lb	records without a primary counterpart (in the row file)	see file	—	R10.md
R10	H1	robustness	D1	h1_ub	records without a primary counterpart (in the row file)	see file	—	R10.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R10	H3	robustness	D1	h3_lb	records without a primary counterpart (in the row file)	see file	—	R10.md
R10	H3	robustness	D1	h3_ub	records without a primary counterpart (in the row file)	see file	—	R10.md
R11	H5	robustness	D4	market	slope of annualised FF5+UMD alpha on z_delegation	1.94	1.96	R11.md
R11	H5b	robustness	D4	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.42	3.46	R11.md
R11	RQ2	robustness	D4	rq2	spearman z_eloundou vs z_usage	0.773	0.768	R11.md
R11	RQ2	robustness	D4	rq2	spearman z_eloundou vs z_delegation	0.721	0.716	R11.md
R11	RQ2	robustness	D4	rq2	spearman z_eloundou vs z_augmentation	0.773	0.783	R11.md
R11	RQ2	robustness	D4	rq2	spearman z_eloundou_gpt4 vs z_usage	0.843	0.843	R11.md
R11	RQ2	robustness	D4	rq2	spearman z_eloundou_gpt4 vs z_delegation	0.815	0.815	R11.md
R11	RQ2	robustness	D4	rq2	spearman z_eloundou_gpt4 vs z_augmentation	0.844	0.850	R11.md
R11	RQ2	robustness	D4	rq2	spearman z_aioe vs z_usage	0.810	0.809	R11.md
R11	RQ2	robustness	D4	rq2	spearman z_aioe vs z_delegation	0.762	0.763	R11.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R11	RQ2	robustness	D4	rq2	spearman z_aioe vs z_ augmentation	0.817	0.825	R11.md
R12	H5	co-primary reporting	D5	confirmatory script	H5 b, R12 (ranks)	-1.065	1.958	confirmatory_ h5.md
R12	H5b	co-primary reporting	D5	confirmatory script	H5b b, R12 (ranks)	-2.386	3.461	confirmatory_ h5.md
R13	H5	co-primary reporting; co-condition	D5	confirmatory script	H5 b, R13 (drop Softw)	1.655	1.958	confirmatory_ h5.md
R13	H5b	co-primary reporting; co-condition	D5	confirmatory script	H5b b, R13 (drop Softw)	3.634	3.461	confirmatory_ h5.md
R14	H5b	robustness	D6	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_ aioe	3.46	3.46	R14.md
R14	RQ2	robustness	D6	rq2	spearman z_ eloundou vs z_ usage	0.768	0.768	R14.md
R14	RQ2	robustness	D6	rq2	spearman z_ eloundou vs z_ delegation	0.716	0.716	R14.md
R14	RQ2	robustness	D6	rq2	spearman z_ eloundou vs z_ augmentation	0.783	0.783	R14.md
R14	RQ2	robustness	D6	rq2	spearman z_ eloundou_gpt4 vs z_usage	0.843	0.843	R14.md
R14	RQ2	robustness	D6	rq2	spearman z_ eloundou_gpt4 vs z_delegation	0.815	0.815	R14.md
R14	RQ2	robustness	D6	rq2	spearman z_ eloundou_gpt4 vs z_ augmentation	0.850	0.850	R14.md
R14	RQ2	robustness	D6	rq2	spearman z_aioe vs z_usage	0.811	0.809	R14.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R14	RQ2	robustness	D6	rq2	spearman z_aioe vs z_delegation	0.766	0.763	R14.md
R14	RQ2	robustness	D6	rq2	spearman z_aioe vs z_ augmentation	0.826	0.825	R14.md
R15	H5	robustness	D7	market	slope of annualised FF5+UMD alpha on z_delegation	1.96	1.96	R15.md
R15	H5b	robustness	D7	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_ aioe	3.46	3.46	R15.md
R15	RQ2	robustness	D7	rq2	spearman z_ eloundou vs z_ usage	0.768	0.768	R15.md
R15	RQ2	robustness	D7	rq2	spearman z_ eloundou vs z_ delegation	0.716	0.716	R15.md
R15	RQ2	robustness	D7	rq2	spearman z_ eloundou vs z_ augmentation	0.783	0.783	R15.md
R15	RQ2	robustness	D7	rq2	spearman z_ eloundou_gpt4 vs z_usage	0.843	0.843	R15.md
R15	RQ2	robustness	D7	rq2	spearman z_ eloundou_gpt4 vs z_delegation	0.815	0.815	R15.md
R15	RQ2	robustness	D7	rq2	spearman z_ eloundou_gpt4 vs z_ augmentation	0.850	0.850	R15.md
R15	RQ2	robustness	D7	rq2	spearman z_aioe vs z_usage	0.809	0.809	R15.md
R15	RQ2	robustness	D7	rq2	spearman z_aioe vs z_delegation	0.763	0.763	R15.md
R15	RQ2	robustness	D7	rq2	spearman z_aioe vs z_ augmentation	0.825	0.825	R15.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R16	RQ2	robustness; exploratory	D9	rq2	spearman z_eloundou vs z_usage	0.847	0.846	R16.md
R16	RQ2	robustness; exploratory	D9	rq2	spearman z_eloundou vs z_delegation	0.773	0.770	R16.md
R16	RQ2	robustness; exploratory	D9	rq2	spearman z_eloundou vs z_augmentation	0.820	0.815	R16.md
R16	RQ2	robustness; exploratory	D9	rq2	spearman z_eloundou_gpt4 vs z_usage	0.886	0.888	R16.md
R16	RQ2	robustness; exploratory	D9	rq2	spearman z_eloundou_gpt4 vs z_delegation	0.856	0.855	R16.md
R16	RQ2	robustness; exploratory	D9	rq2	spearman z_eloundou_gpt4 vs z_augmentation	0.881	0.876	R16.md
R16	RQ2	robustness; exploratory	D9	rq2	spearman z_aioe vs z_usage	0.868	0.867	R16.md
R16	RQ2	robustness; exploratory	D9	rq2	spearman z_aioe vs z_delegation	0.815	0.813	R16.md
R16	RQ2	robustness; exploratory	D9	rq2	spearman z_aioe vs z_augmentation	0.861	0.858	R16.md
R16	A_pooled	robustness; exploratory	D9	index	records without a primary counterpart (in the row file)	see file	—	R16.md
R16	terciles	robustness; exploratory	D9	terciles	alpha_CAPM A_pooled cap	4.60	4.60	R16.md
R16	terciles	robustness; exploratory	D9	terciles	alpha_FF3 A_pooled cap	5.07	5.07	R16.md
R16	terciles	robustness; exploratory	D9	terciles	alpha_FF5+UMD A_pooled cap	3.66	3.66	R16.md
R16	terciles	robustness; exploratory	D9	terciles	mean A_pooled cap	7.92	7.92	R16.md
R16	terciles	robustness; exploratory	D9	terciles	sharpe A_pooled cap	0.82	0.82	R16.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R16	terciles	robustness; exploratory	D9	terciles	psr A_pooled cap	0.95	0.95	R16.md
R16	terciles	robustness; exploratory	D9	terciles	dsr A_pooled cap	0.90	0.68	R16.md
R16	terciles	robustness; exploratory	D9	terciles	alpha_CAPM A_ pooled ew	0.52	0.52	R16.md
R16	terciles	robustness; exploratory	D9	terciles	alpha_FF3 A_ pooled ew	0.86	0.86	R16.md
R16	terciles	robustness; exploratory	D9	terciles	alpha_ FF5+UMD A_ pooled ew	-0.41	-0.41	R16.md
R16	terciles	robustness; exploratory	D9	terciles	mean A_pooled ew	2.66	2.66	R16.md
R16	terciles	robustness; exploratory	D9	terciles	sharpe A_pooled ew	0.32	0.32	R16.md
R16	terciles	robustness; exploratory	D9	terciles	psr A_pooled ew	0.72	0.72	R16.md
R16	terciles	robustness; exploratory	D9	terciles	dsr A_pooled ew	0.60	0.30	R16.md
R16	Fama-MacBeth	robustness; exploratory	D9	fm	FM z A_pooled	0.16	0.17	R16.md
R16	Fama-MacBeth	robustness; exploratory	D9	fm	FM log_size A_ pooled	-0.07	-0.07	R16.md
R16	Fama-MacBeth	robustness; exploratory	D9	fm	FM log_beme A_ pooled	0.00	0.00	R16.md
R16	Fama-MacBeth	robustness; exploratory	D9	fm	FM mom A_ pooled	0.25	0.24	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope ChatGPT h5 A_ pooled	-0.39	-0.39	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope ChatGPT h20 A_ pooled	0.27	0.26	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope ChatGPT h9 none [z_aioe in z_aioe]	-0.13	-0.13	R16.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R16	event studies	robustness; exploratory	D9	events	CAR slope ChatGPT h9 A_ pooled [z_aioe in z_aioe+z_ delegation]	0.02	0.01	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope ChatGPT h9 A_ pooled [z_ delegation in z_ aioe+z_ delegation]	-0.22	-0.21	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope GPT-4 h5 A_pooled	0.66	0.66	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope GPT-4 h20 A_pooled	0.41	0.41	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope PaLM 2 h5 A_pooled	1.23	1.22	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope PaLM 2 h20 A_pooled	2.18	2.19	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Claude 2 h5 A_ pooled	0.45	0.45	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Claude 2 h20 A_ pooled	0.39	0.39	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Claude 2.1 h5 A_ pooled	-0.04	-0.04	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Claude 2.1 h20 A_pooled	0.12	0.12	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Gemini 1.0 h5 A_ pooled	0.15	0.15	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Gemini 1.0 h20 A_pooled	-0.04	-0.04	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Gemini 1.5 h5 A_ pooled	-0.68	-0.68	R16.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R16	event studies	robustness; exploratory	D9	events	CAR slope Gemini 1.5 h20 A_pooled	-1.10	-1.09	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Claude 3 h5 A_ pooled	-0.25	-0.24	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Claude 3 h20 A_ pooled	-0.72	-0.70	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Claude 3.5 h5 A_ pooled	0.41	0.40	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Claude 3.5 h20 A_pooled	-0.22	-0.22	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope OpenAI o1 h5 A_ pooled	-0.06	-0.06	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope OpenAI o1 h20 A_pooled	-0.14	-0.14	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Gemini 2.0 h5 A_ pooled	-0.20	-0.19	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Gemini 2.0 h20 A_pooled	-0.22	-0.22	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Claude 3.7 h5 A_ pooled	-0.21	-0.22	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Claude 3.7 h20 A_pooled	-0.41	-0.40	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope GPT-4.5 h5 A_ pooled	0.13	0.12	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope GPT-4.5 h20 A_ pooled	-0.64	-0.63	R16.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R16	event studies	robustness; exploratory	D9	events	CAR slope Gemini 2.5 h5 A_ pooled	-0.46	-0.46	R16.md
R16	event studies	robustness; exploratory	D9	events	CAR slope Gemini 2.5 h20 A_pooled	-1.12	-1.12	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 1 directive_pct	28.90	28.07	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 1 feedback_loop_ pct	7.95	8.58	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 1 task_iteration_ pct	17.29	17.26	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 1 validation_pct	2.48	2.48	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 1 learning_pct	43.37	43.61	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 1 automation_pct	35.97	36.65	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 1 augmentation_ pct	64.03	63.35	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 2 directive_pct	40.56	40.45	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 2 feedback_loop_ pct	14.28	14.03	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 2 task_iteration_ pct	22.54	23.09	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 2 validation_pct	3.30	3.24	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 2 learning_pct	19.32	19.19	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 2 automation_pct	54.83	54.48	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 2 augmentation_ pct	45.17	45.52	R16.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 3 directive_pct	26.48	26.63	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 3 feedback_loop_pct	12.65	12.76	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 3 task_iteration_pct	27.20	27.43	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 3 validation_pct	3.19	3.17	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 3 learning_pct	30.48	30.02	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 3 automation_pct	39.13	39.38	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 3 augmentation_pct	60.87	60.62	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 4 directive_pct	46.59	46.05	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 4 feedback_loop_pct	6.93	6.87	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 4 task_iteration_pct	27.02	26.06	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 4 validation_pct	2.39	2.50	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 4 learning_pct	17.08	18.52	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 4 automation_pct	53.52	52.92	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 4 augmentation_pct	46.48	47.08	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 5 directive_pct	41.22	40.62	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 5 feedback_loop_pct	9.32	9.27	R16.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 5 task_iteration_pct	34.01	32.66	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 5 validation_pct	3.69	3.57	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 5 learning_pct	11.77	13.88	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 5 automation_pct	50.54	49.89	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 5 augmentation_pct	49.46	50.11	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 6 directive_pct	34.45	34.37	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 6 feedback_loop_pct	13.92	14.03	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 6 task_iteration_pct	26.45	26.42	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 6 validation_pct	5.52	5.50	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 6 learning_pct	19.65	19.66	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 6 automation_pct	48.38	48.41	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 6 augmentation_pct	51.62	51.59	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 7 directive_pct	31.31	31.23	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 7 feedback_loop_pct	14.61	14.69	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 7 task_iteration_pct	30.26	30.47	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 7 validation_pct	2.86	3.04	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 7 learning_pct	20.96	20.57	R16.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 7 automation_pct	45.91	45.92	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 7 augmentation_pct	54.09	54.08	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 8 directive_pct	30.61	30.08	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 8 feedback_loop_pct	10.70	10.56	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 8 task_iteration_pct	47.14	46.42	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 8 validation_pct	3.17	3.36	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 8 learning_pct	8.38	9.59	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 8 automation_pct	41.31	40.64	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 8 augmentation_pct	58.69	59.36	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 9 directive_pct	26.74	27.11	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 9 feedback_loop_pct	23.68	22.81	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 9 task_iteration_pct	34.83	33.73	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 9 validation_pct	3.61	3.40	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 9 learning_pct	11.14	12.94	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 9 automation_pct	50.42	49.92	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 9 augmentation_pct	49.58	50.08	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 10 directive_pct	34.17	33.51	R16.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 10 feedback_loop_pct	22.71	22.77	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 10 task_iteration_pct	24.58	24.39	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 10 validation_pct	2.32	2.37	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 10 learning_pct	16.21	16.95	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 10 automation_pct	56.89	56.29	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile 10 augmentation_pct	43.11	43.71	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile no wage directive_pct	26.87	27.78	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile no wage feedback_loop_pct	21.95	23.52	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile no wage task_iteration_pct	50.11	47.29	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile no wage validation_pct	0.36	0.40	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile no wage learning_pct	0.71	1.02	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile no wage automation_pct	48.82	51.30	R16.md
R16	r6 descriptives	robustness; exploratory	D9	r6_descriptives	r6 wage decile no wage augmentation_pct	51.18	48.70	R16.md
R17	RQ2	robustness; descriptive	D8	rq2	spearman z_eloundou_gpt4 vs z_usage	0.843	0.843	R17.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R17	RQ2	robustness; descriptive	D8	rq2	spearman z_eloundou_gpt4 vs z_delegation	0.815	0.815	R17.md
R17	RQ2	robustness; descriptive	D8	rq2	spearman z_eloundou_gpt4 vs z_augmentation	0.850	0.850	R17.md
R17	RQ2	robustness; descriptive	D8	rq2	spearman z_eloundou vs z_usage	0.768	0.768	R17.md
R17	RQ2	robustness; descriptive	D8	rq2	spearman z_eloundou vs z_delegation	0.716	0.716	R17.md
R17	RQ2	robustness; descriptive	D8	rq2	spearman z_eloundou vs z_augmentation	0.783	0.783	R17.md
R17	RQ2	robustness; descriptive	D8	rq2	spearman z_aioe vs z_usage	0.809	0.809	R17.md
R17	RQ2	robustness; descriptive	D8	rq2	spearman z_aioe vs z_delegation	0.763	0.763	R17.md
R17	RQ2	robustness; descriptive	D8	rq2	spearman z_aioe vs z_augmentation	0.825	0.825	R17.md
R18	H5	robustness	D10	market	slope of annualised FF5+UMD alpha on z_delegation	1.96	1.96	R18.md
R18	H5b	robustness	D10	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.46	3.46	R18.md
R19	H5	robustness	D10	market	slope of annualised FF5+UMD alpha on z_delegation	1.96	1.96	R19.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R19	H5b	robustness	D10	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.46	3.46	R19.md
R20	H5	robustness	Task 10 ruling	market	slope of annualised FF5+UMD alpha on z_delegation	2.43	1.96	R20.md
R20	H5b	robustness	Task 10 ruling	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.74	3.46	R20.md
R20	RQ2	robustness	Task 10 ruling	rq2	spearman z_eloundou vs z_usage	0.751	0.768	R20.md
R20	RQ2	robustness	Task 10 ruling	rq2	spearman z_eloundou vs z_delegation	0.714	0.716	R20.md
R20	RQ2	robustness	Task 10 ruling	rq2	spearman z_eloundou vs z_augmentation	0.761	0.783	R20.md
R20	RQ2	robustness	Task 10 ruling	rq2	spearman z_eloundou_gpt4 vs z_usage	0.836	0.843	R20.md
R20	RQ2	robustness	Task 10 ruling	rq2	spearman z_eloundou_gpt4 vs z_delegation	0.807	0.815	R20.md
R20	RQ2	robustness	Task 10 ruling	rq2	spearman z_eloundou_gpt4 vs z_augmentation	0.842	0.850	R20.md
R20	RQ2	robustness	Task 10 ruling	rq2	spearman z_aioe vs z_usage	0.797	0.809	R20.md
R20	RQ2	robustness	Task 10 ruling	rq2	spearman z_aioe vs z_delegation	0.760	0.763	R20.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R20	RQ2	robustness	Task 10 ruling	rq2	spearman z_aioe vs z_ augmentation	0.811	0.825	R20.md
R21	H5	robustness; no error control	spec §5.4.1	usage	slope of annualised FF5+UMD alpha on z_usage	2.12	1.96	R21.md
R21	H5b	robustness; no error control	spec §5.4.1	usage	slope of annualised FF5+UMD alpha on z_usage, controlling for z_ aioe	4.16	3.46	R21.md
R21	H5	robustness; no error control	spec §5.4.1	augmentation	slope of annualised FF5+UMD alpha on z_ augmentation	2.44	1.96	R21.md
R21	H5b	robustness; no error control	spec §5.4.1	augmentation	slope of annualised FF5+UMD alpha on z_ augmentation, controlling for z_ aioe	4.87	3.46	R21.md
R22	H2	robustness; confounded with the r6 suppression-regime change	D2	all_r2_r6	beta (pp per year)	7.31	4.98	R22.md
R22	H2	robustness	D2	all_r2_r6_ r6dummy	beta (pp per year)	5.47	4.98	R22.md
R22	H2	robustness	D2	unbalanced	beta (pp per year)	5.00	4.98	R22.md
R22	H2	robustness	D2	augmentation_ pct	records without a primary counterpart (in the row file)	see file	—	R22.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R22	H2	robustness	D2	directive_pct	records without a primary counterpart (in the row file)	see file	—	R22.md
R22	H2	robustness	D2	feedback_loop_pct	records without a primary counterpart (in the row file)	see file	—	R22.md
R22	H2	robustness	D2	task_iteration_pct	records without a primary counterpart (in the row file)	see file	—	R22.md
R22	H2	robustness	D2	validation_pct	records without a primary counterpart (in the row file)	see file	—	R22.md
R22	H2	robustness	D2	learning_pct	records without a primary counterpart (in the row file)	see file	—	R22.md
R22	H2	robustness	D2	api_r3_r5	beta (pp per year)	-9.51	4.98	R22.md
R23	H1	robustness	spec §5.4.1	h1	records without a primary counterpart (in the row file)	see file	—	R23.md
R23	H2	robustness; confounded with the r6 suppression-regime change	spec §5.4.1	h2	records without a primary counterpart (in the row file)	see file	—	R23.md
R24	H1	robustness	D13	r4	human_education_years_mean	-0.73	-0.53	R24.md
R24	H1	robustness	D13	r4	log_wage	-3.60	4.40	R24.md
R24	H3	robustness	D13	r4	error_cost	-0.96	-0.18	R24.md
R24	H1	robustness	D13	r5	human_education_years_mean	-1.11	-0.53	R24.md
R24	H1	robustness	D13	r5	log_wage	-5.55	4.40	R24.md
R24	H3	robustness	D13	r5	error_cost	0.40	-0.18	R24.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R24	H1	robustness	D13	r6_1p_api	human_ education_years_ mean	0.38	-0.53	R24.md
R24	H1	robustness	D13	r6_1p_api	log_wage	1.36	4.40	R24.md
R24	H3	robustness	D13	r6_1p_api	error_cost	-0.15	-0.18	R24.md
R25	H1	robustness	D17	H1a	human_ education_years_ mean	-0.53	-0.53	R25.md
R25	H1	robustness	D17	H1b	log_wage	4.40	4.40	R25.md
R25	H3	robustness	D17	H3	error_cost	-0.18	-0.18	R25.md
R25	H4	robustness	D17	H4a	log_aui	-1.19	-1.19	R25.md
R25	H4	robustness	D17	H4b	log_income	-0.92	-0.92	R25.md
R26	H1	robustness	D14	gwa	human_ education_years_ mean	0.23	-0.53	R26.md
R26	H1	robustness	D14	gwa	log_wage	5.57	4.40	R26.md
R26	H3	robustness	D14	gwa	error_cost	-0.24	-0.18	R26.md
R26	H1	robustness	D14	dwa	human_ education_years_ mean	-0.52	-0.53	R26.md
R26	H1	robustness	D14	dwa	log_wage	2.15	4.40	R26.md
R26	H3	robustness	D14	dwa	error_cost	0.01	-0.18	R26.md
R27	H3	robustness	D15	coe	coe	-0.16	-0.18	R27.md
R28	H4	robustness	D16	us_states	log_aui	-0.71	-1.19	R28.md
R28	H4	robustness	D16	us_states	log_income	-2.61	-0.92	R28.md
R28	H4	robustness	D16	r4_countries	log_aui	3.68	-1.19	R28.md
R28	H4	robustness	D16	r4_countries	log_income	-5.36	-0.92	R28.md
R28	H4	robustness	D16	r5_countries	log_aui	-1.03	-1.19	R28.md
R28	H4	robustness	D16	r5_countries	log_income	-1.96	-0.92	R28.md
R28	H4	robustness	D16	usage_weighted	log_aui	-1.73	-1.19	R28.md
R28	H4	robustness	D16	usage_weighted	log_income	0.09	-0.92	R28.md
R28	H4	robustness	D16	iwa_mix	log_aui	-1.22	-1.19	R28.md
R28	H4	robustness	D16	iwa_mix	log_income	-0.84	-0.92	R28.md
R29	H6 divergence	robustness; exploratory	D18	pre_formation_ 2025	Spearman(T_j, Design B z_ usage) across FF49 industries	0.267	0.361	R29.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R29	H6 gap-return	robustness; exploratory	D18	pre_formation_2025	slope of annualised FF5+UMD alpha on gap_j (Design B window, no control)	-2.15	-2.40	R29.md
R29	H6 divergence	robustness; exploratory	D18	change_2019_2024	Spearman(T_j, Design B z_usage) across FF49 industries	0.081	0.361	R29.md
R29	H6 gap-return	robustness; exploratory	D18	change_2019_2024	slope of annualised FF5+UMD alpha on gap_j (Design B window, no control)	-3.37	-2.40	R29.md
R29	H6 divergence	robustness; exploratory	D18	float_weighted	Spearman(T_j, Design B z_usage) across FF49 industries	0.190	0.361	R29.md
R29	H6 gap-return	robustness; exploratory	D18	float_weighted	slope of annualised FF5+UMD alpha on gap_j (Design B window, no control)	-0.21	-2.40	R29.md
R30	H6 divergence	robustness; exploratory	D12	narrow	Spearman(T_j, Design B z_usage) across FF49 industries	0.361	0.361	R30.md
R30	H6 gap-return	robustness; exploratory	D12	narrow	slope of annualised FF5+UMD alpha on gap_j (Design B window, no control)	-2.40	-2.40	R30.md
R30	H6 divergence	robustness; exploratory	D12	broad	Spearman(T_j, Design B z_usage) across FF49 industries	0.364	0.361	R30.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R30	H6 gap-return	robustness; exploratory	D12	broad	slope of annualised FF5+UMD alpha on gap_j (Design B window, no control)	-1.98	-2.40	R30.md
R31	H3	robustness	P13	covariates	error_cost	-0.20	-0.18	R31.md
R31	H3	robustness	P13	covariates	records without a primary counterpart (in the row file)	see file	—	R31.md
R32	H5	robustness	P15	market	slope of annualised FF5+UMD alpha on z_delegation	2.20	1.96	R32.md
R32	H5b	robustness	P15	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.72	3.46	R32.md
R33	H5	robustness	D23; P17	market	slope of annualised FF5+UMD alpha on z_delegation	1.92	1.96	R33.md
R33	H5b	robustness	D23; P17	market	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.67	3.46	R33.md
R33	RQ2	robustness	D23; P17	rq2	spearman z_eloundou vs z_usage	0.758	0.768	R33.md
R33	RQ2	robustness	D23; P17	rq2	spearman z_eloundou vs z_delegation	0.708	0.716	R33.md
R33	RQ2	robustness	D23; P17	rq2	spearman z_eloundou vs z_augmentation	0.769	0.783	R33.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R33	RQ2	robustness	D23; P17	rq2	spearman z_eloundou_gpt4 vs z_usage	0.821	0.843	R33.md
R33	RQ2	robustness	D23; P17	rq2	spearman z_eloundou_gpt4 vs z_delegation	0.802	0.815	R33.md
R33	RQ2	robustness	D23; P17	rq2	spearman z_eloundou_gpt4 vs z_augmentation	0.829	0.850	R33.md
R33	RQ2	robustness	D23; P17	rq2	spearman z_aioe vs z_usage	0.786	0.809	R33.md
R33	RQ2	robustness	D23; P17	rq2	spearman z_aioe vs z_delegation	0.746	0.763	R33.md
R33	RQ2	robustness	D23; P17	rq2	spearman z_aioe vs z_augmentation	0.804	0.825	R33.md
R34	H5	robustness	P6	from_1969	slope of annualised FF5+UMD alpha on z_delegation	1.96	1.96	R34.md
R34	H5b	robustness	P6	from_1969	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.46	3.46	R34.md
R34	H5	robustness	P6	from_1963	slope of annualised FF5+UMD alpha on z_delegation	1.96	1.96	R34.md
R34	H5b	robustness	P6	from_1963	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.46	3.46	R34.md
R35	H5	robustness	P10	h5	slope of annualised FF5+UMD alpha on z_delegation	1.96	1.96	R35.md

row	applies to	label	PAP source	part	estimand	variant	primary	file
R35	H5b	robustness	P10	h5b	slope of annualised FF5+UMD alpha on z_delegation, controlling for z_aioe	3.46	3.46	R35.md
R36	H2	robustness	P19	publication	beta (pp per year)	4.69	4.98	R36.md
R37	H2	robustness; descriptive	P19	within_r6	records without a primary counterpart (in the row file)	see file	—	R37.md
R38	H1	robustness	P18	oneway_occ	human_education_years_mean	-0.53	-0.53	R38.md
R38	H1	robustness	P18	oneway_occ	log_wage	4.40	4.40	R38.md
R38	H3	robustness	P18	oneway_occ	error_cost	-0.18	-0.18	R38.md

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.2 Implementation choices where the PAP is silent

Generated by `uv run python -m aideleg.robustness --summary`. Do not edit by hand.

Every implementation choice that a robustness row file attributes to one of Plan 4's PAP §11 clarifications (C-56 to C-82), grouped by clarification: under each, the rows that cite it, each with its note as the row file prints it and every §11 entry the note names. Notes marked **[default]** are the defaults the plan fixed where the PAP is silent; a note citing several clarifications appears under each. All of these clarifications were filed after the confirmatory results; the "Results seen" line under each, quoted from its §11 entry, says which results had been seen, including any robustness row already run.

Every market interval of the robustness rows (H5, H5b and the H6 gap–return slope) is the c_0.05 interval $\hat{b} \pm$ c_0.05·s, with s and c recomputed for the row (R35: its own scale and t_{21} critical, PAP §11 C-75); P14 also defines the c_0.025 interval $\hat{b} \pm$ c_0.025·s, which the row files do not print (both criticals are in each market record).

Any prose that calls a market robustness result a non-rejection uses the fixed P14 wording of `docs/results/confirmatory_h5.md` verbatim (PAP §3.3 P14).

D.2.1 C-56 (Unbalanced H2 panel)

Results seen when filed (PAP §11 C-56): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R22 (R22.md): [default] The unbalanced panel keeps every (task, release) row with a non-null outcome for tasks with such rows in at least two of the listed releases (Plan 2's `build_h2_panel(balanced=False)`); there is no other unbalanced rule (PAP §11 C-56). — §11 C-56 (Unbalanced H2 panel).

D.2.2 C-57 (R10 for H3)

Results seen when filed (PAP §11 C-57): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R10 (R10.md): [default] H3 with `validation_share_all_lb` = v·(100 − s)/100 and `_ub` = lb + s in place of `validation_pct` (parts h3_lb, h3_ub), v = `validation_pct` and s = r6's suppression share (`none_pct`), per period and then D13-averaged, as for the automation bounds (PAP §11 C-57). — §11 C-57 (R10 for H3).
- R10 (R10.md): Market parts of R10: the D1 suppression bounds `z_delegation_lb` and `z_delegation_ub` (built from `automation_pct_lb` / `_ub`) in place of `z_delegation`, for H5 and H5b (PAP §8.2 R10, D1). The H1 and H3 parts of R10 are RQ1 rows (PAP §11 C-57). — §11 C-57 (R10 for H3).

D.2.3 C-58 (R27 standardisation)

Results seen when filed (PAP §11 C-58): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R27 (R27.md): [default] H3 with the O*NET 29.3 Work Context element Consequence of Error (`CX`, the mean `Data Value` over the O*NET-SOC codes within soc6) in place of the D15 indicator, z-scored across the distinct soc6 codes of the H3 rows that have a value (population SD), so γ is in pp of validation share per 1 SD (PAP §8.2 R27, §11 C-58). Rows without a value are dropped and reported as `missing_coe`. — §11 C-58 (R27 standardisation).

D.2.4 C-59 (R28 r4/r5 countries)

Results seen when filed (PAP §11 C-59): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R28 (R28.md): [default] r4_countries, r5_countries: Plan 4's `h4_frame_r45` (PAP §11 C-59), because r4 and r5 publish no country usage index and no country GWA rows. log AUI = ln(usage_pct_c / working-age population_c); AEI's index is the usage share over the working-age population share, so the common scale moves only the intercept. s_cg is the country's `onet_task` pct aggregated to GWAs through the D14 files (`task_families("gwa")`), each task split equally over its GWAs, then renormalised over the GWAs with an ā_g. ā_g is the same release's global task `automation_pct` aggregated to GWA with D13 weights (the task's global pct, split equally over its GWAs). Income is 2024 GDP per working-age person (D16, the r3 inputs Plan 2 uses). Each release has one published period, so nothing is averaged over months. Inclusion as D16: overall automation_pct and usage_pct published, income, and a mix. — §11 C-59 (R28 r4/r5 countries).
- R28 (R28.md): [default] In r4_countries and r5_countries a task's GWAs are the union over its candidate Task IDs (its panel Task ID and every Task ID carrying its statement), as Plan 2's H1/H3 task rows apply D14 (PAP §11 C-9, C-59); unmatched task nodes and tasks with no DWA link carry no GWA. — §11 C-9 (Task-to-family union (D14)); C-59 (R28 r4/r5 countries).

D.2.5 C-60 (R9 scope)

Results seen when filed (PAP §11 C-60): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R9 (R09.md): [default] Placebo day-0 dates (PAP §8.2 R9; §11 C-60): each of the 14 D19 events' month and day in 2017, 2018, 2019 and 2021 (56 dates, named " placebo "); 2020 is skipped, as the PAP states. Day 0 is the next trading day when the date is not one (Plan 3's `day_zero`). — §11 C-60 (R9 scope).
- R9 (R09.md): [default] Placebo records: the CAR slope on the primary A_r2 `z_delegation` at horizons 5 and 20, one per placebo date and horizon, as Plan 3's `run_events` returns it (OLS with intercept, HC3, normal reference; C-44). Summary records: per horizon, the mean A_r2 slope over the 14 actual D19 events (a second `run_events` on the same exposure) and its empirical percentile among the 56 placebo slopes, in the stat column; descriptive, no p (PAP §11 C-60). — §11 C-44 (Event-study defaults: returns, HC3 reference, event-window drops); C-60 (R9 scope).

D.2.6 C-61 (R29 denominator)

Results seen when filed (PAP §11 C-61): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R29 (R29.md): [default] pre_formation_2025: the numerator hits (file date) and the denominator filers (Date Filed) are both cut at 2025-03-26, the filings public before Design B's formation date 2025-03-27 (PAP §11 C-61). pre_formation_2025: 2,456 talk filers of 4,262 filers (3,546 mapped to an FF49 industry); T_j undefined: none. — §11 C-61 (R29 denominator).
- R29 (R29.md): [default] float_weighted: $T_j = \Sigma$ float·1[talks] / Σ float over industry j's 2024 filers with a public float in the SEC XBRL frame dei/EntityPublicFloat/USD/CY2024Q2I (the R29 cap proxy, a §11 addition outside §9.4, A-46; PAP §11 C-61); T_j is undefined where that sum is not positive. Filers with a public float: 4,159 of 6,774 2024 filers (61.40%); 4,062 of 5,796 FF49-mapped filers (70.08%); 244 of them report a float of zero and enter with weight zero. T_j undefined: none. — §11 C-61 (R29 denominator).

D.2.7 C-62 (R30 lists)

Results seen when filed (PAP §11 C-62): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R30 (R30.md): [default] narrow: the primary list without items 9–11 (computer vision, reinforcement learning, foundation model(s)): 9 terms, 11 phrases; broad: the 12 primary terms plus 'automation' and 'robotics': 14 terms, 17 phrases, the plural rule applied to the primary list only (PAP §6.1 D12 alternatives; PAP §11 C-62). — §11 C-62 (R30 lists).
- R30 (R30.md): [default] Only T_j (10-Ks filed in 2024) is computed for each list (PAP §11 C-62). — §11 C-62 (R30 lists).

D.2.8 C-63 (R7)

Results seen when filed (PAP §11 C-63): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R7 (R07.md): [default] z not re-standardised after the drop, as for R13 (PAP §11 C-63). — §11 C-63 (R7).

D.2.9 C-64 (R18)

Results seen when filed (PAP §11 C-64): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R18 (R18.md): s = the largest sd of the 17 non-overlapping start offsets; $\bar{b}$ stays the 378-window mean; q is recomputed with the new s (PAP §3.3, §8.2 R18; §11 C-64). — §11 C-64 (R18).

D.2.10 C-65 (Exposure-variant benchmarks)

Results seen when filed (PAP §11 C-65): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R1 (R01.md): Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).
- R2 (R02.md): Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).
- R3 (R03.md): Benchmarks: AIOE and Eloundou stay at their primary values, because this occupation-level variant changes neither the matrix nor the crosswalk (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).
- R4 (R04.md): `1p_api` in place of `claude_ai` for A_pooled, pooled over r3–r6 with equal release weights: the API source starts at r3 (PAP §8.2 R4; §11 C-65). Design B has no API source. For RQ1 this variant is merged into R22 and R24 (PAP §8.2 R4). — §11 C-65 (Exposure-variant benchmarks).
- R4 (R04.md): Benchmarks: AIOE and Eloundou stay at their primary values, because this occupation-level variant changes neither the matrix nor the crosswalk (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).

- R6 (R06.md): Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).
- R11 (R11.md): Benchmarks: AIOE and Eloundou stay at their primary values, because this occupation-level variant changes neither the matrix nor the crosswalk (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).
- R14 (R14.md): Benchmarks: AIOE itself is replaced; Eloundou and the AEI indices keep the primary matrix and crosswalk (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).
- R15 (R15.md): Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).
- R16 (R16.md): Benchmarks: AIOE and Eloundou stay at their primary values, because this occupation-level variant changes neither the matrix nor the crosswalk (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).
- R20 (R20.md): Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).
- R32 (R32.md): Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).
- R33 (R33.md): Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65). — §11 C-65 (Exposure-variant benchmarks).

D.2.11 C-66 (Wage-bill weights (R2))

Results seen when filed (PAP §11 C-66): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R2 (R02.md): Wage-bill weights (OEWS employment × mean wage) in place of employment weights (PAP §8.2 R2). A_MEAN `#` is set to the May 2024 top-code $239,200 and `*`/`**` give a null wage bill left out of the shares; the U_o denominators stay national employment shares (PAP §11 C-66). — §11 C-66 (Wage-bill weights (R2)).

D.2.12 C-67 (Employment-weighted crosswalk (R33))

Results seen when filed (PAP §11 C-67): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R33 (R33.md): Employment-weighted NAICS→SIC crosswalk from OEWS `nat5d_6d` (PAP D23, §8.2 R33). Children without published employment share the residual max(E_4 − Σ published, 0) equally; with no published child, the primary equal split (PAP §11 C-67). — §11 C-67 (Employment-weighted crosswalk (R33)).

D.2.13 C-68 (RQ2 variant bootstrap)

Results seen when filed (PAP §11 C-68): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R1 (R01.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).
- R2 (R02.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).
- R3 (R03.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).
- R4 (R04.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).

- R11 (R11.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).
- R14 (R14.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).
- R15 (R15.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).
- R16 (R16.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).
- R17 (R17.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).
- R20 (R20.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).
- R33 (R33.md): RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive. — §11 C-68 (RQ2 variant bootstrap).

D.2.14 C-70 (R15 is identical to the primary by construction)

Results seen when filed (PAP §11 C-70): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R15 (R15.md): Identical to the primary by construction (PAP §11 C-70): 5511's occupation mix is added to every FF49 industry with the same fraction of its mapped employment, an affine map of each industry index, so the standardised indices, every Spearman rank and every H5, H5b and RQ2 estimate equal the primary's; only coverage and index levels change. The row is still run and reported (PAP §8.1). — §11 C-70 (R15 is identical to the primary by construction).

D.2.15 C-71 (R20 on the 46 industries the majority rule assigns)

Results seen when filed (PAP §11 C-71): RQ1 and H5 confirmatory results and all Plan 3 results seen; no robustness row run

- R20 (R20.md): The majority rule assigns no NAICS code to Guns (26), Gold (27) or Boxes (40): the row reports on the 46 industries it assigns, with z standardised over those 46; the market part removes the three from the window and from the 378-window history before the null scales are computed (PAP §11 C-71). — §11 C-71 (R20 on the 46 industries the majority rule assigns).

D.2.16 C-72 (R2 replaces employment by the wage bill in every industry weight)

Results seen when filed (PAP §11 C-72): RQ1 and H5 confirmatory results and all Plan 3 results seen; no R2 row run. Before this entry the implementation computed max |Δ z_ delegation| against the primary under this rule (0.607) and under the alternative below (0.415); no H5, H5b or RQ2 estimate was computed under either

- R2 (R02.md): The wage bill replaces employment in both industry weights (the occupation shares within each OEWS NAICS code and the code weights that combine codes into an FF49 industry), so w_oj is o's share of industry j's wage bill; emp_covered_share is wage-bill weighted (PAP §11 C-72), while the zero-assigned share (D3) below is employment-based, as in every row. — §11 C-72 (R2 replaces employment by the wage bill in every industry weight).

D.2.17 C-73 (R14 maps AIOE's 19-1020 by its O*NET successor)

Results seen when filed (PAP §11 C-73): RQ1 and H5 confirmatory results and all Plan 3 results seen; no R14 row run

- R14 (R14.md): AIOE's 19-1020 (not a SOC 2010 detailed code, absent from the BLS crosswalk) is mapped by its O*NET-SOC 2010→2019 successor 19-1029; every other AIOE code uses the BLS crosswalk (PAP §11 C-73). — §11 C-73 (R14 maps AIOE's 19-1020 by its O*NET successor).

D.2.18 C-74 (R32's O*NET part is identical to the primary)

Results seen when filed (PAP §11 C-74): RQ1 and H5 confirmatory results and all Plan 3 results seen; no R32 row run

- R32 (R32.md): The O*NET 29.2 Task Statements file is byte-identical to the 29.3 file of the primary, so R32's task index equals the primary's and R32 differs from the primary only through the OEWS May 2023 weights (PAP §11 C-74). — §11 C-74 (R32's O*NET part is identical to the primary).

D.2.19 C-75 (R35 interval)

Results seen when filed (PAP §11 C-75): RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 8 rows including R35 (d2da8f1) seen

- R35 (R35.md): [default] In the confirmatory window SE_Σ is computed the same way; stat = T_stud = $(\hat{b} - b_0)/(\kappa \cdot SE_\Sigma)$ with $b_0 = \bar{b}$ (D21), p from t_{21}, and the t_{21} criticals of §3.3. The scale column is κ·SE_Σ in pp and the interval is $\hat{b} \pm t_{21,0.975} \cdot \kappa \cdot SE_\Sigma$, the §3.3 interval form with this scale, a presentation-only reading (PAP §11 C-75). — §11 C-75 (R35 interval).

D.2.20 C-76 (R9 label and percentile convention; Design A DSR trial counts)

Results seen when filed (PAP §11 C-76): RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 10 rows R4, R8, R9, R16 (92e12c3) seen

- R4 (R04.md): DSR here is against the 2 tercile variants run in this row (the primary used 6, C-37); not comparable with the primary's DSR (PAP §11 C-76). — §11 C-37 (Tercile bootstrap and DSR); C-76 (R9 label and percentile convention; Design A DSR trial counts).
- R8 (R08.md): DSR here is against the 4 tercile variants run in this row (the primary used 6, C-37); not comparable with the primary's DSR (PAP §11 C-76). — §11 C-37 (Tercile bootstrap and DSR); C-76 (R9 label and percentile convention; Design A DSR trial counts).
- R9 (R09.md): [default] Percentile convention: 100 × the share of the 56 placebo slopes ≤ the actual mean slope, per horizon (PAP §11 C-76). — §11 C-76 (R9 label and percentile convention; Design A DSR trial counts).
- R9 (R09.md): [default] mean of the 14 actual D19 slopes at h5; stat = its empirical percentile among the 56 placebo slopes (100 × share of placebo slopes ≤ the mean, out of 56; PAP §11 C-76); n = placebo slopes; descriptive — §11 C-76 (R9 label and percentile convention; Design A DSR trial counts).
- R9 (R09.md): [default] mean of the 14 actual D19 slopes at h20; stat = its empirical percentile among the 56 placebo slopes (100 × share of placebo slopes ≤ the mean, out of 56; PAP §11 C-76); n = placebo slopes; descriptive — §11 C-76 (R9 label and percentile convention; Design A DSR trial counts).
- R16 (R16.md): DSR here is against the 2 tercile variants run in this row (the primary used 6, C-37); not comparable with the primary's DSR (PAP §11 C-76). — §11 C-37 (Tercile bootstrap and DSR); C-76 (R9 label and percentile convention; Design A DSR trial counts).

D.2.21 C-77 (R23's disagreement condition)

Results seen when filed (PAP §11 C-77): RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 11 rows R10, R23, R24, R26, R27, R31, R38 (25480d9) seen

- R23 (R23.md): R23 is reported beside H1 (PAP §4.1): the H1 decision uses `automation_pct` only and is unchanged. The PAP §4.1 disagreement sentence is printed below only if each R23 coefficient (education, log wage) has the sign opposite to the committed primary H1 coefficient (P-RQ1 record) and $p < 0.05$, a mechanical check (PAP §11 C-77). — §11 C-77 (R23's disagreement condition).
- R23 (R23.md): Sign check against the committed primary H1 (P-RQ1): education R23 0.03 vs primary -0.53, opposite sign, $p < 0.001$; log wage R23 0.03 vs primary 4.40, same sign, p 0.177 (PAP §11 C-77) — §11 C-77 (R23's disagreement condition).

D.2.22 C-78 (RQ1 robustness tables label their coefficients)

Results seen when filed (PAP §11 C-78): RQ1 and H5 confirmatory results, all Plan 3 results, and the Task 11 rows R10, R23, R24, R26, R27, R31, R38 (25480d9) seen

- R10 (R10.md): RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of the outcome share per unit of the regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78). — §11 C-78 (RQ1 robustness tables label their coefficients); C-2 (Variable-level drops); C-30 (H1/H3 task unit, zero-weight drop, r6 automation measure, H3 sample).
- R22 (R22.md): H2 parts: Plan 2's `estimate_h2` on Plan 2's `build_h2_panel` (through the adapter), with H2's inference (PAP §4.2, D22, §8.1): WLS of the outcome on t_r and task dummies with fixed task weights (the task's mean `pct` over the listed releases), SEs clustered by task, t with G_task − 1 df; `p` is the two-sided task-clustered p of the trend. The note gives the restricted wild cluster bootstrap p (clusters = releases, Webb weights, every one of the 6^k weight vectors enumerated; PAP D22, §11 C-15) and whether the D22 rule's conditions would hold (both p ≤ 0.0125, compared exactly as PAP §11 C-24, and β > 0): a mechanical statement, not a decision; the release-label randomisation p is descriptive. The smallest attainable wild-bootstrap p is 6/6^k: 3 releases, 6/216 = 0.0278, above 0.0125, so the wild condition cannot hold; 4 releases, 6/1296 = 0.0046; 5 releases, 6/7776 = 0.0008. β is the change in measured within-task outcome per year, including any change in the classifier and platform scope (PAP §5). Coefficients are in pp of the outcome share per year. Each table row's note cell starts with its estimand (PAP §11 C-78). `n` is the number of (task, release) rows; `dropped` shows none and `n_dropped` is not reported for H2 rows, because Plan 2's H2 builder applies the sample rule and records no drops (PAP §11 C-78). — §11 C-15 (H2 wild-bootstrap granularity (disclosure)); C-24 (Exact levels); C-78 (RQ1 robustness tables label their coefficients).
- R23 (R23.md): RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in ai_autonomy_ mean scale points per unit of the regressor (per year of education, per unit of log wage). Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78). — §11 C-78 (RQ1 robustness tables label their coefficients); C-2 (Variable-level drops); C-30 (H1/H3 task unit, zero-weight drop, r6 automation measure, H3 sample).
- R23 (R23.md): H2 parts: Plan 2's `estimate_h2` on Plan 2's `build_h2_panel` (through the adapter), with H2's inference (PAP §4.2, D22, §8.1): WLS of the outcome on t_r and task dummies with fixed task weights (the task's mean `pct` over the listed releases), SEs clustered by task, t with G_task − 1 df; `p` is the two-sided task-clustered p of the trend. The note gives the restricted wild cluster bootstrap p (clusters = releases, Webb weights, every one of the 6^k weight vectors enumerated; PAP D22, §11 C-15) and whether the D22 rule's conditions would hold (both p ≤ 0.0125, compared exactly as PAP §11 C-24, and β > 0): a mechanical statement, not a decision; the release-label randomisation p is descriptive. The smallest attainable wild-bootstrap p is 6/6^k: 3 releases, 6/216 = 0.0278, above 0.0125, so the wild condition cannot hold. β is the change in measured within-task outcome per year, including any change in the classifier and platform scope (PAP §5). Coefficients are in ai_autonomy_mean scale points per year. Each table row's note cell starts with its estimand (PAP §11 C-78). `n` is the number of (task, release) rows; `dropped` shows none and `n_dropped` is not reported for H2 rows, because Plan 2's H2 builder applies the sample rule and records no drops (PAP §11 C-78). — §11 C-15 (H2 wild-bootstrap granularity (disclosure)); C-24 (Exact levels); C-78 (RQ1 robustness tables label their coefficients).
- R24 (R24.md): RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of the outcome share per unit of the regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78). — §11 C-78 (RQ1 robustness tables label their coefficients); C-2 (Variable-level drops); C-30 (H1/H3 task unit, zero-weight drop, r6 automation measure, H3 sample).

- R25 (R25.md): Coefficients are in pp of the outcome share per unit of the regressor (H3: validation share). Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the committed primary sample: (task, occupation, family) rows for H1 and H3, countries for H4; `dropped` shows none and `n_dropped` is not reported, because the samples are the primary ones (PAP §11 C-78). — §11 C-78 (RQ1 robustness tables label their coefficients).
- R26 (R26.md): RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of the outcome share per unit of the regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78). — §11 C-78 (RQ1 robustness tables label their coefficients); C-2 (Variable-level drops); C-30 (H1/H3 task unit, zero-weight drop, r6 automation measure, H3 sample).
- R27 (R27.md): RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of validation share per 1 SD of Consequence of Error. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78). — §11 C-78 (RQ1 robustness tables label their coefficients); C-2 (Variable-level drops); C-30 (H1/H3 task unit, zero-weight drop, r6 automation measure, H3 sample).
- R28 (R28.md): H4 alternatives (PAP §4.4 D16, §8.2 R28), each part ending in Plan 2's `estimate_h4` (through the adapter) with H4's inference (PAP §8.1): OLS of the geography's overall automation_pct on [1, log AUI, log income, mix] with HC3 covariance (WLS in usage_weighted); b_1 (log AUI) and b_2 (log income) carry normal-reference HC3 95% intervals and two-sided p; p_wald (joint Wald) is the joint HC3 Wald F test of $b_1 = b_2 = 0$ with $(2, n - 4)$ df (D17), whose `stat` is F. Coefficients are in pp of the automation share per unit of the log regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of geographies in the fit; `n_dropped` is the number excluded and `dropped` the count by first failing reason, in Plan 2's order (missing_month, missing_income, missing_mix); the excluded geographies are listed below. Reported only, not a decision: the H4 decision is P-RQ1's. — §11 C-78 (RQ1 robustness tables label their coefficients).
- R31 (R31.md): RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of the outcome share per unit of the regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78). — §11 C-78 (RQ1 robustness tables label their coefficients); C-2 (Variable-level drops); C-30 (H1/H3 task unit, zero-weight drop, r6 automation measure, H3 sample).
- R36 (R36.md): H2 parts: Plan 2's `estimate_h2` on Plan 2's `build_h2_panel` (through the adapter), with H2's inference (PAP §4.2, D22, §8.1): WLS of the outcome on t_r and task dummies with fixed task weights (the task's mean `pct` over the listed releases), SEs clustered by task, t with G_task − 1 df; `p` is the two-sided task-clustered p of the trend. The note gives the restricted wild cluster bootstrap p (clusters = releases, Webb weights, every one of the 6^k weight vectors enumerated; PAP D22, §11 C-15) and whether the D22 rule's conditions would hold (both $p \leq 0.0125$, compared exactly as PAP §11 C-24, and $\beta > 0$): a mechanical statement, not a decision; the release-label randomisation p is descriptive. The smallest attainable wild-bootstrap p is 6/6^k: 4 releases, $6/1296 = 0.0046$. β is the change in measured within-task outcome per year, including any change in the classifier and platform scope (PAP §5). Coefficients are in pp of the outcome share per year. Each table row's note cell starts with its estimand (PAP §11 C-78). `n` is the number of (task, release) rows; `dropped` shows none and `n_dropped` is not reported for H2 rows, because Plan 2's H2 builder applies the sample rule and records no drops (PAP §11 C-78). — §11 C-15 (H2 wild-bootstrap granularity (disclosure)); C-24 (Exact levels); C-78 (RQ1 robustness tables label their coefficients).
- R37 (R37.md): `n` is the number of (task, month) rows; `dropped` shows none and `n_dropped` is not reported, as for the H2 rows (PAP §11 C-78). The note cell starts with its estimand (C-78). — §11 C-78 (RQ1 robustness tables label their coefficients).
- R38 (R38.md): RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD

rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of the outcome share per unit of the regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78). — §11 C-78 (RQ1 robustness tables label their coefficients); C-2 (Variable-level drops); C-30 (H1/H3 task unit, zero-weight drop, r6 automation measure, H3 sample).

D.2.23 C-79 (R37 zero-weight tasks; R23 H2 estimand label)

Results seen when filed (PAP §11 C-79): RQ1 and H5 confirmatory results, all Plan 3 results, the Task 11 rows, and the Task 12 rows R22, R23, R36, R37 (f8ea9d1) seen

- R37 (R37.md): 1141 tasks with `automation_pct` in both months have zero weight (r6 task `pct` is published rounded to 0.01; D13, PAP §11 C-30 (b)) and are left out; they would carry no weight in the fit; n_tasks, G_task and df count the tasks with positive weight (PAP §11 C-79). — §11 C-30 (H1/H3 task unit, zero-weight drop, r6 automation measure, H3 sample); C-79 (R37 zero-weight tasks; R23 H2 estimand label).

D.2.24 Reproduction of the row files

- §11 C-80 (Reproduction tolerance for robustness CSVs): a re-run must reproduce every row `.md` byte for byte; a regenerated CSV reproduces the committed one when every non-numeric cell is identical and every numeric cell agrees within relative 1e-9 (absolute 1e-12 near zero), and the committed bytes are then kept (`reproduce.csv_equivalent`, used by `report.write_row`). The last float digits of R15, R28 and R29 move from run to run (summation order); no printed number moves.

D.2.25 Approved deviation affecting code the robustness rows call

- D-2 (hits_from_saved default phrases): approved by Ajai on 2026-10-23. Plan 3's `h6.edgar.hits_from_saved(phrases=None)` now filters to `fts_phrases()`, as its docstring says; the primary H6 outputs are byte-identical. R29 and R30 pass their phrase lists explicitly, so no robustness number changed by it.

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.3 R1: robustness row (PAP §8.2 R1)

Source: PAP §8.2, row R1. Label: robustness; look-ahead.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation; pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.3.1 Notes

- Variant: ExposureSpec(oews_year=2025); every other construction choice is the primary Design B one.
- OEWS May 2025 weights (industry matrix and national shares) in place of May 2024 (PAP §8.2 R1).
- Look-ahead (PAP §8.2 R1): the May 2025 weights were published after the formation date and during the return window.
- Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65).
- Zero-assigned employment share (D3): 12.02%
- Market part: H5 and H5b on ff49_vw with s, $\bar{b}$ and q recomputed for this exposure vector on the same 378 pre-seal windows, the t_{21} rule and the D21 null $b_0 = \bar{b}$ (PAP §3.3, §8.1); z is the variant's own and is not re-standardised after a drop.

- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.

D.3.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	1.96	2.30	[-2.82, 6.74]	0.117	0.908	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.852, p = 0.404
market	H5b	3.51	2.56	[-1.82, 8.84]	0.870	0.394	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.368, p = 0.186
rq2	RQ2	0.779	—	[0.618, 0.878]	—	—	—	—	—	49	none	z_usage vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)
rq2	RQ2	0.724	—	[0.542, 0.846]	—	—	—	—	—	49	none	z_delegation vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.783	—	[0.627, 0.882]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.846	—	[0.722, 0.916]	—	—	—	—	—	49	none	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.818	—	[0.675, 0.906]	—	—	—	—	—	49	none	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.849	—	[0.731, 0.915]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.813	—	[0.666, 0.898]	—	—	—	—	—	49	none	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.768	—	[0.601, 0.870]	—	—	—	—	—	49	none	z_ delegation vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.824	—	[0.687, 0.902]	—	—	—	—	—	49	none	z_ augmentation vs z_aioe; 10000 bootstrap draws used

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.4 R2: robustness row (PAP §8.2 R2)

Source: PAP §8.2, row R2. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation; pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.4.1 Notes

- Variant: ExposureSpec(weights='wage_bill'); every other construction choice is the primary Design B one.
- Wage-bill weights (OEWS employment × mean wage) in place of employment weights (PAP §8.2 R2). A_MEAN # is set to the May 2024 top-code $239,200 and */** give a null wage bill left out of the shares; the U_o denominators stay national employment shares (PAP §11 C-66).
- The wage bill replaces employment in both industry weights (the occupation shares within each OEWS NAICS code and the code weights that combine codes into an FF49 industry), so w_oj is o's share of industry j's wage bill; emp_covered_share is wage-bill weighted (PAP §11 C-72), while the zero-assigned share (D3) below is employment-based, as in every row.
- Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65).

- Zero-assigned employment share (D3): 12.14%
- Market part: H5 and H5b on ff49_vw with s, $\bar{b}$ and q recomputed for this exposure vector on the same 378 pre-seal windows, the t_{21} rule and the D21 null $b_0 = \bar{b}$ (PAP §3.3, §8.1); z is the variant's own and is not re-standardised after a drop.
- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.

D.4.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	1.75	2.33	[-3.09, 6.59]	0.047	0.963	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.751, p = 0.461
market	H5b	3.36	2.74	[-2.34, 9.06]	0.817	0.423	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.227, p = 0.234
rq2	RQ2	0.814	—	[0.669, 0.903]	—	—	—	—	—	49	none	z_usage vs z_ eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.786	—	[0.629, 0.886]	—	—	—	—	—	49	none	z_ delegation vs z_ eloundou; 10000 bootstrap draws used; headline bench-mark (D8)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.822	—	[0.682, 0.908]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.879	—	[0.786, 0.932]	—	—	—	—	—	49	none	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.868	—	[0.762, 0.930]	—	—	—	—	—	49	none	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.878	—	[0.786, 0.928]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.847	—	[0.728, 0.917]	—	—	—	—	—	49	none	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.817	—	[0.680, 0.901]	—	—	—	—	—	49	none	z_ delegation vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.855	—	[0.745, 0.919]	—	—	—	—	—	49	none	z_ augmentation vs z_aioe; 10000 bootstrap draws used

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.5 R3: robustness row (PAP §8.2 R3)

Source: PAP §8.2, row R3. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation; pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.5.1 Notes

- Variant: ExposureSpec(unmatched='drop'); every other construction choice is the primary Design B one.
- Occupations with no AEI match are dropped (`unmatched="drop"`) instead of assigned zero usage (PAP D3 alternative, §8.2 R3).
- Benchmarks: AIOE and Eloundou stay at their primary values, because this occupation-level variant changes neither the matrix nor the crosswalk (PAP §11 C-65).
- Zero-assigned employment share (D3): 0.00%
- Market part: H5 and H5b on ff49_vw with s, $\bar{b}$ and q recomputed for this exposure vector on the same 378 pre-seal windows, the t_{21} rule and the D21 null $b_0 = \bar{b}$ (PAP §3.3, §8.1); z is the variant's own and is not re-standardised after a drop.
- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.

D.5.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	1.55	2.26	[-3.16, 6.26]	-0.024	0.981	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.686, p = 0.500
market	H5b	2.44	2.47	[-2.70, 7.57]	0.525	0.605	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.986, p = 0.335
rq2	RQ2	0.667	—	[0.448, 0.817]	—	—	—	—	—	49	none	z_usage vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)
rq2	RQ2	0.568	—	[0.321, 0.754]	—	—	—	—	—	49	none	z_delegation vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.648	—	[0.425, 0.807]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.771	—	[0.606, 0.880]	—	—	—	—	—	49	none	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.710	—	[0.515, 0.843]	—	—	—	—	—	49	none	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.757	—	[0.585, 0.871]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.727	—	[0.532, 0.851]	—	—	—	—	—	49	none	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.643	—	[0.420, 0.801]	—	—	—	—	—	49	none	z_ delegation vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.720	—	[0.525, 0.850]	—	—	—	—	—	49	none	z_ augmentation vs z_aioe; 10000 bootstrap draws used

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.6 R4: robustness row (PAP §8.2 R4)

Source: PAP §8.2, row R4. Label: robustness; exploratory.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: annualised Sharpe ratio; annualised pp; correlation; pp of CAR per 1 SD; pp per month per 1 SD; pp per month per unit of log BE/ME; pp per month per unit of log size; pp per month per unit of the 12-2 return (decimal); probability. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.6.1 Notes

- Variant: ExposureSpec(design='A_pooled', source='1p_api'); every other construction choice is the primary A_pooled one. A_pooled is labelled look-ahead (PAP §8.3 D19).
- `1p_api` in place of `claude_ai` for A_pooled, pooled over r3–r6 with equal release weights: the API source starts at r3 (PAP §8.2 R4; §11 C-65). Design B has no API source. For RQ1 this variant is merged into R22 and R24 (PAP §8.2 R4).
- Benchmarks: AIOE and Eloundou stay at their primary values, because this occupation-level variant changes neither the matrix nor the crosswalk (PAP §11 C-65).
- Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

- Zero-assigned employment share (D3): 6.89%
- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.
- Index part: the Spearman correlation of the variant A_pooled `z_delegation` with the primary A_pooled `z_delegation` (Plan 3's `data.exposures()`), across the FF49 industries with both values; descriptive.
- Design A parts run Plan 3's own runners on the exposure frame passed: `run_terciles` (fixed-membership H-L terciles, cap and EW legs; C-34, C-35, C-37, C-45), `run_fm` (Fama–MacBeth on [1, z, log size, log BE/ME, 12-2 momentum]; C-38, C-46) and `run_events` (the 14 D19 events, CAR slopes at horizons 5 and 20, HC3 normal reference; C-33, C-44). Records are read from the frames the runners return; their docs are written under the untracked `data/derived/robustness/<row>/<part>` and no figure is written.
- Columns: estimate, interval and p are as Plan 3 returns them; the scale column holds the SE (Newey-West for terciles and Fama–MacBeth, HC3 for event slopes); n is T (months) for terciles and Fama–MacBeth and the number of industries for event slopes. Null 0, nominal and single-test; no error control across records or rows (PAP §8.1). Sharpe, PSR and DSR carry no p; PSR and DSR carry no interval.
- DSR here is against the 2 tercile variants run in this row (the primary used 6, C-37); not comparable with the primary's DSR (PAP §11 C-76).

D.6.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.772	—	[0.610, 0.874]	—	—	—	—	—	49	none	z_usage vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)
rq2	RQ2	0.769	—	[0.603, 0.875]	—	—	—	—	—	49	none	z_delegation vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.777	—	[0.623, 0.872]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)
rq2	RQ2	0.846	—	[0.717, 0.921]	—	—	—	—	—	49	none	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.831	—	[0.688, 0.916]	—	—	—	—	—	49	none	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.858	—	[0.747, 0.926]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.798	—	[0.646, 0.890]	—	—	—	—	—	49	none	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.788	—	[0.622, 0.891]	—	—	—	—	—	49	none	z_ delegation vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.814	—	[0.680, 0.891]	—	—	—	—	—	49	none	z_ augmentation vs z_aioe; 10000 bootstrap draws used
index	A_pooled	0.964	—	—	—	—	—	—	—	49	none	variant vs primary A_pooled z_ delegation (Spearman, FF49 industries with both values); descriptive, no interval

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	4.66	3.48	[-2.35, 11.67]	—	0.187	0	—	—	45	none	alpha_CAPM A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	5.02	3.05	[-1.13, 11.17]	—	0.107	0	—	—	45	none	alpha_FF3 A_pooled cap; A_ pooled (look-ahead); nominal; Bootstrap draws: `run_ terciles` consumes one `default_ rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_ pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	3.79	2.91	[-2.10, 9.67]	—	0.201	0	—	—	45	none	alpha_FF5+UMD A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	7.58	4.47	[-0.47, 15.93]	—	0.097	0	—	—	45	none	mean A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.82	—	[-0.06, 1.66]	—	—	0	—	—	45	none	sharpe A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.95	—	—	—	—	0	—	—	45	none	psr A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.91	—	—	—	—	0	—	—	45	none	dsr A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	2.33	3.41	[-4.55, 9.20]	—	0.499	0	—	—	45	none	alpha_CAPM A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	1.96	2.83	[-3.75, 7.67]	—	0.493	0	—	—	45	none	alpha_FF3 A_pooled ew; A_ pooled (look-ahead); nominal; Bootstrap draws: `run_ terciles` consumes one `default_ rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_ pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.95	3.00	[-5.13, 7.03]	—	0.754	0	—	—	45	none	alpha_FF5+UMD A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	3.02	3.37	[-3.76, 9.32]	—	0.374	0	—	—	45	none	mean A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.39	—	[-0.46, 1.38]	—	—	0	—	—	45	none	sharpe A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.77	—	—	—	—	0	—	—	45	none	psr A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.67	—	—	—	—	0	—	—	45	none	dsr A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
fm	Fama-MacBeth	0.20	0.11	[-0.02, 0.42]	—	0.078	0	—	—	45	none	FM z A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal
fm	Fama-MacBeth	-0.09	0.15	[-0.39, 0.22]	—	0.579	0	—	—	45	none	FM log_size A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal
fm	Fama-MacBeth	-0.01	0.19	[-0.40, 0.39]	—	0.977	0	—	—	45	none	FM log_beme A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
fm	Fama-MacBeth	0.23	0.86	[-1.51, 1.97]	—	0.793	0	—	—	45	none	FM mom A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal
events	event studies	-0.33	0.22	[-0.76, 0.11]	—	0.138	0	—	—	49	none	CAR slope ChatGPT h5 A_pooled; day 0 2022-11-30; A_pooled (look-ahead); nominal
events	event studies	0.43	0.56	[-0.67, 1.53]	—	0.445	0	—	—	49	none	CAR slope ChatGPT h20 A_pooled; day 0 2022-11-30; A_pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-0.13	0.40	[-0.92, 0.65]	—	0.736	0	—	—	49	none	CAR slope ChatGPT h9 none [z_aioe in z_aioe]; day 0 2022-11-30; none; nominal
events	event studies	-0.17	0.54	[-1.23, 0.88]	—	0.746	0	—	—	49	none	CAR slope ChatGPT h9 A_ pooled [z_ aioe in z_ aioe+z_ delegation]; day 0 2022-11-30; A_ pooled (look-ahead); nominal
events	event studies	0.05	0.39	[-0.71, 0.82]	—	0.888	0	—	—	49	none	CAR slope ChatGPT h9 A_ pooled [z_ delegation in z_ aioe+z_ delegation]; day 0 2022-11-30; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	0.65	0.19	[0.28, 1.02]	—	< 0.001	0	—	—	49	none	CAR slope GPT-4 h5 A_pooled; day 0 2023-03-14; A_pooled (look-ahead); nominal
events	event studies	0.46	0.37	[-0.26, 1.17]	—	0.213	0	—	—	49	none	CAR slope GPT-4 h20 A_pooled; day 0 2023-03-14; A_pooled (look-ahead); nominal
events	event studies	1.18	0.23	[0.72, 1.64]	—	< 0.001	0	—	—	49	none	CAR slope PaLM 2 h5 A_pooled; day 0 2023-05-10; A_pooled (look-ahead); nominal
events	event studies	2.04	0.54	[0.98, 3.09]	—	< 0.001	0	—	—	49	none	CAR slope PaLM 2 h20 A_pooled; day 0 2023-05-10; A_pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	0.43	0.43	[-0.42, 1.27]	—	0.324	0	—	—	49	none	CAR slope Claude 2 h5 A_ pooled; day 0 2023-07-11; A_ pooled (look-ahead); nominal
events	event studies	0.52	0.38	[-0.21, 1.26]	—	0.164	0	—	—	49	none	CAR slope Claude 2 h20 A_ pooled; day 0 2023-07-11; A_ pooled (look-ahead); nominal
events	event studies	0.01	0.18	[-0.34, 0.35]	—	0.971	0	—	—	49	none	CAR slope Claude 2.1 h5 A_ pooled; day 0 2023-11-21; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	0.17	0.45	[-0.71, 1.05]	—	0.705	0	—	—	49	none	CAR slope Claude 2.1 h20 A_ pooled; day 0 2023-11-21; A_ pooled (look-ahead); nominal
events	event studies	0.14	0.30	[-0.44, 0.72]	—	0.638	0	—	—	49	none	CAR slope Gemini 1.0 h5 A_ pooled; day 0 2023-12-06; A_ pooled (look-ahead); nominal
events	event studies	0.02	0.36	[-0.68, 0.72]	—	0.952	0	—	—	49	none	CAR slope Gemini 1.0 h20 A_ pooled; day 0 2023-12-06; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-0.63	0.22	[-1.06, -0.21]	—	0.004	0	—	—	49	none	CAR slope Gemini 1.5 h5 A_ pooled; day 0 2024-02-15; A_ pooled (look-ahead); nominal
events	event studies	-0.93	0.49	[-1.89, 0.04]	—	0.059	0	—	—	49	none	CAR slope Gemini 1.5 h20 A_ pooled; day 0 2024-02-15; A_ pooled (look-ahead); nominal
events	event studies	-0.13	0.22	[-0.57, 0.30]	—	0.547	0	—	—	49	none	CAR slope Claude 3 h5 A_ pooled; day 0 2024-03-04; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-0.50	0.51	[-1.51, 0.51]	—	0.329	0	—	—	49	none	CAR slope Claude 3 h20 A_ pooled; day 0 2024-03-04; A_ pooled (look-ahead); nominal
events	event studies	0.40	0.22	[-0.04, 0.84]	—	0.073	0	—	—	49	none	CAR slope Claude 3.5 h5 A_ pooled; day 0 2024-06-20; A_ pooled (look-ahead); nominal
events	event studies	-0.21	0.51	[-1.21, 0.80]	—	0.686	0	—	—	49	none	CAR slope Claude 3.5 h20 A_ pooled; day 0 2024-06-20; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-0.07	0.32	[-0.69, 0.56]	—	0.834	0	—	—	49	none	CAR slope OpenAI o1 h5 A_ pooled; day 0 2024-09-12; A_ pooled (look-ahead); nominal
events	event studies	-0.10	0.47	[-1.03, 0.83]	—	0.832	0	—	—	49	none	CAR slope OpenAI o1 h20 A_ pooled; day 0 2024-09-12; A_ pooled (look-ahead); nominal
events	event studies	-0.07	0.32	[-0.71, 0.56]	—	0.820	0	—	—	49	none	CAR slope Gemini 2.0 h5 A_ pooled; day 0 2025-02-05; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-0.12	0.64	[-1.38, 1.14]	—	0.849	0	—	—	49	none	CAR slope Gemini 2.0 h20 A_ pooled; day 0 2025-02-05; A_ pooled (look-ahead); nominal
events	event studies	-0.20	0.39	[-0.97, 0.56]	—	0.600	0	—	—	49	none	CAR slope Claude 3.7 h5 A_ pooled; day 0 2025-02-24; A_ pooled (look-ahead); nominal
events	event studies	-0.27	0.52	[-1.29, 0.75]	—	0.609	0	—	—	49	none	CAR slope Claude 3.7 h20 A_ pooled; day 0 2025-02-24; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	0.17	0.29	[-0.40, 0.75]	—	0.552	0	—	—	49	none	CAR slope GPT-4.5 h5 A_pooled; day 0 2025-02-27; A_ pooled (look-ahead); nominal
events	event studies	-0.56	0.55	[-1.64, 0.52]	—	0.311	0	—	—	49	none	CAR slope GPT-4.5 h20 A_ pooled; day 0 2025-02-27; A_ pooled (look-ahead); nominal
events	event studies	-0.51	0.22	[-0.94, -0.08]	—	0.020	0	—	—	49	none	CAR slope Gemini 2.5 h5 A_ pooled; day 0 2025-03-25; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-1.20	0.66	[-2.49, 0.09]	—	0.069	0	—	—	49	none	CAR slope Gemini 2.5 h20 A_ pooled; day 0 2025-03-25; A_ pooled (look-ahead); nominal

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.7 R5: robustness row (PAP §8.2 R5)

Source: PAP §8.2, row R5. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.7.1 Notes

- Equal-weighted FF49 daily returns in place of value-weighted, in the confirmatory window and in the 378-window history (PAP §8.2 R5).
- Exposure: the primary Design B vector (`ExposureSpec()`). H5 and H5b with s, $\bar{b}$ and q recomputed for this row by the pre-seal procedure (the same 378 windows unless the row says otherwise, the t_{21} rule and the D21 null $b_0 = \bar{b}$; PAP §3.3, §8.1); industries with a missing return in the confirmatory window are dropped and z is not re-standardised. The secondary b = 0 statistic is in each record's note.
- Zero-assigned employment share (D3): 12.14% (the primary Design B exposure; this row rebuilds no index).

D.7.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	4.88	2.21	[0.30, 9.47]	1.288	0.212	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 2.214, p = 0.038

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5b	4.82	2.95	[-1.32, 10.96]	1.352	0.191	b = bbar	2.5086	2.0796	49	none	secondary b = 0: T = 1.632, p = 0.118

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.8 R6: robustness row (PAP §8.2 R6)

Source: PAP §8.2, row R6. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.8.1 Notes

- Variant ff30: ExposureSpec(industries=30) on ff30_vw (30 industries); every other construction choice is the primary Design B one.
- Variant ff48: ExposureSpec(industries=48) on ff48_vw (48 industries); every other construction choice is the primary Design B one.
- FF30 and FF48 industry granularity (PAP §8.2 R6): the exposure is rebuilt on the 30 / 48 industries of Siccodes30 / Siccodes48 with the same split crosswalk, z re-standardised across those industries; alphas from the 30- / 48-industry daily value-weighted files; s, $\bar{b}$ and q from the same 378 windows on those files.
- FF30/FF48 files retrieved after the freeze; their history may reflect French's later revisions. French's newer download vintage applies to these files only: FF49 and the factors stay the frozen files (PAP §8.2 R6, §9.1).
- Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65).
- ff30: Zero-assigned employment share (D3): 12.14%
- ff48: Zero-assigned employment share (D3): 12.14%
- Market parts: H5 and H5b on each part's return set with s, $\bar{b}$ and q recomputed for the part's exposure vector on the same 378 pre-seal windows with the same code, the critical $\max(t_{21}, q)$ and the D21 null $b_0 = \bar{b}$ (PAP §3.3, §8.1); industries with a missing return in the confirmatory window are dropped and z is not re-standardised after a drop. The secondary b = 0 statistic is in each record's note.

D.8.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
ff30	H5	1.55	2.51	[-3.68, 6.78]	0.225	0.824	b = bbar	2.4138	2.0796	30	none	secondary b = 0: T = 0.617, p = 0.544

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
ff30	H5b	0.72	3.49	[-6.61, 8.05]	0.221	0.827	b = bbar	2.4604	2.0977	30	none	secondary b = 0: T = 0.207, p = 0.838
ff48	H5	2.17	2.58	[-3.20, 7.53]	0.254	0.802	b = bbar	2.4138	2.0796	48	none	secondary b = 0: T = 0.840, p = 0.410
ff48	H5b	3.81	2.98	[-2.38, 10.01]	0.910	0.373	b = bbar	2.4138	2.0796	48	none	secondary b = 0: T = 1.280, p = 0.215

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.9 R7: robustness row (PAP §8.2 R7)

Source: PAP §8.2, row R7. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.9.1 Notes

- Technology industries Hardw, Softw and Chips dropped from the window and the history (PAP §8.2 R7).
- [default] z not re-standardised after the drop, as for R13 (PAP §11 C-63).
- Exposure: the primary Design B vector (`ExposureSpec()`). H5 and H5b with s, $\bar{b}$ and q recomputed for this row by the pre-seal procedure (the same 378 windows unless the row says otherwise, the t_{21} rule and the D21 null $b_0 = \bar{b}$; PAP §3.3, §8.1); industries with a missing return in the confirmatory window are dropped and z is not re-standardised. The secondary b = 0 statistic is in each record's note.
- Zero-assigned employment share (D3): 12.14% (the primary Design B exposure; this row rebuilds no index).

D.9.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	-1.72	3.03	[-8.02, 4.57]	-0.805	0.430	b = bbar	2.4138	2.0796	46	Hardw, Softw, Chips	secondary b = 0: T = -0.569, p = 0.575

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5b	-0.24	3.58	[-7.98, 7.50]	-0.037	0.971	b = bbar	2.4499	2.1640	46	Hardw, Softw, Chips	secondary b = 0: T = -0.066, p = 0.948

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.10 R8: robustness row (PAP §8.2 R8)

Source: PAP §8.2, row R8. Label: robustness; exploratory.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: annualised Sharpe ratio; annualised pp; pp per month per 1 SD; pp per month per unit of log BE/ME; pp per month per unit of log size; pp per month per unit of the 12-2 return (decimal); probability. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.10.1 Notes

- Calendar 2023 excluded (PAP §8.2 R8): the Design A terciles' months and the Fama–MacBeth monthly cross-sections use the 33 month ends of 2022-12 to 2026-08 outside 2023 (2022-12, then 2024-01 to 2026-08); the Newey-West SEs run on those months only. No null scale is involved.
- Not applicable to Design B by design (its window starts 2025-04; PAP §8.2 R8). Event studies are not part of R8.
- Exposure: Plan 3's primary A_r2 and A_pooled `z_delegation` (A_pooled labelled look-ahead, PAP §8.3 D19); this row rebuilds no index.
- Zero-assigned employment share, Design A_r2: 12.14% (D3)
- Zero-assigned employment share, Design A_pooled: 6.34% (D3)
- Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.
- Design A parts run Plan 3's own runners on the exposure frame passed: `run_terciles` (fixed-membership H-L terciles, cap and EW legs; C-34, C-35, C-37, C-45), `run_fm` (Fama–MacBeth on [1, z, log size, log BE/ME, 12-2 momentum]; C-38, C-46) and `run_events` (the 14 D19 events, CAR slopes at horizons 5 and 20, HC3 normal reference; C-33, C-44). Records are read from the frames the runners return; their docs are written under the untracked `data/derived/robustness/<row>/<part>` and no figure is written.
- Columns: estimate, interval and p are as Plan 3 returns them; the scale column holds the SE (Newey-West for terciles and Fama–MacBeth, HC3 for event slopes); n is T (months) for terciles and Fama–MacBeth and the number of industries for event slopes. Null 0, nominal and single-test; no error control across records or rows (PAP §8.1). Sharpe, PSR and DSR carry no p; PSR and DSR carry no interval.
- DSR here is against the 4 tercile variants run in this row (the primary used 6, C-37); not comparable with the primary's DSR (PAP §11 C-76).

D.10.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	3.83	4.50	[-5.35, 13.01]	—	0.401	0	—	—	33	none	alpha_CAPM A_r2 cap; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	6.99	4.70	[-2.62, 16.61]	—	0.148	0	—	—	33	none	alpha_FF3 A_r2 cap; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	5.50	3.39	[-1.47, 12.47]	—	0.117	0	—	—	33	none	alpha_FF5+UMD A_r2 cap; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	7.89	6.30	[-3.34, 19.28]	—	0.219	0	—	—	33	none	mean A_r2 cap; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.73	—	[-0.33, 1.79]	—	—	0	—	—	33	none	sharpe A_r2 cap; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.89	—	—	—	—	0	—	—	33	none	psr A_r2 cap; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.79	—	—	—	—	0	—	—	33	none	dsr A_r2 cap; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	-0.48	5.04	[-10.76, 9.80]	—	0.924	0	—	—	33	none	alpha_CAPM A_r2 ew; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	1.84	4.60	[-7.57, 11.26]	—	0.692	0	—	—	33	none	alpha_FF3 A_r2 ew; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	1.90	4.65	[-7.67, 11.47]	—	0.687	0	—	—	33	none	alpha_FF5+UMD A_r2 ew; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	2.87	5.03	[-7.24, 12.14]	—	0.573	0	—	—	33	none	mean A_r2 ew; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.34	—	[-0.80, 1.69]	—	—	0	—	—	33	none	sharpe A_r2 ew; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.71	—	—	—	—	0	—	—	33	none	psr A_r2 ew; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.55	—	—	—	—	0	—	—	33	none	dsr A_r2 ew; A_r2; nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	3.78	4.48	[-5.35, 12.92]	—	0.405	0	—	—	33	none	alpha_CAPM A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	6.91	4.67	[-2.65, 16.47]	—	0.150	0	—	—	33	none	alpha_FF3 A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	5.43	3.37	[-1.50, 12.36]	—	0.119	0	—	—	33	none	alpha_FF5+UMD A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	7.83	6.27	[-3.60, 19.33]	—	0.221	0	—	—	33	none	mean A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.73	—	[-0.36, 1.80]	—	—	0	—	—	33	none	sharpe A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.89	—	—	—	—	0	—	—	33	none	psr A_ pooled cap; A_ pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.79	—	—	—	—	0	—	—	33	none	dsr A_ pooled cap; A_ pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	-0.47	5.27	[-11.23, 10.28]	—	0.929	0	—	—	33	none	alpha_CAPM A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	1.24	4.70	[-8.38, 10.86]	—	0.794	0	—	—	33	none	alpha_FF3 A_pooled ew; A_ pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_ rng(SEED)` (con-sumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	1.37	4.70	[-8.28, 11.03]	—	0.772	0	—	—	33	none	alpha_FF5+UMD A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	2.51	5.19	[-8.01, 12.25]	—	0.632	0	—	—	33	none	mean A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.28	—	[-0.84, 1.58]	—	—	0	—	—	33	none	sharpe A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.67	—	—	—	—	0	—	—	33	none	psr A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.51	—	—	—	—	0	—	—	33	none	dsr A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: with Design B absent and 33 months, the one `default_rng(SEED)` (consumed in `VARIANTS` order, PAP §11 C-37) gives the A_r2 and A_pooled variants different draws from the primary run, so their bootstrap CIs differ partly by draw noise.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
fm	Fama-MacBeth	0.21	0.16	[-0.11, 0.53]	—	0.184	0	—	—	33	none	FM z A_r2; A_r2; window A; mean n 48.0; 0 months skipped; nominal
fm	Fama-MacBeth	0.08	0.17	[-0.26, 0.42]	—	0.635	0	—	—	33	none	FM log_size A_r2; A_r2; window A; mean n 48.0; 0 months skipped; nominal
fm	Fama-MacBeth	0.22	0.17	[-0.13, 0.57]	—	0.205	0	—	—	33	none	FM log_beme A_r2; A_r2; window A; mean n 48.0; 0 months skipped; nominal
fm	Fama-MacBeth	-0.08	0.92	[-1.94, 1.78]	—	0.931	0	—	—	33	none	FM mom A_r2; A_r2; window A; mean n 48.0; 0 months skipped; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
fm	Fama-MacBeth	0.16	0.15	[-0.14, 0.46]	—	0.278	0	—	—	33	none	FM z A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal
fm	Fama-MacBeth	0.10	0.16	[-0.24, 0.43]	—	0.563	0	—	—	33	none	FM log_size A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal
fm	Fama-MacBeth	0.21	0.17	[-0.15, 0.56]	—	0.246	0	—	—	33	none	FM log_beme A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
fm	Fama-MacBeth	-0.07	0.90	[-1.90, 1.77]	—	0.941	0	—	—	33	none	FM mom A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.11 R9: robustness row (PAP §8.2 R9)

Source: PAP §8.2, row R9. Label: robustness; exploratory.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of CAR per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.11.1 Notes

- [default] Placebo day-0 dates (PAP §8.2 R9; §11 C-60): each of the 14 D19 events' month and day in 2017, 2018, 2019 and 2021 (56 dates, named " placebo "); 2020 is skipped, as the PAP states. Day 0 is the next trading day when the date is not one (Plan 3's `day_zero`).
- The placebo dates are not D19 events. Plan 3's `event_study.md` for this run (in the untracked work directory) prints the D19 rule text whatever events it is given; the records here are read from the frame `run_events` returns, not from that doc. No placebo is named `ChatGPT`, so no Eisfeldt et al. step runs on them.
- [default] Placebo records: the CAR slope on the primary A_r2 `z_delegation` at horizons 5 and 20, one per placebo date and horizon, as Plan 3's `run_events` returns it (OLS with intercept, HC3, normal reference; C-44). Summary records: per horizon, the mean A_r2 slope over the 14 actual D19 events (a second `run_events` on the same exposure) and its empirical percentile among the 56 placebo slopes, in the stat column; descriptive, no p (PAP §11 C-60).
- [default] Percentile convention: 100 × the share of the 56 placebo slopes ≤ the actual mean slope, per horizon (PAP §11 C-76).
- Exposure: Plan 3's primary A_r2 `z_delegation`; this row rebuilds no index.
- Zero-assigned employment share, Design A_r2: 12.14% (D3)
- Columns: placebo records hold the CAR slope as estimate, its HC3 SE in the scale column, and the interval and p as Plan 3's `run_events` returns them, with n the number of industries and null 0, nominal and single-test. Summary records hold the mean actual slope as estimate and its percentile in the stat column, with n the number of placebo slopes and no null, interval or p. No error control across records or rows (PAP §8.1).

D.11.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.40	0.34	[-1.07, 0.27]	—	0.239	0	—	—	49	none	CAR slope ChatGPT placebo 2017 h5 A_r2; day 0 2017-11-30; A_r2; nominal
placebo	event studies	0.08	0.20	[-0.31, 0.47]	—	0.674	0	—	—	49	none	CAR slope ChatGPT placebo 2018 h5 A_r2; day 0 2018-11-30; A_r2; nominal
placebo	event studies	-0.33	0.38	[-1.07, 0.41]	—	0.378	0	—	—	49	none	CAR slope ChatGPT placebo 2019 h5 A_r2; day 0 2019-12-02; A_r2; nominal
placebo	event studies	-0.33	0.29	[-0.89, 0.23]	—	0.252	0	—	—	49	none	CAR slope ChatGPT placebo 2021 h5 A_r2; day 0 2021-11-30; A_r2; nominal
placebo	event studies	-0.05	0.16	[-0.37, 0.27]	—	0.772	0	—	—	49	none	CAR slope GPT-4 placebo 2017 h5 A_r2; day 0 2017-03-14; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.01	0.28	[-0.56, 0.54]	—	0.977	0	—	—	49	none	CAR slope GPT-4 placebo 2018 h5 A_r2; day 0 2018-03-14; A_r2; nominal
placebo	event studies	-0.33	0.19	[-0.70, 0.04]	—	0.083	0	—	—	49	none	CAR slope GPT-4 placebo 2019 h5 A_r2; day 0 2019-03-14; A_r2; nominal
placebo	event studies	-0.04	0.31	[-0.64, 0.57]	—	0.904	0	—	—	49	none	CAR slope GPT-4 placebo 2021 h5 A_r2; day 0 2021-03-15; A_r2; nominal
placebo	event studies	-0.40	0.29	[-0.97, 0.16]	—	0.162	0	—	—	49	none	CAR slope PaLM 2 placebo 2017 h5 A_r2; day 0 2017-05-10; A_r2; nominal
placebo	event studies	-0.36	0.32	[-0.99, 0.27]	—	0.261	0	—	—	49	none	CAR slope PaLM 2 placebo 2018 h5 A_r2; day 0 2018-05-10; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	0.14	0.22	[-0.29, 0.57]	—	0.532	0	—	—	49	none	CAR slope PaLM 2 placebo 2019 h5 A_r2; day 0 2019-05-10; A_r2; nominal
placebo	event studies	-0.36	0.62	[-1.58, 0.86]	—	0.559	0	—	—	49	none	CAR slope PaLM 2 placebo 2021 h5 A_r2; day 0 2021-05-10; A_r2; nominal
placebo	event studies	0.30	0.11	[0.09, 0.51]	—	0.005	0	—	—	49	none	CAR slope Claude 2 placebo 2017 h5 A_r2; day 0 2017-07-11; A_r2; nominal
placebo	event studies	-0.05	0.26	[-0.55, 0.45]	—	0.844	0	—	—	49	none	CAR slope Claude 2 placebo 2018 h5 A_r2; day 0 2018-07-11; A_r2; nominal
placebo	event studies	-0.49	0.25	[-0.98, 0.00]	—	0.052	0	—	—	49	none	CAR slope Claude 2 placebo 2019 h5 A_r2; day 0 2019-07-11; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.02	0.26	[-0.54, 0.49]	—	0.931	0	—	—	49	none	CAR slope Claude 2 placebo 2021 h5 A_r2; day 0 2021-07-12; A_r2; nominal
placebo	event studies	-0.28	0.17	[-0.61, 0.05]	—	0.093	0	—	—	49	none	CAR slope Claude 2.1 placebo 2017 h5 A_r2; day 0 2017-11-21; A_r2; nominal
placebo	event studies	0.33	0.28	[-0.22, 0.87]	—	0.241	0	—	—	49	none	CAR slope Claude 2.1 placebo 2018 h5 A_r2; day 0 2018-11-21; A_r2; nominal
placebo	event studies	-0.31	0.25	[-0.81, 0.18]	—	0.218	0	—	—	49	none	CAR slope Claude 2.1 placebo 2019 h5 A_r2; day 0 2019-11-21; A_r2; nominal
placebo	event studies	0.20	0.57	[-0.93, 1.33]	—	0.727	0	—	—	49	none	CAR slope Claude 2.1 placebo 2021 h5 A_r2; day 0 2021-11-22; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	0.23	0.15	[-0.06, 0.52]	—	0.126	0	—	—	49	none	CAR slope Gemini 1.0 placebo 2017 h5 A_r2; day 0 2017-12-06; A_r2; nominal
placebo	event studies	0.18	0.26	[-0.32, 0.68]	—	0.474	0	—	—	49	none	CAR slope Gemini 1.0 placebo 2018 h5 A_r2; day 0 2018-12-06; A_r2; nominal
placebo	event studies	0.03	0.18	[-0.31, 0.37]	—	0.865	0	—	—	49	none	CAR slope Gemini 1.0 placebo 2019 h5 A_r2; day 0 2019-12-06; A_r2; nominal
placebo	event studies	-0.35	0.53	[-1.39, 0.68]	—	0.506	0	—	—	49	none	CAR slope Gemini 1.0 placebo 2021 h5 A_r2; day 0 2021-12-06; A_r2; nominal
placebo	event studies	0.43	0.45	[-0.45, 1.32]	—	0.339	0	—	—	49	none	CAR slope Gemini 1.5 placebo 2017 h5 A_r2; day 0 2017-02-15; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	0.27	0.15	[-0.03, 0.56]	—	0.080	0	—	—	49	none	CAR slope Gemini 1.5 placebo 2018 h5 A_r2; day 0 2018-02-15; A_r2; nominal
placebo	event studies	-0.02	0.38	[-0.76, 0.72]	—	0.959	0	—	—	49	none	CAR slope Gemini 1.5 placebo 2019 h5 A_r2; day 0 2019-02-15; A_r2; nominal
placebo	event studies	0.04	0.29	[-0.54, 0.61]	—	0.901	0	—	—	49	none	CAR slope Gemini 1.5 placebo 2021 h5 A_r2; day 0 2021-02-16; A_r2; nominal
placebo	event studies	0.18	0.15	[-0.12, 0.47]	—	0.243	0	—	—	49	none	CAR slope Claude 3 placebo 2017 h5 A_r2; day 0 2017-03-06; A_r2; nominal
placebo	event studies	-0.08	0.33	[-0.73, 0.57]	—	0.799	0	—	—	49	none	CAR slope Claude 3 placebo 2018 h5 A_r2; day 0 2018-03-05; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.42	0.50	[-1.40, 0.55]	—	0.395	0	—	—	49	none	CAR slope Claude 3 placebo 2019 h5 A_r2; day 0 2019-03-04; A_r2; nominal
placebo	event studies	0.30	0.29	[-0.26, 0.86]	—	0.298	0	—	—	49	none	CAR slope Claude 3 placebo 2021 h5 A_r2; day 0 2021-03-04; A_r2; nominal
placebo	event studies	-0.28	0.27	[-0.82, 0.25]	—	0.292	0	—	—	49	none	CAR slope Claude 3.5 placebo 2017 h5 A_r2; day 0 2017-06-20; A_r2; nominal
placebo	event studies	-0.05	0.16	[-0.36, 0.25]	—	0.743	0	—	—	49	none	CAR slope Claude 3.5 placebo 2018 h5 A_r2; day 0 2018-06-20; A_r2; nominal
placebo	event studies	-0.12	0.15	[-0.41, 0.17]	—	0.430	0	—	—	49	none	CAR slope Claude 3.5 placebo 2019 h5 A_r2; day 0 2019-06-20; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	0.02	0.21	[-0.39, 0.44]	—	0.917	0	—	—	49	none	CAR slope Claude 3.5 placebo 2021 h5 A_r2; day 0 2021-06-21; A_r2; nominal
placebo	event studies	0.21	0.34	[-0.45, 0.86]	—	0.541	0	—	—	49	none	CAR slope OpenAI o1 placebo 2017 h5 A_r2; day 0 2017-09-12; A_r2; nominal
placebo	event studies	0.30	0.43	[-0.55, 1.14]	—	0.494	0	—	—	49	none	CAR slope OpenAI o1 placebo 2018 h5 A_r2; day 0 2018-09-12; A_r2; nominal
placebo	event studies	-0.02	0.25	[-0.52, 0.48]	—	0.937	0	—	—	49	none	CAR slope OpenAI o1 placebo 2019 h5 A_r2; day 0 2019-09-12; A_r2; nominal
placebo	event studies	0.60	0.42	[-0.22, 1.42]	—	0.150	0	—	—	49	none	CAR slope OpenAI o1 placebo 2021 h5 A_r2; day 0 2021-09-13; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	0.25	0.29	[-0.31, 0.82]	—	0.377	0	—	—	49	none	CAR slope Gemini 2.0 placebo 2017 h5 A_r2; day 0 2017-02-06; A_r2; nominal
placebo	event studies	0.11	0.24	[-0.36, 0.58]	—	0.644	0	—	—	49	none	CAR slope Gemini 2.0 placebo 2018 h5 A_r2; day 0 2018-02-05; A_r2; nominal
placebo	event studies	0.07	0.38	[-0.67, 0.82]	—	0.845	0	—	—	49	none	CAR slope Gemini 2.0 placebo 2019 h5 A_r2; day 0 2019-02-05; A_r2; nominal
placebo	event studies	0.63	0.27	[0.09, 1.16]	—	0.022	0	—	—	49	none	CAR slope Gemini 2.0 placebo 2021 h5 A_r2; day 0 2021-02-05; A_r2; nominal
placebo	event studies	-0.28	0.26	[-0.79, 0.22]	—	0.271	0	—	—	49	none	CAR slope Claude 3.7 placebo 2017 h5 A_r2; day 0 2017-02-24; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	0.01	0.22	[-0.42, 0.44]	—	0.950	0	—	—	49	none	CAR slope Claude 3.7 placebo 2018 h5 A_r2; day 0 2018-02-26; A_r2; nominal
placebo	event studies	-0.05	0.19	[-0.43, 0.33]	—	0.789	0	—	—	49	none	CAR slope Claude 3.7 placebo 2019 h5 A_r2; day 0 2019-02-25; A_r2; nominal
placebo	event studies	0.92	0.49	[-0.04, 1.88]	—	0.060	0	—	—	49	none	CAR slope Claude 3.7 placebo 2021 h5 A_r2; day 0 2021-02-24; A_r2; nominal
placebo	event studies	-0.14	0.25	[-0.64, 0.36]	—	0.587	0	—	—	49	none	CAR slope GPT-4.5 placebo 2017 h5 A_r2; day 0 2017-02-27; A_r2; nominal
placebo	event studies	-0.11	0.21	[-0.53, 0.31]	—	0.620	0	—	—	49	none	CAR slope GPT-4.5 placebo 2018 h5 A_r2; day 0 2018-02-27; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.13	0.22	[-0.55, 0.30]	—	0.565	0	—	—	49	none	CAR slope GPT-4.5 placebo 2019 h5 A_r2; day 0 2019-02-27; A_r2; nominal
placebo	event studies	0.11	0.46	[-0.79, 1.01]	—	0.808	0	—	—	49	none	CAR slope GPT-4.5 placebo 2021 h5 A_r2; day 0 2021-03-01; A_r2; nominal
placebo	event studies	-0.07	0.14	[-0.35, 0.21]	—	0.610	0	—	—	49	none	CAR slope Gemini 2.5 placebo 2017 h5 A_r2; day 0 2017-03-27; A_r2; nominal
placebo	event studies	-0.21	0.31	[-0.82, 0.41]	—	0.505	0	—	—	49	none	CAR slope Gemini 2.5 placebo 2018 h5 A_r2; day 0 2018-03-26; A_r2; nominal
placebo	event studies	0.09	0.25	[-0.40, 0.58]	—	0.727	0	—	—	49	none	CAR slope Gemini 2.5 placebo 2019 h5 A_r2; day 0 2019-03-25; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	0.31	0.22	[-0.11, 0.74]	—	0.152	0	—	—	49	none	CAR slope Gemini 2.5 placebo 2021 h5 A_r2; day 0 2021-03-25; A_r2; nominal
placebo	event studies	-0.42	0.43	[-1.25, 0.42]	—	0.329	0	—	—	49	none	CAR slope ChatGPT placebo 2017 h20 A_r2; day 0 2017-11-30; A_r2; nominal
placebo	event studies	-0.07	0.67	[-1.38, 1.25]	—	0.920	0	—	—	49	none	CAR slope ChatGPT placebo 2018 h20 A_r2; day 0 2018-11-30; A_r2; nominal
placebo	event studies	-0.28	0.61	[-1.48, 0.92]	—	0.645	0	—	—	49	none	CAR slope ChatGPT placebo 2019 h20 A_r2; day 0 2019-12-02; A_r2; nominal
placebo	event studies	-0.59	0.61	[-1.78, 0.60]	—	0.329	0	—	—	49	none	CAR slope ChatGPT placebo 2021 h20 A_r2; day 0 2021-11-30; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.27	0.42	[-1.08, 0.55]	—	0.521	0	—	—	49	none	CAR slope GPT-4 placebo 2017 h20 A_r2; day 0 2017-03-14; A_r2; nominal
placebo	event studies	0.02	0.31	[-0.59, 0.63]	—	0.943	0	—	—	49	none	CAR slope GPT-4 placebo 2018 h20 A_r2; day 0 2018-03-14; A_r2; nominal
placebo	event studies	-0.23	0.28	[-0.79, 0.33]	—	0.416	0	—	—	49	none	CAR slope GPT-4 placebo 2019 h20 A_r2; day 0 2019-03-14; A_r2; nominal
placebo	event studies	-0.29	0.52	[-1.32, 0.74]	—	0.577	0	—	—	49	none	CAR slope GPT-4 placebo 2021 h20 A_r2; day 0 2021-03-15; A_r2; nominal
placebo	event studies	-0.22	0.34	[-0.89, 0.45]	—	0.514	0	—	—	49	none	CAR slope PaLM 2 placebo 2017 h20 A_r2; day 0 2017-05-10; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.56	0.70	[-1.94, 0.82]	—	0.424	0	—	—	49	none	CAR slope PaLM 2 placebo 2018 h20 A_r2; day 0 2018-05-10; A_r2; nominal
placebo	event studies	-0.55	0.50	[-1.53, 0.42]	—	0.266	0	—	—	49	none	CAR slope PaLM 2 placebo 2019 h20 A_r2; day 0 2019-05-10; A_r2; nominal
placebo	event studies	0.65	0.68	[-0.69, 1.98]	—	0.342	0	—	—	49	none	CAR slope PaLM 2 placebo 2021 h20 A_r2; day 0 2021-05-10; A_r2; nominal
placebo	event studies	0.50	0.87	[-1.20, 2.19]	—	0.565	0	—	—	49	none	CAR slope Claude 2 placebo 2017 h20 A_r2; day 0 2017-07-11; A_r2; nominal
placebo	event studies	-0.68	0.84	[-2.33, 0.96]	—	0.414	0	—	—	49	none	CAR slope Claude 2 placebo 2018 h20 A_r2; day 0 2018-07-11; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.18	0.48	[-1.12, 0.75]	—	0.700	0	—	—	49	none	CAR slope Claude 2 placebo 2019 h20 A_r2; day 0 2019-07-11; A_r2; nominal
placebo	event studies	0.04	0.43	[-0.79, 0.88]	—	0.918	0	—	—	49	none	CAR slope Claude 2 placebo 2021 h20 A_r2; day 0 2021-07-12; A_r2; nominal
placebo	event studies	-0.50	0.43	[-1.35, 0.35]	—	0.250	0	—	—	49	none	CAR slope Claude 2.1 placebo 2017 h20 A_r2; day 0 2017-11-21; A_r2; nominal
placebo	event studies	-0.08	0.73	[-1.52, 1.36]	—	0.918	0	—	—	49	none	CAR slope Claude 2.1 placebo 2018 h20 A_r2; day 0 2018-11-21; A_r2; nominal
placebo	event studies	-0.51	0.64	[-1.77, 0.75]	—	0.429	0	—	—	49	none	CAR slope Claude 2.1 placebo 2019 h20 A_r2; day 0 2019-11-21; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.45	0.69	[-1.81, 0.90]	—	0.511	0	—	—	49	none	CAR slope Claude 2.1 placebo 2021 h20 A_r2; day 0 2021-11-22; A_r2; nominal
placebo	event studies	0.24	0.31	[-0.36, 0.84]	—	0.432	0	—	—	49	none	CAR slope Gemini 1.0 placebo 2017 h20 A_r2; day 0 2017-12-06; A_r2; nominal
placebo	event studies	0.52	0.46	[-0.39, 1.44]	—	0.259	0	—	—	49	none	CAR slope Gemini 1.0 placebo 2018 h20 A_r2; day 0 2018-12-06; A_r2; nominal
placebo	event studies	0.33	0.35	[-0.36, 1.01]	—	0.349	0	—	—	49	none	CAR slope Gemini 1.0 placebo 2019 h20 A_r2; day 0 2019-12-06; A_r2; nominal
placebo	event studies	-0.66	0.42	[-1.49, 0.16]	—	0.116	0	—	—	49	none	CAR slope Gemini 1.0 placebo 2021 h20 A_r2; day 0 2021-12-06; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	0.35	0.68	[-0.99, 1.69]	—	0.608	0	—	—	49	none	CAR slope Gemini 1.5 placebo 2017 h20 A_r2; day 0 2017-02-15; A_r2; nominal
placebo	event studies	0.10	0.45	[-0.79, 0.99]	—	0.819	0	—	—	49	none	CAR slope Gemini 1.5 placebo 2018 h20 A_r2; day 0 2018-02-15; A_r2; nominal
placebo	event studies	-0.77	0.43	[-1.61, 0.07]	—	0.071	0	—	—	49	none	CAR slope Gemini 1.5 placebo 2019 h20 A_r2; day 0 2019-02-15; A_r2; nominal
placebo	event studies	1.54	0.72	[0.13, 2.95]	—	0.032	0	—	—	49	none	CAR slope Gemini 1.5 placebo 2021 h20 A_r2; day 0 2021-02-16; A_r2; nominal
placebo	event studies	-0.02	0.28	[-0.56, 0.52]	—	0.937	0	—	—	49	none	CAR slope Claude 3 placebo 2017 h20 A_r2; day 0 2017-03-06; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.30	0.24	[-0.77, 0.17]	—	0.213	0	—	—	49	none	CAR slope Claude 3 placebo 2018 h20 A_r2; day 0 2018-03-05; A_r2; nominal
placebo	event studies	-0.90	0.74	[-2.35, 0.55]	—	0.224	0	—	—	49	none	CAR slope Claude 3 placebo 2019 h20 A_r2; day 0 2019-03-04; A_r2; nominal
placebo	event studies	0.10	0.59	[-1.06, 1.27]	—	0.862	0	—	—	49	none	CAR slope Claude 3 placebo 2021 h20 A_r2; day 0 2021-03-04; A_r2; nominal
placebo	event studies	-0.03	0.28	[-0.57, 0.52]	—	0.918	0	—	—	49	none	CAR slope Claude 3.5 placebo 2017 h20 A_r2; day 0 2017-06-20; A_r2; nominal
placebo	event studies	-0.13	0.35	[-0.80, 0.55]	—	0.714	0	—	—	49	none	CAR slope Claude 3.5 placebo 2018 h20 A_r2; day 0 2018-06-20; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.47	0.41	[-1.29, 0.34]	—	0.253	0	—	—	49	none	CAR slope Claude 3.5 placebo 2019 h20 A_r2; day 0 2019-06-20; A_r2; nominal
placebo	event studies	-0.32	0.36	[-1.03, 0.39]	—	0.381	0	—	—	49	none	CAR slope Claude 3.5 placebo 2021 h20 A_r2; day 0 2021-06-21; A_r2; nominal
placebo	event studies	0.25	0.41	[-0.56, 1.05]	—	0.548	0	—	—	49	none	CAR slope OpenAI o1 placebo 2017 h20 A_r2; day 0 2017-09-12; A_r2; nominal
placebo	event studies	1.28	0.80	[-0.28, 2.85]	—	0.107	0	—	—	49	none	CAR slope OpenAI o1 placebo 2018 h20 A_r2; day 0 2018-09-12; A_r2; nominal
placebo	event studies	-0.21	0.41	[-1.02, 0.60]	—	0.612	0	—	—	49	none	CAR slope OpenAI o1 placebo 2019 h20 A_r2; day 0 2019-09-12; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	1.14	0.79	[-0.41, 2.70]	—	0.150	0	—	—	49	none	CAR slope OpenAI o1 placebo 2021 h20 A_r2; day 0 2021-09-13; A_r2; nominal
placebo	event studies	0.59	0.98	[-1.34, 2.51]	—	0.550	0	—	—	49	none	CAR slope Gemini 2.0 placebo 2017 h20 A_r2; day 0 2017-02-06; A_r2; nominal
placebo	event studies	0.21	0.54	[-0.84, 1.27]	—	0.690	0	—	—	49	none	CAR slope Gemini 2.0 placebo 2018 h20 A_r2; day 0 2018-02-05; A_r2; nominal
placebo	event studies	-0.01	0.57	[-1.13, 1.11]	—	0.983	0	—	—	49	none	CAR slope Gemini 2.0 placebo 2019 h20 A_r2; day 0 2019-02-05; A_r2; nominal
placebo	event studies	1.58	0.60	[0.40, 2.76]	—	0.009	0	—	—	49	none	CAR slope Gemini 2.0 placebo 2021 h20 A_r2; day 0 2021-02-05; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.15	0.35	[-0.83, 0.53]	—	0.663	0	—	—	49	none	CAR slope Claude 3.7 placebo 2017 h20 A_r2; day 0 2017-02-24; A_r2; nominal
placebo	event studies	-0.11	0.32	[-0.75, 0.52]	—	0.731	0	—	—	49	none	CAR slope Claude 3.7 placebo 2018 h20 A_r2; day 0 2018-02-26; A_r2; nominal
placebo	event studies	-0.97	0.73	[-2.40, 0.45]	—	0.181	0	—	—	49	none	CAR slope Claude 3.7 placebo 2019 h20 A_r2; day 0 2019-02-25; A_r2; nominal
placebo	event studies	0.71	0.85	[-0.96, 2.37]	—	0.408	0	—	—	49	none	CAR slope Claude 3.7 placebo 2021 h20 A_r2; day 0 2021-02-24; A_r2; nominal
placebo	event studies	-0.14	0.27	[-0.66, 0.39]	—	0.606	0	—	—	49	none	CAR slope GPT-4.5 placebo 2017 h20 A_r2; day 0 2017-02-27; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	-0.22	0.31	[-0.83, 0.39]	—	0.478	0	—	—	49	none	CAR slope GPT-4.5 placebo 2018 h20 A_r2; day 0 2018-02-27; A_r2; nominal
placebo	event studies	-1.12	0.82	[-2.73, 0.50]	—	0.176	0	—	—	49	none	CAR slope GPT-4.5 placebo 2019 h20 A_r2; day 0 2019-02-27; A_r2; nominal
placebo	event studies	-0.16	0.73	[-1.58, 1.27]	—	0.831	0	—	—	49	none	CAR slope GPT-4.5 placebo 2021 h20 A_r2; day 0 2021-03-01; A_r2; nominal
placebo	event studies	0.06	0.23	[-0.39, 0.50]	—	0.807	0	—	—	49	none	CAR slope Gemini 2.5 placebo 2017 h20 A_r2; day 0 2017-03-27; A_r2; nominal
placebo	event studies	0.22	0.31	[-0.39, 0.82]	—	0.486	0	—	—	49	none	CAR slope Gemini 2.5 placebo 2018 h20 A_r2; day 0 2018-03-26; A_r2; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
placebo	event studies	0.37	0.56	[-0.73, 1.46]	—	0.511	0	—	—	49	none	CAR slope Gemini 2.5 placebo 2019 h20 A_r2; day 0 2019-03-25; A_r2; nominal
placebo	event studies	-0.14	0.37	[-0.86, 0.58]	—	0.699	0	—	—	49	none	CAR slope Gemini 2.5 placebo 2021 h20 A_r2; day 0 2021-03-25; A_r2; nominal
summary	event studies	0.04	—	—	60.714	—	—	—	—	56	none	[default] mean of the 14 actual D19 slopes at h5; stat = its empirical percentile among the 56 placebo slopes (100 × share of placebo slopes ≤ the mean, out of 56; PAP §11 C-76); n = placebo slopes; descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
summary	event studies	-0.05	—	—	57.143	—	—	—	—	56	none	[default] mean of the 14 actual D19 slopes at h20; stat = its empirical percentile among the 56 placebo slopes (100 × share of placebo slopes ≤ the mean, out of 56; PAP §11 C-76); n = placebo slopes; descriptive

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.12 R10: robustness row (PAP §8.2 R10)

Source: PAP §8.2, row R10. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of annualised alpha per 1 SD; pp of the outcome share per unit of the regressor. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.12.1 Notes

- Market parts of R10: the D1 suppression bounds `z_delegation_lb` and `z_delegation_ub` (built from `automation_pct_lb` / `_ub`) in place of `z_delegation`, for H5 and H5b (PAP §8.2 R10, D1). The H1 and H3 parts of R10 are RQ1 rows (PAP §11 C-57).

- Exposure: the primary Design B vector (`ExposureSpec()`). H5 and H5b with s, $\bar{b}$ and q recomputed for this row by the pre-seal procedure (the same 378 windows unless the row says otherwise, the t_{21} rule and the D21 null $b_0 = \bar{b}$; PAP §3.3, §8.1); industries with a missing return in the confirmatory window are dropped and z is not re-standardised. The secondary b = 0 statistic is in each record's note.
- Zero-assigned employment share (D3): 12.14% (the primary Design B exposure; this row rebuilds no index).
- RQ1 parts of R10 (PAP §8.2 R10, D1): H1 with `automation_share_all_lb` / `_ub` in place of `automation_pct` (parts h1_lb, h1_ub); Plan 2's builder computes the bounds per period with r6's suppression component (`none_pct`) and then D13-averages them (PAP §11 C-4).
- [default] H3 with `validation_share_all_lb` = v·(100 − s)/100 and `_ub` = lb + s in place of `validation_pct` (parts h3_lb, h3_ub), v = `validation_pct` and s = r6's suppression share (`none_pct`), per period and then D13-averaged, as for the automation bounds (PAP §11 C-57).
- RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of the outcome share per unit of the regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78).

D.12.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market_lb	H5	1.98	2.27	[-2.74, 6.69]	0.143	0.888	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.872, p = 0.393
market_lb	H5b	3.46	2.54	[-1.83, 8.75]	0.883	0.387	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.361, p = 0.188
market_ub	H5	2.01	2.28	[-2.73, 6.75]	0.151	0.881	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.881, p = 0.388
market_ub	H5b	3.61	2.58	[-1.77, 8.98]	0.915	0.370	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.396, p = 0.177

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
h1_lb	H1	-0.06	0.48	[-0.99, 0.88]	-0.121	0.904	coefficient = 0	—	—	1684	unmatched_task 0.08%; no_family 1.00%; missing_automation_share_all_lb 0.03%; wage_suppressed 0.16%	education: SE two-way cluster (occupation, task); df 387; G_occ 388, G_task 1142
h1_lb	H1	4.61	1.99	[0.71, 8.52]	2.324	0.021	coefficient = 0	—	—	1684	unmatched_task 0.08%; no_family 1.00%; missing_automation_share_all_lb 0.03%; wage_suppressed 0.16%	log wage: SE two-way cluster (occupation, task); df 387; G_occ 388, G_task 1142
h1_lb	H1	—	—	—	—	0.904	β_e = 0 or β_w = 0	—	—	1684	unmatched_task 0.08%; no_family 1.00%; missing_automation_share_all_lb 0.03%; wage_suppressed 0.16%	p_H1: p_H1 = max(p_educ, p_wage), the intersection-union p (D17)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
h1_ub	H1	-0.86	0.66	[-2.15, 0.43]	-1.310	0.191	coefficient = 0	—	—	1684	unmatched_task 0.08%; no_family 1.00%; missing_automation_share_all_ub 0.03%; wage_suppressed 0.16%	education: SE two-way cluster (occupation, task); df 387; G_occ 388, G_task 1142
h1_ub	H1	3.98	2.26	[-0.48, 8.43]	1.756	0.080	coefficient = 0	—	—	1684	unmatched_task 0.08%; no_family 1.00%; missing_automation_share_all_ub 0.03%; wage_suppressed 0.16%	log wage: SE two-way cluster (occupation, task); df 387; G_occ 388, G_task 1142
h1_ub	H1	—	—	—	—	0.191	β_e = 0 or β_w = 0	—	—	1684	unmatched_task 0.08%; no_family 1.00%; missing_automation_share_all_ub 0.03%; wage_suppressed 0.16%	p_H1: p_H1 = max(p_educ, p_wage), the intersection-union p (D17)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
h3_lb	H3	-0.16	0.47	[-1.09, 0.77]	-0.342	0.733	coefficient = 0	—	—	1689	unmatched_task 0.08%; no_family 1.00%	error cost: SE two-way cluster (occupation, task); df 390; G_occ 391, G_task 1147
h3_ub	H3	-2.52	1.90	[-6.26, 1.21]	-1.328	0.185	coefficient = 0	—	—	1689	unmatched_task 0.08%; no_family 1.00%	error cost: SE two-way cluster (occupation, task); df 390; G_occ 391, G_task 1147

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.13 R11: robustness row (PAP §8.2 R11)

Source: PAP §8.2, row R11. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation; pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.13.1 Notes

- Variant: ExposureSpec(allocation='employment'); every other construction choice is the primary Design B one.
- Employment-weighted task allocation in place of the equal split (PAP D4 alternative, §8.2 R11): a task's usage goes to its candidate occupations in proportion to their national OEWS employment. It applies to the task-basis tables only; r6's published SOC rows stay as published.
- PAP D4: this makes U equal across occupations sharing a task by construction, so it removes rather than tests the within-group dispersion.
- Benchmarks: AIOE and Eloundou stay at their primary values, because this occupation-level variant changes neither the matrix nor the crosswalk (PAP §11 C-65).
- Zero-assigned employment share (D3): 12.14%
- Market part: H5 and H5b on ff49_vw with s, $\bar{b}$ and q recomputed for this exposure vector on the same 378 pre-seal windows, the t_{21} rule and the D21 null $b_0 = \bar{b}$ (PAP §3.3, §8.1); z is the variant's own and is not re-standardised after a drop.

- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.

D.13.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	1.94	2.27	[-2.78, 6.66]	0.139	0.891	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.856, p = 0.402
market	H5b	3.42	2.56	[-1.89, 8.74]	0.879	0.389	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.339, p = 0.195
rq2	RQ2	0.773	—	[0.606, 0.876]	—	—	—	—	—	49	none	z_usage vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)
rq2	RQ2	0.721	—	[0.533, 0.845]	—	—	—	—	—	49	none	z_delegation vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.773	—	[0.611, 0.875]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.843	—	[0.721, 0.914]	—	—	—	—	—	49	none	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.815	—	[0.674, 0.902]	—	—	—	—	—	49	none	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.844	—	[0.727, 0.912]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.810	—	[0.660, 0.898]	—	—	—	—	—	49	none	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.762	—	[0.591, 0.869]	—	—	—	—	—	49	none	z_ delegation vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.817	—	[0.680, 0.897]	—	—	—	—	—	49	none	z_ augmentation vs z_aioe; 10000 bootstrap draws used

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.14 R14: robustness row (PAP §8.2 R14)

Source: PAP §8.2, row R14. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation; pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.14.1 Notes

- Variant: ExposureSpec(aioe_crosswalk='bls'); every other construction choice is the primary Design B one.
- AIOE through the BLS SOC 2010→2018 crosswalk in place of the O*NET-detail union (PAP D6 alternative, §8.2 R14). The row applies to H5b and RQ2: H5 does not use AIOE, so the market part reports H5b only.
- AIOE's 19-1020 (not a SOC 2010 detailed code, absent from the BLS crosswalk) is mapped by its O*NET-SOC 2010→2019 successor 19-1029; every other AIOE code uses the BLS crosswalk (PAP §11 C-73).
- Benchmarks: AIOE itself is replaced; Eloundou and the AEI indices keep the primary matrix and crosswalk (PAP §11 C-65).
- Zero-assigned employment share (D3): 12.14%

- Market part: H5 and H5b on ff49_vw with s, $\bar{b}$ and q recomputed for this exposure vector on the same 378 pre-seal windows, the t_{21} rule and the D21 null $b_0 = \bar{b}$ (PAP §3.3, §8.1); z is the variant's own and is not re-standardised after a drop.
- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.

D.14.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5b	3.46	2.55	[-1.85, 8.77]	0.882	0.388	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.356, p = 0.190
rq2	RQ2	0.768	—	[0.600, 0.873]	—	—	—	—	—	49	none	z_usage vs z_ eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.716	—	[0.527, 0.839]	—	—	—	—	—	49	none	z_ delegation vs z_ eloundou; 10000 bootstrap draws used; headline bench-mark (D8)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.783	—	[0.624, 0.883]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.843	—	[0.721, 0.913]	—	—	—	—	—	49	none	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.815	—	[0.676, 0.900]	—	—	—	—	—	49	none	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.850	—	[0.736, 0.915]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.811	—	[0.663, 0.898]	—	—	—	—	—	49	none	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.766	—	[0.597, 0.867]	—	—	—	—	—	49	none	z_ delegation vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.826	—	[0.694, 0.903]	—	—	—	—	—	49	none	z_ augmentation vs z_aioe; 10000 bootstrap draws used

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.15 R15: robustness row (PAP §8.2 R15)

Source: PAP §8.2, row R15. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation; pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.15.1 Notes

- Variant: ExposureSpec(hq='redistribute'); every other construction choice is the primary Design B one.
- NAICS 5511 redistributed (`hq="redistribute"`, PAP D7 alternative, §8.2 R15).
- Identical to the primary by construction (PAP §11 C-70): 5511's occupation mix is added to every FF49 industry with the same fraction of its mapped employment, an affine map of each industry index, so the standardised indices, every Spearman rank and every H5, H5b and RQ2 estimate equal the primary's; only coverage and index levels change. The row is still run and reported (PAP §8.1).
- Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65).
- Zero-assigned employment share (D3): 12.14%

- Market part: H5 and H5b on ff49_vw with s, $\bar{b}$ and q recomputed for this exposure vector on the same 378 pre-seal windows, the t_{21} rule and the D21 null $b_0 = \bar{b}$ (PAP §3.3, §8.1); z is the variant's own and is not re-standardised after a drop.
- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.

D.15.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	1.96	2.27	[-2.76, 6.68]	0.136	0.893	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.862, p = 0.398
market	H5b	3.46	2.56	[-1.86, 8.79]	0.877	0.391	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.352, p = 0.191
rq2	RQ2	0.768	—	[0.600, 0.873]	—	—	—	—	—	49	none	z_usage vs z_ eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.716	—	[0.527, 0.839]	—	—	—	—	—	49	none	z_ delegation vs z_ eloundou; 10000 bootstrap draws used; headline bench-mark (D8)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.783	—	[0.624, 0.883]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.843	—	[0.721, 0.913]	—	—	—	—	—	49	none	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.815	—	[0.676, 0.900]	—	—	—	—	—	49	none	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.850	—	[0.736, 0.915]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.809	—	[0.661, 0.895]	—	—	—	—	—	49	none	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.763	—	[0.594, 0.866]	—	—	—	—	—	49	none	z_ delegation vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.825	—	[0.692, 0.901]	—	—	—	—	—	49	none	z_ augmentation vs z_aioe; 10000 bootstrap draws used

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.16 R16: robustness row (PAP §8.2 R16)

Source: PAP §8.2, row R16. Label: robustness; exploratory.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: annualised Sharpe ratio; annualised pp; correlation; percent (usage-weighted mean pattern share); pp of CAR per 1 SD; pp per month per 1 SD; pp per month per unit of log BE/ME; pp per month per unit of log size; pp per month per unit of the 12-2 return (decimal); probability. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.16.1 Notes

- Variant: ExposureSpec(design='A_pooled', r6_basis='tasks'); every other construction choice is the primary A_pooled one. A_pooled is labelled look-ahead (PAP §8.3 D19).
- r6 occupation tables on the task-allocated basis (`basis="tasks"`) in place of AEI's published SOC rows, in the A_pooled pool r2–r6 (PAP D9 alternative, §8.2 R16).
- Benchmarks: AIOE and Eloundou stay at their primary values, because this occupation-level variant changes neither the matrix nor the crosswalk (PAP §11 C-65).
- Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

- r6 descriptives: Plan 2's `by_wage_decile` (r6, `claude_ai`) on `basis="tasks"` beside `basis="published"` with the same inputs; one record per (decile, pattern) cell, the estimate the task-basis value and the published value in the note.
- Zero-assigned employment share (D3): 6.77%
- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.
- Index part: the Spearman correlation of the variant A_pooled `z_delegation` with the primary A_pooled `z_delegation` (Plan 3's `data.exposures()`), across the FF49 industries with both values; descriptive.
- Design A parts run Plan 3's own runners on the exposure frame passed: `run_terciles` (fixed-membership H-L terciles, cap and EW legs; C-34, C-35, C-37, C-45), `run_fm` (Fama–MacBeth on [1, z, log size, log BE/ME, 12-2 momentum]; C-38, C-46) and `run_events` (the 14 D19 events, CAR slopes at horizons 5 and 20, HC3 normal reference; C-33, C-44). Records are read from the frames the runners return; their docs are written under the untracked `data/derived/robustness/<row>/<part>` and no figure is written.
- Columns: estimate, interval and p are as Plan 3 returns them; the scale column holds the SE (Newey-West for terciles and Fama–MacBeth, HC3 for event slopes); n is T (months) for terciles and Fama–MacBeth and the number of industries for event slopes. Null 0, nominal and single-test; no error control across records or rows (PAP §8.1). Sharpe, PSR and DSR carry no p; PSR and DSR carry no interval.
- DSR here is against the 2 tercile variants run in this row (the primary used 6, C-37); not comparable with the primary's DSR (PAP §11 C-76).

D.16.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.847	—	[0.728, 0.917]	—	—	—	—	—	49	none	z_usage vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)
rq2	RQ2	0.773	—	[0.603, 0.876]	—	—	—	—	—	49	none	z_delegation vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.820	—	[0.688, 0.899]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.886	—	[0.789, 0.937]	—	—	—	—	—	49	none	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.856	—	[0.739, 0.923]	—	—	—	—	—	49	none	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.881	—	[0.788, 0.933]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.868	—	[0.760, 0.930]	—	—	—	—	—	49	none	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.815	—	[0.671, 0.900]	—	—	—	—	—	49	none	z_ delegation vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.861	—	[0.755, 0.921]	—	—	—	—	—	49	none	z_ augmentation vs z_aioe; 10000 bootstrap draws used
index	A_pooled	1.000	—	—	—	—	—	—	—	49	none	variant vs primary A_pooled z_ delegation (Spearman, FF49 industries with both values); descriptive, no interval

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	4.60	3.64	[-2.75, 11.95]	—	0.214	0	—	—	45	none	alpha_CAPM A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	5.07	3.10	[-1.19, 11.34]	—	0.110	0	—	—	45	none	alpha_FF3 A_pooled cap; A_ pooled (look-ahead); nominal; Bootstrap draws: `run_ terciles` consumes one `default_ rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_ pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	3.66	3.01	[-2.42, 9.75]	—	0.230	0	—	—	45	none	alpha_FF5+UMD A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	7.92	4.66	[-0.52, 16.51]	—	0.096	0	—	—	45	none	mean A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.82	—	[-0.06, 1.67]	—	—	0	—	—	45	none	sharpe A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.95	—	—	—	—	0	—	—	45	none	psr A_pooled cap; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.90	—	—	—	—	0	—	—	45	none	dsr A_ pooled cap; A_ pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_ pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.52	4.10	[-7.74, 8.78]	—	0.899	0	—	—	45	none	alpha_CAPM A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.86	3.71	[-6.63, 8.34]	—	0.818	0	—	—	45	none	alpha_FF3 A_pooled ew; A_ pooled (look-ahead); nominal; Bootstrap draws: `run_ terciles` consumes one `default_ rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_ pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	-0.41	3.88	[-8.26, 7.45]	—	0.917	0	—	—	45	none	alpha_FF5+UMD A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	2.66	3.99	[-5.36, 10.25]	—	0.508	0	—	—	45	none	mean A_ pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_ pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.32	—	[-0.60, 1.37]	—	—	0	—	—	45	none	sharpe A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.72	—	—	—	—	0	—	—	45	none	psr A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
terciles	terciles	0.60	—	—	—	—	0	—	—	45	none	dsr A_pooled ew; A_pooled (look-ahead); nominal; Bootstrap draws: `run_terciles` consumes one `default_rng(SEED)` in `VARIANTS` order (PAP §11 C-37); with only A_pooled passed its variants draw first, so the mean and Sharpe bootstrap CIs use different draws from the primary run's A_pooled rows. A CI difference against the primary is partly draw noise, not only the exposure change.

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
fm	Fama-MacBeth	0.16	0.11	[-0.07, 0.39]	—	0.158	0	—	—	45	none	FM z A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal
fm	Fama-MacBeth	-0.07	0.15	[-0.37, 0.24]	—	0.665	0	—	—	45	none	FM log_size A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal
fm	Fama-MacBeth	0.00	0.19	[-0.39, 0.38]	—	0.988	0	—	—	45	none	FM log_beme A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
fm	Fama-MacBeth	0.25	0.86	[-1.48, 1.97]	—	0.773	0	—	—	45	none	FM mom A_pooled; A_pooled (look-ahead); window A; mean n 48.0; 0 months skipped; nominal
events	event studies	-0.39	0.24	[-0.85, 0.07]	—	0.100	0	—	—	49	none	CAR slope ChatGPT h5 A_pooled; day 0 2022-11-30; A_pooled (look-ahead); nominal
events	event studies	0.27	0.55	[-0.80, 1.33]	—	0.626	0	—	—	49	none	CAR slope ChatGPT h20 A_pooled; day 0 2022-11-30; A_pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-0.13	0.40	[-0.92, 0.65]	—	0.736	0	—	—	49	none	CAR slope ChatGPT h9 none [z_aioe in z_aioe]; day 0 2022-11-30; none; nominal
events	event studies	0.02	0.59	[-1.13, 1.18]	—	0.969	0	—	—	49	none	CAR slope ChatGPT h9 A_pooled [z_aioe in z_aioe+z_delegation]; day 0 2022-11-30; A_pooled (look-ahead); nominal
events	event studies	-0.22	0.44	[-1.09, 0.64]	—	0.610	0	—	—	49	none	CAR slope ChatGPT h9 A_pooled [z_delegation in z_aioe+z_delegation]; day 0 2022-11-30; A_pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	0.66	0.18	[0.31, 1.01]	—	< 0.001	0	—	—	49	none	CAR slope GPT-4 h5 A_pooled; day 0 2023-03-14; A_pooled (look-ahead); nominal
events	event studies	0.41	0.42	[-0.41, 1.22]	—	0.328	0	—	—	49	none	CAR slope GPT-4 h20 A_pooled; day 0 2023-03-14; A_pooled (look-ahead); nominal
events	event studies	1.23	0.23	[0.77, 1.68]	—	< 0.001	0	—	—	49	none	CAR slope PaLM 2 h5 A_pooled; day 0 2023-05-10; A_pooled (look-ahead); nominal
events	event studies	2.18	0.56	[1.08, 3.28]	—	< 0.001	0	—	—	49	none	CAR slope PaLM 2 h20 A_pooled; day 0 2023-05-10; A_pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	0.45	0.43	[-0.39, 1.28]	—	0.292	0	—	—	49	none	CAR slope Claude 2 h5 A_ pooled; day 0 2023-07-11; A_ pooled (look-ahead); nominal
events	event studies	0.39	0.33	[-0.25, 1.03]	—	0.229	0	—	—	49	none	CAR slope Claude 2 h20 A_ pooled; day 0 2023-07-11; A_ pooled (look-ahead); nominal
events	event studies	-0.04	0.22	[-0.46, 0.38]	—	0.853	0	—	—	49	none	CAR slope Claude 2.1 h5 A_ pooled; day 0 2023-11-21; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	0.12	0.48	[-0.83, 1.07]	—	0.810	0	—	—	49	none	CAR slope Claude 2.1 h20 A_ pooled; day 0 2023-11-21; A_ pooled (look-ahead); nominal
events	event studies	0.15	0.32	[-0.48, 0.78]	—	0.644	0	—	—	49	none	CAR slope Gemini 1.0 h5 A_ pooled; day 0 2023-12-06; A_ pooled (look-ahead); nominal
events	event studies	-0.04	0.31	[-0.66, 0.57]	—	0.886	0	—	—	49	none	CAR slope Gemini 1.0 h20 A_ pooled; day 0 2023-12-06; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-0.68	0.20	[-1.06, -0.29]	—	< 0.001	0	—	—	49	none	CAR slope Gemini 1.5 h5 A_ pooled; day 0 2024-02-15; A_ pooled (look-ahead); nominal
events	event studies	-1.10	0.46	[-2.00, -0.21]	—	0.016	0	—	—	49	none	CAR slope Gemini 1.5 h20 A_ pooled; day 0 2024-02-15; A_ pooled (look-ahead); nominal
events	event studies	-0.25	0.24	[-0.71, 0.22]	—	0.300	0	—	—	49	none	CAR slope Claude 3 h5 A_ pooled; day 0 2024-03-04; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-0.72	0.63	[-1.95, 0.51]	—	0.253	0	—	—	49	none	CAR slope Claude 3 h20 A_ pooled; day 0 2024-03-04; A_ pooled (look-ahead); nominal
events	event studies	0.41	0.22	[-0.02, 0.84]	—	0.062	0	—	—	49	none	CAR slope Claude 3.5 h5 A_ pooled; day 0 2024-06-20; A_ pooled (look-ahead); nominal
events	event studies	-0.22	0.46	[-1.13, 0.69]	—	0.632	0	—	—	49	none	CAR slope Claude 3.5 h20 A_ pooled; day 0 2024-06-20; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-0.06	0.31	[-0.66, 0.54]	—	0.851	0	—	—	49	none	CAR slope OpenAI o1 h5 A_ pooled; day 0 2024-09-12; A_ pooled (look-ahead); nominal
events	event studies	-0.14	0.50	[-1.12, 0.83]	—	0.772	0	—	—	49	none	CAR slope OpenAI o1 h20 A_ pooled; day 0 2024-09-12; A_ pooled (look-ahead); nominal
events	event studies	-0.20	0.32	[-0.83, 0.44]	—	0.545	0	—	—	49	none	CAR slope Gemini 2.0 h5 A_ pooled; day 0 2025-02-05; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-0.22	0.56	[-1.33, 0.88]	—	0.693	0	—	—	49	none	CAR slope Gemini 2.0 h20 A_ pooled; day 0 2025-02-05; A_ pooled (look-ahead); nominal
events	event studies	-0.21	0.36	[-0.92, 0.50]	—	0.558	0	—	—	49	none	CAR slope Claude 3.7 h5 A_ pooled; day 0 2025-02-24; A_ pooled (look-ahead); nominal
events	event studies	-0.41	0.52	[-1.41, 0.60]	—	0.431	0	—	—	49	none	CAR slope Claude 3.7 h20 A_ pooled; day 0 2025-02-24; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	0.13	0.32	[-0.49, 0.75]	—	0.688	0	—	—	49	none	CAR slope GPT-4.5 h5 A_pooled; day 0 2025-02-27; A_ pooled (look-ahead); nominal
events	event studies	-0.64	0.57	[-1.75, 0.48]	—	0.263	0	—	—	49	none	CAR slope GPT-4.5 h20 A_ pooled; day 0 2025-02-27; A_ pooled (look-ahead); nominal
events	event studies	-0.46	0.25	[-0.95, 0.03]	—	0.068	0	—	—	49	none	CAR slope Gemini 2.5 h5 A_ pooled; day 0 2025-03-25; A_ pooled (look-ahead); nominal

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
events	event studies	-1.12	0.72	[-2.53, 0.29]	—	0.120	0	—	—	49	none	CAR slope Gemini 2.5 h20 A_ pooled; day 0 2025-03-25; A_ pooled (look-ahead); nominal
r6_ descriptives	r6 descrip-tives	28.90	—	—	—	—	—	—	—	28	none	published: 28.07; decile 1 directive_ pct; task basis (n = OEWS oc-cupations in the decile); descriptive
r6_ descriptives	r6 descrip-tives	7.95	—	—	—	—	—	—	—	28	none	published: 8.58; decile 1 feedback_ loop_pct; task basis (n = OEWS oc-cupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descrip- tives	17.29	—	—	—	—	—	—	—	28	none	published: 17.26; decile 1 task_ iteration_ pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	2.48	—	—	—	—	—	—	—	28	none	published: 2.48; decile 1 valida- tion_pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	43.37	—	—	—	—	—	—	—	28	none	published: 43.61; decile 1 learning_ pct; task basis (n = OEWS oc- cupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_descriptives	r6 descriptives	35.97	—	—	—	—	—	—	—	28	none	published: 36.65; decile 1 automation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	64.03	—	—	—	—	—	—	—	28	none	published: 63.35; decile 1 augmentation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	40.56	—	—	—	—	—	—	—	29	none	published: 40.45; decile 2 directive_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descrip- tives	14.28	—	—	—	—	—	—	—	29	none	published: 14.03; decile 2 feedback_ loop_pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	22.54	—	—	—	—	—	—	—	29	none	published: 23.09; decile 2 task_ iteration_ pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	3.30	—	—	—	—	—	—	—	29	none	published: 3.24; decile 2 valida- tion_pct; task basis (n = OEWS oc- cupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	19.32	—	—	—	—	—	—	—	29	none	published: 19.19; decile 2 learning_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	54.83	—	—	—	—	—	—	—	29	none	published: 54.48; decile 2 automation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	45.17	—	—	—	—	—	—	—	29	none	published: 45.52; decile 2 augmentation_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_descriptives	r6 descriptives	26.48	—	—	—	—	—	—	—	56	none	published: 26.63; decile 3 directive_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	12.65	—	—	—	—	—	—	—	56	none	published: 12.76; decile 3 feedback_loop_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	27.20	—	—	—	—	—	—	—	56	none	published: 27.43; decile 3 task_iteration_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	3.19	—	—	—	—	—	—	—	56	none	published: 3.17; decile 3 validation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	30.48	—	—	—	—	—	—	—	56	none	published: 30.02; decile 3 learning_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	39.13	—	—	—	—	—	—	—	56	none	published: 39.38; decile 3 automation_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_descriptives	r6 descriptives	60.87	—	—	—	—	—	—	—	56	none	published: 60.62; decile 3 augmentation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	46.59	—	—	—	—	—	—	—	63	none	published: 46.05; decile 4 directive_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	6.93	—	—	—	—	—	—	—	63	none	published: 6.87; decile 4 feedback_loop_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	27.02	—	—	—	—	—	—	—	63	none	published: 26.06; decile 4 task_ iteration_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	2.39	—	—	—	—	—	—	—	63	none	published: 2.50; decile 4 validation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	17.08	—	—	—	—	—	—	—	63	none	published: 18.52; decile 4 learning_ pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	53.52	—	—	—	—	—	—	—	63	none	published: 52.92; decile 4 automation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	46.48	—	—	—	—	—	—	—	63	none	published: 47.08; decile 4 augmentation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	41.22	—	—	—	—	—	—	—	96	none	published: 40.62; decile 5 directive_ pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	9.32	—	—	—	—	—	—	—	96	none	published: 9.27; decile 5 feedback_ loop_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	34.01	—	—	—	—	—	—	—	96	none	published: 32.66; decile 5 task_ iteration_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	3.69	—	—	—	—	—	—	—	96	none	published: 3.57; decile 5 validation_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	11.77	—	—	—	—	—	—	—	96	none	published: 13.88; decile 5 learning_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	50.54	—	—	—	—	—	—	—	96	none	published: 49.89; decile 5 automation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	49.46	—	—	—	—	—	—	—	96	none	published: 50.11; decile 5 augmentation_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_descriptives	r6 descriptives	34.45	—	—	—	—	—	—	—	145	none	published: 34.37; decile 6 directive_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	13.92	—	—	—	—	—	—	—	145	none	published: 14.03; decile 6 feedback_loop_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	26.45	—	—	—	—	—	—	—	145	none	published: 26.42; decile 6 task_iteration_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	5.52	—	—	—	—	—	—	—	145	none	published: 5.50; decile 6 validation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	19.65	—	—	—	—	—	—	—	145	none	published: 19.66; decile 6 learning_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	48.38	—	—	—	—	—	—	—	145	none	published: 48.41; decile 6 automation_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	51.62	—	—	—	—	—	—	—	145	none	published: 51.59; decile 6 augmentation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	31.31	—	—	—	—	—	—	—	92	none	published: 31.23; decile 7 directive_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	14.61	—	—	—	—	—	—	—	92	none	published: 14.69; decile 7 feedback_ loop_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descrip- tives	30.26	—	—	—	—	—	—	—	92	none	published: 30.47; decile 7 task_ iteration_ pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	2.86	—	—	—	—	—	—	—	92	none	published: 3.04; decile 7 valida- tion_pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	20.96	—	—	—	—	—	—	—	92	none	published: 20.57; decile 7 learning_ pct; task basis (n = OEWS oc- cupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_descriptives	r6 descriptives	45.91	—	—	—	—	—	—	—	92	none	published: 45.92; decile 7 automation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	54.09	—	—	—	—	—	—	—	92	none	published: 54.08; decile 7 augmentation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	30.61	—	—	—	—	—	—	—	118	none	published: 30.08; decile 8 directive_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descrip- tives	10.70	—	—	—	—	—	—	—	118	none	published: 10.56; decile 8 feedback_ loop_pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	47.14	—	—	—	—	—	—	—	118	none	published: 46.42; decile 8 task_ iteration_ pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	3.17	—	—	—	—	—	—	—	118	none	published: 3.36; decile 8 valida- tion_pct; task basis (n = OEWS oc- cupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	8.38	—	—	—	—	—	—	—	118	none	published: 9.59; decile 8 learning_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	41.31	—	—	—	—	—	—	—	118	none	published: 40.64; decile 8 automation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	58.69	—	—	—	—	—	—	—	118	none	published: 59.36; decile 8 augmentation_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	26.74	—	—	—	—	—	—	—	93	none	published: 27.11; decile 9 directive_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	23.68	—	—	—	—	—	—	—	93	none	published: 22.81; decile 9 feedback_ loop_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	34.83	—	—	—	—	—	—	—	93	none	published: 33.73; decile 9 task_ iteration_ pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	3.61	—	—	—	—	—	—	—	93	none	published: 3.40; decile 9 validation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	11.14	—	—	—	—	—	—	—	93	none	published: 12.94; decile 9 learning_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	50.42	—	—	—	—	—	—	—	93	none	published: 49.92; decile 9 automation_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	49.58	—	—	—	—	—	—	—	93	none	published: 50.08; decile 9 augmentation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	34.17	—	—	—	—	—	—	—	106	none	published: 33.51; decile 10 directive_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	22.71	—	—	—	—	—	—	—	106	none	published: 22.77; decile 10 feedback_ loop_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_descriptives	r6 descriptives	24.58	—	—	—	—	—	—	—	106	none	published: 24.39; decile 10 task_iteration_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	2.32	—	—	—	—	—	—	—	106	none	published: 2.37; decile 10 validation_pct; task basis (n = OEWS occupations in the decile); descriptive
r6_descriptives	r6 descriptives	16.21	—	—	—	—	—	—	—	106	none	published: 16.95; decile 10 learning_pct; task basis (n = OEWS occupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descrip- tives	56.89	—	—	—	—	—	—	—	106	none	published: 56.29; decile 10 automa- tion_pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	43.11	—	—	—	—	—	—	—	106	none	published: 43.71; decile 10 augmenta- tion_pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	26.87	—	—	—	—	—	—	—	5	none	published: 27.78; decile no wage directive_ pct; task basis (n = OEWS oc- cupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descrip- tives	21.95	—	—	—	—	—	—	—	5	none	published: 23.52; decile no wage feedback_ loop_pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	50.11	—	—	—	—	—	—	—	5	none	published: 47.29; decile no wage task_ iteration_ pct; task basis (n = OEWS oc- cupations in the decile); descriptive
r6_ descriptives	r6 descrip- tives	0.36	—	—	—	—	—	—	—	5	none	published: 0.40; decile no wage valida- tion_pct; task basis (n = OEWS oc- cupations in the decile); descriptive

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_ descriptives	r6 descriptives	0.71	—	—	—	—	—	—	—	5	none	published: 1.02; decile no wage learning_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	48.82	—	—	—	—	—	—	—	5	none	published: 51.30; decile no wage automation_ pct; task basis (n = OEWS occupations in the decile); descriptive
r6_ descriptives	r6 descriptives	51.18	—	—	—	—	—	—	—	5	none	published: 48.70; decile no wage augmentation_pct; task basis (n = OEWS occupations in the decile); descriptive

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.17 R17: robustness row (PAP §8.2 R17)

Source: PAP §8.2, row R17. Label: robustness; descriptive.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.17.1 Notes

- Variant: ExposureSpec(); every other construction choice is the primary Design B one.
- Primary Design B exposure and benchmarks; the RQ2 table is ordered with the Eloundou GPT-4 β (`z_eloundou_gpt4`) as headline (PAP D8 alternative, §8.2 R17). Both betas are always shown, so this is a labelling choice (PAP D8).
- Zero-assigned employment share (D3): 12.14%
- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.

D.17.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.843	—	[0.721, 0.913]	—	—	—	—	—	49	none	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used; headline bench-mark (R17)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.815	—	[0.676, 0.900]	—	—	—	—	—	49	none	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used; headline benchmark (R17)
rq2	RQ2	0.850	—	[0.736, 0.915]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used; headline benchmark (R17)
rq2	RQ2	0.768	—	[0.600, 0.873]	—	—	—	—	—	49	none	z_usage vs z_eloundou; 10000 bootstrap draws used
rq2	RQ2	0.716	—	[0.527, 0.839]	—	—	—	—	—	49	none	z_delegation vs z_eloundou; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.783	—	[0.624, 0.883]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou; 10000 bootstrap draws used
rq2	RQ2	0.809	—	[0.661, 0.895]	—	—	—	—	—	49	none	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.763	—	[0.594, 0.866]	—	—	—	—	—	49	none	z_delegation vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.825	—	[0.692, 0.901]	—	—	—	—	—	49	none	z_augmentation vs z_aioe; 10000 bootstrap draws used

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.18 R18: robustness row (PAP §8.2 R18)

Source: PAP §8.2, row R18. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.18.1 Notes

- s = the largest sd of the 17 non-overlapping start offsets; $\bar{b}$ stays the 378-window mean; q is recomputed with the new s (PAP §3.3, §8.2 R18; §11 C-64).
- Exposure: the primary Design B vector (`ExposureSpec()`). H5 and H5b with s, $\bar{b}$ and q recomputed for this row by the pre-seal procedure (the same 378 windows unless the row says otherwise, the t_{21} rule and the D21 null $b_0 = \bar{b}$; PAP §3.3, §8.1); industries with a missing return in the confirmatory window are dropped and z is not re-standardised. The secondary b = 0 statistic is in each record's note.
- Zero-assigned employment share (D3): 12.14% (the primary Design B exposure; this row rebuilds no index).

D.18.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	1.96	2.82	[-3.91, 7.83]	0.109	0.914	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.694, p = 0.495
market	H5b	3.46	3.22	[-3.25, 10.17]	0.696	0.494	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.073, p = 0.295

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.19 R19: robustness row (PAP §8.2 R19)

Source: PAP §8.2, row R19. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.19.1 Notes

- s = the root mean square of the 378 historical slopes around the null value $\bar{b}$, sqrt(mean($(b - \bar{b})^2$)) (PAP §3.3, §8.2 R19).
- q is recomputed with the RMS s: q = the 0.975 / 0.95 quantiles of $|b - \bar{b}|/s$ with this s (PAP §8.1: s, $\bar{b}$ and q recomputed; Plan 4 Task 4 Step 3 rule), and the critical is max(t_{21}, q); q moves a critical only where it exceeds t_{21} (2.4138 / 2.0796), so the c_0.025 and c_0.05 columns show the criticals used.
- Exposure: the primary Design B vector (`ExposureSpec()`). H5 and H5b with s, $\bar{b}$ and q recomputed for this row by the pre-seal procedure (the same 378 windows unless the row says otherwise, the t_{21} rule and the D21 null $b_0 = \bar{b}$; PAP §3.3, §8.1); industries with a missing return in the confirmatory window are dropped and z is not re-standardised. The secondary b = 0 statistic is in each record's note.
- Zero-assigned employment share (D3): 12.14% (the primary Design B exposure; this row rebuilds no index).

D.19.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	1.96	2.27	[-2.76, 6.67]	0.136	0.893	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.864, p = 0.398
market	H5b	3.46	2.56	[-1.86, 8.78]	0.878	0.390	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.354, p = 0.190

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.20 R20: robustness row (PAP §8.2 R20)

Source: PAP §8.2, row R20. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation; pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.20.1 Notes

- Variant: ExposureSpec(crosswalk='majority'); every other construction choice is the primary Design B one.
- NAICS→FF49 crosswalk in `majority` mode instead of `split` (PAP §8.2 R20).
- The majority rule assigns no NAICS code to Guns (26), Gold (27) or Boxes (40): the row reports on the 46 industries it assigns, with z standardised over those 46; the market part removes the three from the window and from the 378-window history before the null scales are computed (PAP §11 C-71).
- Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65).
- Zero-assigned employment share (D3): 12.14%
- Market part: H5 and H5b on ff49_vw with s, $\bar{b}$ and q recomputed for this exposure vector on the same 378 pre-seal windows, the t_{21} rule and the D21 null $b_0 = \bar{b}$ (PAP §3.3, §8.1); z is the variant's own and is not re-standardised after a drop.
- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.

D.20.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	2.43	2.46	[-2.69, 7.55]	0.260	0.797	b = bbar	2.4138	2.0796	46	Guns, Gold, Boxes	secondary b = 0: T = 0.985, p = 0.336
market	H5b	3.74	2.90	[-2.29, 9.78]	0.776	0.446	b = bbar	2.4138	2.0796	46	Guns, Gold, Boxes	secondary b = 0: T = 1.289, p = 0.211
rq2	RQ2	0.751	—	[0.573, 0.860]	—	—	—	—	—	46	Guns, Gold, Boxes	z_usage vs z_ eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.714	—	[0.524, 0.835]	—	—	—	—	—	46	Guns, Gold, Boxes	z_ delegation vs z_ eloundou; 10000 bootstrap draws used; headline bench-mark (D8)
rq2	RQ2	0.761	—	[0.589, 0.865]	—	—	—	—	—	46	Guns, Gold, Boxes	z_ augmentation vs z_ eloundou; 10000 bootstrap draws used; headline bench-mark (D8)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.836	—	[0.703, 0.907]	—	—	—	—	—	46	Guns, Gold, Boxes	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.807	—	[0.657, 0.892]	—	—	—	—	—	46	Guns, Gold, Boxes	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.842	—	[0.719, 0.906]	—	—	—	—	—	46	Guns, Gold, Boxes	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.797	—	[0.648, 0.881]	—	—	—	—	—	46	Guns, Gold, Boxes	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.760	—	[0.596, 0.859]	—	—	—	—	—	46	Guns, Gold, Boxes	z_delegation vs z_aioe; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.811	—	[0.673, 0.884]	—	—	—	—	—	46	Guns, Gold, Boxes	z_augmentation vs z_aioe; 10000 bootstrap draws used

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.21 R21: robustness row (PAP §8.2 R21)

Source: PAP §8.2, row R21. Label: robustness; no error control.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.21.1 Notes

- `z_usage` and `z_augmentation` in place of `z_delegation` (PAP §8.2 R21), each with its own s, $\bar{b}$ and q.
- No error control: R21 carries no error control across its parts or across rows (PAP §8.1).
- Exposure: the primary Design B vector (`ExposureSpec()`). H5 and H5b with s, $\bar{b}$ and q recomputed for this row by the pre-seal procedure (the same 378 windows unless the row says otherwise, the t_{21} rule and the D21 null $b_0 = \bar{b}$; PAP §3.3, §8.1); industries with a missing return in the confirmatory window are dropped and z is not re-standardised. The secondary b = 0 statistic is in each record's note.
- Zero-assigned employment share (D3): 12.14% (the primary Design B exposure; this row rebuilds no index).

D.21.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
usage	H5	2.12	2.26	[-2.59, 6.82]	0.228	0.822	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.936, p = 0.360
usage	H5b	4.16	2.60	[-1.25, 9.57]	1.161	0.259	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.598, p = 0.125

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
augmentation	H5	2.44	2.32	[-2.39, 7.26]	0.330	0.745	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.050, p = 0.305
augmentation	H5b	4.87	2.67	[-0.68, 10.43]	1.345	0.193	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.825, p = 0.082

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.22 R22: robustness row (PAP §8.2 R22)

Source: PAP §8.2, row R22. Label: robustness; part all_r2_r6 is also labelled confounded with the r6 suppression-regime change.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of the outcome share per year. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.22.1 Notes

- D2 alternatives (PAP §5, §8.2 R22): all releases r2–r6, balanced over the five, without an r6 indicator (part all_r2_r6, labelled confounded with the r6 suppression-regime change, D2) and with one (all_r2_r6_r6dummy; under a linear trend the indicator absorbs r6, D2; its coefficient is in the note); the unbalanced r2–r5 panel (unbalanced); each other PAP §5 definition-stable metric as the outcome over the balanced r2–r5 panel (parts named by metric); `1p_api` over r3–r5, balanced (api_r3_r5). t_r is from the `claude_ai` window midpoints in every part (Plan 2's `release_times`).
- [default] The unbalanced panel keeps every (task, release) row with a non-null outcome for tasks with such rows in at least two of the listed releases (Plan 2's `build_h2_panel(balanced=False)`); there is no other unbalanced rule (PAP §11 C-56).
- H2 parts: Plan 2's `estimate_h2` on Plan 2's `build_h2_panel` (through the adapter), with H2's inference (PAP §4.2, D22, §8.1): WLS of the outcome on t_r and task dummies with fixed task weights (the task's mean `pct` over the listed releases), SEs clustered by task, t with G_task − 1 df; p is the two-sided task-clustered p of the trend. The note gives the restricted wild cluster bootstrap p (clusters = releases, Webb weights, every one of the 6^k weight vectors enumerated; PAP D22, §11 C-15) and whether the D22 rule's conditions would hold (both $p \leq 0.0125$, compared exactly as PAP §11 C-24, and $\beta > 0$): a mechanical statement, not a decision; the release-label randomisation p is descriptive. The smallest attainable wild-bootstrap p is 6/6^k: 3 releases, 6/216 = 0.0278, above 0.0125, so the wild condition cannot hold; 4 releases, 6/1296 = 0.0046; 5 releases, 6/7776 = 0.0008. β is the change in measured within-task outcome per year, including any change in the classifier and platform scope (PAP §5). Coefficients are in pp of the outcome share per year. Each table row's note cell starts with its estimand (PAP §11 C-78). `n` is the number of (task, release) rows; `dropped` shows none and `n_dropped` is not reported for H2 rows, because Plan 2's H2 builder applies the sample rule and records no drops (PAP §11 C-78).

D.22.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
all_r2_r6	H2	7.31	2.03	[3.31, 11.31]	3.593	< 0.001	β = 0	—	—	1920	none	trend β: wild-bootstrap p = 0.557; weight vectors = 6^5 = 7776; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 4328 of the 7776 weight vectors at least as extreme as the observed t; SE clustered by task, df 383, G_ task 384; release-label randomi-sation p = 0.517 over 120 orderings (descrip-tive)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
all_r2_r6_r6dummy	H2	5.47	1.77	[1.98, 8.95]	3.083	0.002	β = 0	—	—	1920	none	trend β: wild-bootstrap p = 0.713; weight vectors = 6^5 = 7776; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 5544 of the 7776 weight vectors at least as extreme as the observed t; SE clustered by task, df 383, G_task 384; release-label randomisation p = 0.667 over 120 orderings (descriptive); r6 indicator 2.97 (SE 1.77, p 0.094)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
unbalanced	H2	5.00	1.16	[2.72, 7.27]	4.304	< 0.001	β = 0	—	—	7545	none	trend β: wild-bootstrap p = 0.926; weight vectors = 6^4 = 1296; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 1200 of the 1296 weight vectors at least as extreme as the observed t; SE clustered by task, df 2204, G_ task 2205; release-label randomi-sation p = 0.708 over 24 orderings (descrip-tive)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
augmentation_H2 pct		-4.98	1.16	[-7.25, -2.71]	-4.298	< 0.001	β = 0	—	—	5260	none	trend β: wild-bootstrap p = 0.927; weight vectors = 6^4 = 1296; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 1202 of the 1296 weight vectors at least as extreme as the observed t; SE clustered by task, df 1314, G_task 1315; release-label randomisation p = 0.708 over 24 orderings (descriptive)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
directive_pct	H2	1.42	1.08	[-0.70, 3.55]	1.313	0.189	β = 0	—	—	4436	none	trend β: wild-bootstrap p = 0.940; weight vectors = 6^4 = 1296; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 1218 of the 1296 weight vectors at least as extreme as the observed t; SE clustered by task, df 1108, G_task 1109; release-label randomisation p = 0.875 over 24 orderings (descriptive)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
feedback_loop_pct	H2	5.62	1.69	[2.29, 8.95]	3.323	0.001	β = 0	—	—	996	none	trend β: wild-bootstrap p = 0.222; weight vectors = 6^4 = 1296; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 288 of the 1296 weight vectors at least as extreme as the observed t; SE clustered by task, df 248, G_task 249; release-label randomisation p = 0.667 over 24 orderings (descriptive)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
task_iteration_pct	H2	2.26	0.73	[0.82, 3.70]	3.087	0.002	β = 0	—	—	3396	none	trend β: wild-bootstrap p = 0.201; weight vectors = 6^4 = 1296; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 260 of the 1296 weight vectors at least as extreme as the observed t; SE clustered by task, df 848, G_task 849; release-label randomisation p = 0.375 over 24 orderings (descriptive)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
validation_pct	H2	1.48	0.45	[0.60, 2.36]	3.310	0.001	β = 0	—	—	1128	none	trend β: wild-bootstrap p = 0.069; weight vectors = 6^4 = 1296; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 90 of the 1296 weight vectors at least as extreme as the observed t; SE clustered by task, df 281, G_task 282; release-label randomisation p = 0.292 over 24 orderings (descriptive)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
learning_pct	H2	-8.47	0.97	[-10.36, -6.57]	-8.761	< 0.001	β = 0	—	—	3148	none	trend β: wild-bootstrap p = 0.781; weight vectors = 6^4 = 1296; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 1012 of the 1296 weight vectors at least as extreme as the observed t; SE clustered by task, df 786, G_task 787; release-label randomisation p = 0.500 over 24 orderings (descriptive)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
api_r3_r5	H2	-9.51	1.65	[-12.75, -6.27]	-5.752	< 0.001	β = 0	—	—	3888	none	trend β: wild-bootstrap p = 0.176; weight vectors = 6^3 = 216; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 38 of the 216 weight vectors at least as extreme as the observed t; SE clustered by task, df 1295, G_ task 1296; release-label randomi-sation p = 0.167 over 6 orderings (descrip-tive)

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.23 R23: robustness row (PAP §8.2 R23)

Source: PAP §8.2, row R23. Label: part h1 "robustness"; part h2 "robustness; confounded with the r6 suppression-regime change".

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: ai_autonomy_mean scale points per unit of the regressor; ai_autonomy_mean scale points per year. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.23.1 Notes

- `ai_autonomy_mean` (D13-averaged, r6 `claude_ai`) in place of `automation_pct` as H1's outcome (PAP §8.2 R23, spec §5.4.1); same predictors, sample rule, fixed effects and inference. R23's H2 part is a separate Plan 4 task.
- R23 is reported beside H1 (PAP §4.1): the H1 decision uses `automation_pct` only and is unchanged. The PAP §4.1 disagreement sentence is printed below only if each R23 coefficient (education, log wage) has the sign opposite to the committed primary H1 coefficient (P-RQ1 record) and $p < 0.05$, a mechanical check (PAP §11 C-77).
- Sign check against the committed primary H1 (P-RQ1): education R23 0.03 vs primary -0.53, opposite sign, $p < 0.001$; log wage R23 0.03 vs primary 4.40, same sign, p 0.177 (PAP §11 C-77)
- RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in ai_autonomy_mean scale points per unit of the regressor (per year of education, per unit of log wage). Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78).
- H2 part (part h2; Plan 4 Task 12): `ai_autonomy_mean` as H2's outcome over r4–r6, the releases over which it is definition-stable (PAP §5 D2, §8.2 R23), balanced over the three; the series includes r6 against D2's regime caveat and has no r6 indicator, so it is labelled "confounded with the r6 suppression-regime change", like R22's all-releases series. The H1 part is Task 11's (commit 61b0625), carried verbatim and not re-estimated.
- H2 parts: Plan 2's `estimate_h2` on Plan 2's `build_h2_panel` (through the adapter), with H2's inference (PAP §4.2, D22, §8.1): WLS of the outcome on t_r and task dummies with fixed task weights (the task's mean `pct` over the listed releases), SEs clustered by task, t with G_task − 1 df; `p` is the two-sided task-clustered p of the trend. The note gives the restricted wild cluster bootstrap p (clusters = releases, Webb weights, every one of the 6^k weight vectors enumerated; PAP D22, §11 C-15) and whether the D22 rule's conditions would hold (both $p \leq 0.0125$, compared exactly as PAP §11 C-24, and $\beta > 0$): a mechanical statement, not a decision; the release-label randomisation p is descriptive. The smallest attainable wild-bootstrap p is 6/6^k: 3 releases, $6/216 = 0.0278$, above 0.0125, so the wild condition cannot hold. β is the change in measured within-task outcome per year, including any change in the classifier and platform scope (PAP §5). Coefficients are in ai_autonomy_mean scale points per year. Each table row's note cell starts with its estimand (PAP §11 C-78). `n` is the number of (task, release) rows; `dropped` shows none and `n_dropped` is not reported for H2 rows, because Plan 2's H2 builder applies the sample rule and records no drops (PAP §11 C-78).

D.23.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
h1	H1	0.03	0.01	[0.01, 0.04]	3.374	< 0.001	coefficient = 0	—	—	1686	unmatched_task 0.08%; no_family 1.00%; wage_suppressed 0.16%	education: SE two-way cluster (occupation, task); df 388; G_occ 389, G_task 1144
h1	H1	0.03	0.02	[-0.01, 0.07]	1.352	0.177	coefficient = 0	—	—	1686	unmatched_task 0.08%; no_family 1.00%; wage_suppressed 0.16%	log wage: SE two-way cluster (occupation, task); df 388; G_occ 389, G_task 1144
h1	H1	—	—	—	—	0.177	β_e = 0 or β_w = 0	—	—	1686	unmatched_task 0.08%; no_family 1.00%; wage_suppressed 0.16%	p_H1: p_H1 = max(p_educ, p_wage), the intersection-union p (D17)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
h2	H2	-1.24	0.08	[-1.40, -1.09]	-15.735	< 0.001	β = 0	—	—	1785	none	trend β: wild-bootstrap p = 0.694; weight vectors = 6^3 = 216; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 150 of the 216 weight vectors at least as extreme as the observed t; SE clustered by task, df 594, G_ task 595; release-label randomi-sation p = 0.667 over 6 orderings (descrip-tive)

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.24 R24: robustness row (PAP §8.2 R24)

Source: PAP §8.2, row R24. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of the outcome share per unit of the regressor. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.24.1 Notes

- D13 alternatives (PAP §8.2 R24): release r4 `claude_ai` (part r4), r5 `claude_ai` (r5) and r6 `1p_api` (r6_1p_api), each through Plan 2's builder for that release and source (D13 averaging over the release's periods; the R7 suppression components of that release), H1 and H3 with the primary specification otherwise.
- RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of the outcome share per unit of the regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78).

D.24.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r4	H1	-0.73	0.42	[-1.56, 0.09]	-1.745	0.082	coefficient = 0	—	—	3278	unmatched_task 6.49%; no_family 12.12%; missing_automation_pct 2.34%; wage_suppressed 0.93%	education: SE two-way cluster (occupation, task); df 494; G_occ 495, G_task 1863
r4	H1	-3.60	1.57	[-6.68, -0.52]	-2.294	0.022	coefficient = 0	—	—	3278	unmatched_task 6.49%; no_family 12.12%; missing_automation_pct 2.34%; wage_suppressed 0.93%	log wage: SE two-way cluster (occupation, task); df 494; G_occ 495, G_task 1863

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r4	H1	—	—	—	—	0.082	β_e = 0 or β_w = 0	—	—	3278	unmatched_task 6.49%; no_family 12.12%; missing_automation_pct 2.34%; wage_suppressed 0.93%	p_H1: p_H1 = max(p_educ, p_wage), the intersection-union p (D17)
r4	H3	-0.96	1.34	[-3.61, 1.69]	-0.715	0.476	coefficient = 0	—	—	762	unmatched_task 6.49%; no_family 12.12%; missing_validation_pct 19.36%	error cost: SE two-way cluster (occupation, task); df 211; G_occ 212, G_task 378
r5	H1	-1.11	0.43	[-1.96, -0.26]	-2.561	0.011	coefficient = 0	—	—	3325	unmatched_task 7.03%; no_family 11.42%; missing_automation_pct 2.32%; wage_suppressed 0.63%	education: SE two-way cluster (occupation, task); df 501; G_occ 502, G_task 1967
r5	H1	-5.55	1.59	[-8.67, -2.43]	-3.495	< 0.001	coefficient = 0	—	—	3325	unmatched_task 7.03%; no_family 11.42%; missing_automation_pct 2.32%; wage_suppressed 0.63%	log wage: SE two-way cluster (occupation, task); df 501; G_occ 502, G_task 1967

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r5	H1	—	—	—	—	0.011	β_e = 0 or β_w = 0	—	—	3325	unmatched_task 7.03%; no_family 11.42%; missing_automation_pct 2.32%; wage_suppressed 0.63%	p_H1: p_H1 = max(p_educ, p_wage), the intersection-union p (D17)
r5	H3	0.40	1.02	[-1.61, 2.41]	0.392	0.695	coefficient = 0	—	—	796	unmatched_task 7.03%; no_family 11.42%; missing_validation_pct 20.73%	error cost: SE two-way cluster (occupation, task); df 216; G_occ 217, G_task 398
r6_1p_api	H1	0.38	0.30	[-0.21, 0.96]	1.272	0.204	coefficient = 0	—	—	1452	no_family 3.06%; wage_suppressed 0.34%	education: SE two-way cluster (occupation, task); df 377; G_occ 378, G_task 1048
r6_1p_api	H1	1.36	0.62	[0.13, 2.58]	2.182	0.030	coefficient = 0	—	—	1452	no_family 3.06%; wage_suppressed 0.34%	log wage: SE two-way cluster (occupation, task); df 377; G_occ 378, G_task 1048

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r6_1p_api	H1	—	—	—	—	0.204	β_e = 0 or β_w = 0	—	—	1452	no_family 3.06%; wage_ suppressed 0.34%	p_H1: p_ H1 = max(p_ educ, p_ wage), the intersection-union p (D17)
r6_1p_api	H3	-0.15	0.32	[-0.78, 0.49]	-0.449	0.653	coefficient = 0	—	—	1459	no_family 3.06%	error cost: SE two-way cluster (occupation, task); df 380; G_ occ 381, G_task 1055

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.25 R25: robustness row (PAP §8.2 R25)

Source: PAP §8.2, row R25. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of the outcome share per unit of the regressor. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.25.1 Notes

- R25 (PAP §4.0 D17 alternative, §8.2 R25): H1 and H4 split into separate coefficients, giving the five-member Holm family {H1a: education, H1b: log wage, H3: error cost, H4a: log AUI, H4b: log income}, tested by Plan 2's `holm` at family-wise 0.0375 (D22): the k-th smallest p of the five (ties by name) is compared with $0.0375/(5-k+1)$, step-down, by exact comparison on each float's decimal repr. Each p is the committed P-RQ1 fit's two-sided coefficient p (`confirmatory_rq1.json`): H1's two-way cluster t p-values, H3's indicator p, and H4's normal-reference HC3 p-values of log AUI and log income. Nothing is re-estimated; the estimates, SEs and intervals are the committed ones.
- H2 is outside this family and is tested on its own at 0.0125 (D22). 'rejected under R25' is the Holm step alone, with no sign condition; it is reported only and is not a decision: the RQ1 decisions are P-RQ1's (D17).
- Coefficients are in pp of the outcome share per unit of the regressor (H3: validation share). Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the committed primary sample: (task, occupation, family) rows for H1 and H3, countries for H4; `dropped` shows none and `n_dropped` is not reported, because the samples are the primary ones (PAP §11 C-78).

D.25.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
H1a	H1	-0.53	0.58	[-1.68, 0.62]	-0.914	0.361	coefficient = 0	—	—	1684	none	education: Holm level 0.01875; rejected under R25: no; not a decision; Holm rank 4 of 5
H1b	H1	4.40	2.20	[0.07, 8.72]	1.998	0.046	coefficient = 0	—	—	1684	none	log wage: Holm level 0.0125; rejected under R25: no; not a decision; Holm rank 3 of 5
H3	H3	-0.18	0.47	[-1.10, 0.75]	-0.376	0.707	coefficient = 0	—	—	1689	none	error cost: Holm level 0.0375; rejected under R25: no; not a decision; Holm rank 5 of 5
H4a	H4	-1.19	0.40	[-1.98, -0.40]	-2.950	0.003	coefficient = 0	—	—	113	none	log AUI: Holm level 0.0075; rejected under R25: yes; not a decision; Holm rank 1 of 5

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
H4b	H4	-0.92	0.35	[-1.61, -0.23]	-2.606	0.009	coefficient = 0	—	—	113	none	log income: Holm level 0.009375; rejected under R25: yes; not a decision; Holm rank 2 of 5

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.26 R26: robustness row (PAP §8.2 R26)

Source: PAP §8.2, row R26. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of the outcome share per unit of the regressor. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.26.1 Notes

- GWA (part gwa) and DWA (part dwa) task-family fixed effects in place of IWA (PAP §8.2 R26, D14), from Plan 2's `task_families(level)`; a task linked to several families is split equally across them and the family set is the union over the task's candidate Task IDs, as for IWA (D14, PAP §11 C-9).
- RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of the outcome share per unit of the regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78).

D.26.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
gwa	H1	0.23	0.51	[-0.78, 1.24]	0.444	0.657	coefficient = 0	—	—	1634	unmatched_task 0.08%; no_family 1.00%; missing_automation_pct 0.03%; wage_suppressed 0.16%	education: SE two-way cluster (occupation, task); df 387; G_occ 388, G_task 1142
gwa	H1	5.57	2.75	[0.16, 10.98]	2.024	0.044	coefficient = 0	—	—	1634	unmatched_task 0.08%; no_family 1.00%; missing_automation_pct 0.03%; wage_suppressed 0.16%	log wage: SE two-way cluster (occupation, task); df 387; G_occ 388, G_task 1142
gwa	H1	—	—	—	—	0.657	β_e = 0 or β_w = 0	—	—	1634	unmatched_task 0.08%; no_family 1.00%; missing_automation_pct 0.03%; wage_suppressed 0.16%	p_H1: p_H1 = max(p_educ, p_wage), the intersection-union p (D17)
gwa	H3	-0.24	0.44	[-1.10, 0.62]	-0.545	0.586	coefficient = 0	—	—	1639	unmatched_task 0.08%; no_family 1.00%	error cost: SE two-way cluster (occupation, task); df 390; G_occ 391, G_task 1147

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
dwa	H1	-0.52	0.53	[-1.57, 0.52]	-0.984	0.326	coefficient = 0	—	—	1825	unmatched_task 0.08%; no_family 1.00%; missing_automation_pct 0.03%; wage_suppressed 0.16%	education: SE two-way cluster (occupation, task); df 387; G_occ 388, G_task 1142
dwa	H1	2.15	1.83	[-1.45, 5.75]	1.173	0.242	coefficient = 0	—	—	1825	unmatched_task 0.08%; no_family 1.00%; missing_automation_pct 0.03%; wage_suppressed 0.16%	log wage: SE two-way cluster (occupation, task); df 387; G_occ 388, G_task 1142
dwa	H1	—	—	—	—	0.326	β_e = 0 or β_w = 0	—	—	1825	unmatched_task 0.08%; no_family 1.00%; missing_automation_pct 0.03%; wage_suppressed 0.16%	p_H1: p_H1 = max(p_educ, p_wage), the intersection-union p (D17)
dwa	H3	0.01	0.34	[-0.66, 0.69]	0.042	0.966	coefficient = 0	—	—	1830	unmatched_task 0.08%; no_family 1.00%	error cost: SE two-way cluster (occupation, task); df 390; G_occ 391, G_task 1147

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.27 R27: robustness row (PAP §8.2 R27)

Source: PAP §8.2, row R27. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of the outcome share per unit of the regressor. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.27.1 Notes

- [default] H3 with the O*NET 29.3 Work Context element Consequence of Error (`CX`, the mean `Data Value` over the O*NET-SOC codes within soc6) in place of the D15 indicator, z-scored across the distinct soc6 codes of the H3 rows that have a value (population SD), so γ is in pp of validation share per 1 SD (PAP §8.2 R27, §11 C-58). Rows without a value are dropped and reported as `missing_coe`.
- 22 of the 391 soc6 codes of the H3 rows have no Consequence of Error value.
- RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of validation share per 1 SD of Consequence of Error. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78).

D.27.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
coe	H3	-0.16	0.21	[-0.57, 0.25]	-0.780	0.436	coefficient = 0	—	—	1589	unmatched_task 0.08%; no_family 1.00%; missing_coe 4.42%	Consequence of Error: SE two-way cluster (occupation, task); df 368; G_occ 369, G_task 1096

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.28 R28: robustness row (PAP §8.2 R28)

Source: PAP §8.2, row R28. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of the outcome share per unit of the regressor. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.28.1 Notes

- H4 alternatives (PAP §4.4 D16, §8.2 R28), each part ending in Plan 2's `estimate_h4` (through the adapter) with H4's inference (PAP §8.1): OLS of the geography's overall automation_pct on [1, log AUI, log income, mix] with HC3 covariance (WLS in usage_weighted); b_1 (log AUI) and b_2 (log income) carry normal-reference HC3 95% intervals and two-sided p; p_wald (joint Wald) is the joint HC3 Wald F test of $b_1 = b_2 = 0$ with $(2, n - 4)$ df (D17), whose `stat` is F. Coefficients are in pp of the automation share per unit of the log regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of geographies in the fit; `n_dropped` is the number excluded and `dropped` the count by first failing reason, in Plan 2's order (missing_month, missing_income, missing_mix); the excluded geographies are listed below. Reported only, not a decision: the H4 decision is P-RQ1's.
- us_states: r6 `claude_ai` subregions whose `geo_id` starts `US-`, by Plan 2's `build_geo_sample(geo="us_state")`: the published state `usage_per_capita_index`, 2024 state GDP per working-age person from the AEI r3 state files, and the GWA mix.
- [default] r4_countries, r5_countries: Plan 4's `h4_frame_r45` (PAP §11 C-59), because r4 and r5 publish no country usage index and no country GWA rows. log AUI = ln(usage_pct_c / working-age population_c); AEI's index is the usage share over the working-age population share, so the common scale moves only the intercept. s_cg is the country's `onet_task` pct aggregated to GWAs through the D14 files (`task_families("gwa")`), each task split equally over its GWAs, then renormalised over the GWAs with an ā_g. ā_g is the same release's global task `automation_pct` aggregated to GWA with D13 weights (the task's global pct, split equally over its GWAs). Income is 2024 GDP per working-age person (D16, the r3 inputs Plan 2 uses). Each release has one published period, so nothing is averaged over months. Inclusion as D16: overall automation_pct and usage_pct published, income, and a mix.
- [default] In r4_countries and r5_countries a task's GWAs are the union over its candidate Task IDs (its panel Task ID and every Task ID carrying its statement), as Plan 2's H1/H3 task rows apply D14 (PAP §11 C-9, C-59); unmatched task nodes and tasks with no DWA link carry no GWA.
- In r4_countries and r5_countries the excluded geographies listed below include AEI's non-country buckets (`unmatched:*`), which are counted in `n_dropped` and `dropped` with the countries.
- usage_weighted: the D16 r6 country frame, estimated by WLS with weights = the country's two-month mean overall `usage_pct` (PAP §4.4 'weights = country usage_pct'); the sample is the primary one.
- iwa_mix: the D16 r6 country sample with the task-mix control built over IWA (`onet_iwa`) nodes in place of GWAs, by Plan 2's builder (`mix_level="iwa"`), with the same formula (PAP §4.4 'IWA-level mix control where published').
- The mix control (Plan 2's, and Plan 4's r4/r5 mix built the same way) sums over an unordered group_by, so mix and the estimates reproduce only to about 1e-14 between runs (PAP §11 C-32), far below the printed precision.
- us_states: excluded missing_month 1 (US-PR)
- r4_countries: excluded missing_month 19 (ABW, ATG, BDI, BLZ, BMU, COM, CUW, GUF, GUM, IMN, JEY, LCA, LIE, MYT, NCL, SUR, VCT, VIR, unmatched:not_classified); missing_income 7 (CYM, GLP, MCO, MTQ, PSE, PYF, REU); missing_mix 34 (AND, BHS, BRB, BRN, BTN, BWA, COG, CPV, DJI, FJI, GAB, GIN, GMB, GUY, HTI, ISL, LBR, LSO, MDV, MLT, MNE, MNG, MRT, MUS, NAM, NER, PNG, SLE, SWZ, TCD, TJK, TLS, TTO, XKX)
- r5_countries: excluded missing_month 21 (ABW, ATG, BMU, COM, CPV, GGY, GIB, GMB, GRD, GUF, GUM, IMN, LCA, LIE, LSO, MAF, SMR, TLS, VCT, VIR, unmatched:not_classified); missing_income 11 (CUW, CYM, GLP, JEY, MCO, MTQ, NCL, PSE, PYF, REU, unmatched:NONE); missing_mix 31 (AND, BDI, BHS, BLZ, BRB, BRN, BTN, BWA, COG, DJI, FJI, GIN, GUY, ISL, LAO, LBR, MDV, MLT, MNE, MRT, MWI, NAM, NER, PNG, SLE, SUR, SWZ, TCD, TJK, TTO, XKX)
- usage_weighted: excluded missing_month 7 (BWA, COG, HTI, MOZ, NAM, TGO, TTO); missing_income 1 (PSE)
- iwa_mix: excluded missing_month 7 (BWA, COG, HTI, MOZ, NAM, TGO, TTO); missing_income 1 (PSE)

D.28.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
us_states	H4	-0.71	0.61	[-1.90, 0.48]	-1.174	0.240	coefficient = 0	—	—	51	missing_ month 1	log AUI: SE OLS, HC3, normal reference
us_states	H4	-2.61	1.51	[-5.56, 0.34]	-1.733	0.083	coefficient = 0	—	—	51	missing_ month 1	log income: SE OLS, HC3, normal reference
us_states	H4	—	—	—	9.393	< 0.001	$b_1 = b_2 = 0$	—	—	51	missing_ month 1	joint Wald: joint HC3 Wald F, df $(2, 47)$ (D17); OLS; control mix 3.70 (SE 0.69, $p < 0.001$); not a decision
r4_ countries	H4	3.68	5.02	[-6.16, 13.53]	0.733	0.463	coefficient = 0	—	—	114	missing_ month 19; missing_ income 7; missing_ mix 34	log AUI: SE OLS, HC3, normal reference
r4_ countries	H4	-5.36	3.43	[-12.07, 1.36]	-1.563	0.118	coefficient = 0	—	—	114	missing_ month 19; missing_ income 7; missing_ mix 34	log income: SE OLS, HC3, normal reference

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
r4_ countries	H4	—	—	—	57.661	< 0.001	$b_1 = b_2 = 0$	—	—	114	missing_ month 19; missing_ income 7; missing_ mix 34	joint Wald: joint HC3 Wald F, df (2, 110) (D17); OLS; control mix 0.26 (SE 0.13, p 0.044); not a decision
r5_ countries	H4	-1.03	0.64	[-2.28, 0.22]	-1.617	0.106	coefficient = 0	—	—	115	missing_ month 21; missing_ income 11; missing_ mix 31	log AUI: SE OLS, HC3, normal reference
r5_ countries	H4	-1.96	0.56	[-3.05, -0.86]	-3.498	< 0.001	coefficient = 0	—	—	115	missing_ month 21; missing_ income 11; missing_ mix 31	log income: SE OLS, HC3, normal reference
r5_ countries	H4	—	—	—	102.119	< 0.001	$b_1 = b_2 = 0$	—	—	115	missing_ month 21; missing_ income 11; missing_ mix 31	joint Wald: joint HC3 Wald F, df (2, 111) (D17); OLS; control mix 0.07 (SE 0.06, p 0.240); not a decision
usage_ weighted	H4	-1.73	1.43	[-4.52, 1.07]	-1.210	0.226	coefficient = 0	—	—	113	missing_ month 7; missing_ income 1	log AUI: SE WLS, HC3, normal reference, weights usage_pct

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
usage_weighted	H4	0.09	1.44	[-2.73, 2.91]	0.062	0.950	coefficient = 0	—	—	113	missing_month 7; missing_income 1	log income: SE WLS, HC3, normal reference, weights usage_pct
usage_weighted	H4	—	—	—	4.144	0.018	$b_1 = b_2 = 0$	—	—	113	missing_month 7; missing_income 1	joint Wald: joint HC3 Wald F, df (2, 109) (D17); WLS; control mix 2.23 (SE 0.83, p 0.007); not a decision
iwa_mix	H4	-1.22	0.41	[-2.03, -0.41]	-2.958	0.003	coefficient = 0	—	—	113	missing_month 7; missing_income 1	log AUI: SE OLS, HC3, normal reference
iwa_mix	H4	-0.84	0.36	[-1.56, -0.13]	-2.318	0.020	coefficient = 0	—	—	113	missing_month 7; missing_income 1	log income: SE OLS, HC3, normal reference

iwa_mix	H4	—	—	—	82.629	< 0.001	$b_1 = b_2 = 0$	—	—	113	missing_month 7; missing_income 1	joint Wald: joint HC3 Wald F, df (2, 109) (D17); OLS; control mix 0.38 (SE 0.20, p 0.062); not a decision

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.29 R29: robustness row (PAP §8.2 R29)

Source: PAP §8.2, row R29. Label: robustness; exploratory.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation; pp of annualised alpha per 1 SD of gap. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.29.1 Notes

- [default] pre_formation_2025: the numerator hits (file date) and the denominator filers (Date Filed) are both cut at 2025-03-26, the filings public before Design B's formation date 2025-03-27 (PAP §11 C-61). pre_formation_2025: 2,456 talk filers of 4,262 filers (3,546 mapped to an FF49 industry); T_j undefined: none.
- change_2019_2024: T_j = $\text{talk}_{2024} - \text{talk}_{2019}$, each from the full calendar year's 10-Ks (a difference of two shares of filers); T_j undefined: none. 2024: 2,538 talk filers of 6,774 filers (5,796 mapped to an FF49 industry); T_j undefined: none. 2019: 558 talk filers of 6,793 filers (5,983 mapped to an FF49 industry); T_j undefined: none.
- [default] float_weighted: T_j = Σ float·1[talks] / Σ float over industry j's 2024 filers with a public float in the SEC XBRL frame dei/EntityPublicFloat/USD/CY2024Q2I (the R29 cap proxy, a §11 addition outside §9.4, A-46; PAP §11 C-61); T_j is undefined where that sum is not positive. Filers with a public float: 4,159 of 6,774 2024 filers (61.40%); 4,062 of 5,796 FF49-mapped filers (70.08%); 244 of them report a float of zero and enter with weight zero. T_j undefined: none.
- Each part replaces Plan 3's T_j (talk in 10-Ks filed in 2024, the 15 primary phrases, PAP §6 D18) by the part's T_j; everything after T_j is Plan 3's code (PAP §11 C-51): gap_j = the standardised residual of zT on Design B (r2) z_usage (`h6.talk.gap`); divergence = Spearman(T_j, z_usage) with a 95% percentile CI from 10,000 industry-bootstrap draws (`default_rng(SEED)`); the gap–return association = `h6.talk.gap_returns`: OLS of the Design B window alphas on gap_j with s, $\bar{b}$ and q recomputed for the gap vector on the same pre-seal historical windows, critical $\max(t_{21}, q)$, primary null $b = \bar{b}$ (D21), $b = 0$ secondary (in the note); the interval is $b \pm$ critical × s.
- H6 is exploratory (PAP §6 D18): the p-values are nominal, belong to no testing family and do not revisit the H5/H5b decision. An FF49 industry with T_j undefined is excluded from zT, the OLS, Spearman and the gap–return regression (C-51 (2)); unmapped filers enter no industry (C-49 (5)).
- Hits come from the saved EDGAR full-text-search responses (Plan 3's `hits_from_saved`, strict: a date range the saved leaf ranges do not cover is an error, never zero hits).
- Zero-assigned employment share, Design B: 12.14% (D3)

D.29.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
pre_formation_2025	H6 divergence	0.267	—	[-0.036, 0.525]	—	—	—	—	—	49	none	95% percentile CI from 10,000 industry-bootstrap draws (default_rng(SEED)); 0 undefined draws; dropped = T
pre_formation_2025	H6 gap-return	-2.15	1.70	[-5.68, 1.38]	-1.316	0.202	b = bbar	—	2.0796	49	none	undefined nominal p, in no testing family; $\bar{b}$ = 0.08 pp; q_0.975 = 1.968; secondary b = 0: T = -1.267, p = 0.219; no gap: none; missing return: none

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
change_2019_2024	H6 divergence	0.081	—	[-0.213, 0.368]	—	—	—	—	—	49	none	95% percentile CI from 10,000 industry-bootstrap draws (default_rng(SEED)); 0 undefined draws; dropped = T undefined nominal p, in no testing family; $\bar{b}$ = -0.57 pp; q_0.975 = 2.032; secondary b = 0: T = -1.947, p = 0.065; no gap: none; missing return: none
change_2019_2024	H6 gap-return	-3.37	1.73	[-6.97, 0.23]	-1.619	0.120	b = bbar	—	2.0796	49	none	

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
float_ weighted	H6 divergence	0.190	—	[-0.121, 0.467]	—	—	—	—	—	49	none	95% percentile CI from 10,000 industry-bootstrap draws (default_ rng(SEED)); 0 undefined draws; dropped = T
float_ weighted	H6 gap-return	-0.21	1.96	[-4.28, 3.86]	-0.153	0.880	b = bbar	—	2.0796	49	none	undefined nominal p, in no testing family; $\bar{b}$ = 0.09 pp; q_ 0.975 = 2.027; secondary b = 0: T = -0.106, p = 0.916; no gap: none; missing return: none

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.30 R30: robustness row (PAP §8.2 R30)

Source: PAP §8.2, row R30. Label: robustness; exploratory.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation; pp of annualised alpha per 1 SD of gap. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.30.1 Notes

- [default] narrow: the primary list without items 9–11 (computer vision, reinforcement learning, foundation model(s)): 9 terms, 11 phrases; broad: the 12 primary terms plus 'automation' and 'robotics': 14 terms, 17 phrases, the plural rule applied to the primary list only (PAP §6.1 D12 alternatives; PAP §11 C-62).
- [default] Only T_j (10-Ks filed in 2024) is computed for each list (PAP §11 C-62).
- The responses for the added terms were retrieved for this row through Plan 3's full-text-search client (exact totals, bisection of capped ranges, one MANIFEST record per response) and are logged as the §11 set addition A-47 of `sec_fts/`.
- narrow: 2,537 talk filers of 6,774 filers (5,796 mapped to an FF49 industry); T_j undefined: none.
- broad: 3,093 talk filers of 6,774 filers (5,796 mapped to an FF49 industry); T_j undefined: none.
- Each part replaces Plan 3's T_j (talk in 10-Ks filed in 2024, the 15 primary phrases, PAP §6 D18) by the part's T_j; everything after T_j is Plan 3's code (PAP §11 C-51): gap_j = the standardised residual of zT on Design B (r2) z_usage (`h6.talk.gap`); divergence = Spearman(T_j, z_usage) with a 95% percentile CI from 10,000 industry-bootstrap draws (`default_rng(SEED)`); the gap–return association = `h6.talk.gap_returns`: OLS of the Design B window alphas on gap_j with s, $\bar{b}$ and q recomputed for the gap vector on the same pre-seal historical windows, critical $\max(t_{21}, q)$, primary null $b = \bar{b}$ (D21), $b = 0$ secondary (in the note); the interval is $b \pm$ critical $\times$ s.
- H6 is exploratory (PAP §6 D18): the p-values are nominal, belong to no testing family and do not revisit the H5/H5b decision. An FF49 industry with T_j undefined is excluded from zT, the OLS, Spearman and the gap–return regression (C-51 (2)); unmapped filers enter no industry (C-49 (5)).
- Hits come from the saved EDGAR full-text-search responses (Plan 3's `hits_from_saved`, strict: a date range the saved leaf ranges do not cover is an error, never zero hits).
- Zero-assigned employment share, Design B: 12.14% (D3)

D.30.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
narrow	H6 divergence	0.361	—	[0.050, 0.612]	—	—	—	—	—	49	none	95% percentile CI from 10,000 industry-bootstrap draws (default_ rng(SEED)); 0 undefined draws; dropped = T undefined

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
narrow	H6 gap-return	-2.40	1.96	[-6.48, 1.68]	-1.278	0.215	b = bbar	—	2.0796	49	none	nominal p, in no testing family; b̄ = 0.11 pp; q_0.975 = 1.773; secondary b = 0: T = -1.221, p = 0.236; no gap: none; missing return: none
broad	H6 divergence	0.364	—	[0.055, 0.603]	—	—	—	—	—	49	none	95% percentile CI from 10,000 industry-bootstrap draws (default_rng(SEED)); 0 undefined draws; dropped = T undefined

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
broad	H6 gap-return	-1.98	2.23	[-6.63, 2.66]	-1.011	0.323	b = bbar	—	2.0796	49	none	nominal p, in no testing family; $\bar{b}$ = 0.27 pp; q_0.975 = 1.798; secondary b = 0: T = -0.888, p = 0.384; no gap: none; missing return: none

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.31 R31: robustness row (PAP §8.2 R31)

Source: PAP §8.2, row R31. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of the outcome share per unit of the regressor. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.31.1 Notes

- H3 with H1's covariates `human_education_years_mean` and `log_wage` added (PAP §8.2 R31, P13); the H3 sample then requires both covariates (Plan 2's drop rule on the estimator's own variables, PAP §11 C-12). The non-PSD rule is applied to γ's block alone and the covariate coefficients come from a second call on the same design, so the covariates never change γ's inference (PAP §11 C-14).
- RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of the outcome share per unit of the regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78).

D.31.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
covariates	H3	-0.20	0.42	[-1.02, 0.62]	-0.483	0.629	coefficient = 0	—	—	1686	unmatched_task 0.08%; no_family 1.00%; wage_suppressed 0.16%	error cost: SE two-way cluster (occupation, task); df 388; G_occ 389, G_task 1144
covariates	H3	0.17	0.07	[0.03, 0.31]	2.426	0.016	coefficient = 0	—	—	1686	unmatched_task 0.08%; no_family 1.00%; wage_suppressed 0.16%	education: SE two-way cluster (occupation, task); df 388; G_occ 389, G_task 1144
covariates	H3	-0.81	0.28	[-1.37, -0.25]	-2.858	0.004	coefficient = 0	—	—	1686	unmatched_task 0.08%; no_family 1.00%; wage_suppressed 0.16%	log wage: SE two-way cluster (occupation, task); df 388; G_occ 389, G_task 1144

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.32 R32: robustness row (PAP §8.2 R32)

Source: PAP §8.2, row R32. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.32.1 Notes

- Variant market: ExposureSpec(oews_year=2023, onet='29.2') on ff49_vw (49 industries); every other construction choice is the primary Design B one.
- Strict real-time Design B (PAP §8.2 R32, P15): release r2 with OEWS May 2023 weights (oesm23in4, oesm23nat) and O*NET task statements ≤ 29.2, i.e. inputs public at 2025-03-27; s, $\bar{b}$ and q recomputed.
- The O*NET 29.2 Task Statements file is byte-identical to the 29.3 file of the primary, so R32's task index equals the primary's and R32 differs from the primary only through the OEWS May 2023 weights (PAP §11 C-74).
- Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65).
- market: Zero-assigned employment share (D3): 12.19%
- Market parts: H5 and H5b on each part's return set with s, $\bar{b}$ and q recomputed for the part's exposure vector on the same 378 pre-seal windows with the same code, the critical $\max(t_{21}, q)$ and the D21 null $b_0 = \bar{b}$ (PAP §3.3, §8.1); industries with a missing return in the confirmatory window are dropped and z is not re-standardised after a drop. The secondary b = 0 statistic is in each record's note.

D.32.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	2.20	2.34	[-2.67, 7.08]	0.200	0.843	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.939, p = 0.358
market	H5b	3.72	2.63	[-1.75, 9.19]	0.908	0.374	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.416, p = 0.172

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.33 R33: robustness row (PAP §8.2 R33)

Source: PAP §8.2, row R33. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: correlation; pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.33.1 Notes

- Variant: ExposureSpec(crosswalk='emp_weighted'); every other construction choice is the primary Design B one.
- Employment-weighted NAICS→SIC crosswalk from OEWS `nat5d_6d` (PAP D23, §8.2 R33). Children without published employment share the residual $\max(E_4 - \Sigma$ published, 0) equally; with no published child, the primary equal split (PAP §11 C-67).
- Benchmarks: AIOE and Eloundou are re-aggregated with the variant's matrix, crosswalk and industry set, so both sides of H5b use the same weights (PAP §11 C-65).

- Zero-assigned employment share (D3): 12.14%
- Market part: H5 and H5b on ff49_vw with s, $\bar{b}$ and q recomputed for this exposure vector on the same 378 pre-seal windows, the t_{21} rule and the D21 null $b_0 = \bar{b}$ (PAP §3.3, §8.1); z is the variant's own and is not re-standardised after a drop.
- RQ2 part: Spearman correlations of the variant AEI indices with the variant benchmarks; the intervals are 2.5/97.5 percentile industry-bootstrap intervals on Plan 2's Design B draw set `bootstrap_indices(n)["B"]` (PAP §11 C-68). RQ2 is descriptive.

D.33.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
market	H5	1.92	2.30	[-2.86, 6.70]	0.118	0.907	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.834, p = 0.414
market	H5b	3.67	2.56	[-1.66, 9.01]	0.945	0.356	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.432, p = 0.167
rq2	RQ2	0.758	—	[0.583, 0.863]	—	—	—	—	—	49	none	z_usage vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)
rq2	RQ2	0.708	—	[0.515, 0.829]	—	—	—	—	—	49	none	z_delegation vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.769	—	[0.603, 0.868]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou; 10000 bootstrap draws used; headline benchmark (D8)
rq2	RQ2	0.821	—	[0.684, 0.902]	—	—	—	—	—	49	none	z_usage vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.802	—	[0.661, 0.890]	—	—	—	—	—	49	none	z_delegation vs z_eloundou_gpt4; 10000 bootstrap draws used
rq2	RQ2	0.829	—	[0.699, 0.903]	—	—	—	—	—	49	none	z_augmentation vs z_eloundou_gpt4; 10000 bootstrap draws used

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
rq2	RQ2	0.786	—	[0.621, 0.880]	—	—	—	—	—	49	none	z_usage vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.746	—	[0.573, 0.850]	—	—	—	—	—	49	none	z_ delegation vs z_aioe; 10000 bootstrap draws used
rq2	RQ2	0.804	—	[0.657, 0.889]	—	—	—	—	—	49	none	z_ augmentation vs z_aioe; 10000 bootstrap draws used

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.34 R34: robustness row (PAP §8.2 R34)

Source: PAP §8.2, row R34. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.34.1 Notes

- Null scale, $\bar{b}$ and criticals from overlapping 17-month windows starting 1969-07 (the first month with all 49 industries) and 1963-07 (the first daily FF5 month), ending by 2022-11-29 (PAP §3.3, §7, §8.2 R34).
- 1963-07 rows are not like-for-like: industries missing in early windows. An industry with any missing return in a historical window has no alpha there and is left out of that window's slope only (the power analysis rule, PAP §7); z is not re-standardised. The drop counts below are computed from the history.
- from_1969 (1969-07, 624 windows): 17 industry-window drops in 17 windows (Softw).
- from_1963 (1963-07, 696 windows): 113 industry-window drops in 89 windows (Hlth, Softw).

- Exposure: the primary Design B vector (`ExposureSpec()`). H5 and H5b with s, $\bar{b}$ and q recomputed for this row by the pre-seal procedure (the same 378 windows unless the row says otherwise, the t_{21} rule and the D21 null $b_0 = \bar{b}$; PAP §3.3, §8.1); industries with a missing return in the confirmatory window are dropped and z is not re-standardised. The secondary b = 0 statistic is in each record's note.
- Zero-assigned employment share (D3): 12.14% (the primary Design B exposure; this row rebuilds no index).

D.34.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
from_1969	H5	1.96	2.51	[-3.26, 7.18]	0.400	0.693	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.780, p = 0.444
from_1969	H5b	3.46	3.00	[-2.77, 9.69]	0.877	0.390	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 1.155, p = 0.261
from_1963	H5	1.96	2.72	[-3.70, 7.61]	0.475	0.640	b = bbar	2.4138	2.0796	49	none	secondary b = 0: T = 0.720, p = 0.479
from_1963	H5b	3.46	3.53	[-3.88, 10.81]	0.826	0.418	b = bbar	2.6632	2.0796	49	none	secondary b = 0: T = 0.980, p = 0.338

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.35 R35: robustness row (PAP §8.2 R35)

Source: PAP §8.2, row R35. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of annualised alpha per 1 SD. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.35.1 Notes

- Studentised statistic (PAP §8.2 R35, P10): for each of the 378 pre-seal windows w, Σ_w is the covariance of the industries' daily FF5+UMD residuals (denominator $T_w - 7$), $c_w = [(X_w' X_w)^{-1}]_{11}$ is the intercept element of the common factor design, ω the OLS weight vector with $b = \omega'\alpha$ ([1, z] for H5, [1, z, z_aioe] for H5b), $SE_{\Sigma,w} = 252 \cdot \sqrt{c_w \cdot \omega' \Sigma_w \omega}$, and $\kappa = sd_w(b_w)/median_w(SE_{\Sigma,w})$, the same sd as s.

- [default] In the confirmatory window SE_Σ is computed the same way; stat = T_stud = $(\hat{b} - b_0)/(\kappa \cdot SE_\Sigma)$ with $b_0 = \bar{b}$ (D21), p from t_{21}, and the t_{21} criticals of §3.3. The scale column is κ·SE_Σ in pp and the interval is $\hat{b} \pm t_{21}$,0.975·κ·SE_Σ, the §3.3 interval form with this scale, a presentation-only reading (PAP §11 C-75).
- Exposure: the primary Design B vector (`ExposureSpec()`). H5 and H5b with s, $\bar{b}$ and q recomputed for this row by the pre-seal procedure (the same 378 windows unless the row says otherwise, the t_{21} rule and the D21 null $b_0 = \bar{b}$; PAP §3.3, §8.1); industries with a missing return in the confirmatory window are dropped and z is not re-standardised. The secondary b = 0 statistic is in each record's note.
- Zero-assigned employment share (D3): 12.14% (the primary Design B exposure; this row rebuilds no index).

D.35.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
h5	H5	1.96	3.24	[-4.78, 8.69]	0.095	0.925	b = bbar	2.4138	2.0796	49	none	κ = 1.312; SE_Σ,2025–26 = 2.47 pp; $\bar{b}$ = 1.65 pp; secondary b = 0: T = 0.605, p = 0.552
h5b	H5b	3.46	3.14	[-3.06, 9.98]	0.716	0.482	b = bbar	2.4138	2.0796	49	none	κ = 1.285; SE_Σ,2025–26 = 2.44 pp; $\bar{b}$ = 1.22 pp; secondary b = 0: T = 1.104, p = 0.282

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.36 R36: robustness row (PAP §8.2 R36)

Source: PAP §8.2, row R36. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of the outcome share per year. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.36.1 Notes

- The H2 primary (r2–r5 `claude_ai`, balanced, `automation_pct`) with t_r from the releases' publication dates in place of the window midpoints (PAP §4.2 Time row, §8.2 R36, P19).
- H2 parts: Plan 2's `estimate_h2` on Plan 2's `build_h2_panel` (through the adapter), with H2's inference (PAP §4.2, D22, §8.1): WLS of the outcome on t_r and task dummies with fixed task weights (the task's mean `pct` over the listed releases), SEs clustered by task, t with G_task − 1 df; `p` is the two-sided task-clustered p of the trend. The note gives the restricted wild cluster bootstrap p (clusters = releases, Webb weights, every one of the 6^k weight vectors enumerated; PAP D22, §11 C-15) and whether the D22 rule's conditions would hold (both $p \leq 0.0125$, compared exactly as PAP §11 C-24, and $\beta > 0$): a mechanical statement, not a decision; the release-label randomisation p is descriptive. The smallest attainable wild-bootstrap p is 6/6^k: 4 releases, $6/1296 = 0.0046$. β is the change in measured within-task outcome per year, including any change in the classifier and platform scope (PAP §5). Coefficients are in pp of the outcome share per year. Each table row's note cell starts with its estimand (PAP §11 C-78). `n` is the number of (task, release) rows; `dropped` shows none and `n_dropped` is not reported for H2 rows, because Plan 2's H2 builder applies the sample rule and records no drops (PAP §11 C-78).

D.36.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
publication	H2	4.69	1.10	[2.54, 6.84]	4.273	< 0.001	β = 0	—	—	5260	none	trend β: wild-bootstrap p = 0.932; weight vectors = 6^4 = 1296; would meet the H2 rule (both p <= 0.0125 and beta > 0): no; not a decision; 1208 of the 1296 weight vectors at least as extreme as the observed t; SE clustered by task, df 1314, G_ task 1315; release-label randomi-sation p = 0.708 over 24 orderings (descrip-tive)

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.37 R37: robustness row (PAP §8.2 R37)

Source: PAP §8.2, row R37. Label: robustness; descriptive.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of automation_pct. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.37.1 Notes

- Within-r6 April vs May (PAP §4.2 'Also reported', §8.2 R37, P19), a check on short-run seasonality and noise, by Plan 4's own `within_r6` on the study panel: r6 `claude_ai` global task rows, April t = 0 and May t = 1, tasks with a published `automation_pct` in both months, weight = the task's mean `pct` over the two months (fixed across them); WLS with task fixed effects, SEs clustered by task, t with G_task − 1 df (H2's primary call on two periods). The estimate is the weighted mean within-task May minus April change, in pp of automation_pct. Descriptive: no wild bootstrap and no rule check.
- 1141 tasks with `automation_pct` in both months have zero weight (r6 task `pct` is published rounded to 0.01; D13, PAP §11 C-30 (b)) and are left out; they would carry no weight in the fit; n_tasks, G_task and df count the tasks with positive weight (PAP §11 C-79).
- `n` is the number of (task, month) rows; `dropped` shows none and `n_dropped` is not reported, as for the H2 rows (PAP §11 C-78). The note cell starts with its estimand (C-78).

D.37.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
within_r6	H2	-0.13	0.10	[-0.32, 0.06]	-1.358	0.175	May minus April = 0	—	—	2266	none	May minus April: SE clustered by task, df 1132, G_task 1133; n_tasks 1133

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

D.38 R38: robustness row (PAP §8.2 R38)

Source: PAP §8.2, row R38. Label: robustness.

Every record below is reported as robustness; not a confirmatory decision (PAP §8.1).

Units: pp of the outcome share per unit of the regressor. Estimates and scales are formatted to 2 decimals (correlations to 3), p to 3 decimals; the CSV keeps full precision.

D.38.1 Notes

- SEs clustered by occupation (soc6) only, the spec's §5.2 rule (PAP §4 departure, §8.2 R38, P18): SE and p are Plan 2's one-way diagnostics `se_oneway_occ` / `p_oneway_occ` of the primary H1 and H3 fits, df = G_occ − 1, interval estimate ± t(0.975, G_occ − 1)·SE; point estimates are the primary ones. p_H1 is D17's max of the two one-way p-values. Reported only; used in no decision.
- RQ1 parts: Plan 2's `estimate_h1` / `estimate_h3` on Plan 2's `build_task_rows` (through the adapter), with the inference of the hypothesis varied (PAP §8.1): WLS with IWA fixed effects unless the row says otherwise, two-way cluster-robust SEs by occupation (soc6) and task, t with min(G_occ, G_task) − 1 df and the PAP §4 non-PSD rule (each use is in the record's note). `scale` is the SE, `stat` the t statistic, intervals 95%, p two-sided; p_H1 = max(p_educ, p_wage) (D17). Coefficients are in pp of the outcome share per unit of the regressor. Each table row's note cell starts with its coefficient (PAP §11 C-78). `n` is the number of (task, occupation, family) rows; `dropped` lists the non-zero drop shares in % of total task usage, structural then variable-level (PAP §4.1, §11 C-2, C-30); `n_dropped` is not reported for RQ1 rows, because Plan 2 records drops as usage shares, not row counts (PAP §11 C-2, C-78).

D.38.2 Results

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
oneway_occ	H1	-0.53	0.58	[-1.68, 0.61]	-0.915	0.361	coefficient = 0	—	—	1684	unmatched_task 0.08%; no_family 1.00%; missing_automation_pct 0.03%; wage_suppressed 0.16%	education: SE one-way cluster by occupation (R38); df 387; G_occ 388, G_task 1142
oneway_occ	H1	4.40	2.20	[0.08, 8.71]	2.000	0.046	coefficient = 0	—	—	1684	unmatched_task 0.08%; no_family 1.00%; missing_automation_pct 0.03%; wage_suppressed 0.16%	log wage: SE one-way cluster by occupation (R38); df 387; G_occ 388, G_task 1142

part	applies to	estimate	scale	interval	stat	p	null	c_0.025	c_0.05	n	dropped	note
oneway_occ	H1	—	—	—	—	0.361	β_e = 0 or β_w = 0	—	—	1684	unmatched_task 0.08%; no_family 1.00%; missing_automation_pct 0.03%; wage_suppressed 0.16%	p_H1: p_H1 = max(p_educ, p_wage), the intersection-union p (D17); from the one-way occupation-cluster p-values
oneway_occ	H3	-0.18	0.46	[-1.08, 0.73]	-0.383	0.702	coefficient = 0	—	—	1689	unmatched_task 0.08%; no_family 1.00%	error cost: SE one-way cluster by occupation (R38); df 390; G_occ 391, G_task 1147

AEI data are conversations matched to O*NET tasks. Nothing here says anything about job risk, displacement or reassurance about the labour market.

E Data and replication

E.1 Raw data (`data/raw/MANIFEST.json`)

Every raw file is listed with its URL, retrieval time, size and SHA-256 (PAP §9.1). The raw files are not committed; the MANIFEST is. A key is either frozen at the tag `pap-v1` or added after it under D20 and logged in PAP §11 (the A-entries named). The two bulk sets are collapsed to one row each.

key	URL	retrieved (UTC)	bytes	SHA-256 (first 16)	status
`aei_geo_gdp_country`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_09_15/data/intermediate/gdp_2024_country.csv	2026-10-21T21:13:17Z	4115	e0618d6ba634a612	frozen at `pap-v1`
`aei_geo_gdp_us_state`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_09_15/data/intermediate/gdp_2024_us_state.csv	2026-10-21T21:13:18Z	2179	cb1573e8143bd407	frozen at `pap-v1`
`aei_geo_iso`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_09_15/data/intermediate/iso_country_codes.csv	2026-10-21T21:13:18Z	4564	eee318d4ab2c2c30	frozen at `pap-v1`
`aei_geo_wap_country`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_09_15/data/intermediate/working_age_pop_2024_country.csv	2026-10-21T21:13:17Z	6321	3a9e0edc0453702e	frozen at `pap-v1`
`aei_geo_wap_us_state`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_09_15/data/intermediate/working_age_pop_2024_us_state.csv	2026-10-21T21:13:17Z	1079	e2d0fb7239934f4f	frozen at `pap-v1`
`aei_job_exposure`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/labor_market_impacts/job_exposure.csv	2026-10-21T21:13:16Z	37176	4f0a3adf5feeb2ec	frozen at `pap-v1`
`aei_onet_tasks`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_03_27/onet_task_statements.csv	2026-10-21T21:12:42Z	3592256	82e4c418dc08475e	frozen at `pap-v1`
`aei_r1_collab`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_02_10/automation_vs_augmentation.csv	2026-10-21T21:12:41Z	197	d1e264882b17618d	frozen at `pap-v1`
`aei_r1_task_pct`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_02_10/onet_task_mappings.csv	2026-10-21T21:12:41Z	461306	8de89c99f762146c	frozen at `pap-v1`
`aei_r1_wage`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_02_10/wage_data.csv	2026-10-21T21:12:41Z	128047	4453fc56402c9430	frozen at `pap-v1`
`aei_r2_collab_by_task`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_03_27/automation_vs_augmentation_by_task.csv	2026-10-21T21:12:42Z	561368	f6570714a9af31ae	frozen at `pap-v1`
`aei_r2_task_pct`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_03_27/task_pct_v2.csv	2026-10-21T21:12:42Z	435372	0c20f184cd4a1e3d	frozen at `pap-v1`
`aei_r2_thinking`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_03_27/task_thinking_fractions.csv	2026-10-21T21:12:42Z	372332	35a78bc803b610e4	frozen at `pap-v1`
`aei_r3_api`	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_09_15/data/intermediate/aei_raw_1p_api_2025-08-04_to_2025-08-11.csv	2026-10-21T21:12:44Z	7027019	c0fc061ea7d4feb7	frozen at `pap-v1`

key	URL	retrieved (UTC)	bytes	SHA-256 (first 16)	status
aei_r3_claude_ai	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_09_15/data/intermediate/aei_raw_claude_ai_2025-08-04_to_2025-08-11.csv	2026-10-21T21:12:43Z	18894517	c8ef9c5eee0c42fe	frozen at pap-v1
aei_r3_docs	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_09_15/data_documentation.md	2026-10-21T21:12:44Z	20133	d7e23aa0888628fd	frozen at pap-v1
aei_r4_api	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2026_01_15/data/intermediate/aei_raw_1p_api_2025-11-13_to_2025-11-20.csv	2026-10-21T21:12:52Z	41518256	432f8d14348ba384	frozen at pap-v1
aei_r4_claude_ai	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2026_01_15/data/intermediate/aei_raw_claude_ai_2025-11-13_to_2025-11-20.csv	2026-10-21T21:12:49Z	94086309	70dc16780061b84d	frozen at pap-v1
aei_r4_docs	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2026_01_15/data_documentation.md	2026-10-21T21:12:52Z	18721	50f117f8f9956060	frozen at pap-v1
aei_r5_api	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2026_03_24/data/aei_raw_1p_api_2026-02-05_to_2026-02-12.csv	2026-10-21T21:13:00Z	43957174	b3bcd68e7f6d820f	frozen at pap-v1
aei_r5_claude_ai	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2026_03_24/data/aei_raw_claude_ai_2026-02-05_to_2026-02-12.csv	2026-10-21T21:12:58Z	103287181	69ebf6f9afc1e3be	frozen at pap-v1
aei_r5_docs	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2026_03_24/data_documentation.md	2026-10-21T21:13:01Z	17739	0a194ff1e317567e	frozen at pap-v1
aei_r6_api	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2026_06_26/data/aei_1p_api_2026-06-26.csv	2026-10-21T21:13:16Z	77282477	62197f003e001945	frozen at pap-v1
aei_r6_claude_ai	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2026_06_26/data/aei_claude_ai_2026-06-26.csv	2026-10-21T21:13:12Z	219174671	f974b358bce0e5a8	frozen at pap-v1
aei_r6_docs	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2026_06_26/data_documentation.md	2026-10-21T21:13:16Z	7397	1a01eeb7d34250b1	frozen at pap-v1
aei_soc_structure	https://huggingface.co/datasets/Anthropic/EconomicIndex/resolve/main/release_2025_02_10/SOC_Structure.csv	2026-10-21T21:12:41Z	77176	05eb5e689fb471b7	frozen at pap-v1
aioe	https://raw.githubusercontent.com/AIOE-Data/AIOE/main/AIOE_DataAppendix.xlsx	2026-10-21T21:13:24Z	170359	c123b4c64840aff3	frozen at pap-v1
bls_soc_2010_2018	https://www.bls.gov/soc/2018/soc_2010_to_2018_crosswalk.xlsx	2026-10-23T15:27:35Z	46630	f3a847561562d3e5	added under D20 (A-36)

key	URL	retrieved (UTC)	bytes	SHA-256 (first 16)	status
census_naics02_07	https://www.census.gov/naics/concordances/2002_to_2007_NAICS.xls	2026-10-21T21:13:24Z	161280	a4b00b2195e31842	frozen at pap-v1
census_naics07_12	https://www.census.gov/naics/concordances/2007_to_2012_NAICS.xls	2026-10-21T21:13:25Z	171520	302b8abcd78881f1	frozen at pap-v1
census_naics12_17	https://www.census.gov/naics/concordances/2012_to_2017_NAICS.xlsx	2026-10-21T21:13:25Z	58543	17f463f7e8385df1	frozen at pap-v1
census_naics17_22	https://www.census.gov/naics/concordances/2017_to_2022_NAICS.xlsx	2026-10-21T21:13:25Z	59656	4662cc7ed9e7f3fb	frozen at pap-v1
census_naics97_02	https://www.census.gov/naics/concordances/1997_NAICS_to_2002_NAICS.xls	2026-10-21T21:13:24Z	1282560	c9940c1236a03302	frozen at pap-v1
census_sic87_naics97	https://www.census.gov/naics/concordances/1987_SIC_to_1997_NAICS.xls	2026-10-21T21:13:24Z	367616	ba04e2332bbe5d56	frozen at pap-v1
census_states_20m	https://www2.census.gov/geo/tiger/GENZ2023/shp/cb_2023_us_state_20m.zip	2026-10-22T14:41:41Z	186432	0fd2d6562708ff81	added under D20 (A-2)
eloundou_occ	https://raw.githubusercontent.com/openai/GPTs-are-GPTs/main/data/occ_level.csv	2026-10-21T21:13:24Z	126022	40c74f53de40aec9	frozen at pap-v1
ff30_daily	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/30_Industry_Portfolios_daily_CSV.zip	2026-10-23T15:27:44Z	2597362	04a7a74fcd7c12e9	added under D20 (A-42)
ff3_monthly	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Research_Data_Factors_CSV.zip	2026-10-22T21:55:15Z	13065	593f4fbef03181bc	added under D20 (A-3)
ff48_daily	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/48_Industry_Portfolios_daily_CSV.zip	2026-10-23T15:27:45Z	4098912	95b65f678bfe12be	added under D20 (A-43)
ff49_daily	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/49_Industry_Portfolios_daily_CSV.zip	2026-10-21T21:13:23Z	4173817	8f394fe34bea54d4	frozen at pap-v1
ff49_monthly	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/49_Industry_Portfolios_CSV.zip	2026-10-21T21:13:23Z	489433	16b3f20bf5bd1db8	frozen at pap-v1
ff5_daily	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Research_Data_5_Factors_2x3_daily_CSV.zip	2026-10-21T21:13:23Z	150292	7881223b51929ee0	frozen at pap-v1
ff5_monthly	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Research_Data_5_Factors_2x3_CSV.zip	2026-10-21T21:13:23Z	11964	36756c5c25596485	frozen at pap-v1
ff_siccodes30	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/Siccodes30.zip	2026-10-23T15:27:45Z	9244	eddab10956ef97c8	added under D20 (A-44)
ff_siccodes48	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/Siccodes48.zip	2026-10-23T15:27:45Z	9551	f37edffc024fe7b9	added under D20 (A-45)
ff_siccodes49	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/Siccodes49.zip	2026-10-21T21:13:24Z	9587	36761b4009107e11	frozen at pap-v1
mom_daily	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Momentum_Factor_daily_CSV.zip	2026-10-21T21:13:23Z	85269	ecf82b116e6ec56d	frozen at pap-v1
mom_monthly	https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Momentum_Factor_CSV.zip	2026-10-21T21:13:24Z	5600	c06d1c2e9a5e4f69	frozen at pap-v1

key	URL	retrieved (UTC)	bytes	SHA-256 (first 16)	status
ne_admin0_50m	https://naciscdn.org/naturalearth/50m/cultural/ne_50m_admin_0_countries.zip	2026-10-22T14:41:41Z	799734	5fed433373581fa6	added under D20 (A-1)
oews_2023_ind	https://www.bls.gov/oes/special-requests/oesm23in4.zip	2026-10-23T15:27:44Z	31801148	2da9b5426eba1a3a	added under D20 (A-40)
oews_2023_nat	https://www.bls.gov/oes/special-requests/oesm23nat.zip	2026-10-23T15:27:43Z	271583	8ce5e8277c3ba68a	added under D20 (A-39)
oews_2024_ind	https://www.bls.gov/oes/special-requests/oesm24in4.zip	2026-10-21T21:13:20Z	32425931	0d76287e92de226c	frozen at pap-v1
oews_2024_nat	https://www.bls.gov/oes/special-requests/oesm24nat.zip	2026-10-21T21:13:18Z	282052	5fa368aa034ca784	frozen at pap-v1
oews_2025_ind	https://www.bls.gov/oes/special-requests/oesm25in4.zip	2026-10-21T21:13:22Z	31514940	3ef0e42fab1b42a7	frozen at pap-v1
oews_2025_nat	https://www.bls.gov/oes/special-requests/oesm25nat.zip	2026-10-23T15:27:43Z	279525	b5855a37f3e03e77	added under D20 (A-38)
onet_2010_2019_crosswalk	https://www.onetcenter.org/taxonomy/2019/walk/2010_to_2019_Crosswalk.csv?fmt=csv	2026-10-21T21:54:31Z	108052	8f026a33134bfde5	frozen at pap-v1
onet_29_2_tasks	https://www.onetcenter.org/dl_files/database/db_29_2_text/Task%20Statements.txt	2026-10-23T15:27:43Z	2759559	fecdcda81a4b8a28	added under D20 (A-41)
onet_29_3_dwa_reference	https://www.onetcenter.org/dl_files/database/db_29_3_text/DWA%20Reference.txt	2026-10-22T02:15:59Z	187200	e38a5b529a06d680	frozen at pap-v1
onet_29_3_tasks	https://www.onetcenter.org/dl_files/database/db_29_3_text/Task%20Statements.txt	2026-10-21T21:54:30Z	2759559	fecdcda81a4b8a28	frozen at pap-v1
onet_29_3_tasks_to_dwas	https://www.onetcenter.org/dl_files/database/db_29_3_text/Tasks%20to%20DWAs.txt	2026-10-22T02:15:59Z	1197935	6cb7098b09a6a03c	frozen at pap-v1
onet_29_3_work_context	https://www.onetcenter.org/dl_files/database/db_29_3_text/Work%20Context.txt	2026-10-23T15:27:42Z	35975227	dbe428983db68eef	added under D20 (A-37)
sec_docs_index	SEC EDGAR 10-K primary documents, 47946 documents (https://www.sec.gov/Archives/edgar/data/)	2026-10-23T09:58:41Z	16077303	c416d6bcba7e41cf	added under D20 (A-34, A-35)
sec_formidx_2019q1	https://www.sec.gov/Archives/edgar/full-index/2019/QTR1/form.idx	2026-10-23T01:14:27Z	46348993	88901ad5ec510ebf	added under D20 (A-4)
sec_formidx_2019q2	https://www.sec.gov/Archives/edgar/full-index/2019/QTR2/form.idx	2026-10-23T01:14:28Z	38937900	64a108ce2f798cbf	added under D20 (A-5)
sec_formidx_2019q3	https://www.sec.gov/Archives/edgar/full-index/2019/QTR3/form.idx	2026-10-23T01:14:29Z	30960533	2a2b2ac46d85e10b	added under D20 (A-6)
sec_formidx_2019q4	https://www.sec.gov/Archives/edgar/full-index/2019/QTR4/form.idx	2026-10-23T01:14:29Z	32230060	a8e6d873ff24cd31	added under D20 (A-7)
sec_formidx_2020q1	https://www.sec.gov/Archives/edgar/full-index/2020/QTR1/form.idx	2026-10-23T01:14:30Z	50685481	593cbf371d422fe5	added under D20 (A-8)
sec_formidx_2020q2	https://www.sec.gov/Archives/edgar/full-index/2020/QTR2/form.idx	2026-10-23T01:14:31Z	41419860	038d6eb07e7fca5b	added under D20 (A-9)

key	URL	retrieved (UTC)	bytes	SHA-256 (first 16)	status
`sec_formidx_2020q3`	https://www.sec.gov/Archives/edgar/full-index/2020/QTR3/form.idx	2026-10-23T01:14:32Z	34590653	d378841b0abbe2d2	added under D20 (A-10)
`sec_formidx_2020q4`	https://www.sec.gov/Archives/edgar/full-index/2020/QTR4/form.idx	2026-10-23T01:14:33Z	36414292	84aa7452907f2415	added under D20 (A-11)
`sec_formidx_2021q1`	https://www.sec.gov/Archives/edgar/full-index/2021/QTR1/form.idx	2026-10-23T01:14:34Z	56365441	78b6719feb4d8da2	added under D20 (A-12)
`sec_formidx_2021q2`	https://www.sec.gov/Archives/edgar/full-index/2021/QTR2/form.idx	2026-10-23T01:14:36Z	46825572	9bddf421721ec657	added under D20 (A-13)
`sec_formidx_2021q3`	https://www.sec.gov/Archives/edgar/full-index/2021/QTR3/form.idx	2026-10-23T01:14:37Z	39349433	8c8ef9f8be2acdc3	added under D20 (A-14)
`sec_formidx_2021q4`	https://www.sec.gov/Archives/edgar/full-index/2021/QTR4/form.idx	2026-10-23T01:14:38Z	40299160	025be2086f7e0118	added under D20 (A-15)
`sec_formidx_2022q1`	https://www.sec.gov/Archives/edgar/full-index/2022/QTR1/form.idx	2026-10-23T01:14:39Z	55621633	3fa246248e37398a	added under D20 (A-16)
`sec_formidx_2022q2`	https://www.sec.gov/Archives/edgar/full-index/2022/QTR2/form.idx	2026-10-23T01:14:39Z	45498636	6b72ce7a562225bc	added under D20 (A-17)
`sec_formidx_2022q3`	https://www.sec.gov/Archives/edgar/full-index/2022/QTR3/form.idx	2026-10-23T01:14:40Z	37094765	bce3e54373776e80	added under D20 (A-18)
`sec_formidx_2022q4`	https://www.sec.gov/Archives/edgar/full-index/2022/QTR4/form.idx	2026-10-23T01:14:41Z	35987632	e4fa292fcd6e15eb	added under D20 (A-19)
`sec_formidx_2023q1`	https://www.sec.gov/Archives/edgar/full-index/2023/QTR1/form.idx	2026-10-23T01:14:43Z	54362089	0f11bf75dd8c43d6	added under D20 (A-20)
`sec_formidx_2023q2`	https://www.sec.gov/Archives/edgar/full-index/2023/QTR2/form.idx	2026-10-23T01:14:44Z	48017880	de6bea77f33fd87c	added under D20 (A-21)
`sec_formidx_2023q3`	https://www.sec.gov/Archives/edgar/full-index/2023/QTR3/form.idx	2026-10-23T01:14:45Z	39411989	58e7d1100b2f7f6b	added under D20 (A-22)
`sec_formidx_2023q4`	https://www.sec.gov/Archives/edgar/full-index/2023/QTR4/form.idx	2026-10-23T01:14:46Z	39211372	a6fde59991513226	added under D20 (A-23)
`sec_formidx_2024q1`	https://www.sec.gov/Archives/edgar/full-index/2024/QTR1/form.idx	2026-10-23T01:14:47Z	57770533	5f91755d27d18df0	added under D20 (A-24)
`sec_formidx_2024q2`	https://www.sec.gov/Archives/edgar/full-index/2024/QTR2/form.idx	2026-10-23T01:14:48Z	49302228	a213f3926c23261d	added under D20 (A-25)
`sec_formidx_2024q3`	https://www.sec.gov/Archives/edgar/full-index/2024/QTR3/form.idx	2026-10-23T01:14:49Z	42269441	2c6e48b5c6a26450	added under D20 (A-26)
`sec_formidx_2024q4`	https://www.sec.gov/Archives/edgar/full-index/2024/QTR4/form.idx	2026-10-23T01:14:50Z	44336440	a94001eef171b013	added under D20 (A-27)
`sec_formidx_2025q1`	https://www.sec.gov/Archives/edgar/full-index/2025/QTR1/form.idx	2026-10-23T01:14:52Z	53059957	becca6a54875c22d	added under D20 (A-28)
`sec_formidx_2025q2`	https://www.sec.gov/Archives/edgar/full-index/2025/QTR2/form.idx	2026-10-23T01:14:53Z	51759696	78a2722d7d483bfe	added under D20 (A-29)

key	URL	retrieved (UTC)	bytes	SHA-256 (first 16)	status
`sec_formidx_2025q3`	https://www.sec.gov/Archives/edgar/full-index/2025/QTR3/form.idx	2026-10-23T01:14:54Z	44912705	924d6992695d0337	added under D20 (A-30)
`sec_formidx_2025q4`	https://www.sec.gov/Archives/edgar/full-index/2025/QTR4/form.idx	2026-10-23T01:14:54Z	43155988	8555ad43d2a491e1	added under D20 (A-31)
`sec_fts/`	EDGAR full-text search, 307 records (https://efts.sec.gov/LATEST/search-index)	2026-10-23T01:35:03Z to 2026-10-23T23:36:05Z	13738487	set digest ac5a3a2966e29a47	added under D20 (A-33, A-47)
`sec_public_float_cy2024q2`	https://data.sec.gov/api/xbrl/frames/dei/EntityPublicFloat/USD/CY2024Q2I.json	2026-10-23T15:27:45Z	578973	d5ef310add8635a1	added under D20 (A-46)
`sec_submissions_bulk`	https://www.sec.gov/Archives/edgar/daily-index/bulkdata/submissions.zip	2026-10-23T01:19:58Z	1570500398	b6534bf3efaca6ea	added under D20 (A-32)
`sec_tickers`	https://www.sec.gov/files/company_tickers_exchange.json	2026-10-21T21:13:25Z	523851	6c29e2e3db973d44	frozen at `pap-v1`

- `sec_docs_index`: 47946 documents; SHA-256 of `sec/docs/INDEX.jsonl` `c416d6bcba7e41cf92d042d2ec5b09cea75ecd8bae5861f201044953ce3b5fed`.
- `sec_fts/`: 307 records; set digest (`paplog.set_digest`, SHA-256 of the sorted `<key> <sha256>` lines) `ac5a3a2966e29a4713af1cf83a4f497b7f702dd0dc1f18b69fb60370de501f12`.

E.2 Environment

Python 3.12 (`.python-version`; `uv.lock` requires-python `>=3.12`), with every direct dependency at its `uv.lock` version:

package	version	group
`duckdb`	2.0.0	runtime
`geopandas`	1.2.0	runtime
`matplotlib`	3.11.2	runtime
`numpy`	2.5.4	runtime
`openpyxl`	3.1.5	runtime
`pandas`	3.0.6	runtime
`polars`	2.0.0	runtime
`pyarrow`	26.0.0	runtime
`requests`	2.34.2	runtime
`scipy`	1.18.1	runtime
`statsmodels`	0.15.0	runtime
`xlrd`	2.0.2	runtime
`pypdf`	6.21.2	dev
`pytest`	9.1.1	dev
`ruff`	0.17.1	dev

E.3 Build targets (`Makefile`)

Each `make` target, what it depends on and what it runs, in the order of the Makefile. Network targets (`data`, `h6-data`) and the write-once confirmatory runs (`rq1-confirm`; `aideleg.confirm_h5`, which has no target) are listed as written and are never part of an offline rebuild.

target	depends on	runs
all	lint, confirmatory-check, panel, exposure, benchmarks, audit-check, power-block, results-rq1, results-rq2, rq3, h6, robustness, diagnostics, paper-includes, paper, brief, test	its prerequisites only
test	nothing	uv run pytest -v
lint	nothing	uv run ruff check
data	nothing	uv run python -m aideleg.acquire
confirmatory-check	nothing	uv run python -c "from aideleg.robustness import guards; guards.require_confirmatory_committed(); print('P-RQ1 run commit', guards.require_rq1_run_committed())"
panel	data/derived/aei_panel.parquet	its prerequisites only
data/derived/aei_panel.parquet	$(wildcard src/aideleg/aei/*.py), src/aideleg/aei/metric_map.csv, src/aideleg/config.py, src/aideleg/acquire/sources.py, src/aideleg/market/french.py, data/raw/MANIFEST.json	uv run python -m aideleg.aei
exposure	data/derived/exposure_ff49.parquet	its prerequisites only
data/derived/exposure_ff49.parquet	src/aideleg/exposure.py, src/aideleg/occupations.py, src/aideleg/aei/tasks.py, src/aideleg/config.py, src/aideleg/acquire/sources.py, src/aideleg/market/french.py, $(wildcard src/aideleg/bls/*.py), $(wildcard src/aideleg/crosswalk/*.py), data/raw/MANIFEST.json, data/derived/aei_panel.parquet	uv run python -m aideleg.exposure
benchmarks	data/derived/benchmark_ff49.parquet	its prerequisites only
data/derived/benchmark_ff49.parquet	src/aideleg/benchmarks.py, src/aideleg/exposure.py, src/aideleg/config.py, src/aideleg/acquire/sources.py, src/aideleg/market/french.py, $(wildcard src/aideleg/bls/*.py), $(wildcard src/aideleg/crosswalk/*.py), data/raw/MANIFEST.json, data/derived/exposure_ff49.parquet	uv run python -m aideleg.benchmarks
audit	data/derived/aei_panel.parquet, data/derived/exposure_ff49.parquet, data/derived/benchmark_ff49.parquet	uv run python -m aideleg.audit
audit-check	data/derived/aei_panel.parquet, data/derived/exposure_ff49.parquet, data/derived/benchmark_ff49.parquet	uv run python scripts/audit_check.py
power-unfrozen	nothing	uv run python -c "import sys; from aideleg.market.seal import pap_frozen; sys.exit('pap-v1 exists: the power analysis is frozen, not rerun' if pap_frozen() else 0)"
power	power-unfrozen, data/derived/exposure_ff49.parquet	uv run python -m aideleg.power
power-block	data/derived/exposure_ff49.parquet	uv run python scripts/power_block.py
freeze	nothing	uv run python scripts/pap_freeze.py

target	depends on	runs
rq1-dry-run	nothing	uv run python -m aideleg.rq1.confirm --dry-run $${TMPDIR:-/tmp}/aideleg-rq1-dry
rq1-confirm	nothing	uv run python -m aideleg.rq1.confirm
results-rq1	nothing	uv run python -m aideleg.results.rq1
results-rq2	nothing	uv run python -m aideleg.results.rq2
results	results-rq1, results-rq2	its prerequisites only
rq3	data/derived/exposure_ff49.parquet	uv run python -m aideleg.rq3 all
h6-data	nothing	uv run python -m aideleg.h6 acquire
h6	data/derived/exposure_ff49.parquet	uv run python -m aideleg.h6 all
robustness	data/derived/exposure_ff49.parquet	uv run python -m aideleg.robustness --offline; uv run python -m aideleg.robustness --summary
diagnostics	data/derived/exposure_ff49.parquet	uv run python -m aideleg.robustness --diagnostics
figures-check	nothing	uv run python -c "from pathlib import Path; d = Path('docs/paper/figures'); pdf = {p.stem for p in d.glob('*.pdf')}; png = {p.stem for p in d.glob('*.png')}; odd = sorted(pdf ^ png); print('figures without a PDF/PNG twin:', odd) if odd else None; raise SystemExit(1 if odd else 0)"
paper-includes	nothing	uv run python -m aideleg.paper
paper	paper-includes	quarto render docs/paper --to pdf; quarto render docs/paper --to html
brief	paper-includes	quarto render docs/paper/brief.qmd --to pdf

E.4 Seed

The seed is `aideleg.config.SEED`, equal to 20261021. Every random draw comes from `numpy.random.default_rng(SEED)` (PAP §0.2).

E.5 Replication

The steps to rebuild every result, table, figure and this paper from a clone (or an archive without `.git`) are in `README.md` at the repository root.

F Supplementary tables and figures

This appendix prints the generated tables and figures that the body cites but does not set in the text, so that the body keeps only its headline tables and the figures that carry a headline or confirmatory result. They are grouped by the section that discusses them (Section 3, Section 5, Section 6, Section 7, Section 9) and, within each group, follow the order in which that section first cites them. Each table and figure is printed unedited from its source file, under the label the body cites; the figures follow the tables, in the same grouping.

F.1 Data and measurement

Table 5: AEI releases: publication dates, sources, data windows, geography levels and taxonomies (data audit; coverage only)

release	published	source	period_start	period_end	window	geo_levels	taxonomies	metrics	artifact_ metrics	rows
r1	2025-02-10	claude_ai	2024-12-01	2025-01-01	DEFAULT (README states none)	global	onet_task, overall	9	0	3522
r2	2025-03-27	claude_ai	2025-02-24	2025-03-04	DEFAULT (README states none)	global	onet_task	9	0	30277
r3	2025-09-15	1p_api	2025-08-04	2025-08-11	as published	global	onet_task, overall, request	11	0	20495
r3	2025-09-15	claude_ai	2025-08-04	2025-08-11	as published	country, global, subregion	onet_task, overall, request	12	0	97754
r4	2026-01-15	1p_api	2025-11-13	2025-11-20	as published	global	onet_task, overall, request	21	0	42066
r4	2026-01-15	claude_ai	2025-11-13	2025-11-20	as published	country, global, subregion	onet_task, overall, request	22	0	199956
r5	2026-03-24	1p_api	2026-02-05	2026-02-12	as published	global	onet_task, overall, request	21	0	43891
r5	2026-03-24	claude_ai	2026-02-05	2026-02-12	as published	country, global, subregion	onet_task, overall, request	22	0	205847
r6	2026-06-26	1p_api	2026-04-01	2026-05-01	as published	global	onet_dwa, onet_gwa, onet_iwa, onet_task, overall, request, soc_ detailed, soc_ major	53	32	234919

release	published	source	period_start	period_end	window	geo_levels	taxonomies	metrics	artifact_ metrics	rows
r6	2026-06-26	1p_api	2026-05-01	2026-06-01	as published	global	onet_dwa, onet_gwa, onet_iwa, onet_task, overall, request, soc_ detailed, soc_ major	53	32	256786
r6	2026-06-26	claude_ai	2026-04-01	2026-05-01	as published	country, global, subregion	onet_dwa, onet_gwa, onet_iwa, onet_task, overall, request, soc_ detailed, soc_ major	53	32	743769
r6	2026-06-26	claude_ai	2026-05-01	2026-06-01	as published	country, global, subregion	onet_dwa, onet_gwa, onet_iwa, onet_task, overall, request, soc_ detailed, soc_ major	53	32	892804

Source: `docs/audit/tables/releases.md`, generated by `make audit`; the numbers are as printed there.

Table 6: AEI measurement regimes by release and source: platform scope, classifier, collaboration taxonomy, suppression and occupation basis (data audit; coverage only)

release	source	platform_scope	platform_source	classifier_model	collab_taxonomy	suppression_ construct	occupation_basis
r1	claude_ai	Claude.ai Free and Pro	release report (Handa et al. 2025); README and files state none	not stated in release files	collab_v1	no task-level collaboration; overall `none` share only	task-derived (O*NET tasks -> SOC)

release	source	platform_scope	platform_source	classifier_model	collab_taxonomy	suppression_construct	occupation_basis
r2	claude_ai	Claude.ai Free and Pro	release report (2025-03-27); README and files state none	not stated in release files	collab_v2	per-task `filtered` share (patterns renormalised over the rest)	task-derived (O*NET tasks -> SOC)
r3	1p_api	1P API	data/raw/aei/r3/data_documentation.md:116	not stated in release files	collab_v2	intersection cells < 15 conversations folded into `not_classified`	task-derived (O*NET tasks -> SOC)
r3	claude_ai	Claude AI (Free and Pro)	data/raw/aei/r3/data_documentation.md:25	not stated in release files	collab_v2	intersection cells < 15 conversations folded into `not_classified`	task-derived (O*NET tasks -> SOC)
r4	1p_api	1P API	data/raw/aei/r4/data_documentation.md:172	not stated in release files	collab_v2	intersection cells < 15 conversations folded into `not_classified`	task-derived (O*NET tasks -> SOC)
r4	claude_ai	Claude AI (Free and Pro)	data/raw/aei/r4/data_documentation.md:24	not stated in release files	collab_v2	intersection cells < 15 conversations folded into `not_classified`	task-derived (O*NET tasks -> SOC)
r5	1p_api	1P API	data/raw/aei/r5/data_documentation.md:170	not stated in release files	collab_v2	intersection cells < 15 conversations folded into `not_classified`	task-derived (O*NET tasks -> SOC)
r5	claude_ai	Claude AI (Free, Pro, and Max)	data/raw/aei/r5/data_documentation.md:24	not stated in release files	collab_v2	intersection cells < 15 conversations folded into `not_classified`	task-derived (O*NET tasks -> SOC)
r6	1p_api	Anthropic 1P API calls, excluding Claude Code.	data/raw/aei/r6/data_documentation.md:10	not stated in release files	collab_v2	cells below thresholds are absent; no `not_classified_pct`, only `none_pct`	published SOC (soc_detailed)
r6	claude_ai	Claude chat and Cowork (Free, Pro, and Max plans)	data/raw/aei/r6/data_documentation.md:9	not stated in release files	collab_v2	cells below thresholds are absent; no `not_classified_pct`, only `none_pct`	published SOC (soc_detailed)

Source: `docs/audit/tables/releases.md`, generated by `make audit`; as printed there.

Table 7: FF49 exposure coverage and the distribution of the industry `z_delegation` scores by design, with their skewness (data audit; coverage only)

design	industries	covered_min	covered_max	covered_ median	z_del_max	z_del_max_ industry	z_del_2nd	z_del_min	z_del_ skewness	z_usage_ skewness
A_pooled	49	0.555	1.000	0.865	4.320	Softw	2.855	-0.891	2.312	2.212
A_r2	49	0.680	0.988	0.890	4.739	Softw	1.977	-0.834	2.571	2.284
B	49	0.680	0.988	0.890	4.739	Softw	1.977	-0.834	2.571	2.284

Source: `docs/audit/tables/exposure_coverage.md`, generated by `make audit`; the numbers are as printed there.

F.2 Behaviour (RQ1)

Table 8: Collaboration patterns by occupation major group, r6 claude_ai (descriptive; conversations matched to O*NET tasks)

major	name	usage_share	directive_pct	feedback_ loop_pct	task_iteration_ pct	validation_pct	learning_pct	automation_ pct	augmentation_ pct
11	Management	5.96	31.20	12.73	39.54	2.70	13.84	43.92	56.08
13	Business and Financial Operations	5.81	30.31	12.48	34.11	3.01	20.09	42.80	57.20
15	Computer and Mathematical	24.70	33.37	30.52	22.96	1.70	11.44	63.89	36.11
17	Architecture and Engineering	3.68	23.39	21.22	25.06	5.90	24.42	44.62	55.38
19	Life, Physical, and Social Science	4.68	26.45	14.03	25.99	4.27	29.25	40.48	59.52
21	Community and Social Service	2.59	14.43	25.50	20.22	7.70	32.14	39.93	60.07
23	Legal	1.00	26.96	7.98	42.06	3.62	19.39	34.93	65.07
25	Educational Instruction and Library	12.77	37.37	5.55	26.07	3.36	27.66	42.91	57.09

major	name	usage_share	directive_pct	feedback_loop_pct	task_iteration_pct	validation_pct	learning_pct	automation_pct	augmentation_pct
27	Arts, Design, Entertainment, Sports, and Media	13.73	31.59	4.40	57.69	3.81	2.52	36.00	64.00
29	Healthcare Practitioners and Technical	3.23	20.30	21.27	13.32	3.22	41.89	41.57	58.43
31	Healthcare Support	0.61	30.32	8.23	10.56	2.83	48.07	38.54	61.46
33	Protective Service	0.35	63.05	12.37	10.01	1.79	12.79	75.42	24.58
35	Food Preparation and Serving Related	0.46	39.51	9.04	26.51	1.40	23.54	48.55	51.45
37	Building and Grounds Cleaning and Maintenance	0.13	29.87	17.31	37.33	3.40	12.09	47.18	52.82
39	Personal Care and Service	1.24	43.50	12.76	25.76	3.48	14.50	56.26	43.74
41	Sales and Related	9.02	26.77	16.74	28.09	2.71	25.69	43.51	56.49
43	Office and Administrative Support	7.89	47.36	9.21	32.60	3.37	7.46	56.56	43.44
45	Farming, Fishing, and Forestry	0.04	48.63	17.53	9.31	2.33	22.20	66.16	33.84
47	Construction and Extraction	0.10	55.51	11.25	26.27	1.38	5.59	66.75	33.25
49	Installation, Maintenance, and Repair	0.61	43.84	24.92	11.38	1.94	17.92	68.76	31.24
51	Production	1.07	19.49	44.76	21.25	5.24	9.27	64.24	35.76
53	Transportation and Material Moving	0.34	38.83	12.48	27.43	9.93	11.32	51.32	48.68

Source: `docs/results/rq1/anatomy.md`, committed unedited; the numbers are as printed there.

Table 9: Collaboration patterns by wage decile, r6 claude_ai, published basis (descriptive)

decile	emp_share	usage_share	n_occ	directive_pct	feedback_loop_pct	task_iteration_pct	validation_pct	learning_pct	automation_pct	augmentation_pct
1	9.98	2.32	28	28.07	8.58	17.26	2.48	43.61	36.65	63.35
2	9.89	1.33	29	40.45	14.03	23.09	3.24	19.19	54.48	45.52
3	10.11	5.92	56	26.63	12.76	27.43	3.17	30.02	39.38	60.62
4	9.85	4.99	63	46.05	6.87	26.06	2.50	18.52	52.92	47.08
5	10.18	5.19	96	40.62	9.27	32.66	3.57	13.88	49.89	50.11
6	9.94	8.36	145	34.37	14.03	26.42	5.50	19.66	48.41	51.59
7	10.20	14.47	92	31.23	14.69	30.47	3.04	20.57	45.92	54.08
8	9.88	15.83	118	30.08	10.56	46.42	3.36	9.59	40.64	59.36
9	10.82	14.28	93	27.11	22.81	33.73	3.40	12.94	49.92	50.08
10	9.07	27.13	106	33.51	22.77	24.39	2.37	16.95	56.29	43.71
—	0.07	0.18	5	27.78	23.52	47.29	0.40	1.02	51.30	48.70

Source: `docs/results/rq1/anatomy.md`, committed unedited; the numbers are as printed there.

Table 10: Collaboration patterns by required education, r6 claude_ai (descriptive)

bin	usage_share	n_tasks	directive_pct	feedback_loop_pct	task_iteration_pct	validation_pct	learning_pct	automation_pct	augmentation_pct
<12	45.38	432	33.15	15.17	27.94	2.69	21.06	48.28	51.72
12–14	38.45	388	31.09	16.79	34.57	3.03	14.53	47.88	52.12
14–16	14.66	299	35.56	20.30	31.36	4.17	8.61	55.86	44.14
16–18	1.49	51	26.86	10.25	41.94	8.73	12.22	37.11	62.89
≥18	0.02	2	16.93	27.18	35.73	12.20	7.96	44.11	55.89
zero published usage	0.00	1649	—	—	—	—	—	—	—

Source: `docs/results/rq1/anatomy.md`, committed unedited; the numbers are as printed there.

Table 11: Concentration of usage across tasks by release and source, as published (descriptive)

release	source	n_tasks	gini	top1_ share	top5_ share	top10_ share	n_tasks_ 50pct	n_zero	n_tasks_ pos	gini_pos	top1_ share_ pos	top5_ share_ pos	top10_ share_ pos
r1	claude_ai	3513	0.7754	37.0296	59.3315	70.7963	97	0	3513	0.7754	37.0296	59.3315	70.7963
r2	claude_ai	3364	0.8015	38.7585	62.7696	73.7679	73	0	3364	0.8015	38.7585	62.7696	73.7679
r3	claude_ai	2616	0.8225	38.5743	65.8054	76.6016	53	0	2616	0.8225	38.5743	65.8054	76.6016
r4	claude_ai	3168	0.8234	43.5899	67.4744	77.2345	50	0	3168	0.8234	43.5899	67.4744	77.2345
r5	claude_ai	3258	0.8078	38.2223	64.0534	75.1454	68	0	3258	0.8078	38.2223	64.0534	75.1454
r6	claude_ai	2821	0.8912	38.0292	69.4254	82.3301	54	1649	1172	0.7381	24.1223	51.9034	65.6899
r3	1p_api	2054	0.8421	42.4094	68.7397	79.5107	34	0	2054	0.8421	42.4094	68.7397	79.5107
r4	1p_api	2251	0.8483	47.7783	71.2131	80.7384	28	0	2251	0.8483	47.7783	71.2131	80.7384
r5	1p_api	2297	0.8515	47.6683	71.3635	80.9984	28	0	2297	0.8515	47.6683	71.3635	80.9984
r6	1p_api	2426	0.8926	40.2336	69.9994	82.9832	43	1349	1077	0.7582	25.2934	54.4212	67.6880

Source: `docs/results/rq1/anatomy.md`, committed unedited; the numbers are as printed there.

Table 12: Concentration of usage across tasks by release and source, on the harmonised task set (descriptive)

release	source	n_tasks	gini	top1_share	top5_share	top10_share	n_tasks_50pct
r1	claude_ai	1996	0.6861	30.8194	52.0174	63.1595	88
r2	claude_ai	1771	0.7078	31.3791	54.6827	65.5270	67
r3	claude_ai	1384	0.7349	29.8124	56.5766	68.4569	50
r4	claude_ai	1557	0.7348	34.1458	58.5971	69.3063	45
r5	claude_ai	1664	0.7143	29.3432	55.0921	66.4633	62
r6	claude_ai	1172	0.7381	24.1223	51.9034	65.6899	54
r3	1p_api	1102	0.7646	34.6846	59.2984	71.4485	32
r4	1p_api	1194	0.7769	39.5459	62.8287	73.7171	26
r5	1p_api	1166	0.7784	39.4504	63.0252	73.4613	25
r6	1p_api	1077	0.7582	25.2934	54.4212	67.6880	43

Source: `docs/results/rq1/anatomy.md`, committed unedited; the numbers are as printed there.

Table 13: H1 influence: leave-out by SOC major group and by the top occupations, a labelled post-result diagnostic (PAP §11 deviations)

part	dropped	dropped weight (%)	n rows	n occ	β_e	p_e	β_w	p_w	p_H1
primary	none	0.00	1684	388	-0.53	0.361	4.40	0.046	0.361
leave_out_soc_major	11	5.82	1535	357	-0.55	0.361	4.86	0.062	0.361
leave_out_soc_major	13	5.90	1517	360	-0.43	0.498	4.23	0.063	0.498
leave_out_soc_major	15	25.81	1367	368	-0.01	0.978	0.34	0.802	0.978
leave_out_soc_major	17	3.08	1573	362	-0.22	0.719	5.00	0.023	0.719
leave_out_soc_major	19	4.02	1598	360	-0.42	0.495	4.09	0.051	0.495
leave_out_soc_major	21	2.40	1647	378	-0.54	0.366	4.45	0.046	0.366
leave_out_soc_major	23	0.91	1667	382	-0.53	0.369	4.46	0.045	0.369
leave_out_soc_major	25	12.93	1410	326	-1.10	0.029	2.77	0.086	0.086
leave_out_soc_major	27	14.27	1582	365	-0.83	0.223	4.79	0.034	0.223
leave_out_soc_major	29	2.85	1599	357	-0.53	0.388	5.35	0.023	0.388
leave_out_soc_major	31	0.60	1675	384	-0.57	0.333	4.35	0.053	0.333
leave_out_soc_major	33	0.11	1676	383	-0.54	0.360	4.45	0.044	0.360
leave_out_soc_major	35	0.39	1676	381	-0.55	0.349	4.38	0.048	0.349
leave_out_soc_major	37	0.12	1681	386	-0.53	0.362	4.43	0.046	0.362
leave_out_soc_major	39	1.19	1648	372	-0.21	0.690	4.86	0.027	0.690
leave_out_soc_major	41	9.58	1594	371	-0.50	0.432	3.89	0.164	0.432
leave_out_soc_major	43	8.17	1556	358	-0.99	0.048	5.81	0.012	0.048
leave_out_soc_major	45	0.03	1680	385	-0.53	0.371	4.41	0.046	0.371

part	dropped	dropped weight (%)	n rows	n occ	β_e	p_e	β_w	p_w	p_H1
leave_out_soc_major	47	0.08	1676	383	-0.54	0.362	4.42	0.045	0.362
leave_out_soc_major	49	0.59	1668	374	-0.55	0.353	4.65	0.036	0.353
leave_out_soc_major	51	0.86	1669	376	-0.52	0.383	4.43	0.048	0.383
leave_out_soc_major	53	0.28	1670	380	-0.52	0.379	4.43	0.045	0.379
leave_out_top_1	15-1299	8.49	1612	387	-0.49	0.366	1.50	0.367	0.367
leave_out_top_3	15-1299, 25-4022, 27-3043	17.10	1593	385	-1.06	0.050	2.75	0.096	0.096

Source: `docs/results/diagnostics/h1_influence.md`, committed unedited; the numbers are as printed there.

Table 14: H2 influence: leave-one-release-out and the balanced-panel drop, a labelled post-result diagnostic (PAP §11 deviations)

part	dropped	releases	n tasks	n rows	usage share (%)	β	p (task-clustered)	p (wild)	note
primary	none	r2 r3 r4 r5	1315	5260	—	4.98	< 0.001	0.927	smallest attainable wild bootstrap p = 6/1296
leave_out_release	r2	r3 r4 r5	1315	3945	—	-6.55	< 0.001	0.204	smallest attainable wild bootstrap p = 6/216
leave_out_release	r3	r2 r4 r5	1315	3945	—	5.95	< 0.001	0.130	smallest attainable wild bootstrap p = 6/216
leave_out_release	r4	r2 r3 r5	1315	3945	—	5.22	< 0.001	0.806	smallest attainable wild bootstrap p = 6/216

part	dropped	releases	n tasks	n rows	usage share (%)	β	p (task-clustered)	p (wild)	note
leave_out_release	r5	r2 r3 r4	1315	3945	—	8.44	< 0.001	0.648	smallest attainable wild bootstrap p = 6/216
balancing_removed	tasks with a value in some but not all of the releases	r2 r3 r4 r5	890	2285	5.39	—	—	—	of 2205 tasks and 7545 rows in build_h2_panel(balanced=False); usage share = their fixed task weight in percent of that panel's task weight

Source: `docs/results/diagnostics/h2_influence.md`, committed unedited; the numbers are as printed there.

F.3 Bridge (RQ2)

Table 15: Top and bottom industries by `z_usage` with their benchmark exposures, Design B

rank	side	ff49_short	z_usage	z_aioe	z_eloundou	z_eloundou_gpt4	emp_covered_share
1	top	Softw	4.407	2.149	2.291	2.798	0.911
2	top	Books	2.154	1.043	1.236	1.386	0.890
3	top	Aero	1.963	1.194	0.965	1.151	0.906
4	top	Guns	1.945	1.139	0.929	1.106	0.905
5	top	Hardw	1.679	1.234	1.085	1.487	0.871
6	top	Telcm	1.270	1.308	1.607	1.571	0.900
7	top	Chips	0.958	0.729	0.332	0.833	0.808
8	top	LabEq	0.672	0.752	0.531	0.843	0.850
9	top	BusSv	0.606	0.586	0.750	0.634	0.890
10	top	Drugs	0.449	0.413	0.277	0.206	0.817
40	bottom	Paper	-0.660	-0.865	-0.914	-0.819	0.861
41	bottom	Trans	-0.691	-0.783	-0.319	-0.295	0.842
42	bottom	Boxes	-0.701	-0.946	-0.999	-0.926	0.835
43	bottom	Food	-0.706	-1.132	-1.201	-1.306	0.887
44	bottom	Other	-0.769	-1.035	-0.839	-0.968	0.906

rank	side	ff49_short	z_usage	z_aioe	z_eloundou	z_eloundou_gpt4	emp_covered_share
45	bottom	Smoke	-0.785	-0.543	-0.559	-0.700	0.848
46	bottom	Coal	-0.790	-1.181	-1.112	-1.160	0.923
47	bottom	Cnstr	-0.805	-0.910	-0.858	-0.816	0.884
48	bottom	Rtail	-0.808	-0.337	0.136	-0.465	0.953
49	bottom	Meals	-0.852	-1.087	-1.096	-1.443	0.916

Source: `docs/results/rq2/top_bottom.md`, committed unedited; the numbers are as printed there.

Table 16: Top and bottom industries by `z_delegation` with their benchmark exposures, Design B

rank	side	ff49_short	z_delegation	z_aioe	z_eloundou	z_eloundou_gpt4	emp_covered_share
1	top	Softw	4.739	2.149	2.291	2.798	0.911
2	top	Books	1.977	1.043	1.236	1.386	0.890
3	top	Hardw	1.768	1.234	1.085	1.487	0.871
4	top	Aero	1.656	1.194	0.965	1.151	0.906
5	top	Guns	1.618	1.139	0.929	1.106	0.905
6	top	Telcm	1.238	1.308	1.607	1.571	0.900
7	top	Chips	1.106	0.729	0.332	0.833	0.808
8	top	LabEq	0.807	0.752	0.531	0.843	0.850
9	top	BusSv	0.596	0.586	0.750	0.634	0.890
10	top	Fun	0.387	-0.142	0.226	-0.327	0.908
40	bottom	Soda	-0.657	-0.770	-0.664	-0.922	0.897
41	bottom	Beer	-0.676	-0.756	-0.552	-0.817	0.908
42	bottom	Trans	-0.678	-0.783	-0.319	-0.295	0.842
43	bottom	Food	-0.706	-1.132	-1.201	-1.306	0.887
44	bottom	Other	-0.769	-1.035	-0.839	-0.968	0.906
45	bottom	Smoke	-0.770	-0.543	-0.559	-0.700	0.848
46	bottom	Coal	-0.792	-1.181	-1.112	-1.160	0.923
47	bottom	Cnstr	-0.796	-0.910	-0.858	-0.816	0.884
48	bottom	Rtail	-0.810	-0.337	0.136	-0.465	0.953
49	bottom	Meals	-0.834	-1.087	-1.096	-1.443	0.916

Source: `docs/results/rq2/top_bottom.md`, committed unedited; the numbers are as printed there.

Table 17: Top and bottom industries by `z_augmentation` with their benchmark exposures, Design B

rank	side	ff49_short	z_augmentation	z_aioe	z_eloundou	z_eloundou_gpt4	emp_covered_share
1	top	Softw	4.044	2.149	2.291	2.798	0.911
2	top	Books	2.159	1.043	1.236	1.386	0.890
3	top	Aero	2.096	1.194	0.965	1.151	0.906
4	top	Guns	2.087	1.139	0.929	1.106	0.905
5	top	Hardw	1.752	1.234	1.085	1.487	0.871
6	top	Chips	1.240	0.729	0.332	0.833	0.808
7	top	Telcm	1.219	1.308	1.607	1.571	0.900
8	top	Drugs	0.837	0.413	0.277	0.206	0.817
9	top	LabEq	0.739	0.752	0.531	0.843	0.850
10	top	BusSv	0.582	0.586	0.750	0.634	0.890
40	bottom	Trans	-0.703	-0.783	-0.319	-0.295	0.842
41	bottom	Paper	-0.703	-0.865	-0.914	-0.819	0.861
42	bottom	Food	-0.725	-1.132	-1.201	-1.306	0.887
43	bottom	Boxes	-0.738	-0.946	-0.999	-0.926	0.835
44	bottom	Other	-0.795	-1.035	-0.839	-0.968	0.906
45	bottom	Smoke	-0.808	-0.543	-0.559	-0.700	0.848
46	bottom	Coal	-0.832	-1.181	-1.112	-1.160	0.923
47	bottom	Cnstr	-0.843	-0.910	-0.858	-0.816	0.884
48	bottom	Rtail	-0.846	-0.337	0.136	-0.465	0.953
49	bottom	Meals	-0.902	-1.087	-1.096	-1.443	0.916

Source: `docs/results/rq2/top_bottom.md`, committed unedited; the numbers are as printed there.

Table 18: Rank gaps, `z_delegation` against `z_eloundou`, the headline benchmark, Design B

ff49_short	rank_does	rank_could	gap
Rtail	48	20	28
RlEst	33	8	25
Whlsl	31	11	20
Hlth	36	18	18
Trans	42	24	18
Smoke	45	31	14
Fin	14	1	13
Banks	15	4	11
Beer	41	30	11
Insur	13	2	11
Boxes	38	46	-8

ff49_short	rank_does	rank_could	gap
Clths	25	33	-8
Agric	23	32	-9
MedEq	12	21	-9
BldMt	27	37	-10
Mach	17	27	-10
Steel	34	44	-10
FabPr	30	41	-11
Autos	28	45	-17
Txtls	16	43	-27

Source: `docs/results/rq2/rank_gaps.md`, committed unedited; the numbers are as printed there.

Table 19: Rank gaps, `z_delegation` against `z_eloundou_gpt4`, Design B

ff49_short	rank_does	rank_could	gap
RlEst	33	12	21
Rtail	48	29	19
Hlth	36	18	18
Trans	42	24	18
Whlsl	31	14	17
Banks	15	3	12
Insur	13	2	11
Smoke	45	34	11
Fin	14	4	10
Cnstr	47	40	7
Drugs	11	16	-5
Food	43	48	-5
Paper	37	42	-5
Steel	34	39	-5
Boxes	38	44	-6
Gold	39	46	-7
Ships	29	37	-8
Agric	23	35	-12
Fun	10	25	-15
Txtls	16	38	-22

Source: `docs/results/rq2/rank_gaps.md`, committed unedited; the numbers are as printed there.

Table 20: Rank gaps, `z_delegation` against `z_aioe`, Design B

ff49_short	rank_does	rank_could	gap
RlEst	33	10	23
Rtail	48	26	22
Hlth	36	15	21
Smoke	45	30	15
Banks	15	3	12
Fin	14	2	12
Insur	13	1	12
Whlsl	31	19	12
Beer	41	32	9
PerSv	20	11	9
FabPr	30	36	-6
MedEq	12	18	-6
Autos	28	35	-7
Books	2	9	-7
Steel	34	41	-7
Mach	17	27	-10
Fun	10	22	-12
BldMt	27	40	-13
Agric	23	39	-16
Txtls	16	45	-29

Source: `docs/results/rq2/rank_gaps.md`, committed unedited; the numbers are as printed there.

Table 21: Delegation exposure with its suppression bounds by industry, Design B (PAP §2)

ff49_short	E_delegation	E_delegation_lb	E_delegation_ub	width	width_pct_of_E
Other	0.083	0.073	0.108	0.035	41.686
Rtail	0.058	0.053	0.077	0.024	41.418
Coal	0.069	0.061	0.085	0.024	35.137
Cnstr	0.067	0.061	0.082	0.021	31.481
Ships	0.289	0.262	0.348	0.086	29.883
Gold	0.159	0.145	0.189	0.044	27.667
Meals	0.043	0.038	0.050	0.012	27.414
Smoke	0.083	0.075	0.098	0.023	27.213
Hlth	0.186	0.172	0.221	0.049	26.368
Trans	0.140	0.128	0.164	0.037	26.074
Beer	0.142	0.130	0.165	0.035	24.860

ff49_short	E_delegation	E_delegation_lb	E_delegation_ub	width	width_pct_of_E
RlEst	0.216	0.196	0.247	0.051	23.578
Oil	0.271	0.250	0.312	0.062	22.987
Soda	0.153	0.143	0.177	0.034	22.421
Util	0.473	0.432	0.535	0.103	21.763
Food	0.123	0.115	0.140	0.025	20.254
Mines	0.312	0.292	0.355	0.063	20.090
Agric	0.378	0.355	0.424	0.069	18.287
Chems	0.417	0.391	0.464	0.073	17.556
Drugs	0.754	0.718	0.846	0.128	16.923
Toys	0.489	0.458	0.539	0.081	16.534
Rubbr	0.209	0.197	0.230	0.033	15.783
Whlsl	0.272	0.257	0.300	0.042	15.605
PerSv	0.467	0.441	0.513	0.072	15.454
Fun	0.808	0.746	0.867	0.121	15.006
Steel	0.214	0.203	0.235	0.032	14.829
Paper	0.180	0.170	0.196	0.026	14.412
ElcEq	0.450	0.428	0.493	0.064	14.270
Banks	0.614	0.584	0.670	0.085	13.904
Fin	0.635	0.604	0.692	0.087	13.732
Boxes	0.167	0.160	0.181	0.022	13.120
MedEq	0.690	0.661	0.750	0.089	12.833
Guns	1.579	1.511	1.709	0.198	12.549
Autos	0.290	0.278	0.314	0.036	12.482
Hshld	0.333	0.317	0.359	0.041	12.440
Clths	0.331	0.313	0.354	0.041	12.430
Aero	1.603	1.534	1.734	0.199	12.427
Telcm	1.340	1.274	1.438	0.165	12.279
Insur	0.638	0.607	0.681	0.074	11.645
FabPr	0.272	0.261	0.291	0.030	11.050
LabEq	1.071	1.030	1.146	0.116	10.852
Chips	1.258	1.214	1.343	0.129	10.281
Mach	0.557	0.537	0.592	0.056	10.025
BusSv	0.938	0.903	0.996	0.093	9.959
BldMt	0.297	0.285	0.314	0.029	9.901
Books	1.804	1.738	1.892	0.154	8.524
Hardw	1.673	1.622	1.762	0.139	8.322
Txtls	0.566	0.551	0.585	0.034	6.044
Softw	3.534	3.453	3.654	0.202	5.707

Source: `docs/results/rq2/bounds.md`, committed unedited; the numbers are as printed there.

Table 22: Could versus does across FF49 industries, Design `A_pooled` (exploratory; look-ahead, PAP §10 S3)

aei	benchmark	n	spearman	ci_low	ci_high	n_draws_used
z_usage	z_aioe	49	0.867	0.754	0.930	10000
z_usage	z_eloundou	49	0.846	0.723	0.914	10000
z_usage	z_eloundou_gpt4	49	0.888	0.789	0.939	10000
z_delegation	z_aioe	49	0.813	0.666	0.897	10000
z_delegation	z_eloundou	49	0.770	0.597	0.872	10000
z_delegation	z_eloundou_gpt4	49	0.855	0.734	0.923	10000
z_augmentation	z_aioe	49	0.858	0.747	0.919	10000
z_augmentation	z_eloundou	49	0.815	0.674	0.896	10000
z_augmentation	z_eloundou_gpt4	49	0.876	0.776	0.931	10000

Source: `docs/results/rq2/could_vs_does.md`, committed unedited; the numbers are as printed there.

F.4 Market evidence (RQ3 and H6)

Table 23: Influence of Softw on the confirmatory H5 and H5b fits (PAP §3.3 P11; not part of the decision)

specification	DFBETAS on b	Cook's D	leverage
H5	0.1475	0.0116	0.4788
H5b	-0.0621	0.0017	0.5185

Source: `docs/results/rq3/influence.md`, committed unedited; the numbers are as printed there.

Table 24: Industries with the largest Cook's distance per specification (PAP §3.3 P11; not part of the decision)

specification	rank	industry	Cook's D	DFBETAS on b	leverage
H5	1	Gold	0.1912	-0.3903	0.0290
H5	2	Hardw	0.1714	0.5254	0.0842
H5	3	Coal	0.0664	-0.2338	0.0332
H5	4	Clths	0.0495	0.1142	0.0233
H5	5	Smoke	0.0349	-0.1632	0.0325
H5b	1	Gold	0.1644	-0.0110	0.0387
H5b	2	Hardw	0.1145	0.3771	0.0843
H5b	3	Coal	0.0599	0.0002	0.0489
H5b	4	Insur	0.0591	0.2568	0.2187

specification	rank	industry	Cook's D	DFBETAS on b	leverage
H5b	5	Clths	0.0389	0.0059	0.0263

Source: `docs/results/rq3/influence.md`, committed unedited; the numbers are as printed there.

Table 25: Event dates under the D19 rule (exploratory, PAP §8.3)

event	announced
ChatGPT	2022-11-30
GPT-4	2023-03-14
PaLM 2	2023-05-10
Claude 2	2023-07-11
Claude 2.1	2023-11-21
Gemini 1.0	2023-12-06
Gemini 1.5	2024-02-15
Claude 3	2024-03-04
Claude 3.5	2024-06-20
OpenAI o1	2024-09-12
Gemini 2.0	2025-02-05
Claude 3.7	2025-02-24
GPT-4.5	2025-02-27
Gemini 2.5	2025-03-25

Source: `docs/results/rq3/event_study.md`, committed unedited; the numbers are as printed there.

Table 26: Event days: announcement, day zero, moved dates and overlaps (D19 disclosure)

event	announced	day 0	moved	overlaps
ChatGPT	2022-11-30	2022-11-30	no	-
GPT-4	2023-03-14	2023-03-14	no	-
PaLM 2	2023-05-10	2023-05-10	no	-
Claude 2	2023-07-11	2023-07-11	no	-
Claude 2.1	2023-11-21	2023-11-21	no	Gemini 1.0
Gemini 1.0	2023-12-06	2023-12-06	no	-
Gemini 1.5	2024-02-15	2024-02-15	no	Claude 3
Claude 3	2024-03-04	2024-03-04	no	-
Claude 3.5	2024-06-20	2024-06-20	no	-

event	announced	day 0	moved	overlaps
OpenAI o1	2024-09-12	2024-09-12	no	-
Gemini 2.0	2025-02-05	2025-02-05	no	Claude 3.7, GPT-4.5
Claude 3.7	2025-02-24	2025-02-24	no	GPT-4.5
GPT-4.5	2025-02-27	2025-02-27	no	Gemini 2.5
Gemini 2.5	2025-03-25	2025-03-25	no	-

Source: `docs/results/rq3/event_study.md`, committed unedited; the numbers are as printed there.

Table 27: Event-study slopes of cumulative abnormal returns on exposure at both horizons, HC3 standard errors (exploratory, PAP §8.3 D19)

event	day 0	horizon k	design	regressors	regressor	b (pp per 1 SD)	HC3 SE	95% CI	p (nominal)	n
ChatGPT	2022-11-30	5	A_r2	z_delegation	z_delegation	-0.317	0.243	[-0.793, 0.159]	0.1923	49
ChatGPT	2022-11-30	5	A_pooled (look-ahead)	z_delegation	z_delegation	-0.387	0.236	[-0.849, 0.075]	0.1010	49
ChatGPT	2022-11-30	20	A_r2	z_delegation	z_delegation	0.510	0.554	[-0.575, 1.595]	0.3569	49
ChatGPT	2022-11-30	20	A_pooled (look-ahead)	z_delegation	z_delegation	0.260	0.545	[-0.808, 1.329]	0.6332	49
GPT-4	2023-03-14	5	A_r2	z_delegation	z_delegation	0.595	0.182	[0.239, 0.952]	0.0011	49
GPT-4	2023-03-14	5	A_pooled (look-ahead)	z_delegation	z_delegation	0.662	0.179	[0.311, 1.013]	0.0002	49
GPT-4	2023-03-14	20	A_r2	z_delegation	z_delegation	0.313	0.381	[-0.434, 1.059]	0.4118	49
GPT-4	2023-03-14	20	A_pooled (look-ahead)	z_delegation	z_delegation	0.410	0.411	[-0.397, 1.216]	0.3195	49
PaLM 2	2023-05-10	5	A_r2	z_delegation	z_delegation	1.158	0.229	[0.710, 1.607]	< 0.0001	49
PaLM 2	2023-05-10	5	A_pooled (look-ahead)	z_delegation	z_delegation	1.219	0.234	[0.761, 1.678]	< 0.0001	49
PaLM 2	2023-05-10	20	A_r2	z_delegation	z_delegation	2.092	0.657	[0.804, 3.380]	0.0015	49
PaLM 2	2023-05-10	20	A_pooled (look-ahead)	z_delegation	z_delegation	2.188	0.563	[1.084, 3.293]	0.0001	49
Claude 2	2023-07-11	5	A_r2	z_delegation	z_delegation	0.396	0.472	[-0.529, 1.322]	0.4011	49
Claude 2	2023-07-11	5	A_pooled (look-ahead)	z_delegation	z_delegation	0.446	0.426	[-0.389, 1.281]	0.2948	49

event	day 0	horizon k	design	regressors	regressor	b (pp per 1 SD)	HC3 SE	95% CI	p (nominal)	n
Claude 2	2023-07-11	20	A_r2	z_delegation	z_delegation	0.317	0.352	[-0.373, 1.008]	0.3674	49
Claude 2	2023-07-11	20	A_pooled (look-ahead)	z_delegation	z_delegation	0.386	0.329	[-0.258, 1.031]	0.2400	49
Claude 2.1	2023-11-21	5	A_r2	z_delegation	z_delegation	-0.048	0.200	[-0.439, 0.344]	0.8119	49
Claude 2.1	2023-11-21	5	A_pooled (look-ahead)	z_delegation	z_delegation	-0.036	0.212	[-0.451, 0.380]	0.8668	49
Claude 2.1	2023-11-21	20	A_r2	z_delegation	z_delegation	0.276	0.409	[-0.526, 1.077]	0.5001	49
Claude 2.1	2023-11-21	20	A_pooled (look-ahead)	z_delegation	z_delegation	0.123	0.480	[-0.818, 1.065]	0.7975	49
Gemini 1.0	2023-12-06	5	A_r2	z_delegation	z_delegation	0.240	0.366	[-0.477, 0.958]	0.5117	49
Gemini 1.0	2023-12-06	5	A_pooled (look-ahead)	z_delegation	z_delegation	0.149	0.320	[-0.478, 0.776]	0.6416	49
Gemini 1.0	2023-12-06	20	A_r2	z_delegation	z_delegation	0.051	0.378	[-0.691, 0.793]	0.8930	49
Gemini 1.0	2023-12-06	20	A_pooled (look-ahead)	z_delegation	z_delegation	-0.044	0.313	[-0.657, 0.568]	0.8874	49
Gemini 1.5	2024-02-15	5	A_r2	z_delegation	z_delegation	-0.692	0.174	[-1.033, -0.351]	< 0.0001	49
Gemini 1.5	2024-02-15	5	A_pooled (look-ahead)	z_delegation	z_delegation	-0.680	0.195	[-1.063, -0.297]	0.0005	49
Gemini 1.5	2024-02-15	20	A_r2	z_delegation	z_delegation	-0.908	0.494	[-1.875, 0.060]	0.0660	49
Gemini 1.5	2024-02-15	20	A_pooled (look-ahead)	z_delegation	z_delegation	-1.093	0.459	[-1.993, -0.192]	0.0174	49
Claude 3	2024-03-04	5	A_r2	z_delegation	z_delegation	-0.161	0.238	[-0.627, 0.306]	0.4994	49
Claude 3	2024-03-04	5	A_pooled (look-ahead)	z_delegation	z_delegation	-0.236	0.233	[-0.692, 0.220]	0.3103	49
Claude 3	2024-03-04	20	A_r2	z_delegation	z_delegation	-0.559	0.614	[-1.761, 0.644]	0.3624	49
Claude 3	2024-03-04	20	A_pooled (look-ahead)	z_delegation	z_delegation	-0.704	0.619	[-1.917, 0.508]	0.2550	49
Claude 3.5	2024-06-20	5	A_r2	z_delegation	z_delegation	0.309	0.286	[-0.252, 0.870]	0.2810	49
Claude 3.5	2024-06-20	5	A_pooled (look-ahead)	z_delegation	z_delegation	0.403	0.223	[-0.035, 0.840]	0.0714	49

event	day 0	horizon k	design	regressors	regressor	b (pp per 1 SD)	HC3 SE	95% CI	p (nominal)	n
Claude 3.5	2024-06-20	20	A_r2	z_delegation	z_delegation	-0.467	0.641	[-1.724, 0.789]	0.4658	49
Claude 3.5	2024-06-20	20	A_pooled (look-ahead)	z_delegation	z_delegation	-0.223	0.465	[-1.134, 0.688]	0.6317	49
OpenAI o1	2024-09-12	5	A_r2	z_delegation	z_delegation	-0.064	0.372	[-0.794, 0.665]	0.8626	49
OpenAI o1	2024-09-12	5	A_pooled (look-ahead)	z_delegation	z_delegation	-0.060	0.307	[-0.662, 0.542]	0.8460	49
OpenAI o1	2024-09-12	20	A_r2	z_delegation	z_delegation	-0.028	0.472	[-0.953, 0.898]	0.9533	49
OpenAI o1	2024-09-12	20	A_pooled (look-ahead)	z_delegation	z_delegation	-0.136	0.496	[-1.108, 0.835]	0.7833	49
Gemini 2.0	2025-02-05	5	A_r2	z_delegation	z_delegation	-0.102	0.331	[-0.751, 0.546]	0.7572	49
Gemini 2.0	2025-02-05	5	A_pooled (look-ahead)	z_delegation	z_delegation	-0.189	0.320	[-0.816, 0.439]	0.5554	49
Gemini 2.0	2025-02-05	20	A_r2	z_delegation	z_delegation	-0.064	0.607	[-1.254, 1.126]	0.9162	49
Gemini 2.0	2025-02-05	20	A_pooled (look-ahead)	z_delegation	z_delegation	-0.225	0.567	[-1.336, 0.887]	0.6919	49
Claude 3.7	2025-02-24	5	A_r2	z_delegation	z_delegation	-0.244	0.372	[-0.973, 0.485]	0.5122	49
Claude 3.7	2025-02-24	5	A_pooled (look-ahead)	z_delegation	z_delegation	-0.225	0.361	[-0.933, 0.483]	0.5336	49
Claude 3.7	2025-02-24	20	A_r2	z_delegation	z_delegation	-0.342	0.525	[-1.372, 0.688]	0.5148	49
Claude 3.7	2025-02-24	20	A_pooled (look-ahead)	z_delegation	z_delegation	-0.403	0.516	[-1.414, 0.608]	0.4343	49
GPT-4.5	2025-02-27	5	A_r2	z_delegation	z_delegation	0.075	0.264	[-0.443, 0.593]	0.7774	49
GPT-4.5	2025-02-27	5	A_pooled (look-ahead)	z_delegation	z_delegation	0.124	0.314	[-0.491, 0.740]	0.6922	49
GPT-4.5	2025-02-27	20	A_r2	z_delegation	z_delegation	-0.654	0.614	[-1.858, 0.550]	0.2869	49
GPT-4.5	2025-02-27	20	A_pooled (look-ahead)	z_delegation	z_delegation	-0.630	0.565	[-1.737, 0.477]	0.2649	49
Gemini 2.5	2025-03-25	5	A_r2	z_delegation	z_delegation	-0.577	0.240	[-1.047, -0.107]	0.0162	49
Gemini 2.5	2025-03-25	5	A_pooled (look-ahead)	z_delegation	z_delegation	-0.464	0.249	[-0.952, 0.023]	0.0621	49

event	day 0	horizon k	design	regressors	regressor	b (pp per 1 SD)	HC3 SE	95% CI	p (nominal)	n
Gemini 2.5	2025-03-25	20	A_r2	z_delegation	z_delegation	-1.246	0.765	[-2.747, 0.254]	0.1034	49
Gemini 2.5	2025-03-25	20	A_pooled (look-ahead)	z_delegation	z_delegation	-1.121	0.712	[-2.516, 0.274]	0.1153	49

Source: `docs/results/rq3/event_study.md`, committed unedited; the numbers are as printed there.

Table 28: Tercile high-minus-low portfolio on `z_delegation`, Design B (secondary), cap-weighted legs: means and alphas (PAP §8.3 D19)

statistic	estimate	NW SE	95% CI	p (nominal)	T
alpha CAPM (annualised pp)	2.393	7.794	[-14.218, 19.005]	0.7630	17
alpha FF3 (annualised pp)	12.906	8.720	[-5.933, 31.745]	0.1627	17
alpha FF5+UMD (annualised pp)	0.889	4.340	[-8.782, 10.560]	0.8419	17
mean H-L (annualised pp)	15.489	10.784	[-5.182, 34.083] (bootstrap)	0.1702	17
Sharpe (annualised)	1.168	-	[-0.403, 3.170] (bootstrap)	-	17
PSR(0)	0.9120	-	-	-	17

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 29: Tercile high-minus-low portfolio after one-way trading costs, Design B (secondary), cap-weighted legs

cost (bps one-way)	mean H-L (annualised pp)	Sharpe (annualised)
0	15.489	1.168
5	15.418	1.162
15	15.277	1.150
30	15.065	1.131

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 30: Tercile high-minus-low portfolio on `z_delegation`, Design B (secondary), equal-weighted legs: means and alphas (PAP §8.3 D19)

statistic	estimate	NW SE	95% CI	p (nominal)	T
alpha CAPM (annualised pp)	-6.653	7.293	[-22.199, 8.893]	0.3761	17
alpha FF3 (annualised pp)	-1.720	5.918	[-14.506, 11.065]	0.7759	17
alpha FF5+UMD (annualised pp)	-5.193	5.533	[-17.523, 7.136]	0.3701	17
mean H-L (annualised pp)	-0.838	7.859	[-16.004, 12.977] (bootstrap)	0.9164	17
Sharpe (annualised)	-0.093	-	[-1.699, 1.945] (bootstrap)	-	17
PSR(0)	0.4571	-	-	-	17

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 31: Tercile high-minus-low portfolio after one-way trading costs, Design B (secondary), equal-weighted legs

cost (bps one-way)	mean H-L (annualised pp)	Sharpe (annualised)
0	-0.838	-0.093
5	-0.908	-0.100
15	-1.050	-0.116
30	-1.261	-0.139

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 32: Tercile high-minus-low portfolio on `z_delegation`, Design `A_r2` (exploratory), cap-weighted legs: means and alphas (PAP §8.3 D19)

statistic	estimate	NW SE	95% CI	p (nominal)	T
alpha CAPM (annualised pp)	4.642	3.661	[-2.741, 12.025]	0.2116	45
alpha FF3 (annualised pp)	5.123	3.120	[-1.177, 11.423]	0.1082	45
alpha FF5+UMD (annualised pp)	3.710	3.016	[-2.396, 9.816]	0.2262	45
mean H-L (annualised pp)	7.975	4.678	[-0.805, 16.680] (bootstrap)	0.0953	45
Sharpe (annualised)	0.822	-	[-0.091, 1.705] (bootstrap)	-	45

statistic	estimate	NW SE	95% CI	p (nominal)	T
PSR(0)	0.9515	-	-	-	45

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 33: Tercile high-minus-low portfolio after one-way trading costs, Design `A_r2` (exploratory), cap-weighted legs

cost (bps one-way)	mean H-L (annualised pp)	Sharpe (annualised)
0	7.975	0.822
5	7.948	0.819
15	7.895	0.813
30	7.815	0.803

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 34: Tercile high-minus-low portfolio on `z_delegation`, Design `A_r2` (exploratory), equal-weighted legs: means and alphas (PAP §8.3 D19)

statistic	estimate	NW SE	95% CI	p (nominal)	T
alpha CAPM (annualised pp)	-0.004	3.873	[-7.814, 7.806]	0.9991	45
alpha FF3 (annualised pp)	0.857	3.592	[-6.396, 8.111]	0.8125	45
alpha FF5+UMD (annualised pp)	-0.140	3.798	[-7.828, 7.548]	0.9708	45
mean H-L (annualised pp)	2.649	3.885	[-4.942, 9.842] (bootstrap)	0.4989	45
Sharpe (annualised)	0.336	-	[-0.596, 1.386] (bootstrap)	-	45
PSR(0)	0.7355	-	-	-	45

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 35: Tercile high-minus-low portfolio after one-way trading costs, Design `A_r2` (exploratory), equal-weighted legs

cost (bps one-way)	mean H-L (annualised pp)	Sharpe (annualised)
0	2.649	0.336
5	2.623	0.332

cost (bps one-way)	mean H-L (annualised pp)	Sharpe (annualised)
15	2.569	0.325
30	2.489	0.315

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 36: Tercile high-minus-low portfolio on `z_delegation`, Design `A_pooled` (exploratory; look-ahead), cap-weighted legs: means and alphas (PAP §8.3 D19)

statistic	estimate	NW SE	95% CI	p (nominal)	T
alpha CAPM (annualised pp)	4.599	3.644	[-2.749, 11.948]	0.2137	45
alpha FF3 (annualised pp)	5.075	3.103	[-1.191, 11.341]	0.1096	45
alpha FF5+UMD (annualised pp)	3.664	3.006	[-2.421, 9.749]	0.2304	45
mean H-L (annualised pp)	7.921	4.656	[-0.427, 16.378] (bootstrap)	0.0960	45
Sharpe (annualised)	0.820	-	[-0.048, 1.675] (bootstrap)	-	45
PSR(0)	0.9511	-	-	-	45

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 37: Tercile high-minus-low portfolio after one-way trading costs, Design `A_pooled` (exploratory; look-ahead), cap-weighted legs

cost (bps one-way)	mean H-L (annualised pp)	Sharpe (annualised)
0	7.921	0.820
5	7.894	0.817
15	7.841	0.811
30	7.761	0.802

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 38: Tercile high-minus-low portfolio on `z_delegation`, Design `A_pooled` (exploratory; look-ahead), equal-weighted legs: means and alphas (PAP §8.3 D19)

statistic	estimate	NW SE	95% CI	p (nominal)	T
alpha CAPM (annualised pp)	0.523	4.096	[-7.738, 8.784]	0.8990	45
alpha FF3 (annualised pp)	0.857	3.705	[-6.626, 8.340]	0.8182	45
alpha FF5+UMD (annualised pp)	-0.409	3.880	[-8.263, 7.445]	0.9166	45
mean H-L (annualised pp)	2.662	3.988	[-5.329, 10.327] (bootstrap)	0.5079	45
Sharpe (annualised)	0.317	-	[-0.605, 1.382] (bootstrap)	-	45
PSR(0)	0.7246	-	-	-	45

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 39: Tercile high-minus-low portfolio after one-way trading costs, Design `A_pooled` (exploratory; look-ahead), equal-weighted legs

cost (bps one-way)	mean H-L (annualised pp)	Sharpe (annualised)
0	2.662	0.317
5	2.636	0.314
15	2.582	0.308
30	2.502	0.298

Source: `docs/results/rq3/terciles.md`, committed unedited; the numbers are as printed there.

Table 40: Fama–MacBeth slopes, Design B (exploratory, PAP §8.3)

regressor	mean slope (pp per month)	x 12 (annualised pp)	NW SE	95% CI	p (nominal)	T	mean n	skipped months
z	0.233	2.797	0.300	[-0.402, 0.868]	0.4479	17	48.0	0
log_size	-0.271	-	0.196	[-0.687, 0.145]	0.1867	17	48.0	0
log_beme	0.077	-	0.300	[-0.559, 0.712]	0.8011	17	48.0	0
mom	2.004	-	1.586	[-1.357, 5.366]	0.2244	17	48.0	0

Source: `docs/results/rq3/fama_macbeth.md`, committed unedited; the numbers are as printed there.

Table 41: Fama–MacBeth slopes, Design `A_r2` (exploratory)

regressor	mean slope (pp per month)	x 12 (annualised pp)	NW SE	95% CI	p (nominal)	T	mean n	skipped months
z	0.208	2.499	0.119	[-0.032, 0.448]	0.0874	45	48.0	0
log_size	-0.081	-	0.156	[-0.395, 0.233]	0.6052	45	48.0	0
log_beme	0.011	-	0.190	[-0.372, 0.393]	0.9545	45	48.0	0
mom	0.221	-	0.858	[-1.508, 1.950]	0.7981	45	48.0	0

Source: `docs/results/rq3/fama_macbeth.md`, committed unedited; the numbers are as printed there.

Table 42: Fama–MacBeth slopes, Design `A_pooled` (exploratory; look-ahead)

regressor	mean slope (pp per month)	x 12 (annualised pp)	NW SE	95% CI	p (nominal)	T	mean n	skipped months
z	0.171	2.046	0.114	[-0.060, 0.401]	0.1435	45	48.0	0
log_size	-0.069	-	0.153	[-0.377, 0.239]	0.6552	45	48.0	0
log_beme	-0.003	-	0.192	[-0.391, 0.385]	0.9884	45	48.0	0
mom	0.245	-	0.857	[-1.482, 1.972]	0.7764	45	48.0	0

Source: `docs/results/rq3/fama_macbeth.md`, committed unedited; the numbers are as printed there.

Table 43: Eisfeldt, Schubert and Zhang replication, step one: cumulative abnormal return on `z_aioe` alone (exploratory)

event	day 0	horizon k	design	regressors	regressor	b (pp per 1 SD)	HC3 SE	95% CI	p (nominal)	n
ChatGPT	2022-11-30	9	none (z_aioe only)	z_aioe	z_aioe	-0.135	0.399	[-0.917, 0.647]	0.7357	49

Source: `docs/results/rq3/eisfeldt_replication.md`, committed unedited; the numbers are as printed there.

Table 44: Eisfeldt, Schubert and Zhang replication, step two: `z_aioe` with `z_delegation` (exploratory)

event	day 0	horizon k	design	regressors	regressor	b (pp per 1 SD)	HC3 SE	95% CI	p (nominal)	n
ChatGPT	2022-11-30	9	A_r2	z_aioe+z_delegation	z_aioe	-0.240	0.491	[-1.202, 0.722]	0.6251	49
ChatGPT	2022-11-30	9	A_r2	z_aioe+z_delegation	z_delegation	0.157	0.349	[-0.527, 0.841]	0.6533	49
ChatGPT	2022-11-30	9	A_pooled (look-ahead)	z_aioe+z_delegation	z_aioe	0.015	0.591	[-1.144, 1.174]	0.9799	49
ChatGPT	2022-11-30	9	A_pooled (look-ahead)	z_aioe+z_delegation	z_delegation	-0.213	0.440	[-1.075, 0.649]	0.6283	49

Source: `docs/results/rq3/eisfeldt_replication.md`, committed unedited; the numbers are as printed there.

Table 45: H6 talk index pooled across filers, by filing year (exploratory, PAP §6 D18)

year	filers	talk filers	talk, all filers	talk, mapped filers	unmapped filers (no SIC / SIC outside Siccodes49)	unmapped share	exhibit-only talk filers	hit CIKs not in the year's filers
2019	6793	558	0.082	0.093	810 (94 / 716)	11.92%	7	0
2020	6627	668	0.101	0.116	856 (90 / 766)	12.92%	3	0
2021	7001	927	0.132	0.152	883 (99 / 784)	12.61%	2	0
2022	7667	1338	0.175	0.198	922 (117 / 805)	12.03%	3	0
2023	7255	1438	0.198	0.227	947 (133 / 814)	13.05%	4	0
2024	6774	2538	0.375	0.429	978 (153 / 825)	14.44%	5	0
2025	6565	3386	0.516	0.591	1038 (168 / 870)	15.81%	6	0

Source: `docs/results/h6/talk_index.md`, committed unedited; the numbers are as printed there.

Table 46: H6 talk–use divergence: Spearman correlation of the talk index with Design B `z_usage` across FF49 industries, with its industry-bootstrap interval (exploratory, PAP §6 D18)

rho	95% CI	industries	bootstrap draws	undefined draws
0.361	[0.050, 0.612]	49	10000	0

Source: `docs/results/h6/gap.md`, committed unedited; the numbers are as printed there.

Table 47: H6 influence: leave-one-industry-out Spearman correlation with a nominal p-value, a labelled post-result diagnostic (PAP §11 deviations)

part	dropped	n	ρ	95% CI	p (nominal)
primary	none	49	0.361	[0.050, 0.612]	0.011
leave_out_industry	Aero	48	0.327	[0.007, 0.595]	—
leave_out_industry	Agric	48	0.380	[0.067, 0.632]	—
leave_out_industry	Autos	48	0.369	[0.054, 0.624]	—
leave_out_industry	Banks	48	0.364	[0.054, 0.624]	—
leave_out_industry	Beer	48	0.362	[0.047, 0.621]	—
leave_out_industry	BldMt	48	0.366	[0.059, 0.623]	—
leave_out_industry	Books	48	0.322	[0.003, 0.587]	—
leave_out_industry	Boxes	48	0.387	[0.074, 0.635]	—
leave_out_industry	BusSv	48	0.323	[-0.002, 0.590]	—
leave_out_industry	Chems	48	0.369	[0.057, 0.625]	—
leave_out_industry	Chips	48	0.322	[0.002, 0.588]	—
leave_out_industry	Clths	48	0.366	[0.053, 0.619]	—
leave_out_industry	Cnstr	48	0.348	[0.029, 0.614]	—
leave_out_industry	Coal	48	0.389	[0.079, 0.638]	—
leave_out_industry	Drugs	48	0.379	[0.064, 0.630]	—
leave_out_industry	ElcEq	48	0.366	[0.051, 0.620]	—
leave_out_industry	FabPr	48	0.357	[0.040, 0.613]	—
leave_out_industry	Fin	48	0.387	[0.077, 0.635]	—
leave_out_industry	Food	48	0.400	[0.081, 0.642]	—
leave_out_industry	Fun	48	0.379	[0.068, 0.632]	—
leave_out_industry	Gold	48	0.356	[0.042, 0.614]	—
leave_out_industry	Guns	48	0.323	[0.004, 0.596]	—
leave_out_industry	Hardw	48	0.322	[-0.006, 0.588]	—
leave_out_industry	Hlth	48	0.367	[0.043, 0.621]	—
leave_out_industry	Hshld	48	0.367	[0.051, 0.622]	—
leave_out_industry	Insur	48	0.337	[0.005, 0.605]	—
leave_out_industry	LabEq	48	0.325	[0.006, 0.593]	—
leave_out_industry	Mach	48	0.361	[0.045, 0.619]	—
leave_out_industry	Meals	48	0.385	[0.067, 0.635]	—
leave_out_industry	MedEq	48	0.362	[0.052, 0.622]	—
leave_out_industry	Mines	48	0.373	[0.058, 0.627]	—
leave_out_industry	Oil	48	0.370	[0.057, 0.624]	—
leave_out_industry	Other	48	0.384	[0.067, 0.631]	—
leave_out_industry	Paper	48	0.371	[0.055, 0.628]	—
leave_out_industry	PerSv	48	0.381	[0.068, 0.629]	—
leave_out_industry	RlEst	48	0.365	[0.050, 0.622]	—
leave_out_industry	Rtail	48	0.383	[0.064, 0.631]	—

part	dropped	n	ρ	95% CI	p (nominal)
leave_out_industry	Rubbr	48	0.377	[0.057, 0.626]	—
leave_out_industry	Ships	48	0.366	[0.046, 0.620]	—
leave_out_industry	Smoke	48	0.362	[0.042, 0.622]	—
leave_out_industry	Soda	48	0.356	[0.039, 0.616]	—
leave_out_industry	Softw	48	0.320	[0.001, 0.585]	—
leave_out_industry	Steel	48	0.352	[0.036, 0.610]	—
leave_out_industry	Telcm	48	0.322	[0.001, 0.587]	—
leave_out_industry	Toys	48	0.369	[0.055, 0.625]	—
leave_out_industry	Trans	48	0.387	[0.076, 0.635]	—
leave_out_industry	Txtls	48	0.353	[0.033, 0.613]	—
leave_out_industry	Util	48	0.340	[0.022, 0.599]	—
leave_out_industry	Whlsl	48	0.368	[0.054, 0.624]	—

Source: `docs/results/diagnostics/h6_influence.md`, committed unedited; the numbers are as printed there.

Table 48: H6 AI-washing gap by industry, highest first (exploratory, PAP §6 D18)

rank	industry	talk filers 2024	filers 2024	T	zT	z_usage	gap
1	Insur	99	141	0.702	1.939	0.198	2.416
2	Hardw	40	48	0.833	2.809	1.679	2.265
3	Food	30	60	0.500	0.599	-0.706	1.427
4	BusSv	242	384	0.630	1.463	0.606	1.417
5	Coal	5	11	0.455	0.298	-0.790	1.098
6	Trans	45	101	0.446	0.238	-0.691	0.931
7	Boxes	4	9	0.444	0.231	-0.701	0.930
8	Meals	38	89	0.427	0.115	-0.852	0.909
9	Telcm	58	91	0.637	1.510	1.270	0.892
10	Hlth	46	100	0.460	0.334	-0.494	0.885
11	Rubbr	8	18	0.444	0.231	-0.616	0.855
12	Other	9	21	0.429	0.126	-0.769	0.850
13	Ships	5	11	0.455	0.298	-0.488	0.831
14	Rtail	84	203	0.414	0.028	-0.808	0.754
15	Chips	108	184	0.587	1.176	0.958	0.722
16	Util	101	220	0.459	0.328	-0.081	0.510
17	Clths	15	36	0.417	0.047	-0.426	0.441
18	Books	8	12	0.667	1.704	2.154	0.369
19	ElcEq	29	71	0.408	-0.007	-0.353	0.303
20	Txtls	3	7	0.429	0.126	-0.123	0.277
21	Mach	50	122	0.410	0.002	-0.160	0.144

rank	industry	talk filers 2024	filers 2024	T	zT	z_usage	gap
22	Hshld	25	66	0.379	-0.204	-0.439	0.117
23	Paper	7	20	0.350	-0.395	-0.660	0.057
24	Autos	29	79	0.367	-0.282	-0.478	0.047
25	RlEst	37	102	0.363	-0.310	-0.509	0.037
26	Smoke	3	9	0.333	-0.505	-0.785	0.020
27	LabEq	40	84	0.476	0.442	0.672	-0.005
28	Cnstr	21	65	0.323	-0.573	-0.805	-0.053
29	BldMt	18	53	0.340	-0.464	-0.527	-0.153
30	MedEq	92	227	0.405	-0.028	0.162	-0.182
31	Banks	213	531	0.401	-0.056	0.224	-0.273
32	Guns	8	14	0.571	1.073	1.945	-0.290
33	Beer	8	26	0.308	-0.675	-0.660	-0.318
34	Chems	34	105	0.324	-0.569	-0.337	-0.461
35	Toys	9	27	0.333	-0.505	-0.169	-0.525
36	Softw	297	376	0.790	2.521	4.407	-0.537
37	Whlsl	34	115	0.296	-0.755	-0.458	-0.603
38	Soda	2	8	0.250	-1.058	-0.637	-0.849
39	Fun	26	79	0.329	-0.533	0.327	-1.002
40	Oil	43	174	0.247	-1.077	-0.437	-1.052
41	Aero	13	27	0.481	0.477	1.963	-1.102
42	Drugs	250	768	0.326	-0.557	0.449	-1.142
43	PerSv	19	68	0.279	-0.863	0.027	-1.177
44	Fin	200	654	0.306	-0.688	0.334	-1.215
45	FabPr	2	11	0.182	-1.510	-0.567	-1.514
46	Steel	6	35	0.171	-1.579	-0.621	-1.559
47	Gold	6	35	0.171	-1.579	-0.613	-1.566
48	Mines	11	64	0.172	-1.576	-0.357	-1.789
49	Agric	5	35	0.143	-1.768	-0.254	-2.137

Source: `docs/results/h6/gap.md`, committed unedited; the numbers are as printed there.

Table 49: H6 gap–return association with nominal p-values, in no testing family (exploratory, PAP §6 D18)

statistic	value
b (pp of annualised alpha per 1 SD of gap)	-2.398
s (pp)	1.963
$\bar{b}$ (pp)	0.113
q_0.975 (empirical centred quantile)	1.773
critical, max(t_21, q)	2.080

statistic	value
T, H0: b = $\bar{b}$ (D21, primary null)	-1.279
p, H0: b = $\bar{b}$ (nominal)	0.2147
T, H0: b = 0 (secondary null)	-1.222
p, H0: b = 0 (nominal)	0.2353
95% CI for b, b $\pm$ critical × s (pp)	[-6.480, 1.684]
industries	49

Source: `docs/results/h6/gap_returns.md`, committed unedited; the numbers are as printed there.

F.5 Limitations

Table 50: Width of the suppression bounds across occupations, Design B (PAP §2)

measure	n_occupations	p25	p50	p75
automation_share_all_ub - automation_share_all_lb	512	2.6529	3.9696	7.3088
D_ub - D_lb	512	0.1602	0.5428	1.1475

Source: `docs/results/rq2/bounds.md`, committed unedited; the numbers are as printed there.

F.6 Supplementary figures

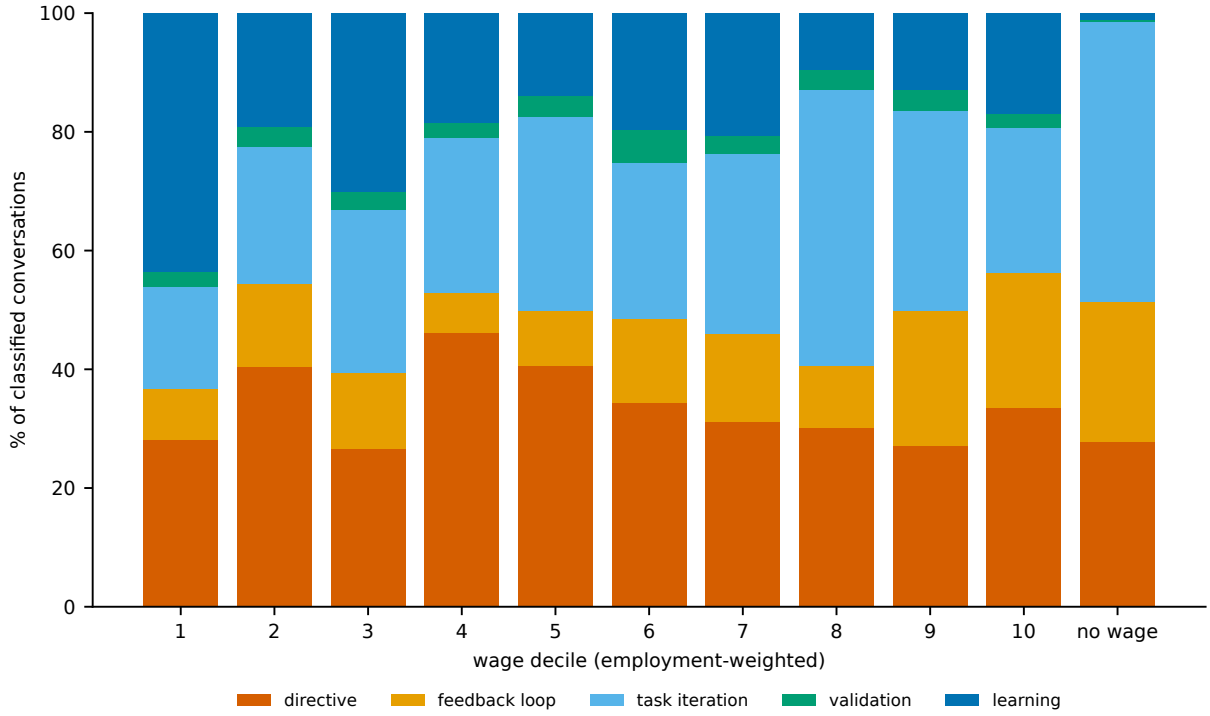


Figure 9: Collaboration patterns by wage decile, r6 claude_ai (descriptive).

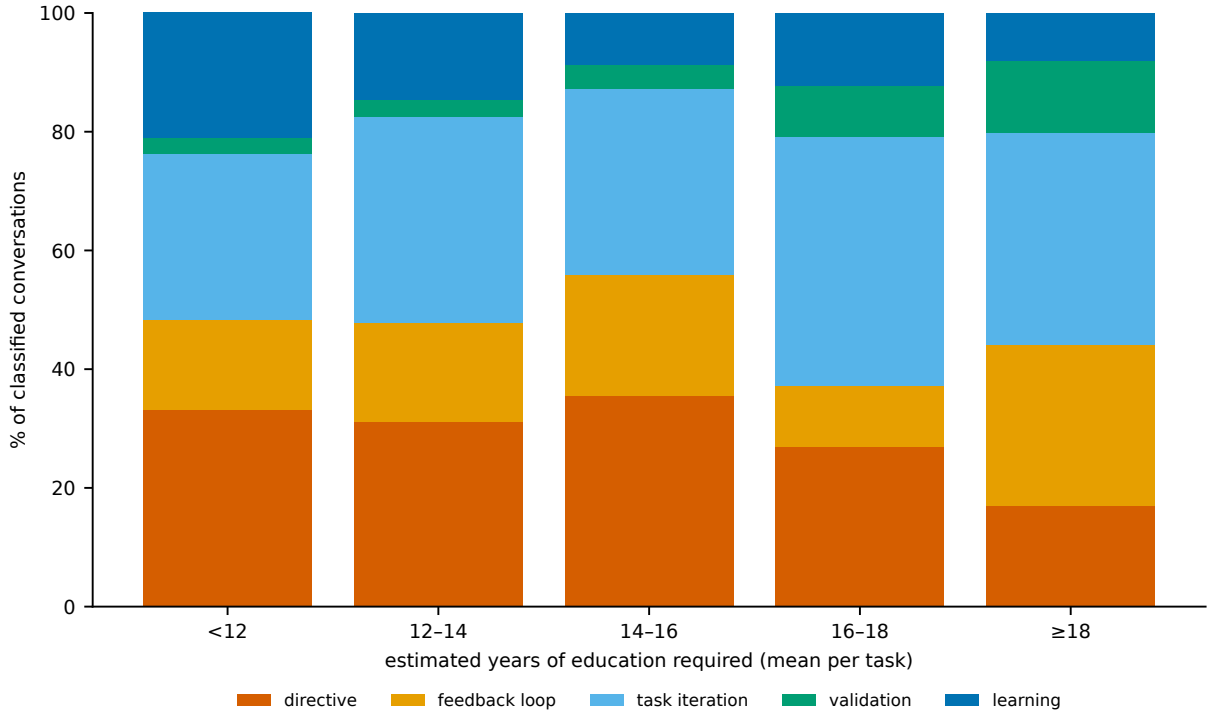


Figure 10: Collaboration patterns by required education, r6 claude_ai (descriptive).

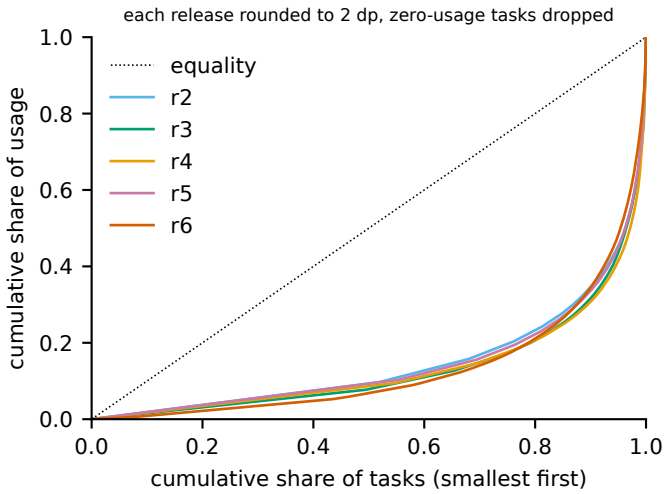


Figure 11: Concentration of usage across tasks by release: Lorenz curves (descriptive).

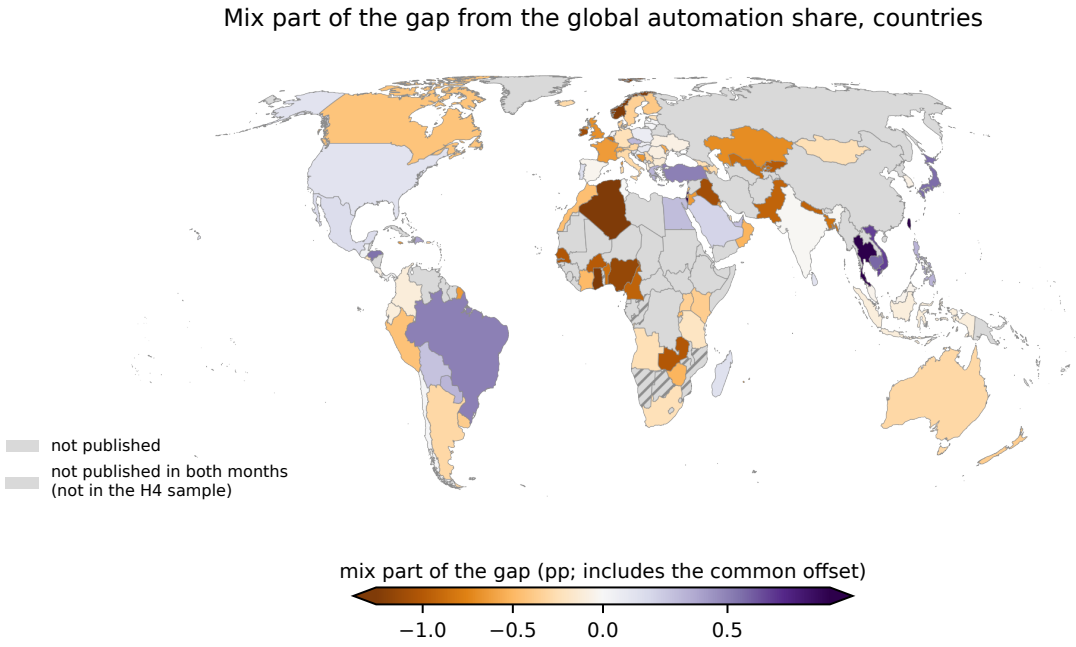


Figure 12: Mix part of the gap from the global automation share, by country (descriptive).

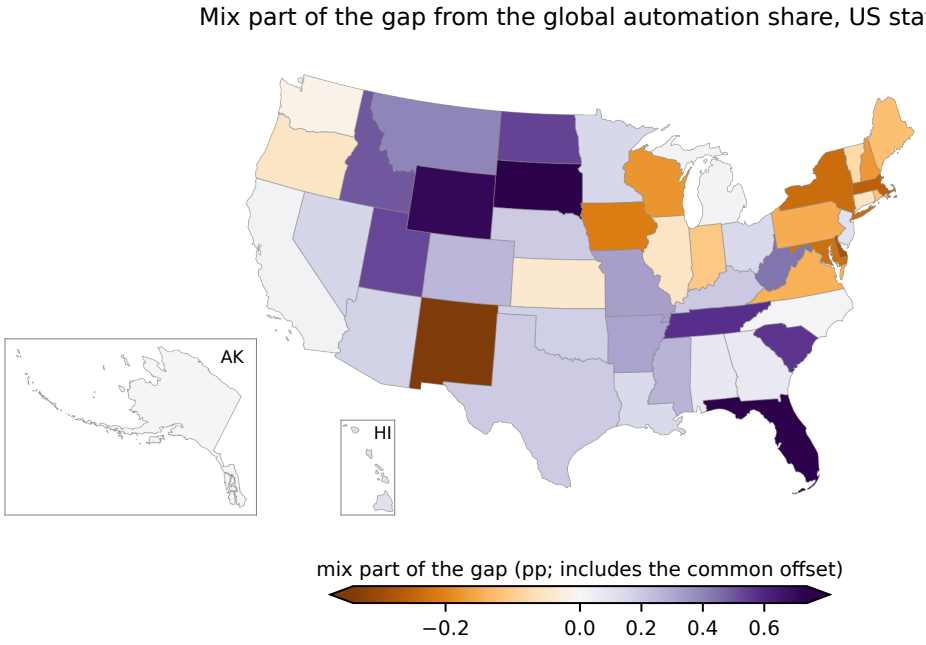


Figure 13: Mix part of the gap from the national automation share, by US state (descriptive).

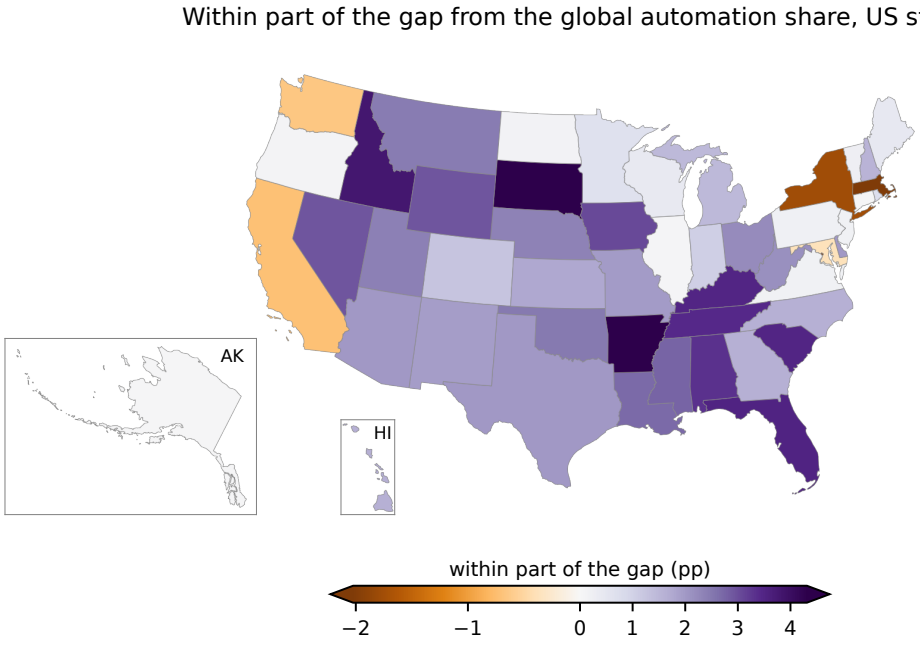


Figure 14: Within part of the gap from the national automation share, by US state (descriptive).

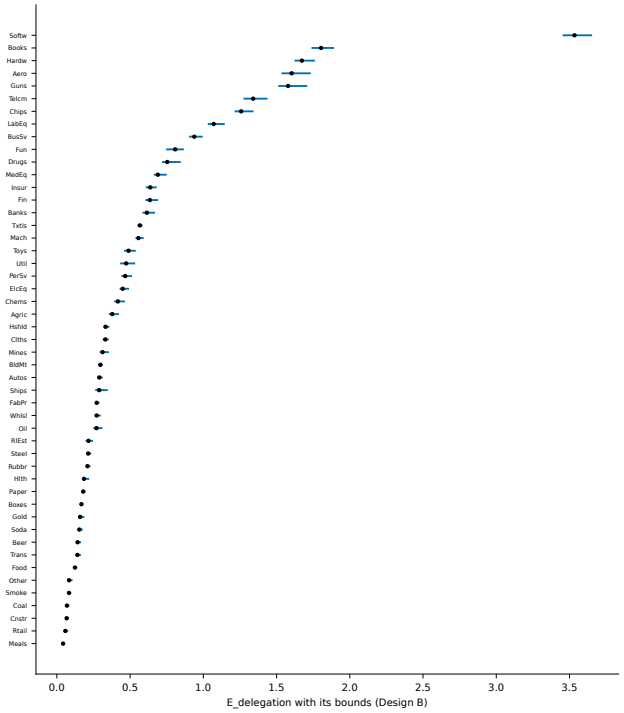


Figure 15: Delegation exposure with its suppression bounds by industry, Design B (PAP §2).

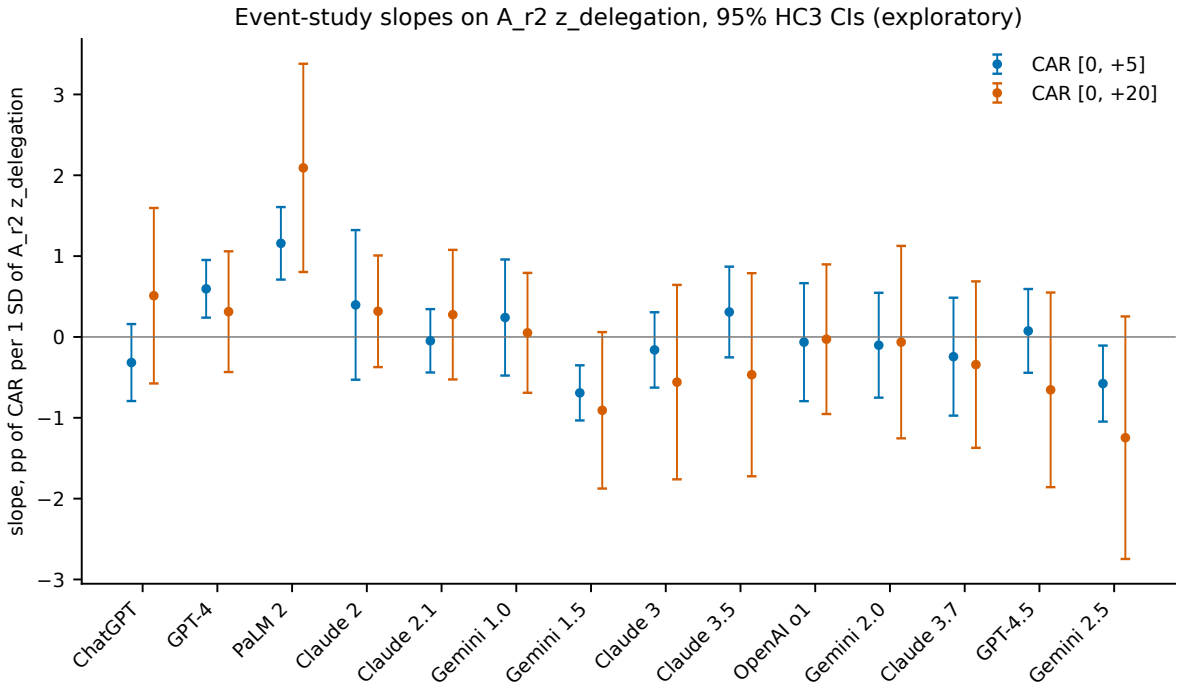


Figure 16: Event-study slopes on Design `A_r2` `z_delegation` with HC3 confidence intervals (exploratory, PAP §8.3 D19).

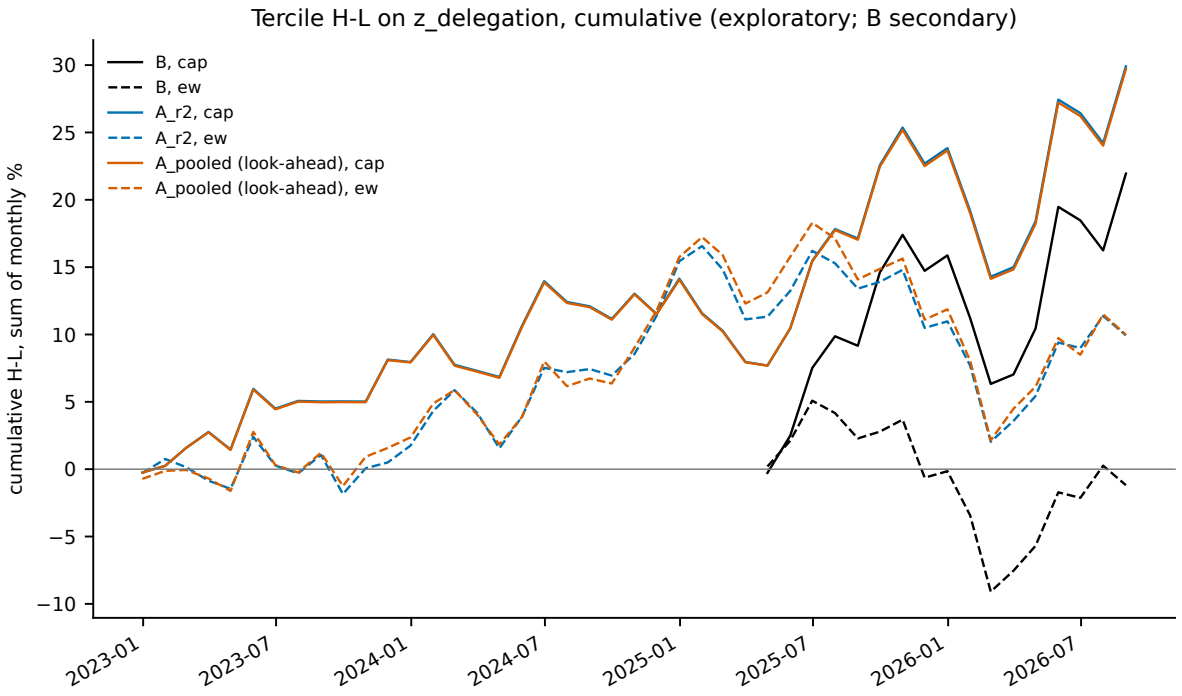


Figure 17: Cumulative return of the tercile high-minus-low portfolio on `z_delegation` (exploratory, PAP §8.3 D19; Design B secondary).

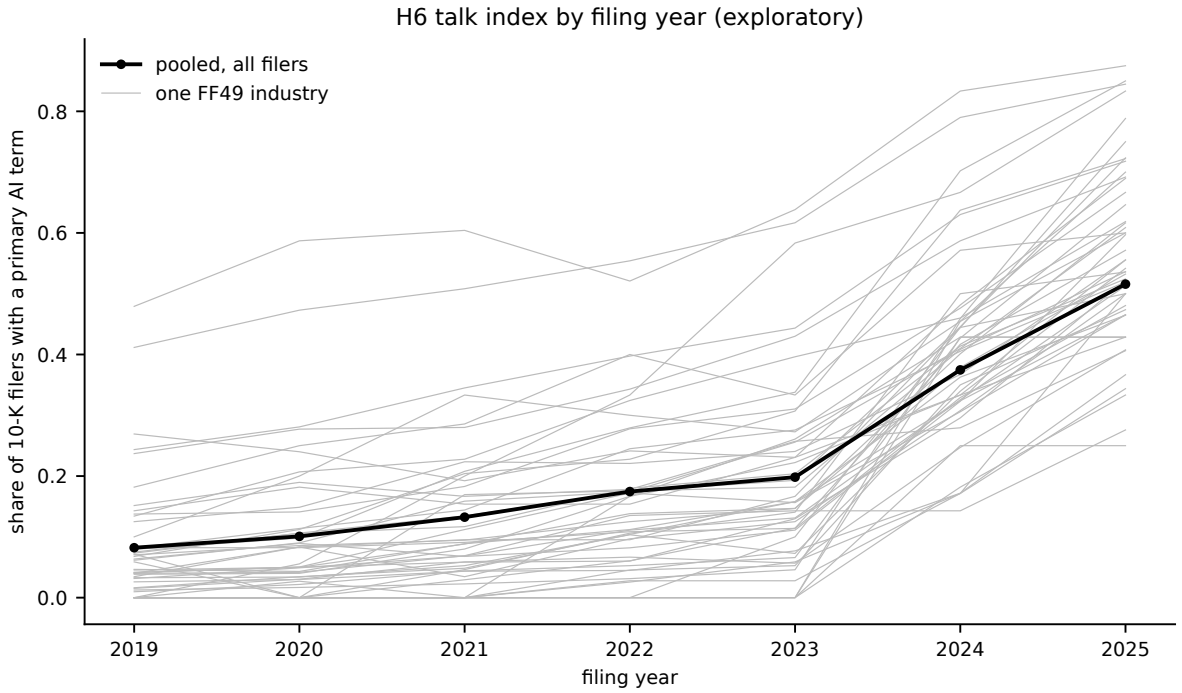


Figure 18: H6 talk index by filing year (exploratory, PAP §6 D18).